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Technical Report

3rd Generation Partnership Project; Technical Specification Group Radio Access Network; Solutions for NR to support non-terrestrial networks (NTN): Non-terrestrial networks (NTN) related RF and co-existence aspects (Release 19)



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3GPP

Postal address

3GPP support office address

650 Route des Lucioles - Sophia Antipolis
Valbonne - FRANCE
Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Internet

<http://www.3gpp.org>

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Foreword

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In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document covers the RF and co-existence aspects of the work item “Solutions for NR to support non-terrestrial networks (NTN)” [2]

The objectives for the study are the following:

- Study and specify adjacent channel co-existence scenarios of Non-terrestrial networks (NTN).
- Study and specify needed generic RF core requirements for the network and the UE such that adjacent channel co-existence scenarios are met.

NOTE: Satellite operation is used as complementarily to Terrestrial Network operation, and in separate location when satellite operates in an adjacent channel/band of terrestrial network. For instance, NTN satellite UE is assumed to connect and transmit to the satellite when it is outside a Terrestrial Network operating in an adjacent channel/band. The satellite beam might still overlap the terrestrial network.

2 Reference

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

- [1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".
- [2] 3GPP RP-213691: "Revised WID: Solutions for NR to support non-terrestrial networks (NTN)".
- [3] ITU-R Radio Regulations, 2020 Edition
- [4] ECC Decision 06(09): "Designation of the bands 1980-2010 MHz and 2170-2200 MHz for use by systems in the Mobile-Satellite Service including those supplemented by a Complementary Ground Component (CGC)", Approved 01 December 2006, Amended 05 September 2007,
- [5] 3GPP TR 38.811: "Study on New Radio (NR) to support non-terrestrial networks".
- [6] 3GPP TR 38.821: "Solutions for NR to support Non-Terrestrial Networks (NTN)".
- [7] 3GPP RP-152284: "Revised Work Item: Narrowband IoT".
- [8] 3GPP TR 36.942: "Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Frequency (RF) system scenarios".

- [9] 3GPP TR 45.820: "Cellular system support for ultra-low complexity and low throughput Internet of Things (CIoT)".
- [10] 3GPP TR 38.901: "Study on channel model for frequencies from 0.5 to 100 GHz".
- [11] Report ITU-R M.2292: "Characteristics of terrestrial IMT-Advanced systems for frequency sharing/interference analyses".
- [12] 3GPP RP-200559: "LS on Parameters of terrestrial component of IMT for sharing and compatibility studies in preparation for WRC-23 (below 5 GHz) ".
- [13] ITU-R Annex 4.4 to Document 5D/716-E.
- [14] 3GPP TR 38.921: "Study on International Mobile Telecommunications (IMT) parameters for 6.425 - 7.025 GHz, 7.025 - 7.125 GHz and 10.0 - 10.5 GHz".
- [15] 3GPP TR 36.802: "Evolved Universal Terrestrial Radio Access (E-UTRA); NB-IOT; Technical Report for BS and UE radio transmission and reception".
- [16] 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception".
- [17] 3GPP TS 38.101-1: "NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone".
- [18] 3GPP R4-2111460: "On the Rx Parameters and Rx Testing Setup for NTN gNB".
- [19] 3GPP R4-2108099: "WF on [312] NTN_Solutions_Part1".
- [20] FCC Online Table of Frequency Allocations, 47 C.F.R § 2.106, February 1, 2021.
- [21] FCC Order 03-15, Report and Order and Notice of Proposed Rule Making, January 29, 2003.
- [22] FCC Order 20-48, Order and Authorization, April 19, 2020.
- [23] FCC Code of Federal Regulations.
- [24] SoftBank, Loon LLC, Nokia, Ericsson, "Proposed deployment and system characteristics of HIBS in the working document towards a preliminary draft new Report ITU-R M.[HIBS-CHARACTERISTICS]," ITU WP-5D contribution, Sep. 28, 2020.
- [25] 3GPP TS 38.181: "NR; Satellite Node conformance testing".
- [26] 3GPP TS 38.108: "NR; Satellite Node radio transmission and reception".
- [27] 3GPP R4-213618: "Moderator's summary of discussion [94e-41-R17-NRNTN-RAN4Spec] ".
- [28] ERC Recommendation 74-01: "Unwanted emissions in the spurious domain".
- [29] Recommendation ITU-R SM.1541-6: "Unwanted emissions in the out-of-band domain".
- [30] Recommendation ITU-R SM.329: "Unwanted emissions in the spurious domain".
- [31] 3GPP TR 38.817-02: "General aspects for Base Station (BS) Radio Frequency (RF) for NR".
- [32] 3GPP TR 38.817-01: " General aspects for User Equipment (UE) Radio Frequency (RF) for NR".
- [33] FCC 16-89, REPORT AND ORDER AND FURTHER NOTICE OF PROPOSED RULEMAKING
- [34] FCC 20-159, REPORT AND ORDER
- [35] FCC 20-166, SECOND REPORT AND ORDER IN IB DOCKET NO. 17-95 AND REPORT AND ORDER IN IB DOCKET NO. 18-315 AND FURTHER NOTICE OF PROPOSED RULEMAKING
- [36] ECC Decision(05)01, The use of the band 27.5-29.5 GHz by the Fixed Service and uncoordinated Earth stations of the Fixed-Satellite Service (Earth-to-space)

- [37] ECC Decision(13)01, The harmonised use, free circulation and exemption from individual licensing of Earth Stations On Mobile Platforms (ESOMPs) within the frequency bands 17.3-20.2 GHz and 27.5-30.0 GHz

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms and definitions given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

Feeder link: Wireless link between NTN-Gateway and satellite

Geostationary Earth Orbit: Circular orbit at 35,786 km above the Earth's equator and following the direction of the Earth's rotation. An object in such an orbit has an orbital period equal to the Earth's rotational period and thus appears motionless, at a fixed position in the sky, to ground observers.

Geosynchronous Orbit: Earth-centered orbit at approximately 35786 kilometres above Earth's surface and synchronised with Earth's rotation. A geostationary orbit is a non-inclined geosynchronous orbit, i.e. in the Earth's equator plane.

Low Earth Orbit: Orbit around the Earth with an altitude between 300 km, and 1500 km.

Minimum Elevation angle: Minimum angle under which the satellite or HAPS can be seen by a UE.

Non-Geostationary Satellites: Satellites (LEO and MEO) orbiting around the Earth with a period that varies approximately between 1.5 hour and 10 hours. It is necessary to have a constellation of several Non-Geostationary satellites associated with handover mechanisms to ensure a service continuity.

Non-terrestrial networks: Networks, or segments of networks, using an airborne or space-borne vehicle to embark a transmission equipment relay node or base station.

NTN-Gateway: An earth station or gateway is located at the surface of Earth, and providing sufficient RF power and RF sensitivity for accessing to the satellite (resp. HAPS).

Satellite: A space-borne vehicle embarking a bent pipe payload or a regenerative payload telecommunication transmitter, placed into Low-Earth Orbit (LEO), Medium-Earth Orbit (MEO), or Geostationary Earth Orbit (GEO).

Service link: Radio link between satellite and UE

Transparent payload: Payload that changes the frequency carrier of the UL/DL RF signal, filters and amplifies it before transmitting it on the DL/UL, respectively.

UE transmission bandwidth configuration: Set of resource blocks located within the UE channel bandwidth which may be used for transmitting or receiving by the UE.

User Throughput: data rate provided to a terminal

3.2 Symbols

For the purposes of the present document, the following symbols apply:

ΔF_{Raster}	Band dependent channel raster granularity
ΔF_{Global}	Granularity of the global frequency raster
ΔF_{Raster}	Band dependent channel raster granularity
BW_{Channel}	Channel bandwidth
$BW_{\text{interferer}}$	Bandwidth of the interferer
$F_{\text{DL_low}}$	The lowest frequency of the downlink <i>operating band</i>
$F_{\text{DL_high}}$	The highest frequency of the downlink <i>operating band</i>

F_{UL_low}	The lowest frequency of the uplink <i>operating band</i>
F_{UL_high}	The highest frequency of the uplink <i>operating band</i>
$F_{Interferer}$	Frequency of the interferer
$F_{Interferer} \text{ (offset)}$	Frequency offset of the interferer (between the center frequency of the interferer and the carrier frequency of the carrier measured)
F_{offset}	Frequency offset of the interferer (between the center frequency of the interferer and the closest edge of the carrier measured)
F_{OOB}	The boundary between the NR out of band emission and spurious emission domains
F_{REF}	RF reference frequency
$F_{REF-Offs}$	Offset used for calculating F_{REF}
$F_{uw} \text{ (offset)}$	The frequency separation of the center frequency of the carrier closest to the interferer and the center frequency of the interferer
N_{RB}	Transmission bandwidth configuration, expressed in units of resource blocks N_{REF} NR Absolute Radio Frequency Channel Number (NR-ARFCN)
N_{REF}	NR Absolute Radio Frequency Channel Number (NR-ARFCN)
$N_{REF-Offs}$	Offset used for calculating N_{REF}
$P_{Interferer}$	Modulated mean power of the interferer
P_{uw}	Power of an unwanted DL signal

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

ACLR	Adjacent Channel Leakage Ratio
ACS	Adjacent Channel Selectivity
A-MPR	Additional Maximum Power Reduction
BS	Base Station
BW	Bandwidth
BWP	Bandwidth Part
CG	Carrier Group
CP-OFDM	Cyclic Prefix-OFDM
CW	Continuous Wave
DFT-s-OFDM	Discrete Fourier Transform-spread-OFDM
DM-RS	Demodulation Reference Signal
DTX	Discontinuous Transmission
EIRP	Equivalent Isotropically Radiated Power
EVM	Error Vector Magnitude
FR	Frequency Range
FRC	Fixed Reference Channel
FRF	Frequency Reuse Factor
FSS	Fixed Satellite Services
FWA	Fixed Wireless Access
GEO	Geostationary Earth Orbiting
gNB	next Generation Node B
GW	Gateway
GSCN	Global Synchronization Channel Number
HAPS	High Altitude Platform Station
HIPS	HAPS as IMT Base Stations
IBB	In-band Blocking
IDFT	Inverse Discrete Fourier Transformation
ISL	Inter-Satellite Links
ITU-R	Radiocommunication Sector of the International Telecommunication Union
LEO	Low Earth Orbiting
Mbps	Mega bit per second
MBW	Measurement bandwidth defined for the protected band
MCG	Master Cell Group
MEO	Medium Earth Orbiting
MOP	Maximum Output Power
MPR	Allowed maximum power reduction
MS	Mobile Services

MSD	Maximum Sensitivity Degradation
MSS	Mobile Satellite Services
NGEO	Non-Geostationary Earth Orbiting
NR	New Radio
NR-ARFCN	NR Absolute Radio Frequency Channel Number
NS	Network Signalling
NTN	Non-Terrestrial Network
OCNG	OFDMA Channel Noise Generator
OOB	Out-of-band
P-MPR	Power Management Maximum Power Reduction
PRB	Physical Resource Block
PSCCH	Physical Sidelink Control CHannel
PSSCH	Physical Sidelink Shared CHannel
QAM	Quadrature Amplitude Modulation
RAN	Radio Access Network
RE	Resource Element
REFSENS	Reference Sensitivity
RF	Radio Frequency
RMS	Root Mean Square (value)
RSRP	Reference Signal Receiving Power
Rx	Receiver
SAN	Satellite Access Node
SC	Single Carrier
SCG	Secondary Cell Group
SCS	Subcarrier spacing
SEM	Spectrum Emission Mask
SNR	Signal-to-Noise Ratio
SRS	Sounding Reference Symbol
SS	Synchronization Symbol
TAE	Time Alignment Error
TAG	Timing Advance Group
Tx	Transmitter
TxD	Tx Diversity
UE	User Equipment
ULFPTx	Uplink Full Power Transmission

4 General aspects

4.1 Work item objectives

The Work item objectives are captured in [2].

5 Regulatory aspects

5.1 ITU-R

The following services are among those defined in the ITU-R Radio Regulations [3]:

- Fixed (1.20): A radiocommunication service between specified fixed points.
- Fixed satellite (1.21) : A radiocommunication service between earth stations at given positions, when one or more satellites are used; the given position may be a specified fixed point or any fixed point within specified areas; in some cases this service includes satellite-to-satellite links, which may also be operated in the inter-satellite service; the fixed-satellite service may also include feeder links for other space radiocommunication services.
- Mobile (1.24): radiocommunication service between mobile and land stations, or between mobile stations (CV).

- Mobile satellite (1.25): A radiocommunication service:
 - between mobile earth stations and one or more space stations, or between space stations used by this service; or
 - between mobile earth stations by means of one or more space stations.

This service may also include feeder links necessary for its operation.

Based on the ITU-R Radio Regulations [3], the following frequency ranges are allocated to MSS and have been identified as first candidate bands for NTN satellite operations:

- S-band: UL: 1980–2010 MHz / DL: 2170–2200 MHz.
- L-band: UL: 1626.5–1660.5 MHz / DL: 1525–1559 MHz.

For providing mobile services through HAPS, current ITU-R Radio Regulations [3] allow the use of frequency ranges 1885–1980 MHz, 2010–2025 MHz and 2110–2170 MHz by HAPS. Additional spectrum may be allocated for HAPS in 2023 (see clause 5.3).

The following additional frequency ranges allocated to MSS based on the ITU-R Radio Regulations [3] have been identified as additional bands for NTN satellite operations:

- Extended L-band: UL: 1668–1675 MHz / DL: 1518–1525 MHz.

5.2 Regulatory aspects for NTN satellites

5.2.1 NTN Satellite band: UL 1980–2010 MHz / DL 2170–2200 MHz

The ITU-R Radio Regulations [3] specify the following service allocation for those frequency ranges, shown in Table 5.2.1-1:

Table 5.2.1-1: Allocation of 1980–2010 MHz and 2170–2200 MHz in the ITU-R Radio Regulations

Allocation to services					
Region 1		Region 2		Region 3	
2 170-2 200	FIXED MOBILE MOBILE-SATELLITE (space-to-Earth) 5.351A 5.388 5.389A 5.389F				
1 980-2 010	FIXED MOBILE MOBILE-SATELLITE (Earth-to-space) 5.351A 5.388 5.389A 5.389B 5.389F				
5.351A: For the use of the bands 1 518-1 544 MHz, 1 545-1 559 MHz, 1 610-1 645.5 MHz, 1 646.5-1 660.5 MHz, 1 668-1 675 MHz, 1 980-2 010 MHz, 2 170-2 200 MHz, 2 483.5-2 520 MHz and 2 670-2 690 MHz by the mobile satellite service, see Resolutions 212 (Rev.WRC-07)* and 225 (Rev.WRC-07)**. * This Resolution was revised by WRC-15 and WRC-19. ** This Resolution was revised by WRC-12.					
5.388: The frequency bands 1 885-2 025 MHz and 2 110-2 200 MHz are intended for use, on a worldwide basis, by administrations wishing to implement International Mobile Telecommunications (IMT). Such use does not preclude the use of these frequency bands by other services to which they are allocated. The frequency bands should be made available for IMT in accordance with Resolution 212 (Rev.WRC-15)* (see also Resolution 223 (Rev.WRC-15)*). (WRC-15) * This Resolution was revised by WRC-19.					
5.389A: The use of the bands 1 980-2 010 MHz and 2 170-2 200 MHz by the mobile-satellite service is subject to coordination under No. 9.11A and to the provisions of Resolution 716 (Rev.WRC-2000)**. ** This Resolution was revised by WRC-12.					
5.389B: The use of the frequency band 1 980-1 990 MHz by the mobile-satellite service shall not cause harmful interference to or constrain the development of the fixed and mobile services in Argentina, Brazil, Canada, Chile, Ecuador, the United States, Honduras, Jamaica, Mexico, Paraguay, Peru, Suriname, Trinidad and Tobago, Uruguay and Venezuela. (WRC-19)					
5.389F: In Algeria, Cape Verde, Egypt, Iran (Islamic Republic of), Mali, Syrian Arab Republic and Tunisia, the use of the frequency bands 1 980-2 010 MHz and 2 170-2 200 MHz by the mobile-satellite service shall neither cause harmful interference to the fixed and mobile services, nor hamper the development of those services prior to 1 January 2005, nor shall the former service request protection from the latter services. (WRC-19)					

Following Figure 5.2.1-1 gives an overview of the NR TN bands adjacent to the NTN n256 band.

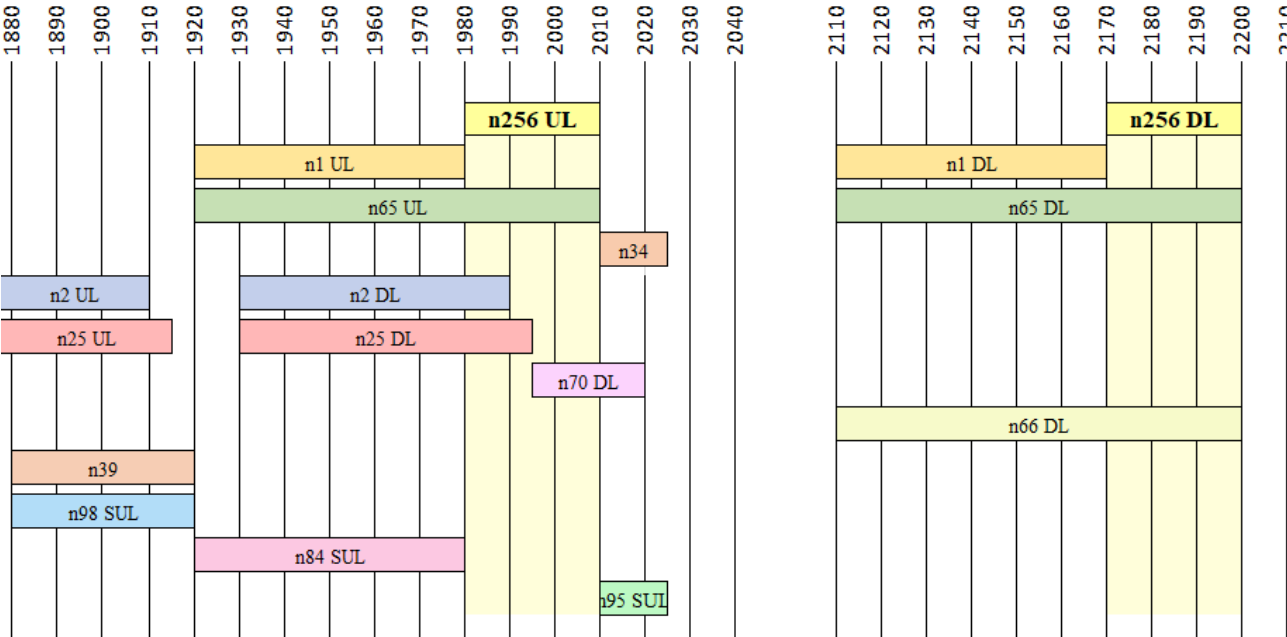


Figure 5.2.1-1: Adjacent TN bands to NTN band n256

As shown in Figure 5.2.1-1, the NTN satellite band is adjacent to NR bands n1 (FDD) and n34 (TDD). The bands need to be protected and co-existence analysis is required to determine the Adjacent Channel Interference Ratio (ACIR) and in-band power levels.

Also, the NTN band n256 will fully be overlapped by TN NR bands n65 and partly overlapped by TN NR bands n2, n25, n70 and n66, limiting band n256 deployment accordingly to countries where n2, n25 and/or n70 are not deployed or regional regulation applies.

Per ECC Decision 06(09) [4], n256 and Complementary Ground Component (CGC) could operate simultaneously in the upper 30 MHz portion of n65 in CEPT countries.

5.2.2 NTN Satellite band: UL 1626.5–1660.5 MHz / DL 1525–1559 MHz

The ITU-R Radio Regulations [3] specify the following service allocation for frequency ranges 1525 – 1559 MHz and 1626.5 – 1660.5 MHz as shown in Table 5.2.2-1.

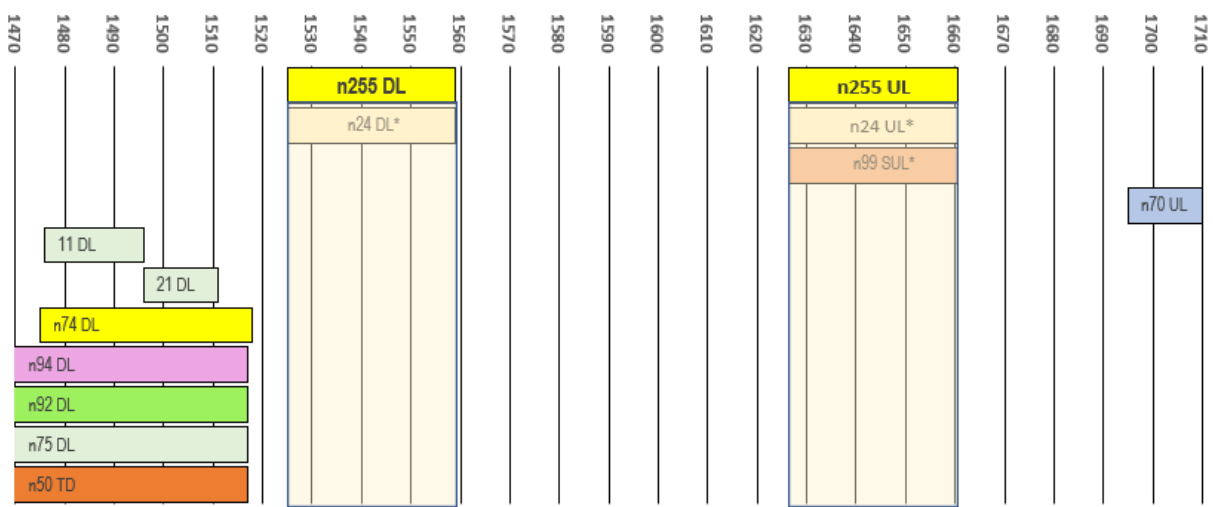
Table 5.2.2-1: Allocation of 1525 –1559 MHz and 1626.5 – 1660.5 MHz in the ITU-R Radio Regulations

Allocation to services		
Region 1	Region 2	Region 3
1 525-1 530 SPACE OPERATION (space-to-Earth) FIXED MOBILE-SATELLITE (space-to-Earth) 5.208B 5.351A Earth exploration-satellite Mobile except aeronautical mobile 5.349 5.341 5.342 5.350 5.351 5.352A 5.354	1 525-1 530 SPACE OPERATION (space-to-Earth) MOBILE-SATELLITE (space-to-Earth) 5.208B 5.351A Earth exploration-satellite Fixed Mobile 5.343 5.341 5.351 5.354	1 525-1 530 SPACE OPERATION (space-to-Earth) FIXED MOBILE-SATELLITE (space-to-Earth) 5.208B 5.351A Earth exploration-satellite Mobile 5.349 5.341 5.351 5.352A 5.354
1 530-1 535 SPACE OPERATION (space-to-Earth) MOBILE-SATELLITE (space-to-Earth) 5.208B 5.351A 5.353A Earth exploration-satellite Fixed Mobile except aeronautical mobile 5.341 5.342 5.351 5.354	1 530-1 535 SPACE OPERATION (space-to-Earth) MOBILE-SATELLITE (space-to-Earth) 5.208B 5.351A 5.353A Earth exploration-satellite Fixed Mobile 5.343 5.341 5.351 5.354	
1 535-1 559	MOBILE-SATELLITE (space-to-Earth) 5.208B 5.351A 5.341 5.351 5.353A 5.354 5.355 5.356 5.357 5.357A 5.359 5.362A	
1 626.5-1 660	MOBILE-SATELLITE (Earth-to-space) 5.351A 5.341 5.351 5.353A 5.354 5.355 5.357A 5.359 5.362A 5.374 5.375 5.376	
1 660-1 660.5	MOBILE-SATELLITE (Earth-to-space) 5.351A RADIO ASTRONOMY 5.149 5.341 5.354 5.362A 5.376A	
The details of the notes 5.208B, 5.341, 5.342, 5.350, 5.351, 5.351A, 5.352A, 5.353A, 5.354, 5.355, 5.356, 5.357, 5.357A, 5.359, 5.362A, 5.374, 5.375 and 5.376A related to MOBILE-SATELLITE service can be found in [3].		

FCC's specifies the following service allocation for frequency ranges 1525 – 1559 MHz and 1626.5 – 1660.5 MHz in USA[20] as shown in Table 5.2.2-2.

Table 5.2.2-2: Allocation of 1525 –1559 MHz and 1626.5 – 1660.5 MHz in the US

Allocation to services	
Federal Table	Non-Federal Table
1525-1535 MOBILE-SATELLITE (space-to-Earth) US315 US380 5.341 5.351	Satellite Communications (25) Maritime (80)
1535-1559 MOBILE-SATELLITE (space-to-Earth) US308 US309 US315 US380 5.341 5.351 5.356	Satellite Communications (25) Maritime (80) Aviation (87)
1626.5-1660 MOBILE-SATELLITE (Earth-to-space) US308 US309 US315 US380 5.341 5.351 5.375	Satellite Communications (25) Maritime (80) Aviation (87)
1660-1660.5 MOBILE-SATELLITE (Earth-to-space) US308 US309 US380 RADIO ASTRONOMY 5.341 5.351 US342	Satellite Communications (25) Aviation (87)
The details of the notes 5.341, 5.351, 5.356, 5.375, US308, US309, US315 and US342 related to MOBILE-SATELLITE service can be found in [20]. US380 In the bands 1525-1544 MHz, 1545-1559 MHz, 1610-1645.5 MHz, 1646.5-1660.5 MHz, and 2483.5-2500 MHz, a non-Federal licensee in the mobile-satellite service (MSS) may also operate an ancillary terrestrial component in conjunction with its MSS network, subject to the Commission's rules for ancillary terrestrial components and subject to all applicable conditions and provisions of its MSS authorization.	



NOTE: DL operation in band n24 is restricted to 1526-1536MHz and UL operation is restricted to 1627.5-1637.5MHz and 1646.5-1656.5MHz

Figure 5.2.2-1: Adjacent TN bands to NTN band n255

As shown in Figure 5.2.2-1, the NTN satellite band is not adjacent to any TN NR or LTE bands. The edge-to-edge separation for nearest TN band, n74 DL, is 7 MHz.

Also, the NTN band n255 fully overlaps with TN NR bands n24 and SUL NR band n99. Per note US380 in Table 5.2.2-2 simultaneous operation of n24/n99 and n255 is allowed in USA. Commission's authorization and technical rules for TN can be found in FCC Orders 03-15 [21] and 20-48 [22] respectively. Commission's technical rules for its MSS authorization can be found in FCC 47 C.F.R Part 25.

Technical rules for NTN SAN and UEs operating in 1 to 3 GHz from FCC's 47 C.F.R. § 25.202 [23] are summarized below:

47 C.F.R. § 25.202:

(d) *Frequency tolerance, Earth stations.* The carrier frequency of each earth station transmitter authorized in these services shall be maintained within 0.001 percent of the reference frequency.

(e) *Frequency tolerance, space stations.* The carrier frequency of each space station transmitter authorized in these services shall be maintained within 0.002 percent of the reference frequency.

47 C.F.R. § 25.202 (f):

Emission limitations. Except for SDARS terrestrial repeaters and as provided for in paragraph (i), the mean power of emissions shall be attenuated below the mean output power of the transmitter in accordance with the schedule set forth in paragraphs (f)(1) through (f)(4) of this section. The out-of-band emissions of SDARS terrestrial repeaters shall be attenuated in accordance with the schedule set forth in paragraph (h) of this section.

- (1) In any 4 kHz band, the center frequency of which is removed from the assigned frequency by more than 50 percent up to and including 100 percent of the authorized bandwidth: 25 dB;
- (2) In any 4 kHz band, the center frequency of which is removed from the assigned frequency by more than 100 percent up to and including 250 percent of the authorized bandwidth: 35 dB;
- (3) In any 4 kHz band, the center frequency of which is removed from the assigned frequency by more than 250 percent of the authorized bandwidth: An amount equal to 43 dB plus 10 times the logarithm (to the base 10) of the transmitter power in watts;
- (4) In any event, when an emission outside of the authorized bandwidth causes harmful interference, the Commission may, at its discretion, require greater attenuation than specified in paragraphs (f) (1), (2) and (3) of this section.

47 C.F.R. § 25.216:

- (a) The e.i.r.p. density of emissions from mobile earth stations placed in service on or before July 21, 2002 with assigned uplink frequencies between 1610 MHz and 1660.5 MHz shall not exceed -70 dBW/MHz, averaged over any 2 millisecond active transmission interval, in the band 1559-1587.42 MHz. The e.i.r.p. of discrete emissions of less than 700 Hz bandwidth generated by such stations shall not exceed -80 dBW, averaged over any 2 millisecond active transmission interval, in that band.
- (b) The e.i.r.p. density of emissions from mobile earth stations placed in service on or before July 21, 2002 with assigned uplink frequencies between 1610 MHz and 1626.5 MHz shall not exceed -64 dBW/MHz, averaged over any 2 millisecond active transmission interval, in the band 1587.42-1605 MHz. The e.i.r.p. of discrete emissions of less than 700 Hz bandwidth generated by such stations shall not exceed -74 dBW, averaged over any 2 millisecond active transmission interval, in the 1587.42-1605 MHz band.
- (c) The e.i.r.p. density of emissions from mobile earth stations placed in service after July 21, 2002 with assigned uplink frequencies between 1610 MHz and 1660.5 MHz shall not exceed -70 dBW/MHz, averaged over any 2 millisecond active transmission interval, in the band 1559-1605 MHz. The e.i.r.p. of discrete emissions of less than 700 Hz bandwidth from such stations shall not exceed -80 dBW, averaged over any 2 millisecond active transmission interval, in the 1559-1605 MHz band.
- (d) As of January 1, 2005, the e.i.r.p. density of emissions from mobile earth stations placed in service on or before July 21, 2002 with assigned uplink frequencies between 1610 MHz and 1660.5 MHz (except Standard A and B Inmarsat terminals used as Global Maritime Distress and Safety System ship earth stations) shall not exceed -70dBW/MHz, averaged over any 2 millisecond active transmission interval, in the 1559-1605 MHz band. The e.i.r.p. of discrete emissions of less than 700 Hz bandwidth from such stations shall not exceed -80 dBW, averaged over any 2 millisecond active transmission interval, in the 1559-1605 MHz band. Standard A Inmarsat terminals used as Global Maritime Distress and Safety System ship earth stations that do not meet the e.i.r.p. density limits specified in this paragraph may continue operation until December 31, 2007. Inmarsat-B terminals manufactured more than six months after Federal Register publication of the rule changes adopted in FCC 03-283 must meet these limits. Inmarsat B terminals manufactured before then are temporarily grandfathered under the condition that no interference is caused by these terminals to aeronautical satellite radio-navigation systems. The full-compliance deadline for grandfathered Inmarsat-B terminals is December 31, 2012.
- (e) The e.i.r.p density of emissions from mobile earth stations with assigned uplink frequencies between 1990 MHz and 2025 MHz shall not exceed -70 dBW/MHz, averaged over any 2 millisecond active transmission interval, in frequencies between 1559 MHz and 1610 MHz. The e.i.r.p. of discrete emissions of less than 700 Hz bandwidth from such stations between 1559 MHz and 1605 MHz shall not exceed -80 dBW, averaged over any 2

millisecond active transmission interval. The e.i.r.p. of discrete emissions of less than 700 Hz bandwidth from such stations between 1605 MHz and 1610 MHz manufactured more than six months after Federal Register publication of the rule changes adopted in FCC 03-283 shall not exceed -80 dBW, averaged over any 2 millisecond active transmission interval.

- (f) Mobile earth stations placed in service after July 21, 2002 with assigned uplink frequencies in the 1610-1660.5 MHz band shall suppress the power density of emissions in the 1605-1610 MHz band to an extent determined by linear interpolation from -70 dBW/MHz at 1605 MHz to -10 dBW/MHz at 1610 MHz.
- (g) Mobile earth stations manufactured more than six months after Federal Register publication of the rule changes adopted in FCC 03-283 with assigned uplink frequencies in the 1610-1626.5 MHz band shall suppress the power density of emissions in the 1605-1610 MHz band-segment to an extent determined by linear interpolation from -70 dBW/MHz at 1605 MHz to -10 dBW/MHz at 1610 MHz averaged over any 2 millisecond active transmission interval. The e.i.r.p. of discrete emissions of less than 700 Hz bandwidth from such stations shall not exceed a level determined by linear interpolation from -80 dBW at 1605 MHz to -20 dBW at 1610 MHz, averaged over any 2 millisecond active transmission interval.
- (h) Mobile earth stations manufactured more than six months after Federal Register publication of the rule changes adopted in FCC 03-283 with assigned uplink frequencies in the 1626.5-1660.5 MHz band shall suppress the power density of emissions in the 1605-1610 MHz band-segment to an extent determined by linear interpolation from -70 dBW/MHz at 1605 MHz to -46 dBW/MHz at 1610 MHz, averaged over any 2 millisecond active transmission interval. The e.i.r.p. of discrete emissions of less than 700 Hz bandwidth from such stations shall not exceed a level determined by linear interpolation from -80 dBW at 1605 MHz to -56 dBW at 1610 MHz, averaged over any 2 millisecond active transmission interval.
- (i) The e.i.r.p. density of carrier-off state emissions from mobile earth stations manufactured more than six months after Federal Register publication of the rule changes adopted in FCC 03-283 with assigned uplink frequencies between 1 and 3 GHz shall not exceed -80 dBW/MHz in the 1559-1610 MHz band averaged over any two millisecond interval.
- (j) A Root-Mean-Square detector shall be used for all power density measurements.

For L-band NTN satellite operation in the US, additional spectrum emission requirements in 47 C.F.R. § 25.216 (h) need to be specified for L-band NTN satellite UE. This can be captured as part of additional spurious emission requirements which allows inclusion of frequency ranges that are less than F_{OoB} (MHz) from the edge of the channel bandwidth.

5.2.3 NTN Satellite band: UL 27.5-30.0 GHz / DL 17.3-20.2 GHz

5.2.3.1 ITU-R

The ITU-R Radio Regulations [3] specify the following service allocation for those frequency ranges, shown in Table 5.2.3.1-1:

Table 5.2.3.1-1: Allocation of 17.3-20.2 GHz and 27.5-30.0 GHz in the ITU-R Radio Regulations

Allocation to services		
Region 1	Region 2	Region 3
17.3-17.7 FIXED-SATELLITE (Earth-to-space) 5.516 (space-to-Earth) 5.516A 5.516B Radiolocation 5.514	17.3-17.7 FIXED-SATELLITE (Earth-to-space) 5.516 BROADCASTING-SATELLITE Radiolocation 5.514 5.515	17.3-17.7 FIXED-SATELLITE (Earth-to-space) 5.516 Radiolocation 5.514
17.7-18.1 FIXED FIXED-SATELLITE (space-to-Earth) 5.484A 5.517A (Earth-to-space) 5.516 MOBILE	17.7-17.8 FIXED FIXED-SATELLITE (space-to-Earth) 5.517 5.517A (Earth-to-space) 5.516 BROADCASTING-SATELLITE Mobile 5.515	17.7-18.1 FIXED FIXED-SATELLITE (space-to-Earth) 5.484A 5.517A (Earth-to-space) 5.516 MOBILE
	17.8-18.1 FIXED FIXED-SATELLITE (space-to-Earth) 5.484A 5.517A (Earth-to-space) 5.516 MOBILE 5.519	
18.1-18.4 FIXED FIXED-SATELLITE (space-to-Earth) 5.484A 5.516B 5.517A (Earth-to-space) 5.520 MOBILE 5.519 5.521		
18.4-18.6 FIXED FIXED-SATELLITE (space-to-Earth) 5.484A 5.516B 5.517A MOBILE		
18.6-18.8 EARTH EXPLORATION-SATELLITE (passive) FIXED FIXED-SATELLITE (space-to-Earth) 5.517A 5.522B MOBILE except aeronautical mobile Space research (passive) 5.522A 5.522C	18.6-18.8 EARTH EXPLORATION- SATELLITE (passive) FIXED FIXED-SATELLITE (space-to-Earth) 5.516B 5.517A 5.522B MOBILE except aeronautical mobile SPACE RESEARCH (passive) 5.522A	18.6-18.8 EARTH EXPLORATION-SATELLITE (passive) FIXED FIXED-SATELLITE (space-to-Earth) 5.517A 5.522B MOBILE except aeronautical mobile Space research (passive) 5.522A
18.8-19.3 FIXED FIXED-SATELLITE (space-to-Earth) 5.516B 5.517A 5.523A MOBILE		
19.3-19.7 FIXED FIXED-SATELLITE (space-to-Earth) (Earth-to-space) 5.517A 5.523B 5.523C 5.523D 5.523E MOBILE		
19.7-20.1 FIXED-SATELLITE (space-to-Earth) 5.484A 5.484B 5.516B 5.527A Mobile-satellite (space-to-Earth)	19.7-20.1 FIXED-SATELLITE (space-to-Earth) 5.484A 5.484B 5.516B 5.527A MOBILE-SATELLITE (space-to-Earth)	19.7-20.1 FIXED-SATELLITE (space-to-Earth) 5.484A 5.484B 5.516B 5.527A Mobile-satellite (space-to-Earth)
5.524	5.524 5.525 5.526 5.527 5.528 5.529	5.524
20.1-20.2 FIXED-SATELLITE (space-to-Earth) 5.484A 5.484B 5.516B 5.527A MOBILE-SATELLITE (space-to-Earth) 5.524 5.525 5.526 5.527 5.528		
27.5-28.5 FIXED 5.537A FIXED-SATELLITE (Earth-to-space) 5.484A 5.516B 5.517A 5.539 MOBILE 5.538 5.540		

28.5-29.1 FIXED FIXED-SATELLITE (Earth-to-space) 5.484A 5.516B 5.517A 5.523A 5.539 MOBILE Earth exploration-satellite (Earth-to-space) 5.541 5.540		
29.1-29.5 FIXED FIXED-SATELLITE (Earth-to-space) 5.516B 5.517A 5.523C 5.523E 5.535A 5.539 5.541A MOBILE Earth exploration-satellite (Earth-to-space) 5.541 5.540		
29.5-29.9 FIXED-SATELLITE (Earth-to-space) 5.484A 5.484B 5.516B 5.527A 5.539 Earth exploration-satellite (Earth-to-space) 5.541 Mobile-satellite (Earth-to-space) 5.540 5.542	29.5-29.9 FIXED-SATELLITE (Earth-to-space) 5.484A 5.484B 5.516B 5.527A 5.539 MOBILE-SATELLITE (Earth-to-space) Earth exploration-satellite (Earth-to-space) 5.541 5.525 5.526 5.527 5.529 5.540	29.5-29.9 FIXED-SATELLITE (Earth-to-space) 5.484A 5.484B 5.516B 5.527A 5.539 Earth exploration-satellite (Earth-to-space) 5.541 Mobile-satellite (Earth-to-space) 5.540 5.542
29.9-30.0 FIXED-SATELLITE (Earth-to-space) 5.484A 5.484B 5.516B 5.527A 5.539 MOBILE-SATELLITE (Earth-to-space) Earth exploration-satellite (Earth-to-space) 5.541 5.543 5.526 5.527 5.538 5.540 5.542		

In addition:

- WRC-15 authorized ESIM operations in 19.7-20.2 GHz (space-to-Earth) and 29.5-30.0 GHz (Earth-to-space) under certain conditions, and for GSO systems only.
- WRC-19 authorized ESIM operations in 17.7-19.7 GHz (space-to-Earth) and 27.5-29.5 GHz (Earth-to-space) under certain conditions, and for GSO systems only (Resolution 173). *Editor's note: This TR would be updated with WRC-23 outcomes (Agenda Item 1.16 on NGSO ESIM Ka band).*

5.2.3.2 FCC

In 2016, FCC issued Report and Order FCC 16-89 [33] allocating the 27.50-28.35 GHz frequency range for terrestrial licensed mobile operations (IMT 5G). This frequency range corresponds to the 3GPP band n261.

In 2020, FCC issued Report and Order FCC 20-159 [34] which streamlines part 25 rules that govern the licensing of satellite space and earth stations in a satellite system through a unified license. The unified licensing framework specified frequency bands which includes:

- Space-to-Earth:
 - For GSO: 18.3-18.8 GHz and 19.7-20.2 GHz
 - For NGSO: 17.8-18.6 GHz, 18.8-19.4 GHz and 19.6-20.2 GHz
- Earth-to-space:
 - For GSO: 28.35-29.1 GHz and 29.25-30.0 GHz.
 - For NGSO: 28.35-29.1 GHz and 29.5-30.0 GHz.

In 2020 as well, FCC issued the Report and Order FCC-20-66 [35] which facilitates the deployment of, and reduce the regulatory burdens on, Earth Stations in Motion (ESIMs). It allows ESIMs to communicate in additional frequency bands with GSO satellites operating in FSS bands and adopts rules for ESIMs to communicate with NGSO satellites in specific FSS bands. Those FSS bands include:

- Space-to-Earth:
 - For GSO: 17.8-19.4 GHz, 19.6-20.2 GHz.

- For NGSO: 17.8-18.6 GHz, 18.8-19.4 GHz, 19.6-20.2 GHz
- Earth-to-space:
 - For GSO: 28.35-29.1 GHz and 29.25-30.0 GHz.
 - For NGSO: 28.4-29.1 GHz and 29.5-30.0 GHz.

5.2.3.3 CEPT

The ECC Decision (05)01 [36] addresses the use of band 27.5-29.5GHz, regulated for Fixed Service and Fixed Satellite Service:

- The bands 27.5-27.8285 GHz, 28.4445-28.8365 GHz and 29.4525-29.5 GHz are designated for the use of uncoordinated FSS earth stations.
- The band 28.8365-28.9485 GHz is designated for the use of uncoordinated FSS earth stations, considering, until 1 January 2020, the legacy FS systems licensed in this band in some countries before 18 March 2005.
- The bands 27.8285-28.4445 GHz and 28.9485-29.4525 GHz are designated for the use of FS systems.

Also, this ECC Decision(05)01 states that the CEPT administrations shall not authorize the deployment of uncoordinated FSS earth stations in the bands designed for the use of FS systems.

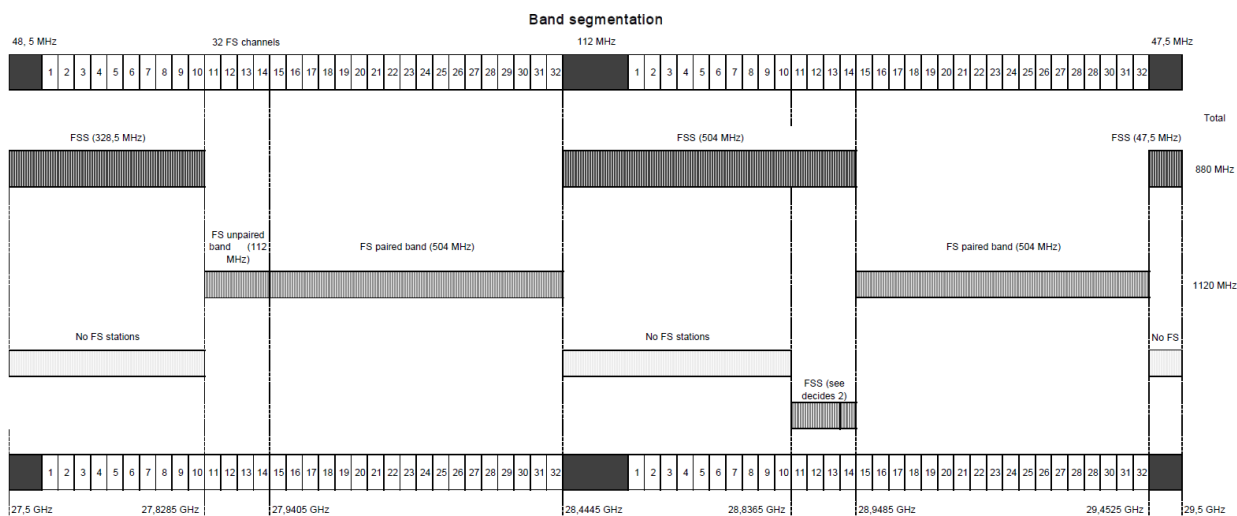


Figure 5.2.3.3-1: ECC Decision(05)01 - 27.5-29.5 band segmentation

The ECC Decision(13)01 [37] specifies the harmonized use, free circulation, and exemption from individual licensing of Earth stations on mobile platforms (ESOMPs) within the ranges 17.3-20.2 GHz and 27.5-30.0 GHz, in the portion identified for their use within the territory of operation, and under the conditions specified in Annexes 1,2 and 3 of this ECC Decision. ESOMPs are terminals with small directional antennas for the provision of broadband communication services. The terminals may be mounted on aircraft, ships or land vehicles or may be transportable devices used in motion or at temporary halts. This ECC Decision also specifies the conditions for those ESOMP devices to operate.

5.2.4 NTN Satellite band: UL 2000–2020 MHz / DL 2180–2200 MHz

Table 5.2.4-1 shows the service allocation for the frequency range in the ITU-R Radio Regulations [3] and Table 5.2.4-2 shows the FCC frequency allocation table for the frequency range [20]. From Table 5.2.4-1 and Table 5.2.4-2, there

is some deviation in how MSS S-band is allocated in region 2 versus in all other regions. Per ITU, the Region 2 MSS S-Band had been allocated to UL 1980-2025 MHz and DL 2160-2200 MHz. In the case of North America region (US, Canada, Mexico) the MSS band is allocated to UL 2000 MHz -2020 MHz and DL 2180 MHz -2200 MHz which is not exactly aligned to n256 definition, i.e., UL 1980 MHz – 2010 MHz, DL 2170 MHz – 2200 MHz. The 3GPP NTN S-band n252 (UL 2000 MHz -2020 MHz and DL 2180 MHz -2200) is defined for consistency across North America and Region 2 as a whole.

Table 5.2.4-1: Allocation of 2000-2020 MHz and 2180–2200 MHz in the ITU-R Radio Regulations

Allocation to services		
Region 1	Region 2	Region 3
1 980-2 010	FIXED MOBILE MOBILE-SATELLITE (Earth-to-space) 5.351A 5.388 5.389A 5.389B 5.389	
2 010-2 025 FIXED MOBILE 5.388A 5.388B 5.388	2 010-2 025 FIXED MOBILE-SATELLITE (Earth-to-space) 5.388 5.389C 5.389E	2 010-2 025 FIXED MOBILE 5.388A 5.388B 5.388
2 170-2 200	FIXED MOBILE MOBILE-SATELLITE (space-to-Earth) 5.351A 5.388 5.389A 5.389F	
5.388:	The frequency bands 1 885-2 025 MHz and 2 110-2 200 MHz are intended for use, on a worldwide basis, by administrations wishing to implement International Mobile Telecommunications (IMT). Such use does not preclude the use of these frequency bands by other services to which they are allocated. The frequency bands should be made available for IMT in accordance with Resolution 212 (Rev.WRC-15)* (see also Resolution 223 (Rev.WRC-15)*).	
5.389A:	The use of the bands 1 980-2 010 MHz and 2 170-2 200 MHz by the mobile-satellite service is subject to coordination under No. 9.11A and to the provisions of Resolution 716 (Rev.WRC-2000)**	
5.389B:	The use of the frequency band 1 980-1 990 MHz by the mobile-satellite service shall not cause harmful interference to or constrain the development of the fixed and mobile services in Argentina, Brazil, Canada, Chile, Ecuador, the United States, Honduras, Jamaica, Mexico, Paraguay, Peru, Suriname, Trinidad and Tobago, Uruguay and Venezuela.	
5.389F:	In Algeria, Cape Verde, Egypt, Iran (Islamic Republic of), Mali, Syrian Arab Republic and Tunisia, the use of the frequency bands 1 980-2 010 MHz and 2 170-2 200 MHz by the mobile-satellite service shall neither cause harmful interference to the fixed and mobile services, nor hamper the development of those services prior to 1 January 2005, nor shall the former service request protection from the latter services.	

Table 5.2.4-2: FCC frequency allocation for 2000-2020 MHz and 2180–2200 MHz

Table of Frequency Allocations						
International Table			United States Table		FCC Rule Part(s)	Canadian Table
Region 1 Table	Region 2 Table	Region 3 Table	Federal Table	Non-Federal Table		
1980-2010 FIXED MOBILE MOBILE-SATELLITE (Earth-to-space) 5.351A 5.388 5.389A 5.389B 5.389F						
2010-2025 FIXED MOBILE 5.388A 5.388B	2010-2025 FIXED MOBILE MOBILE-SATELLITE (Earth-to-space)	2010-2025 FIXED MOBILE 5.388A 5.388B		2000-2020 FIXED MOBILE MOBILE-SATELLITE (Earth-to-space)	Satellite Communications (25) Wireless Communications (27)	2000-2020 MOBILE MOBILE-SATELLITE (Earth-to-space) 5.351A 5.388 5.389A 5.389C 5.389E C36
5.388	5.388 5.389C 5.389E	5.388		2020-2025 FIXED MOBILE		2020-2025 FIXED MOBILE 5.388 C37

Table of Frequency Allocations						
International Table			United States Table		FCC Rule Part(s)	Canadian Table
Region 1 Table	Region 2 Table	Region 3 Table	Federal Table	Non-Federal Table		
2120-2170 FIXED MOBILE 5.388A 5.388B	2120-2160 FIXED MOBILE 5.388A 5.388B Mobile-satellite (space-to-Earth) 5.388 2160-2170 FIXED MOBILE MOBILE-SATELLITE (space-to-Earth) 5.388 5.389C 5.389E	2120-2170 FIXED MOBILE 5.388A 5.388B	2120-2200	2120-2180 FIXED MOBILE	Public Mobile (22) Wireless Communications (27) Fixed Microwave (101)	2120 - 2180 FIXED MOBILE 5.388A 5.388
5.388		5.388				
2180-2200 FIXED MOBILE MOBILE-SATELLITE (space-to-Earth) 5.351A 5.388 5.389A 5.389F				2180-2200 FIXED MOBILE MOBILE-SATELLITE (space-to-Earth)	Satellite Communications (25) Wireless Communications (27)	2180-2200 MOBILE MOBILE-SATELLITE (space-to-Earth) 5.351A 5.388 5.389A C36

The following are the applicable FCC regulatory requirements:

47 CFR § 25.202

According to paragraph (a)(4)(ii), the following frequencies are available for use by the 2 GHz Mobile-Satellite Service:

1. 2000-2020 MHz: User-to-Satellite Link;
2. 2180-2200 MHz: Satellite-to-User Link.

According to paragraph (f) on emission limitations, except for SDARS terrestrial repeaters and as provided for in paragraph (i), the mean power of emissions shall be attenuated below the mean output power of the transmitter in accordance with the schedule set forth in paragraphs (f)(1) through (f)(4) of this section:

- (1) In any 4 kHz band, the center frequency of which is removed from the assigned frequency by more than 50 percent up to and including 100 percent of the authorized bandwidth: 25 dB;
- (2) In any 4 kHz band, the center frequency of which is removed from the assigned frequency by more than 100 percent up to and including 250 percent of the authorized bandwidth: 35 dB;
- (3) In any 4 kHz band, the center frequency of which is removed from the assigned frequency by more than 250 percent of the authorized bandwidth: An amount equal to 43 dB plus 10 times the logarithm (to the base 10) of the transmitter power in watts;
- (4) In any event, when an emission outside of the authorized bandwidth causes harmful interference, the Commission may, at its discretion, require greater attenuation than specified in paragraphs (f) (1), (2) and (3) of this section.

47 CFR § 25.216

According to paragraph (e) on limits on emissions from mobile earth stations for protection of aeronautical radionavigation-satellite service, the e.i.r.p density of emissions from mobile earth stations with assigned uplink frequencies between 1990 MHz and 2025 MHz shall not exceed –70 dBW/MHz, averaged over any 2 millisecond active transmission interval, in frequencies between 1559 MHz and 1610 MHz. The e.i.r.p. of discrete emissions of less than 700 Hz bandwidth from such stations between 1559 MHz and 1605 MHz shall not exceed –80 dBW, averaged

over any 2 millisecond active transmission interval. The e.i.r.p. of discrete emissions of less than 700 Hz bandwidth from such stations between 1605 MHz and 1610 MHz manufactured more than six months after Federal Register publication of the rule changes adopted in FCC 03-283 shall not exceed -80 dBW, averaged over any 2 millisecond active transmission interval.

It should be noted that for the TN Band 23, which is studied and documented in [xx], the FCC regulation part 25.252 on ATC (Ancillary Terrestrial Component) for satellite service was referred to derive the RF requirements. However, since n252 is defined as an NTN band, ATC related requirements are not applicable for this band.

The following figure 5.2.4-1 shows an overview of the NR TN bands adjacent to the NTN n252 band.

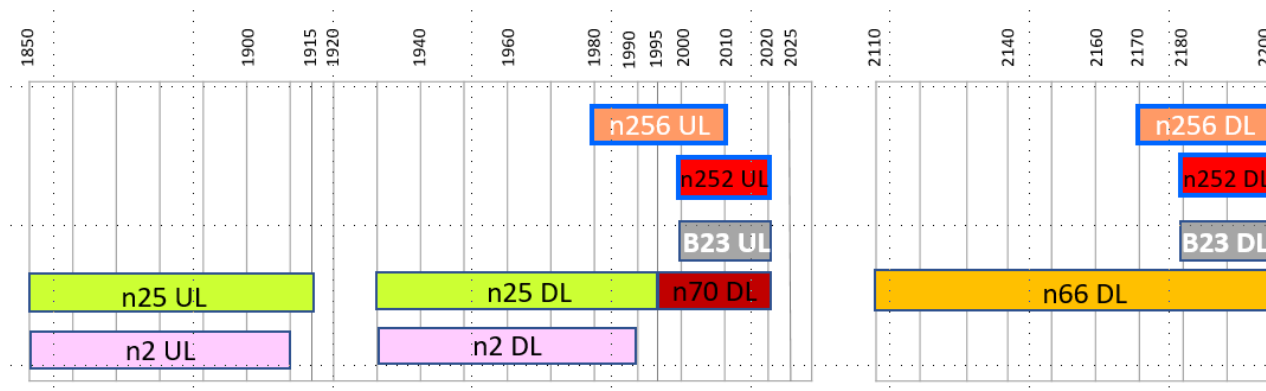


Figure 5.2.4-1: Adjacent TN bands to NTN band n252 in North America

As shown in figure 5.2.4-1, the n252 is adjacent to NR bands n25 DL and n2 DL, is overlap with n70 DL and n66 DL. Section 7.4.2.2.10.2 describes the RAN4 agreement on UE-to-UE coexistence requirement to protect the n25 DL, n2 DL, and n66 DL. The same section also describes the agreement for n70 DL.

5.2.5 NTN Satellite band: UL 1668-1675 MHz / DL 1518-1525 MHz)

The purpose of this section is to collect information from existing regulatory documents concerning regulatory rules applicable to this NTN Extended L-band.

NOTE1: As per agreement in RAN4#108, the following assumption is captured:

For the time being, re-use TN-NTN coexistence assumptions from TR 38.863, which would yield a total separation distance of 5250 m from TN BS site and NTN UE, at least for low-gain land-based NTN UE, compatible with smartphone and IoT use cases.

NOTE2: At RAN#108-bis Xiamen, an LS Reply was received from ETSI SES SCN (R4-2317932), with the following content:

“ETSI TC SES would like to thank 3GPP RAN4 for its LS on the 3GPP NTN requirements and the potential impact on the specification work in ETSI TC SES.

ETSI TC SES want to clarify that the requirements in the ETSI harmonised standards for earth stations are defined based on and referenced in CEPT regulatory framework or in accordance with ITU recommendations. Current TC-SES ETSI harmonised standards have not yet considered 3GPP NTN access technology and updates may be needed.

Therefore, it is proposed to create a new work item (NWI) in ETSI in order to accommodate NTN requirements in harmonised standards. There are already ongoing discussions between ETSI TC SES and ETSI TC MSG groups to find a way forward to capture the 3GPP NTN specifications in ETSI standards.

ETSI TC SES would remain at the disposal of 3GPP RAN4 for any additional information.”

NOTE3: ETSI requirements from EN 301 681 and ECC requirements from ECC Report 263 are captured for information, but have not presently been included in the specification.

As per the agreements in RAN4#109, the introduction of additional blocking requirements from ETSI EN 301 681 and recommendations from ECC Report 263 has been put on hold. Any ETSI- or ECC-specific requirements will be addressed after further feedback from ETSI is received.

5.2.5.1 ITU-R

The ITU-R Radio Regulations [3] specify the following service allocation for the frequency ranges 1518 – 1525 MHz and 1668 – 1675 MHz, shown in Table 5.2.5.1-1:

Table 5.2.5.1-1: Allocation of 1518-1525 MHz and 1668-1675 MHz in the ITU-R Radio Regulations

Allocation to services		
Region 1	Region 2	Region 3
1 518-1 525 FIXED MOBILE except aeronautical mobile MOBILE-SATELLITE (space-to-Earth) 5.348 5.348A 5.348B 5.351A	1 518-1 525 FIXED MOBILE 5.343 MOBILE-SATELLITE (space-to-Earth) 5.348 5.348A 5.348B 5.351A	1 518-1 525 FIXED MOBILE MOBILE-SATELLITE (space-to-Earth) 5.348 5.348A 5.348B 5.351A
5.341 5.342	5.341 5.344	5.341
1 668-1 668.4	MOBILE-SATELLITE (Earth-to-space) 5.351A 5.379B 5.379C RADIO ASTRONOMY SPACE RESEARCH (passive) Fixed Mobile except aeronautical mobile 5.149 5.341 5.379 5.379A	
1 668.4-1 670	METEOROLOGICAL AIDS FIXED MOBILE except aeronautical mobile MOBILE-SATELLITE (Earth-to-space) 5.351A 5.379B 5.379C RADIO ASTRONOMY 5.149 5.341 5.379D 5.379E	
1 670-1 675	METEOROLOGICAL AIDS FIXED METEOROLOGICAL-SATELLITE (space-to-Earth) MOBILE MOBILE-SATELLITE (Earth-to-space) 5.351A 5.379B 5.341 5.379D 5.379E 5.380A	

5.149 In making assignments to stations of other services to which the bands:

13 360-13 410 kHz,	4 950-4 990 MHz,	102-109.5 GHz,
25 550-25 670 kHz,	4 990-5 000 MHz,	111.8-114.25 GHz,
37.5-38.25 MHz,	6 650-6 675.2 MHz,	128.33-128.59 GHz,
73-74.6 MHz in Regions 1 and 3,	10.6-10.68 GHz,	129.23-129.49 GHz,
150.05-153 MHz in Region 1,	14.47-14.5 GHz,	130-134 GHz,
322-328.6 MHz,	22.01-22.21 GHz,	136-148.5 GHz,
406.1-410 MHz,	22.21-22.5 GHz,	151.5-158.5 GHz,
608-614 MHz in Regions 1 and 3,	22.81-22.86 GHz,	168.59-168.93 GHz,
1 330-1 400 MHz,	23.07-23.12 GHz,	171.11-171.45 GHz,
1 610.6-1 613.8 MHz,	31.2-31.3 GHz,	172.31-172.65 GHz,
1 660-1 670 MHz,	31.5-31.8 GHz in Regions 1 and 3,	173.52-173.85 GHz,
1 718.8-1 722.2 MHz,	36.43-36.5 GHz,	195.75-196.15 GHz,
2 655-2 690 MHz,	42.5-43.5 GHz,	209-226 GHz,
3 260-3 267 MHz,	48.94-49.04 GHz,	241-250 GHz,
3 332-3 339 MHz,	76-86 GHz,	252-275 GHz
3 345.8-3 352.5 MHz,	92-94 GHz,	
4 825-4 835 MHz,	94.1-100 GHz,	

are allocated, administrations are urged to take all practicable steps to protect the radio astronomy service from harmful interference. Emissions from spaceborne or airborne stations can be particularly serious sources of interference to the radio astronomy service (see Nos. **4.5** and **4.6** and Article **29**). (WRC-07)

5.341 In the bands 1 400-1 727 MHz, 101-120 GHz and 197-220 GHz, passive research is being conducted by some countries in a programme for the search for intentional emissions of extraterrestrial origin.

5.342 *Additional allocation:* in Armenia, Azerbaijan, Belarus, the Russian Federation, Uzbekistan, Kyrgyzstan and Ukraine, the frequency band 1 429-1 535 MHz is also allocated to the aeronautical mobile service on a primary basis, exclusively for the purposes of aeronautical telemetry within the national territory. As of 1 April 2007, the use of the frequency band 1 452-1 492 MHz is subject to agreement between the administrations concerned. (WRC-15)

5.348 The use of the band 1 518-1 525 MHz by the mobile-satellite service is subject to coordination under No. 9.11A. In the band 1 518-1 525 MHz stations in the mobile-satellite service shall not claim protection from the stations in the fixed service. No. **5.43A** does not apply. (WRC-03)

5.348A In the band 1 518-1 525 MHz, the coordination threshold in terms of the power flux-density levels at the surface of the Earth in application of No. 9.11A for space stations in the mobile-satellite (space-to-Earth) service, with respect to the land mobile service use for specialized mobile radios or used in conjunction with public switched telecommunication networks (PSTN) operating within the territory of Japan, shall be – 150 dB(W/m²) in any 4 kHz band for all angles of arrival, instead of those given in Table 5-2 of Appendix 5. In the band 1 518-1 525 MHz stations in the mobile-satellite service shall not claim protection from stations in the mobile service in the territory of Japan. No. **5.43A** does not apply. (WRC-03)

5.348B In the band 1 518-1 525 MHz, stations in the mobile-satellite service shall not claim protection from aeronautical mobile telemetry stations in the mobile service in the territory of the United States (see Nos. **5.343** and **5.344**) and in the countries listed in No. **5.342**. No. **5.43A** does not apply. (WRC-03)

5.351A For the use of the frequency bands 1 518-1 544 MHz, 1 545-1 559 MHz, 1 610-1 645.5 MHz, 1 646.5-1 660.5 MHz, 1 668-1 675 MHz, 1 980-2 010 MHz, 2 170-2 200 MHz, 2 483.5-2 520 MHz and 2 670-2 690 MHz by the mobile-satellite service, see Resolutions **212 (Rev.WRC-23)** and **225 (Rev.WRC-23)**. (WRC-23)

5.379 *Additional allocation:* in Bangladesh, India, Indonesia, Nigeria and Pakistan, the band 1 660.5-1 668.4 MHz is also allocated to the meteorological aids service on a secondary basis.

5.379A Administrations are urged to give all practicable protection in the band 1 660.5-1 668.4 MHz for future research in radio astronomy, particularly by eliminating air-to-ground transmissions in the meteorological aids service in the band 1 664.4-1 668.4 MHz as soon as practicable.

5.379B The use of the frequency band 1 668-1 675 MHz by the mobile-satellite service is subject to coordination under No. **9.11A**. (WRC-23)

5.379C In order to protect the radio astronomy service in the band 1 668-1 670 MHz, the aggregate power flux-density values produced by mobile earth stations in a network of the mobile-satellite service operating in this band shall not exceed -181 dB(W/m²) in 10 MHz and -194 dB(W/m²) in any 20 kHz at any radio astronomy station recorded in the Master International Frequency Register, for more than 2% of integration periods of 2 000 s. (WRC-03)

5.379D For sharing of the frequency band 1 668.4-1 675 MHz between the mobile-satellite service and the fixed and mobile services, Resolution **744 (Rev.WRC-23)** shall apply. (WRC-23)

5.379E In the band 1 668.4-1 675 MHz, stations in the mobile-satellite service shall not cause harmful interference to stations in the meteorological aids service in China, Iran (Islamic Republic of), Japan and Uzbekistan. In the band 1 668.4-1 675 MHz, administrations are urged not to implement new systems in the meteorological aids service and are encouraged to migrate existing meteorological aids service operations to other bands as soon as practicable. (WRC-03)

5.380A In the band 1 670-1 675 MHz, stations in the mobile-satellite service shall not cause harmful interference to, nor constrain the development of, existing earth stations in the meteorological-satellite service notified before 1 January 2004. Any new assignment to these earth stations in this band shall also be protected from harmful interference from stations in the mobile-satellite service. (WRC-07)

5.2.5.2 FCC

The FCC Rules [23] specify the following service allocations and technical rules for the frequency ranges 1518 – 1525 MHz and 1668 – 1675 MHz.

Table of Frequency Allocations			1626.5-2110 MHz (UHF)		Page 35
International Table			United States Table		FCC Rule Part(s)
Region 1 Table	Region 2 Table	Region 3 Table	Federal Table	Non-Federal Table	
1668-1668.4			1660.5-1668.4		
MOBILE-SATELLITE (Earth-to-space) 5.351A 5.379B 5.379C			RADIO ASTRONOMY US74		
RADIO ASTRONOMY			SPACE RESEARCH (passive)		
SPACE RESEARCH (passive)					
Fixed					
Mobile except aeronautical mobile					
5.149 5.341 5.379 5.379A					
1668.4-1670			5.341 US246		

METEOROLOGICAL AIDS		
FIXED		
MOBILE except aeronautical mobile		
MOBILE-SATELLITE (Earth-to-space) 5.351A 5.379B 5.379C		
RADIO ASTRONOMY		
5.149 5.341 5.379D 5.379E		
1670-1675	1668.4-1670	
METEOROLOGICAL AIDS	METEOROLOGICAL AIDS (radiosonde)	
FIXED	RADIO ASTRONOMY US74	
METEOROLOGICAL-SATELLITE (space-to-Earth)		
MOBILE		
MOBILE-SATELLITE (Earth-to-space) 5.351A 5.379B		
5.341 5.379D 5.379E 5.380A	5.341 US99 US342	

47 CFR 2.106(b)(341)

5.341 In the bands 1400-1727 MHz, 101-120 GHz and 197-220 GHz, passive research is being conducted by some countries in a programme for the search for intentional emissions of extraterrestrial origin.

47 CFR 2.106(b) (348)

5.348 The use of the band 1518-1525 MHz by the mobile-satellite service is subject to coordination under No. 9.11A. In the band 1518-1525 MHz stations in the mobile-satellite service shall not claim protection from the stations in the fixed service. No. 5.43A does not apply.

(i) 5.348A In the band 1518-1525 MHz, the coordination threshold in terms of the power flux-density levels at the surface of the Earth in application of No. 9.11A for space stations in the mobile-satellite (space-to-Earth) service, with respect to the land mobile service use for specialized mobile radios or used in conjunction with public switched telecommunication networks (PSTN) operating within the territory of Japan, shall be -150 dB(W/m²) in any 4 kHz band for all angles of arrival, instead of those given in Table 5-2 of Appendix 5. In the band 1518-1525 MHz stations in the mobile-satellite service shall not claim protection from stations in the mobile service in the territory of Japan. No. 5.43A does not apply.

(ii) 5.348B In the band 1518-1525 MHz, stations in the mobile-satellite service shall not claim protection from aeronautical mobile telemetry stations in the mobile service in the territory of the United States (see paragraphs (b)(343) and (344) of this section) and in the countries listed in paragraph (b)(342) of this section. No. 5.43A does not apply.

47 CFR 2.106(b)(351)

(i) 5.351A For the use of the bands 1518-1544 MHz, 1545-1559 MHz, 1610-1645.5 MHz, 1646.5-1660.5 MHz, 1668-1675 MHz, 1980-2010 MHz, 2170-2200 MHz, 2483.5-2520 MHz and 2670-2690 MHz by the mobile-satellite service, see Resolutions 212 (Rev.WRC-19) and 225 (Rev.WRC-12).

47 CFR 2.106(b)(379)

5.379 Additional allocation: in Bangladesh, India, Indonesia, Nigeria and Pakistan, the band 1660.5-1668.4 MHz is also allocated to the meteorological aids service on a secondary basis.

(i) 5.379A Administrations are urged to give all practicable protection in the band 1660.5-1668.4 MHz for future research in radio astronomy, particularly by eliminating air-to-ground transmissions in the meteorological aids service in the band 1664.4-1668.4 MHz as soon as practicable.

(ii) 5.379B The use of the band 1668-1675 MHz by the mobile-satellite service is subject to coordination under No. 9.11A. In the band 1668-1668.4 MHz, Resolution 904 (WRC-07) shall apply.

(iii) 5.379C In order to protect the radio astronomy service in the band 1668-1670 MHz, the aggregate power flux-density values produced by mobile earth stations in a network of the mobile-satellite service operating in this band shall not exceed -181 dB(W/m²) in 10 MHz and -194 dB(W/m²) in any 20 kHz at any radio astronomy station recorded in the Master International Frequency Register, for more than 2% of integration periods of 2000 s.

(iv) 5.379D For sharing of the band 1668.4-1675 MHz between the mobile-satellite service and the fixed and mobile services, Resolution 744 (Rev.WRC-07) shall apply.

(v) 5.379E In the band 1668.4-1675 MHz, stations in the mobile-satellite service shall not cause harmful interference to stations in the meteorological aids service in China, Iran (Islamic Republic of), Japan and Uzbekistan. In the band 1668.4-1675 MHz, administrations are urged not to implement new systems in the meteorological aids service and are encouraged to migrate existing meteorological aids service operations to other bands as soon as practicable.

47 CFR 2.106(b)(380)

5.380A In the band 1670-1675 MHz, stations in the mobile-satellite service shall not cause harmful interference to, nor constrain the development of, existing earth stations in the meteorological-satellite service notified before 1 January 2004. Any new assignment to these earth stations in this band shall also be protected from harmful interference from stations in the mobile-satellite service.

US211: In the bands 1670-1690, 5000-5250 MHz and 10.7-11.7, 15.1365-15.35, 15.4-15.7, 22.5-22.55, 24-24.05, 31.0-31.3, 31.8-32.0, 40.5-42.5, 116-122.25, 123-130, 158.5-164, 167-168, 191.8-200, and 252-265 GHz, applicants for airborne or space station assignments are urged to take all practicable steps to protect radio astronomy observations in the adjacent bands from harmful interference; however, US74 applies.

US362: The band 1670-1675 MHz is allocated to the meteorological-satellite service (space-to-Earth) on a primary basis for Federal use. Earth station use of this allocation is limited to Wallops Island, VA (37° 56' 44" N, 75° 27' 37" W), Fairbanks, AK (64° 58' 22" N, 147° 30' 04" W), and Greenbelt, MD (39° 00' 02" N, 76° 50' 29" W). Applicants for non-Federal stations within 100 kilometers of the Wallops Island or Fairbanks coordinates and within 65 kilometers of the Greenbelt coordinates shall notify NOAA in accordance with the procedures specified in 47 CFR 1.924.

5.2.5.3 CEPT/ECC

5.2.5.3.1 CEPT

This section is reserved.

5.2.5.3.2 ECC Report 263

NOTE: The coexistence assumptions used in ECC Report 263 are different compared to the assumptions used by 3GPP in the TN-NTN coexistence analysis captured in TR 38.863.

The following section contains some excerpts from ECC Report 263.

The report has established the technical characteristics of the IMT and the MSS system and determined the relevant scenarios. It has also determined the appropriate propagation models for these scenarios according to the environment where the equipment is used and further the protection criteria has been established.

The report has, from these characteristics and parameters, developed an MCL analysis with the resulting required separation distances for the 3 different frequency separations (1 MHz, 3 MHz and 6 MHz frequency separations).

The frequency separation allocation is not addressed by this report.

Further the report makes use of the MCL to establish interference arising in the first MES channel above the 3 different frequency separations investigated for an area with IMT coverage.

The results of the simulations show that there will be some interference irrespective of the selected frequency separation.

MES terminals currently on the market which have characteristics similar to those selected for this study, may experience interference problems because of the susceptibility of the MES receiver to the wanted signals from the IMT systems. As there are currently no available technical characteristics which outline at what frequency this effect starts to occur, blocking may also be experienced from IMT transmitters more than 6 MHz away into the IMT band below 1518 MHz.

Based on the findings of this report, the following mitigation techniques could further improve the compatibility between IMT and MSS around 1518 MHz:

- The interference due to IMT OOB emissions can be reduced by improved filtering on the IMT base station.
- The interference due to blocking can be reduced by improving the MES resilience to LTE blocking signals in the adjacent band.
- Either adding location based frequency allocation to MSS to avoid the use the lower couple of MHz and/or, implementing interference avoidance which would in addition allow for a better frequency utilisation of the lower part of the 1518-1559 MHz frequency band for MSS. The feasibility and impact of these techniques have not been assessed.

The following values have been used to examine the impact of enhanced MES receiver performance.

Table 7.1.2.2-1: Assumed blocking level for enhanced MES receivers

Frequency separation between channel edges	Interference level (at output of receiving antenna)
1 MHz	–55 to –45 dBm
3 MHz	–35 to –30 dBm
6 MHz	–30 to –25 dBm

Based on the final results of its compatibility studies, it is concluded that:

- the minimum in-band blocking characteristic for land mobile earth stations receivers from a 5 MHz broadband signal interferer (LTE) operating below 1518 MHz shall be –30dBm above 1520 MHz (Note: when the MES operates above 1520 MHz),
- the base station unwanted emission limits EIRP for a broadband signal interferer (LTE) operating below 1518 MHz shall be –30dBm/MHz above 1520 MHz. This figure is 10 dB more stringent than ECC Decision (13)03 due to a different service in the adjacent band.

It is noted that the IMT block ends at 1517 MHz.

5.2.5.4 ETSI

5.2.5.4.1 ETSI EN 301 681

5.2.5.4.1.1 In-band and Out-of-band Emission Requirements

The following section includes excerpts from ETSI EN 301 681.

ETSI EN 301 681 is applicable to Mobile Earth Stations (SES) with both transmit and receive capabilities for operation in Satellite Personal Communications Networks (S-PCN) in any combination of all or any part of the Mobile Satellite Service (MSS) frequency bands sub-band 1 and sub-band 2 defined in table 5.2.5.4.1.1-1.

Table 5.2.5.4.1.1-1: Mobile Satellite Service (MSS) frequency band

Sub-band	Transmission path	MSS frequency band
1	MESs transmit 1	1 626,5 MHz to 1 660,5 MHz
	MESs receive 1	1 525 MHz to 1 559 MHz
2	MESs transmit 2	1 668,0 MHz to 1 675,0 MHz
	MESs receive 2	1 518,0 MHz to 1 525,0 MHz

Unwanted emissions from Mobile Earth Stations (LMESs) outside the band 1 626,5 MHz to 1 660,5 MHz and outside the band 1 668 MHz to 1 675 MHz shall be below the following limits.

Unless otherwise stated the specification in this clause shall apply to all types of MESs notwithstanding their transmitting capabilities within the frequency bands as defined in table 5.2.5.4.1.1-1:

- For MES that are capable of transmitting within only the sub-band 1 frequency band as defined in table 5.2.5.4.1.1-1, the maximum EIRP spectral density of the unwanted emissions from the MES outside the band 1 626,5 MHz to 1 660,5 MHz shall not exceed the limits in either table 5.2.5.4.1.1-2 or table 5.2.5.4.1.1-3. The applicant shall declare which alternative shall be used.
- For MES that are capable of transmitting within only sub-band 2 frequency band or within both the sub-band 1 and sub-band 2 frequency bands, the maximum EIRP spectral density of the unwanted emissions from the MES outside the bands 1 626,5 MHz to 1 660,5 MHz and 1 668,0 MHz to 1 675,0 MHz shall not exceed the limits in table 5.2.5.4.1.1-3.

Table 5.2.5.4.1.1-2: Unwanted emissions outside the band 1 626,5 MHz to 1 660,5 MHz for MES only capable of transmitting within sub-band 1 frequency band as defined in table 5.2.5.4.1.1-1

Frequency (MHz)	Carrier - on state		
	EIRP (dBW)	Measurement bandwidth	Measurement method (see note 6)
30 to 1 000	-66	100 kHz	Peak Hold
1 000 to 1 559	-61	1 MHz	Average
1 559 to 1 605,0	-70	1 MHz (see note 3)	Average (see note 2)
1 605,0 to 1 612,5	-70 to -58,5 (see note 4)	1 MHz (see note 3)	Average
1 612,5 to 1 616,5	-55 to -50 (see note 4)	1 MHz	Average
1 616,5 to 1 621,5	-50 to -46 (see note 4)	1 MHz	Average
1 621,5 to 1 624,5	-60	30 kHz	Average
1 624,5 to 1 625,0	-60 to -57,5 (see notes 4, 5, 7)	30 kHz	Average
1 625,0 to 1 625,125	-57,5 to -57,2 (see notes 4, 5, 7)	30 kHz	Average
1 625,125 to 1 625,8	-57,2 to -50 (see notes 4, 5, 7)	30 kHz	Average
1 625,8 to 1 626,0	-50 to -47 (see notes 4, 5, 7)	30 kHz	Average
1 626,0 to 1 626,2	-47 to -40 (see notes 4, 5, 7)	30 kHz	Average
1 626,2 to 1 626,5	-40 (see notes 5, 7)	30 kHz	Average
1 626,5 to 1 660,5	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
1 660,5 to 1 662,5	The levels in table 4a shall apply from 1 660,5 MHz to 1 662,5 MHz		
1 662,5 to 1 665,5	-60	30 kHz	Average
1 665,5 to 1 670,5	-60	100 kHz	Average
1 670,5 to 1 680,5	-60	300 kHz	Average
1 680,5 to 1 690,5	-60	1 MHz	Average
1 690,5 to 2 250	-60	3 MHz	Average
2 250 to 12 750 (see note 1)	-60	3 MHz	Peak Hold

NOTE 1: In the band 3 253,0 MHz to 3 321,0 MHz the maximum EIRP in one, and only one, 3 MHz measurement bandwidth shall not exceed -38 dBW. Elsewhere in this band the power limit in table 3 shall be applied.

In each of the bands 4 879,5 MHz to 4 981,5 MHz, 6 506,0 MHz to 6 642,0 MHz and 8 132,5 MHz to 8 302,5 MHz the maximum EIRP in one, and only one, 3 MHz measurement bandwidth shall not exceed -48 dBW. Elsewhere in this band the power limit in table 3 shall be applied.

NOTE 2: The average measurement method defined in clause 5.2.2.3 shall apply except that an averaging period of 20 ms shall be used in the sub-band 1 573,42 MHz to 1 580,42 MHz.

NOTE 3: Measurement bandwidths less than 1 MHz are allowable provided the power in the narrower bandwidth is integrated over 1 MHz.

NOTE 4: Linearly interpolated in dBW vs. Frequency.

NOTE 5: The power limits specified in the band 1 624,5 MHz to 1 626,5 MHz require further study. This study is important to determine whether less stringent limits may enhance spectrum efficiency and utilization immediately above 1 626,5 MHz.

NOTE 6: Peak Hold and Average measurements shall be performed as specified in clauses 5.2.2.2 and 5.2.2.3.

NOTE 7: For systems employing CDMA, the EIRP limits shall be decreased by $10 \log(N)$ dB, where N is the maximum number of MESs in the receive beam of the satellite to which these MESs are communicating and which are expected to transmit simultaneously in the same frequency band within that same beam. This number shall be declared by the manufacturer ($N = 1$ in a TDMA system).

Table 5.2.5.4.1.1-3: Unwanted emissions outside the bands 1 626,5 MHz to 1 660,5 MHz and 1 668,0 MHz to 1 675,0 MHz for MES capable of transmitting within all or any part of sub-band 1 and sub-band 2 frequency bands as defined in table 5.2.5.4.1.1-1

Frequency (MHz)	Carrier - on state		
	EIRP (dBW)	Measurement bandwidth	Measurement method (see note 8)
30 to 1 000	-66	100 kHz	Peak Hold
1 000 to 1 559	-61	1 MHz	Average
1 559 to 1 605,0	-70	1 MHz (see note 5)	Average (see note 2)
1 605,0 to 1 612,5	-70 to -58,5 (see note 6)	1 MHz (see note 5)	Average
1 612,5 to 1 616,5	-55 to -50 (see note 6)	1 MHz	Average
1 616,5 to 1 621,5	-50 to -46 (see note 6)	1 MHz	Average
1 621,5 to 1 624,5	-60	30 kHz	Average
1 624,5 to 1 625,0	-60 to -57,5 (see notes 6, 7, 9)	30 kHz	Average
1 625,0 to 1 625,125	-57,5 to -57,2 (see notes 6, 7, 9)	30 kHz	Average
1 625,125 to 1 625,8	-57,2 to -50 (see notes 6, 7, 9)	30 kHz	Average
1 625,8 to 1 626,0	-50 to -47 (see notes 6, 7, 9)	30 kHz	Average
1 626,0 to 1 626,2	-47 to -40 (see notes 6, 7, 9)	30 kHz	Average
1 626,2 to 1 626,5	-40 (see notes 7, 9)	30 kHz	Average
1 626,5 to 1 660,5	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
1 660,5 to 1 662,5	note 1	note 1	note 1
1 662,5 to 1 666,0	-55	30 kHz	Average
1 666,0 to 1 668,0	note 2	note 2	note 2
1 668,0 to 1 675,0	NOT APPLICABLE	NOT APPLICABLE	NOT APPLICABLE
1 675,0 to 1 677,0	note 2	note 2	note 2
1 677,0 to 1 680,0	-60	30 kHz	Average
1 680,0 to 1 685,0	-60	100 kHz	Average
1 685,0 to 1 695,0	-60	300 kHz	Average
1 695,0 to 1 705,0	-60	1 MHz	Average
1 705,0 to 2 250	-60	3 MHz	Average
2 250 to 12 750 (see note 3)	-60	3 MHz	Peak Hold

NOTE 1:	For an MES transmitting in sub-band 1 (as defined in table 1) the limits defined in table 4a shall apply for the band 1 660,5 MHz to 1 662,5 MHz. For an MES transmitting in sub-band 2 (as defined in table 1) a limit of -55 dBW in 30 kHz shall apply for the band 1 660,5 MHz to 1 662,5 MHz.
NOTE 2:	For an MES transmitting in sub-band 2 (as defined in table 1) the limits defined in table 4a shall apply for the bands 1 666,0 MHz to 1 668,0 MHz and 1 675,0 MHz to 1 677,0 MHz. For an MES transmitting in sub-band 1 (as defined in table 1) a limit of -55 dBW in 30 kHz shall apply for the bands 1 666,0 MHz to 1 668,0 MHz and 1 675,0 MHz to 1 677,0 MHz.
NOTE 3:	In the bands 3 253,0 MHz to 3 321,0 MHz and 3 336,0 MHz to 3 350,0 MHz the maximum EIRP in one, and only one, 3 MHz measurement bandwidth shall not exceed -38 dBW. Elsewhere in this band the power limit in table 3a shall be applied. In the bands 4 879,5 MHz to 4 981,5 MHz and 5 004,0 MHz to 5 025,0 MHz the maximum EIRP in one, and only one, 3 MHz measurement bandwidth shall not exceed -48 dBW. Elsewhere in this band the power limit in table 3a shall be applied. In the bands 6 506,0 MHz to 6 642,0 MHz and 6 672,0 MHz to 6 700,0 MHz the maximum EIRP in one, and only one, 3 MHz measurement bandwidth shall not exceed -48 dBW. Elsewhere in this band the power limit in table 3a shall be applied. In the bands 8 132,5 MHz to 8 302,5 MHz and 8 340,0 MHz to 8 375,0 MHz the maximum EIRP in one, and only one, 3 MHz measurement bandwidth shall not exceed -48 dBW. Elsewhere in this band the power limit in table 3a shall be applied.
NOTE 4:	The average measurement method defined in clause 5.2.2.3 shall apply except that an averaging period of 20 ms shall be used in the sub-band 1 573,42 MHz to 1 580,42 MHz.
NOTE 5:	Measurement bandwidths less than 1 MHz are allowable provided the power in the narrower bandwidth is integrated over 1 MHz.
NOTE 6:	Linearly interpolated in dBW vs. Frequency.
NOTE 7:	The power limits specified in the band 1 624,5 MHz to 1 626,5 MHz require further study. This study is important to determine whether less stringent limits may enhance spectrum efficiency and utilization immediately above 1 626,5 MHz.
NOTE 8:	Peak Hold and Average measurements shall be performed as specified in clauses 5.2.2.2 and 5.2.2.3.
NOTE 9:	For systems employing CDMA, the EIRP limits shall be decreased by $10 \log(N)$ dB, where N is the maximum number of MESs in the receive beam of the satellite to which these MESs are communicating and which are expected to transmit simultaneously in the same frequency band within that same beam. This number shall be declared by the manufacturer ($N = 1$ in a TDMA system).

Table 5.2.5.4.1.1-4: Maximum unwanted emissions in the bands 1 626,5 MHz to 1 660,5 MHz and 1 660,5 MHz to 1 662,5 MHz caused by S-PCN MESs transmitting in the band 1 626,5 MHz to 1 660,5 MHz; and maximum unwanted emissions in the bands 1 666,0 MHz to 1 668,0 MHz, 1 668,0 MHz to 1 675,0 MHz and 1 675,0 MHz to 1 677,0 MHz caused by S-PCN MESs transmitting in the band 1 668,0 MHz to 1 675,0 MHz

Frequency offset (kHz) (see note 1)	Carrier-on state		
	EIRP (dBW) (see note 2)	Measurement bandwidth (kHz)	Measurement method
0 to 25	0 to -15	3	Average
25 to 125	-15 to -50	3	Average
125 to 425	-50	3	Average
425 to 1 500	-50 to -65	3	Average
1 500 to 36 000	-55	30	Average
NOTE 1: Frequency offset is determined from the edge of the nominated bandwidth. NOTE 2: Linearly interpolated in dBW vs. Frequency offset.			

Table 5.2.5.4.1.1-5: Maximum unwanted emissions in the bands 1 626,5 MHz to 1 660,5 MHz caused by S-PCN MESSs transmitting in the band 1 626,5 MHz to 1 660,5 MHz; and maximum unwanted emissions in the bands 1 668,0 MHz to 1 675,0 MHz caused by S-PCN MESSs transmitting in the band 1 668,0 MHz to 1 675,0 MHz

Frequency offset (kHz)	Carrier-on state		
	EIRP (dBW) (see note 2)	Measurement bandwidth (kHz)	Measurement method
0 to 25	0 to -15	3	Average
25 to 55	-15 to -25 (see note 4)	3	Average
55 to AB	-25 (see note 4)	3	Average
AB to (AB + 0,35 x B3dB)	-25 to -40 (see note 4)	3	Average
(AB + 0,35 x B3dB) to CD	-40	3	Average
CD to (CD + 0,25 x B3dB)	-40 to -50	3	Average
(CD + 0,25 x B3dB) to EF	-50	3	Average
EF to 1 500	-50 to -65	3	Average
1 500 to 36 000	-55	30	Average

NOTE 1: Frequency offset is determined from the edge of the nominated bandwidth.
NOTE 2: Linearly interpolated in dBW vs. Frequency offset.
NOTE 3: The parameters AB, CD, EF are defined below.
NOTE 4: The limit of -25 dBW in this table is determined on the assumption that the adjacent channel interference results from a single interferer. This limit shall apply to MESSs that are designed for operation in a network where the occurrence of two (or more) interferers, all transmitting with the maximum permitted level of unwanted emissions, does not exceed 0,1 % of the time; otherwise a limit of -30 dBW shall apply.

The parameters AB, CD and EF are defined as a proportion of the 3 dB Bandwidth as follows:

AB = (55) or (100 % of the B3dB), whichever is the greater; • CD = (95) or (200 % of the B3dB), whichever is the greater;

EF = (125) or (300 % of the B3dB), whichever is the greater.

5.2.5.4.1.2 Additional measures to protect Radioastronomy (RA) stations

ETSI EN 301 681 harmonized standard specifies requirements for the protection of radioastronomy bands. In section 4.2.6 of the document, the protection of radioastronomy services is mandated by disabling devices in the vicinity of radioastronomy stations recorded in the ITU master international frequency register.

To protect the radio astronomy service in the 1 660 MHz to 1 660,5 MHz band and the 1 668,0 MHz to 1 670,0 MHz band from emissions produced by MESSs the transmissions in these frequency bands shall be capable of being disabled in the vicinity of RA stations recorded in the ITU Master International Frequency Register.

Requirements are also set out for the activation of this disable function:

“An MES which is in the carrier-on state shall not continue transmissions for a period of time longer than 1s after receipt of a transmission disable command.”

Spectrum emission requirements are further specified when a transmission disable command is issued, in the form of carrier-off state emissions:

Table 5.2.5.4.1.2-1 EN 301 681 Carrier-off state requirements

Frequency (MHz)	EIRP (dBW)	Measurement bandwidth	Measurement method
30 to 1 000	-87	100 kHz	peak hold
1 000 to 1 525	-87	100 kHz	peak hold

1 525 to 1 559	-97 (see note 3)	100 kHz	average
1 559 to 1 610	-70	1 MHz (see note 1)	average (see note 2)
1 610 to 12 750	-87	100 kHz	peak hold
NOTE 1: Measurement bandwidths less than 1 MHz are allowable provided the power in the narrower bandwidth is integrated over 1 MHz. NOTE 2: The average measurement method defined in clause 5.2.2.3 shall apply except that a measurement time of 20 ms shall be used in the sub-band 1 573,42 MHz to 1 580,42 MHz. NOTE 3: If MES antenna gain is higher than 8 dBi, the unwanted emission limit in the band 1 525 MHz to 1 559 MHz shall be considered to be -90 dBW in 100 kHz.			

The emissions limit for 1610-12750MHz captures the carrier-off requirements for protection of radioastronomy bands within 1660-1660.5MHz and 1668-1670MHz, mandating a limit of -87dBW/100kHz, measured in any time period as a peak hold measurement.

5.2.6 NTN Satellite band: UL 13.75-14.5 GHz / DL 10.7-12.75 GHz

5.2.6.1 ITU-R

The ITU-R Radio Regulations [3] specify the following service allocation for those frequency ranges, shown in Table 5.2.6.1-1 (10.7-12.75 GHz) and Table : 5.2.6.1-2 (13.75-14.5 GHz).

Table 5.2.6-1: Allocation of 10.7 – 12.75 GHz in the ITU-R Radio Regulations

Allocation to services		
Region 1	Region 2	Region 3
10.7-10.95 FIXED FIXED-SATELLITE (space-to-Earth) 5.441 (Earth-to-space) 5.484 MOBILE except aeronautical mobile	10.7-10.95 FIXED FIXED-SATELLITE (space-to-Earth) 5.441 MOBILE except aeronautical mobile	
10.95-11.2 FIXED FIXED-SATELLITE (space-to-Earth) 5.484A 5.484B (Earth-to-space) 5.484 MOBILE except aeronautical mobile	10.95-11.2 FIXED FIXED-SATELLITE (space-to-Earth) 5.484A 5.484B MOBILE except aeronautical mobile	
11.2-11.45 FIXED FIXED-SATELLITE (space-to-Earth) 5.441 (Earth-to-space) 5.484 MOBILE except aeronautical mobile	11.2-11.45 FIXED FIXED-SATELLITE (space-to-Earth) 5.441 MOBILE except aeronautical mobile	
11.45-11.7 FIXED FIXED-SATELLITE (space-to-Earth) 5.484A 5.484B (Earth-to-space) 5.484 MOBILE except aeronautical mobile	11.45-11.7 FIXED FIXED-SATELLITE (space-to-Earth) 5.484A 5.484B MOBILE except aeronautical mobile	
11.7-12.5 FIXED MOBILE except aeronautical mobile BROADCASTING BROADCASTING-SATELLITE 5.492 5.487 5.487A	11.7-12.1 FIXED 5.486 FIXED-SATELLITE (space-to-Earth) 5.484A 5.484B 5.488 Mobile except aeronautical mobile 5.485	11.7-12.2 FIXED MOBILE except aeronautical mobile BROADCASTING BROADCASTING-SATELLITE 5.492 5.487 5.487A
	12.1-12.2 FIXED-SATELLITE (space-to-Earth) 5.484A 5.484B 5.488 5.485 5.489	
	12.2-12.7 FIXED MOBILE except aeronautical mobile BROADCASTING BROADCASTING-SATELLITE 5.492	12.2-12.5 FIXED FIXED-SATELLITE (space-to-Earth) 5.484A 5.484B MOBILE except aeronautical mobile BROADCASTING 5.487
	5.487A 5.488 5.490 12.7-12.75 FIXED FIXED-SATELLITE (Earth-to-space) MOBILE except aeronautical mobile	12.5-12.75 FIXED FIXED-SATELLITE (space-to-Earth) 5.484A 5.484B MOBILE except aeronautical mobile BROADCASTING-SATELLITE 5.493
12.5-12.75 FIXED-SATELLITE (space-to-Earth) 5.484A 5.484B (Earth-to-space) 5.494 5.495 5.496		

Table 5.2.6-2: Allocation of 13.75 – 14.5 GHz in the ITU-R Radio Regulations

Allocation to services		
Region 1	Region 2	Region 3
13.75-14 FIXED-SATELLITE (Earth-to-space) 5.484A RADIOLOCATION Earth exploration-satellite Standard frequency and time signal-satellite (Earth-to-space) Space research 5.499 5.500 5.501 5.502 5.503		
14-14.25 FIXED-SATELLITE (Earth-to-space) 5.457A 5.457B 5.484A 5.484B 5.506 5.506B RADIONAVIGATION 5.504 Mobile-satellite (Earth-to-space) 5.504B 5.504C 5.506A Space research 5.504A 5.505		
14.25-14.3 FIXED-SATELLITE (Earth-to-space) 5.457A 5.457B 5.484A 5.484B 5.506 5.506B RADIONAVIGATION 5.504 Mobile-satellite (Earth-to-space) 5.504B 5.506A 5.508A Space research 5.504A 5.505 5.508		
14.3-14.4 FIXED FIXED-SATELLITE (Earth-to-space) 5.457A 5.457B 5.484A 5.484B 5.506 5.506B MOBILE except aeronautical mobile Mobile-satellite (Earth-to-space) 5.504B 5.506A 5.509A Radionavigation-satellite 5.504A	14.3-14.4 FIXED-SATELLITE (Earth-to-space) 5.457A 5.484A 5.484B 5.506 5.506B Mobile-satellite (Earth-to-space) 5.506A Radionavigation-satellite 5.504A	14.3-14.4 FIXED FIXED-SATELLITE (Earth-to-space) 5.457A 5.484A 5.484B 5.506 5.506B MOBILE except aeronautical mobile Mobile-satellite (Earth-to-space) 5.504B 5.506A 5.509A Radionavigation-satellite 5.504A
14.4-14.47 FIXED FIXED-SATELLITE (Earth-to-space) 5.457A 5.457B 5.484A 5.484B 5.506 5.506B MOBILE except aeronautical mobile Mobile-satellite (Earth-to-space) 5.504B 5.506A 5.509A Space research (space-to-Earth) 5.504A		
14.47-14.5 FIXED FIXED-SATELLITE (Earth-to-space) 5.457A 5.457B 5.484A 5.506 5.506B MOBILE except aeronautical mobile Mobile-satellite (Earth-to-space) 5.504B 5.506A 5.509A Radio astronomy 5.149 5.504A		

5.2.6.2 FCC

The FCC specifies the following service allocation for the 10.7-12.75 GHz and 13.75-14.5 GHz frequency ranges, as shown in Table 5.2.6.2-1 (10.7-12.75 GHz) and Table 5.2.6.2-2 (13.75-14.5 GHz).

Table 5.2.6.2-1 FCC allocation of 10.7 – 12.75 GHz

United States Table		FCC Rule Part(s)
Federal Table	Non-Federal Table	
10.7-11.7 US131 US211	10.7-11.7 FIXED FIXED-SATELLITE (space-to-Earth) 5.441 US131 US211 NG52	Satellite Communications (25) Fixed Microwave (101)
11.7-12.2	11.7-12.2 FIXED-SATELLITE (space-to-Earth) 5.485 5.488 NG143 NG527A	Satellite Communications (25)
12.2-12.75	12.2-12.7 FIXED BROADCASTING-SATELLITE 5.487A 5.488 5.490	Satellite Communications (25) Fixed Microwave (101)
	12.7-12.75 FIXED NG118 FIXED-SATELLITE (Earth-to-space) MOBILE	TV Broadcast Auxiliary (74F) Cable TV Relay (78) Fixed Microwave (101)

Table 5.2.6.2-2 FCC allocation of 13.75 – 14.5 GHz

United States Table		FCC Rule Part(s)
Federal Table	Non-Federal Table	
13.75-14 RADIOLOCATION G59 Standard frequency and time signal-satellite (Earth-to-space) Space research US337 US356 US357	13.75-14 FIXED-SATELLITE (Earth-to-space) US337 Standard frequency and time signal-satellite (Earth-to-space) Space research Radiolocation US356 US357	Satellite Communications (25) Private Land Mobile (90)
14-14.2 Space research US133	14-14.2 FIXED-SATELLITE (Earth-to-space) NG527A Mobile-satellite (Earth-to-space) Space research US133	Satellite Communications (25)
14.2-14.4	14.2-14.47 FIXED-SATELLITE (Earth-to-space) NG527A Mobile-satellite (Earth-to-space)	Satellite Communications (25)
14.4-14.47 Fixed Mobile		
14.47-14.5 Fixed Mobile US113 US133 US342	14.47-14.5 FIXED-SATELLITE (Earth-to-space) NG527A Mobile-satellite (Earth-to-space) US113 US133 US342	

Also, the 13.75-14 GHz frequency range is submitted to additional restrictions on minimum antenna diameter, minimum and maximum EIRP.

(356) US356 In the band 13.75-14 GHz, an earth station in the fixed-satellite service shall have a minimum antenna diameter of 4.5 m and the e.i.r.p. of any emission should be at least 68 dBW and should not exceed 85 dBW. In addition the e.i.r.p., averaged over one second, radiated by a station in the radiolocation service shall not exceed 59 dBW. Receiving space stations in the fixed-satellite service shall not claim protection from radiolocation

transmitting stations operating in accordance with the United States Table of Frequency Allocations in this section. ITU Radio Regulation No. 5.43A does not apply.

ESIM operations are allowed in the following frequency ranges (CFR 47 25.202):

- 10.7-11.7 GHz (s-to-E) for both GSO and NGSO.
- 11.7-12.2 GHz (s-to-E) for both GSO and NGSO.
- 14.0-14.5 GHz (E-to-s) for both GSO and NGSO.

But under the condition of footnote NG527A, which stipulates that ESIM operation in 10.7-11.7 GHz frequency range may be authorized under the conditions that those earth stations don't claim protection from fixed service.

(527) NG527A Earth Stations in Motion (ESIMs), as regulated under part 25 of this chapter, are an application of the fixed-satellite service (FSS) and the following provisions shall apply:

(i) In the bands 10.7-11.7 GHz, 19.3-19.4 GHz, and 19.6-19.7 GHz, ESIMs may be authorized for the reception of FSS emissions from geostationary and non-geostationary satellites, subject to the conditions that these earth stations may not claim protection from transmissions of non-Federal stations in the fixed service and that non-geostationary-satellite systems not cause unacceptable interference to, or claim protection from, geostationary-satellite networks.

(ii) In the bands 11.7-12.2 GHz (space-to-Earth), 14.0-14.5 GHz (Earth-to-space), 18.3-18.8 GHz (space-to-Earth), 19.7-20.2 GHz (space-to-Earth), 28.35-28.6 GHz (Earth-to-space), and 29.25-30.0 GHz (Earth-to-space), ESIMs may be authorized to communicate with geostationary satellites on a primary basis.

(iii) In the bands 11.7-12.2 GHz (space-to-Earth), 14.0-14.5 GHz (Earth-to-space), 18.3-18.6 GHz (space-to-Earth), 19.7-20.2 GHz (space-to-Earth), 28.4-28.6 GHz (Earth-to-space), and 29.5-30.0 GHz (Earth-to-space), ESIMs may be authorized to communicate with non-geostationary satellites, subject to the condition that non-geostationary-satellite systems may not cause unacceptable interference to, or claim protection from, geostationary-satellite networks.

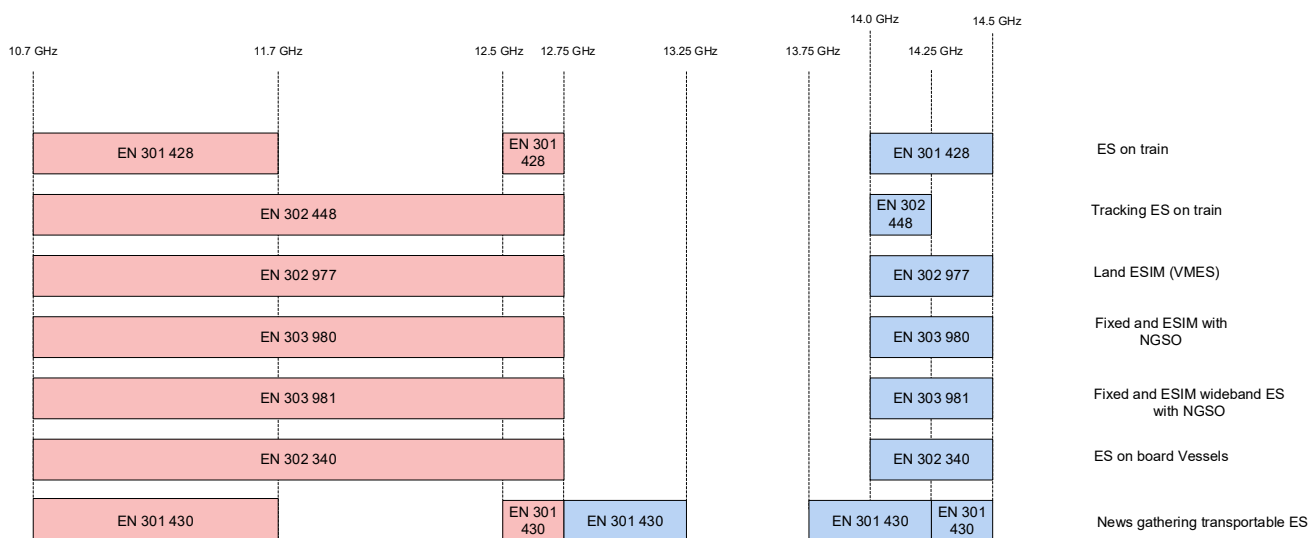
5.2.6.3 CEPT

In the considered frequency ranges, the ECC Decisions captured in Table 5.2.6.3-1 are relevant for in the scope of the NTN specification work.

Table 5.2.6.3-1 Relevant ECC Decisions

ECC Decision	Frequency Range	Title
ECC Decision(18)04	10.7 - 12.75 GHz (S-to-E) 14.0 – 14.5 GHz (E-to-S)	The harmonised use, exemption from individual licensing and free circulation and use of land based Earth Stations In-Motion (ESIM) operating with GSO FSS satellite systems in the frequency bands 10.7-12.75 GHz and 14.0- 14.5 GHz
ECC Decision(18)05	10.7 - 12.75 GHz (S-to-E) 14.0 – 14.5 GHz (E-to-S)	The harmonised use, exemption from individual licensing and free circulation and use of Earth Stations In-Motion (ESIM) operating with NGSO FSS satellite systems in the frequency bands 10.7-12.75 GHz and 14.0-14.5 GHz
ECC Decision(19)04	10.7 - 12.75 GHz (S-to-E) 12.75 – 13.25 GHz (E-to-S)	The harmonised use of spectrum, free circulation and use of earth stations on-board aircraft operating with GSO FSS networks and NGSO FSS systems in the frequency bands 12.75-13.25 GHz (Earth-to-space) and 10.7-12.75 GHz (space-to-Earth)
ECC Decision(05)10	10.7 – 11.7 GHz and 12.5 – 12.75 GHz (S-to-E) 14.0 - 14.5 GHz (E-to-S)	The free circulation and use of Earth Stations on board Vessels operating in fixed satellite service networks in the frequency bands 14-14.5 GHz
ECC Decision(17)04	10.7 - 12.75 GHz (S-to-E) 14.0 – 14.5 GHz (E-to-S)	The harmonised use and exemption from individual licensing of fixed earth stations operating with NGSO FSS satellite systems in the frequency bands 10.7-12.75 GHz and 14.0-14.5 GHz
ECC Decision(03)04	10.7 – 11.7 GHz (S-to-E) 14.25 – 14.5 GHz (E-to-S)	Exemption from Individual Licensing of Very Small Aperture Terminals (VSAT) operating in the frequency bands 14.25-14.50 GHz Earth-to-space and 10.70-11.70 GHz space-to-Earth
ECC Decision(06)03	10.7 – 12.75 GHz (S-to-E) 14.0 – 14.25 GHz (E-to-S)	Exemption from Individual Licensing of high e.i.r.p. satellite terminals (HEST) operating with geostationary satellites and in the frequency bands 10.70-12.75 GHz or 19.70-20.20 GHz space-to-Earth and 14.00-14.25 GHz or 29.50-30.00 GHz Earth-to-space
ERC Decision(00)08	10.7 – 12.5 GHz (S-to-E)	ERC Decision of 19 October 2000 on the use of the band 10.7 – 12.5 GHz by the fixed service and Earth stations of the broadcasting-satellite and fixed-satellite service (space-to-Earth)

In the considered frequency ranges, Figure 5.2.6.3-1 captures the relevant ETSI Harmonized Standards in the scope of the NTN specification work.

Figure 5.2.6.3-1 Relevant ETSI Harmonized Standards

5.3 Regulatory aspects for HAPS

ITU-R began to study HAPS in the 1990s initially for fixed services. The telecommunications ecosystem and technology enablers for HAPS have evolved significantly since then. At WRC-2000, the bands 1885 – 1980 MHz, 2010 – 2025 MHz and 2110 – 2170 MHz were identified for HAPS operating as IMT Base Stations, in accordance with Resolution 221 (Rev.WRC-07) [3]. With increasing interest in HAPS to offer mobile services, the WRC-19 agreed to study certain frequency bands which have been already identified for IMT below 2.7 GHz for HAPS as IMT Base Stations (HIBS). ITU-R Working Party 5D is currently studying co-existence requirements for HIBS in the additional three bands listed in Table 5.4-1 to support spectrum identification decisions in WRC-23 in accordance with Resolution 247 (WRC-19) [3].

NOTE: In the current ITU terminology, the use of HAPS to implement IMT (i.e. offer mobile wireless services) is referred to as “HAPS as IMT Base Stations” or HIBS.

Table 5.4-1: Frequencies for HAPS as IMT Base Stations (HIBS).

Region	Spectrum	Remarks
Region 1 and 3	1885 – 1980 MHz 2010 – 2025 MHz 2110 – 2170 MHz	Identified HIBS designations at WRC-2000
Region 2	1885 – 1980 MHz ³ 2110 – 2160 MHz ⁴	
Global	694 – 960 MHz 1710 – 1885 MHz ¹ 2500 – 2690 MHz ²	Under study, for decision at WRC-23
NOTE 1: 1710-1815 MHz to be used for uplink only in Region 3. NOTE 2: 2500-2535 MHz to be used for uplink only in Region 3, except 2655--2690 MHz in Region 3). NOTE 3: In most of Region 2 1885-1910 is used for Mobile uplink and 1930-1980 is used for Mobile downlink. In the United States, 1910-1915 and 1915-1920 are also used for Mobile uplink. NOTE 4: In Region 2 2110-2160 is used for Mobile downlink.		

6 Co-existence study

6.1 Co-existence simulation scenario

Scenarios for coexistence study are listed in Table 6.1-1.

Table 6.1-1: Scenarios for NTN-NTN/TN co-existence

FR1: 2GHz		NTN ^{1,4,5}						
		Set 1			Set 2 ²			HAPS
		GEO ³	LEO 600km	LEO 1200km	GEO	LEO 600km	LEO 1200km	
NR / NB-IoT	Rural	X	X	X	X	X	X	X
	Urban macro	X	X	X	X	X	X	X
	Dense Urban ⁶	N/A	N/A	N/A	N/A	N/A	N/A	N/A
HAPS		N/A	N/A	N/A	N/A	N/A	N/A	X
NOTE 1: Earth Fixed beam has been considered for co-existence studies.								
NOTE 2: Use Set 1 satellite antenna has been used as the starting point for co-existence studies.								
NOTE 3: GEO and LEO only operate at adjacent channel.								
NOTE 4: Use GEO and LEO@600km when TN is victim.								
NOTE 5: The satellite-to-satellite coexistence scenarios are not in the scope of this study considering this is already addressed by ITU (ITU RR Article 9 etc.) and regional regulations (e.g. FCC rules).								
NOTE 6: Dense Urban is not considered as it is expected NTN UE will connect to terrestrial networks rather than satellite networks in such scenario.								

The aggressor and victim combinations are listed in Table 6.1-2.

Table 6.1-2: Aggressor and victim

No.	Combination	Aggressor	Victim	Notes	Study Phase
1	TN with NTN	TN DL	NTN DL	Applicable for satellite operating in e.g. S-band, for e.g. coexistence with n1 FDD.	Phase 1
2	TN with NTN	TN UL	NTN UL	Applicable for satellite operating in e.g. S-band, for e.g. coexistence with n1 FDD.	Phase 1
3	TN with NTN	NTN DL	TN DL	Applicable for satellite operating in e.g. S-band, for e.g. coexistence with n1 FDD.	Phase 1
4	TN with NTN	NTN UL	TN UL	Applicable for satellite operating in e.g. S-band, for e.g. coexistence with n1 FDD.	Phase 1
5	TN with NTN	NTN UL	TN DL	Applicable for satellite operating in S-band, for e.g. coexistence with n34 TDD.	Phase 1
6	TN with NTN	TN DL	NTN UL	Applicable for satellite operating in S-band, for e.g. coexistence with n34 TDD.	Phase 1
7	NTN with NTN	NTN DL	NTN DL	HAPS-HAPS	Phase 2
		NTN UL	NTN UL	HAPS-HAPS	Phase 2

For further visualisation, the interference charts with respect to scenarios 1 to 6 are further represented with respect to NTN-TN coexistence scenarios in Figure 6.1-1 (scenarios 1 to 4) and Figure 6.1-2 (scenarios 5 and 6).

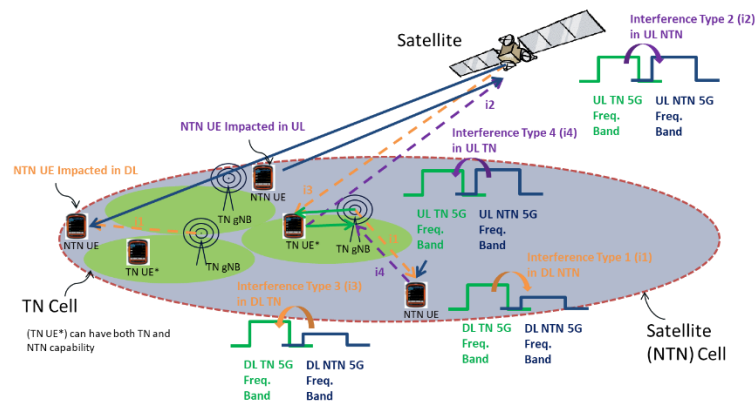


Figure 6.1-1: S-band NTN-TN adjacent band coexistence scenarios with TN in FDD mode

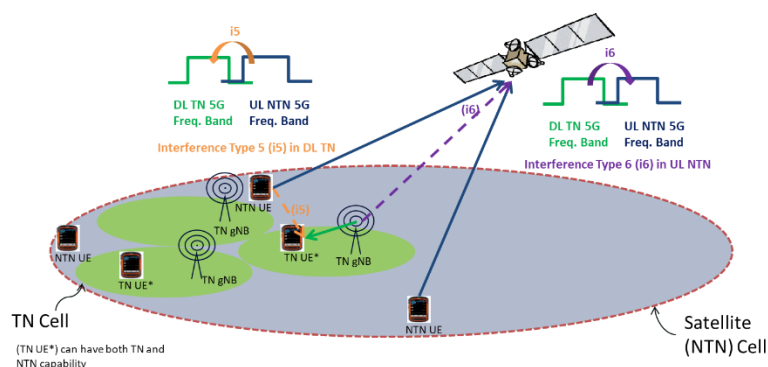


Figure 6.1-2: S-band NTN-TN adjacent band coexistence scenarios with TN in TDD mode (e.g. n34)

The frequency and bandwidth are listed in Table 6.1-3.

Table 6.1-3: Proposed frequency and bandwidth for co-existence study

	Frequency	Bandwidth	Duplex mode	Frequency reuse factor
TN Rural	2 GHz	20MHz	FDD, TDD	1
TN Urban macro	2 GHz	20MHz	FDD, TDD	1
GEO	2 GHz	5/10/15/20 MHz for FR1	FDD	1, 3 ¹
LEO	2 GHz	5/10/15/20 MHz for FR1	FDD	1, 3 ¹
HAPS	2 GHz	TBD	FDD	1
NOTE: Only FRF=1 has been used in co-existence studies for simplification.				

6.2 Co-existence simulation assumption

6.2.1 Network layout model

6.2.1.1 Co-existence between NTN satellite and TN

Cellular cell structure is considered for both NTN satellite and TN network layout.

Referring to TR 38.811[5] Section 6.3 and Annex A, a 3D global coordinate system is considered (Earth-Centred Earth Fixed) for simulating NTN satellite beams direction and location on the earth surface. It means the NTN satellite beam location, TN randomly dropping location are generated with a set of three parameters (x,y,z).

Deployment of NTN satellite and TN cells and UEs for co-existence study is listed in Table 6.2.1.1-1.

NOTE: The NTN UE(s) shall be dropped at the edge of the “central 19 TN cells (cluster)”. For Case 1 (Urban scenario), an Isolation distance of 1500m as $2 \times \text{ISD}$ is considered to reflect the NTN-TN selection algorithm at the border, assuming TN connection will always be prioritized over NTN.

As defined in Figure 6.2.1.1-1, isolation distance is the distance between the blue-dotted line which represents TN cell boarder and the red line. No UEs deployed in the isolation region is assumed to reduce the calculation complexity.

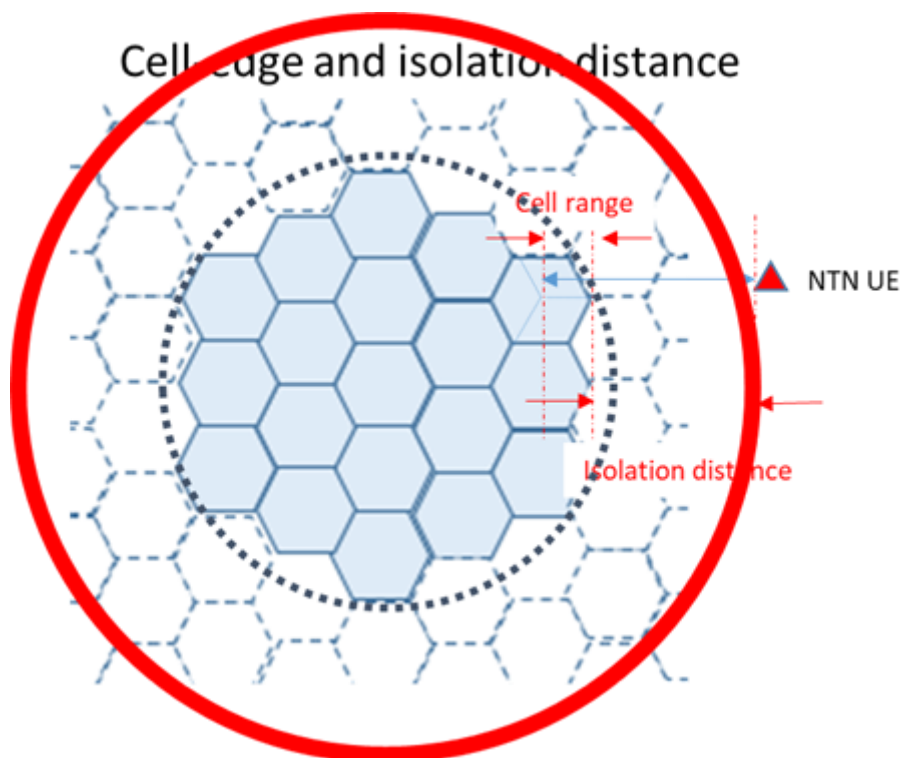
**Figure 6.2.1.1-1: Isolation distance for Case 1**

Table 6.2.1.1-1: Network and UE deployment

No.	Combination	Aggressor	Victim	Which NTN cell/UE to observe?	Which TN/UE to observe?	Which TN cells in a TN to observe?
1	TN with NTN satellite	TN DL	NTN DL	NTN cell: Observe NTN central beam for SINR, 6 adjacent beams for inter-beam interference. NTN UE: NTN UEs dropped at the edge of TN clusters	One cluster with 19 TN cells (57 sectors) randomly placed in the central NTN beam	All active TN clusters which has the NTN UE(s) at its edge.
2	TN with NTN satellite	TN UL	NTN UL	NTN cell: Observe NTN central beam for SINR, 6 adjacent beams for inter-beam interference. NTN UE: NTN UEs dropped at the edge of TN clusters	Consider an active rate of 20% for Rural and Urban of TN.	All active TN cells in central NTN beam
3	TN with NTN satellite	NTN DL	TN DL	NTN cell: Nadir point. NTN UE: NTN UEs dropped outside or at the edge of TN clusters	TN clusters randomly placed in this NTN beam	All in central NTN beam
				NTN cell: NTN cell with satellite at low elevation (45° for GEO and LEO Interested companies can bring analysis and results for other values) NTN UE: NTN UEs dropped outside or at the edge of TN clusters	TN clusters randomly placed in this NTN beam	
4	TN with NTN satellite	NTN UL	TN UL	NTN cell: Nadir point. NTN UE: NTN UEs dropped at the edge of TN clusters	TN randomly placed in this NTN beam	Option 1: All active TN clusters which has the NTN UE(s) at its edge. Option 2: Only the TN sectors which have NTN UE(s) at their edges. Option 1 is the baseline and it is not precluded companies can follow Option 2 to bring results

No.	Combination	Aggressor	Victim	Which NTN cell/UE to observe?	Which TN/UE to observe?	Which TN cells in a TN to observe?
5	TN with NTN satellite	NTN UL	TN DL	NTN cell: Nadir point NTN UE: NTN UEs dropped at the edge of TN clusters	TN clusters randomly placed in this NTN beam	All active TN clusters which has the NTN UE(s) at its edge
				NTN cell: NTN cell with satellite at low elevation (45° for GEO and LEO Interested companies can bring analysis and results for other values). NTN UE: NTN UEs dropped at the edge of TN clusters	TN clusters randomly placed in this NTN beam	All active TN clusters which has the NTN UE(s) at its edge.
6	TN with NTN satellite	TN DL	NTN UL	NTN cell: Observe NTN central beam for SINR, 6 adjacent beams for inter-beam interference. NTN cell with satellite at low elevation to be further investigated. NTN UE: NTN UEs dropped outside or at the edge of TN clusters	Consider the active rate of 20% for Rural and Urban of TN. The Urban TN deployment for GEO in Case 6 is a mixture of urban and rural TN deployment. A representative percentage of urban TN could be considered for further study.	All active TN cells in central NTN beam

6.2.1.2 Co-existence between HAPS and TN

For simulations of HAPS and TN co-existence, a cluster of TN cells is randomly dropped in the HAPS coverage area as shown in Figure 6.2.1.2-1. The TN cell cluster consists of 19 sites, 57 sectors, with the same assumption described in clause 6.2.2.4. HAPS coverage and cell layout are described in clause 6.2.2.3.

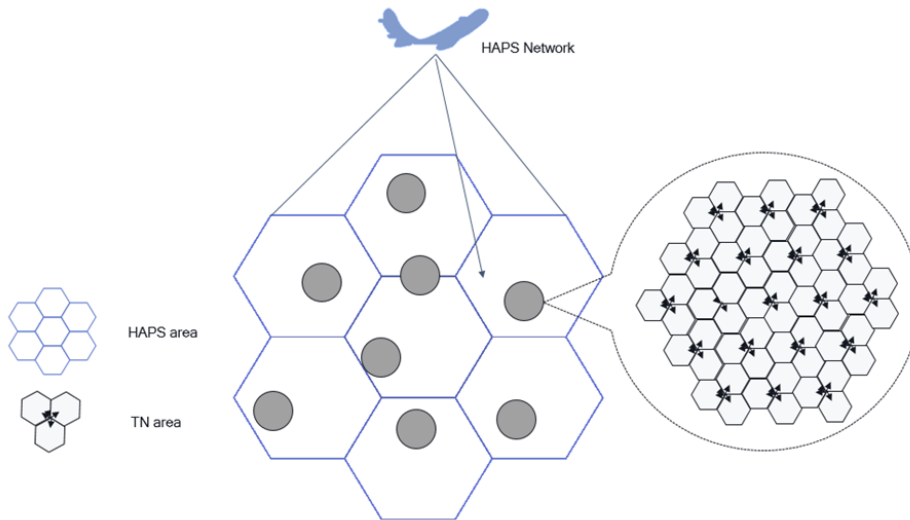


Figure 6.2.1.2-1: Simulation layout between HAPS and TN

6.2.1.3 Co-existence between HAPS and HAPS

HAPS and HAPS co-existence can be characterized by “center-to-center” inter-system distance (i.e., the distance between two coverage centers) as depicted in Figure 6.2.1.3-1. Since HAPS location is at the center of its coverage area, the inter-system distance is the separation of the two HAPS. Different inter-system distances may be simulated to evaluate the interference caused by antenna gain variation in the elevation domain.

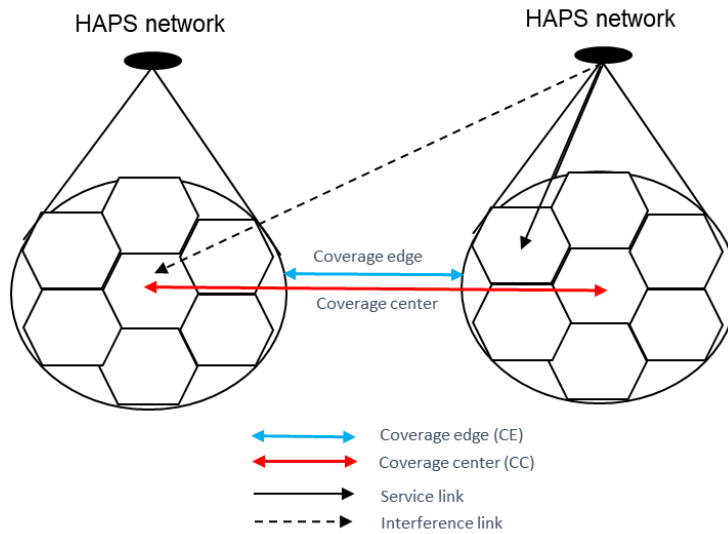


Figure 6.2.1.3-1: Simulation layout between HAPS and HAPS

6.2.2 System parameters

6.2.2.1 Satellite parameters

Two sets of satellite parameters are listed in Table 6.2.2.1-2 and Table 6.2.2.1-3 according to TR 38.821[6].

The satellite max Tx power can be calculated by the equation as below:

$$\max Tx \text{ power}[dBm] = EIRP \text{ density} \left[\frac{dBW}{MHz} \right] + 30 + 10 \log_{10}(N_{RB} * SCS[MHz] * 12) - \text{Max Gain}[dBi]$$

Table 6.2.2.1-1: N_{RB} configuration per BandWidth size and SCS

Configuration FR1 S-band	N_{RB} (5MHz BW)	N_{RB} (10MHz BW)	N_{RB} (15MHz BW)	N_{RB} (20MHz BW)
SCS 15 kHz	25	52	79	106
SCS 30 kHz	11	24	38	51

Table 6.2.2.1-2: Set-1 satellite parameters for co-existence study

Satellite orbit			GEO				LEO-1200				LEO-600			
Satellite altitude			35786 km				1200 km				600 km			
Payload characteristics for DL transmissions														
Satellite EIRP density		2GHz	59 dBW/MHz				40 dBW/MHz				34 dBW/MHz			
Satellite max TX power in dBm	BW (MHz)		5	10	15	20	5	10	15	20	5	10	15	20
	SCS 15kHz		44.53	47.71	49.53	50.81	46.53	49.71	51.53	52.81	40.53	43.71	45.53	46.81
	SCS 30kHz		43.98	47.37	49.36	50.64	45.98	49.37	51.36	52.64	39.98	43.37	45.36	46.64
Satellite Tx max Gain			51 dBi				30 dBi				30 dBi			
Channel bandwidth			5/10/15/20MHz				5/10/15/20MHz				5/10/15/20MHz			
3dB beamwidth or HPBW (Half-Power BandWidth) of main central beam			0.4011 deg				4.4127 deg				4.4127 deg			
ABS (Adjacent Beam Spacing) of adjacent beams from the central beam			0.3474 deg				3.8206 deg				3.8206 deg			
Satellite beam diameter			250 km				90 km				50 km			
Payload characteristics for UL transmissions														
G/T		2 GHz	19 dB K ⁻¹				1.1 dB K ⁻¹				1.1 dB K ⁻¹			
Satellite Rx max Gain			51 dBi				30 dBi				30 dBi			

Table 6.2.2.1-3: Set-2 satellite parameters for co-existence study

Satellite orbit		GEO		LEO-1200		LEO-600	
Satellite altitude		35786 km		1200 km		600 km	
Payload characteristics for DL transmissions							
Satellite EIRP density	2GHz	53.5 dBW/MHz		34 dBW/MHz		28 dBW/MHz	
Satellite Tx max Gain		45.5 dBi		24 dBi		24	
Channel bandwidth		5/10/15/20MHz		5/10/15/20MHz		5/10/15/20MHz	
3dB beamwidth		0.7353 deg		8.8320 deg		8.8320 deg	
Satellite beam diameter		450 km		190 km		90 km	
Payload characteristics for UL transmissions							
G/T	2 GHz	14 dB K ⁻¹		-4.9 dB K ⁻¹		-4.9 dB K ⁻¹	
Satellite Rx max Gain		45.5 dBi		24 dBi		24 dBi	

Table 6.2.2.1-4: Other parameters for NTN satellite

Parameters	NTN satellite	Remark
Carrier frequency	2GHz	
The number of active UE (UL)	9 UEs and 2RBs per UE for GEO and LEO ¹	NOTE
The number of active UE (DL)	1	Same with TN
Traffic model	Full buffer	
DL power control	NO	
UL power control	See Session 6.2.6.2	
NTN satellite Noise Figure in dB	See Table 2.3-3-1	
Handover margin	3dB	

NOTE: UEs are equally splitted inside the channel bandwidth into ACIR 3 regions. Scheduled PRB position for UE1 per satellite beam should be also fully aligned to simulate the worst case for co-channel interference and this is also aligned with full buffer case.

Table 6.2.2.1-5: NTN satellite Noise Figure in dB

Satellite	GEO	LEO 600	LEO 1200
G/T (dB K ⁻¹)	19	1.1	1.1
G Rx (dBi)	51	30	30
NF (dB)	7.4	4.3	4.3

6.2.2.2 NTN UE parameters

NTN UE parameters are given in Table 6.2.2.2-1

Table 6.2.2.2-1: NTN UE characteristics for system level simulations

Characteristics	Handheld
Frequency band	S band (i.e. 2 GHz)
Antenna type and configuration	(1, 1, 2) with omni-directional antenna element
Polarisation	Linear: +/-45°X-pol
Rx Antenna gain	0 dBi per element
Antenna temperature	290 K
Noise figure	9 dB
Tx transmit power	200 mW (23 dBm)
Tx antenna gain	0 dBi per element

6.2.2.3 HAPS parameters

Using the antenna model in Section 6.2.3.3, HAPS deployed at 20 km altitude has a coverage radius of 100 km in a 7-cell layout. Two implementations of the layout are shown in Figure 6.2.2.3-1. The UEs in the HAPS system are assumed to be outdoor and uniformly distributed in the coverage area in the rural environment. The serving cell is selected by the strongest RSRP or least coupling loss. (Note: Calibration data show these two implementations yield similar results). HAPS system parameters for the co-existence study are listed in Table 6.2.2.3-1.

For DL transmission, one UE is scheduled for the full bandwidth. For UL transmission, 9 UEs are scheduled, each using a bandwidth of 6 RBs. The scheduled UEs are evenly distributed in the frequency domain as shown in Figure 6.2.2.3-2. Note that the scheduled resources are aligned among different cells to model co-channel effect. Since the UE uses only part of the channel bandwidth in UL, the ACI impact may differ in the frequency domain of the adjacent channel. The ACIR model described in Section 6.2.4 should be used to model the ACI from aggressor UEs in UL, taking into account the UE's transmission bandwidth.

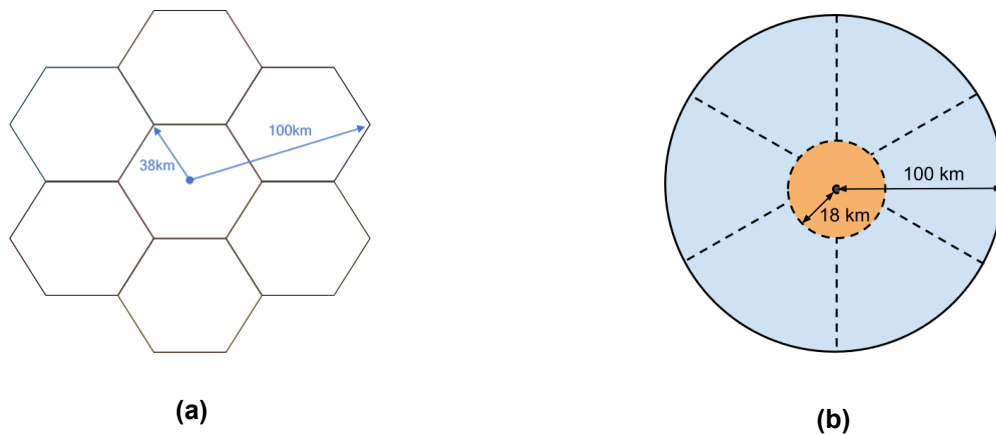
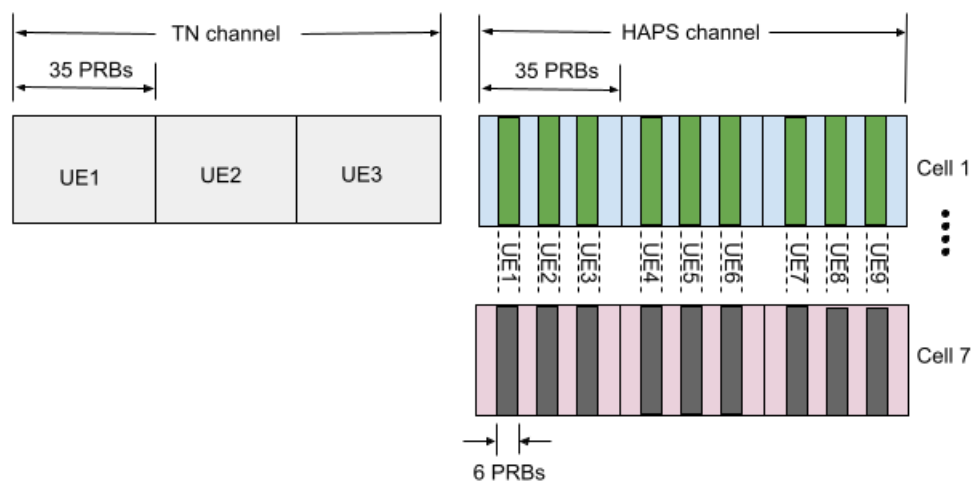


Figure 6.2.2.3-1: HAPS cell layout

Table 6.2.2.3-1: HAPS system parameters

HAPS altitude	20 Km
Carrier frequency	2 GHz
Frequency reuse factor	1
Duplex scheme	FDD
Channel bandwidth	20 MHz
Subcarrier spacing (SCS)	15 KHz
Number of cells	7
Coverage area (7 cells combined)	A 100 Km radius circular area centered by the serving HAPS
Frequency reuse	1
Environment ¹	Rural
UE distribution	Uniformly distributed in the coverage area
Indoor UE percentage	0%
Number of DL scheduled UEs per cell	1
Number of UL scheduled UEs per cell	9
DL scheduled bandwidth per UE	20 MHz
UL scheduled bandwidth per UE	6 RBs
DL power control	No
UL power control	See section 6.2.6.3
HAPS antenna	See section 6.2.3.3
HAPS noise figure	5 dB
HAPS-UE pathloss model	See section 6.2.5.4
HAPS UE assumption	Same as TN UE in Table 6.2.2.4-1
NOTE 1: HAPS is assumed to serve UEs in the rural environment, but the co-existed TN may be in the rural or urban macro environment.	

**Figure 6.2.2.3-2: UL scheduled UE bandwidth allocation**

6.2.2.4 TN parameters

TN parameters for co-existence study are given in Table 6.2.2.4-1, 6.2.2.4-2 and 6.2.2.4-3.

Table 6.2.2.4-1: Simulation assumptions of TN respectively based on NB-IoT and NR

	NB-IoT standalone	NR
Carrier frequency in GHz	2	2
Size of each nominal channel BW in MHz	0.2	20
Transmission bandwidth in MHz	0.18	N/A
Environment	Urban macro Rural	Deployment scenario related, see Table 6.2.2.4-2.
Network layout	19-sites [57 sectors] with wrap-around	19-sites 57 sectors with wrap-around
Inter-site distance in meter	500 for 2GHz band for UMA TBD For Rural	Deployment scenario related, see Table 6.2.2.4-2
System loading and activity	Full buffer 100%	See Table 6.2.1.1-1
Network location	TBD	See Table 6.2.1.1-1
DL subcarrier spacing	15kHz	15kHz
UL	See RP-152284[7]	OFDMA
DL power control	No	No
UL power control	TR 36.942[8] section 5.1.1.6 (set 1) by bandwidth scale, target SNR at BS is 15 dB	TR 36.942[8]
Frequency reuse	1	1
Number of scheduled UE per cell (DL)	1	1
Number of scheduled UE per cell (UL)	3 for multi-tone (60kHz per UE), 12 for 15kHz single-tone, 48 for 3.75kHz single-tone	3
UE antenna height in meter	1.5	1.5m
UE TX power in dBm	-40 to 23	-40 to 23
UE antenna gain in dBi	0	0
Building penetration loss	TR 45.820[9] Annex D.1	In pathloss model, TR 38.901[10]
Cell selection margin in dB	3	3
BS-MS min distance in meters	35	35
BS noise figure in dB	5	5
UE noise figure in dB	9	9
BS-UE path-loss model	TR 36.942[8] macro urban	TR 38.901[10]
Standard deviation of BS-UE log-normal shadow fading in dB	10	Deployment scenario related, referring to TR 38.901[10]
Shadowing correlation	Inter-cell 0.5 Intra-cell 1	Inter-cell 0.5 Intra-cell 1
Link-level performance model		See Section 6.2.8 Throughput-SINR mapping
UE distribution		Uniform
Evaluation metrics	SINR vs ACS (as victim)	See Section 6.2.8 Throughput or SNR loss criteria

Table 6.2.2.4-2: Deployment-related parameters of TN (2 GHz)

	Urban Macro	Rural Macro	Remarks
ISD in meters	750	7500	ITU-R Report M.2292[11]
BS Antenna height in meters	25	30	
UE Outdoor/indoor	100% Outdoor		
UE height in meter	1.5	1.5	RP-200559 [12] 3GPP LS to ITU-R WP5D and ITU-R WP5D [IMT_Parameters] [13]

Table 6.2.2.4-3: ACLR/ACS for TN (2GHz)

		NR	NB-IOT
		45 dB	40 dB
BS	ACLR	45 dB	40 dB
	ACS	46 dB	46 dB
UE	ACLR	30dB (ACLR1) 43dB (ACLR2)	37
	ACS	33	28

6.2.3 Antenna and beamforming pattern modelling

6.2.3.1 Satellite and UE Antenna and beam forming pattern modelling

Satellite and UE Antenna and beam forming pattern modelling of satellite could be referred to section 6.4.1 in TR 38.811[5].

The following normalized antenna gain pattern, corresponding to a typical reflector antenna with a circular aperture, is considered.

$$\begin{aligned}
 &1 && \text{for } \theta = 0 \\
 &4 \left| \frac{J_1(ka \sin \theta)}{ka \sin \theta} \right|^2 && \text{for } 0 < |\theta| \leq 90^\circ
 \end{aligned}$$

where:

- $J_1(x)$ is the Bessel function of the first kind and first order with argument 'x', where x is $ka \sin \theta$;
- a, is the radius of the antenna's circular aperture;
- $k = 2\pi f/c$ is the wave number;
- f is the frequency of operation;
- c is the speed of light in a vacuum and θ is the angle measured from the bore sight of the antenna's main beam.

Note that ka equals to the number of wavelengths on the circumference of the aperture and is independent of the operating frequency.

The antenna patterns for LEO 600km, 1200km and GEO are shown in Figure 6.2.3.1-1 and 6.2.3.1-2.

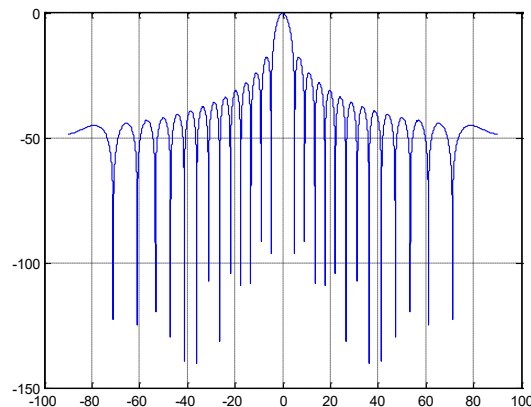


Figure 6.2.3.1-1: Antenna pattern for LEO 600KM and 1200KM (4.4127 deg for 3dB beamwidth)

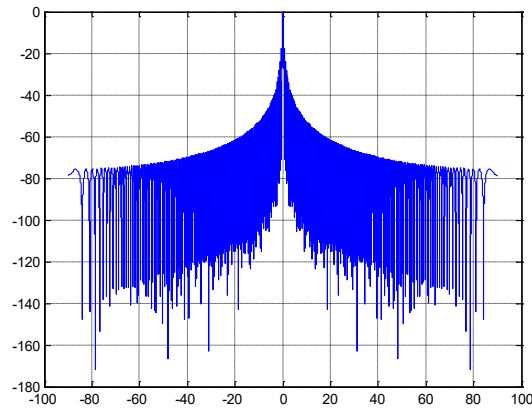
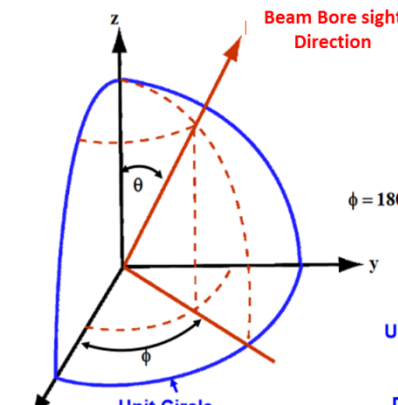
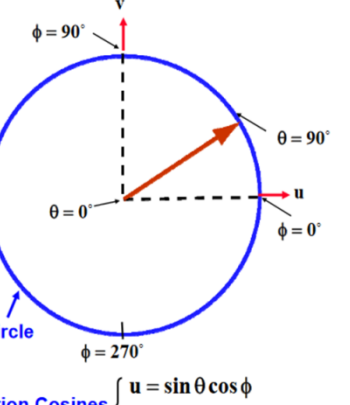
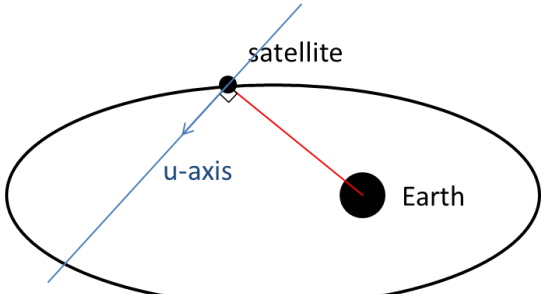
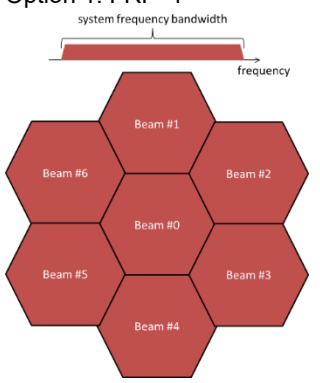
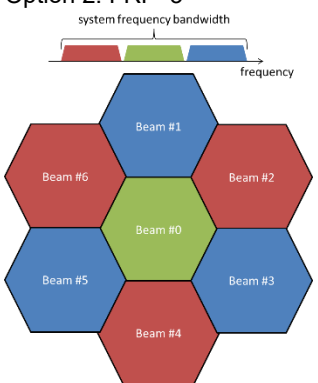
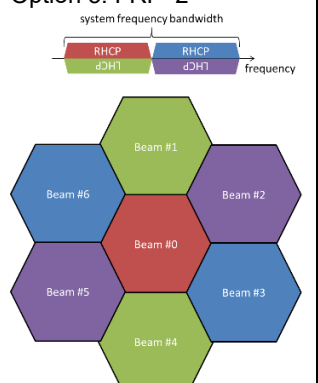


Figure 6.2.3.1-2: Antenna pattern for antenna aperture of GEO (0.4011 deg for 3dB beamwidth)

The beam layout definition for a single satellite simulation in S-Band is defined in Table 6.2.3.1-1.

Table 6.2.3.1-1: Beam layout definition for single satellite simulation

Beam layout definition	Baseline: Hexagonal mapping of the beam bore sight directions on UV plane defined in the satellite reference frame. Only the 3dB beam width parameters should be used. The beam diameter and beam spacing values can be computed directly from the 3 dB beam width assumptions and should be considered as informative.		
Number of beams	Baseline: 7-beam layout (i.e. 6 co-frequency beams surrounding the central beam)		
UV plane illustration (extracted from [19])	Spherical Coordinate System  Projection of Coordinate System On the X-Y Plane (view from above Z-axis)  Direction Cosines $\begin{cases} u = \sin \theta \cos \phi \\ v = \sin \theta \sin \phi \end{cases}$		
UV plane convention	U axis is defined as the perpendicular line to the satellite-earth line on the orbital plane as illustrated here after:  The straight line being orthogonal to UV plane is pointing towards the Earth centre. UV coordinates of the nadir of the reference satellite is (0,0)		
Adjacent beam spacing on UV plane	Baseline: Adjacent beam spacing computation based on 3dB beam width of the satellite antenna pattern: $ABS[rad] = \sqrt{3} \times \sin(HPBW[degrees]/2)$ or $ABS[rad] = \sqrt{3} \times \sin(HPBW[rad]/2)$ with $ABS [degree] = 180/\pi \times ABS[rad]$ and with HPBW the Half-Power BandWidth of the main lobe from the satellite antenna pattern.		
Central beam bore sight direction definition	Baseline: Case 1: Central beam center is considered at nadir point Case 2: 45° for GEO and LEO		
	Option 1: FRF=1 	Option 2: FRF=3 	Option 3: FRF=2 
Polarization re-use	Option 1: Disable Option 2: Enable Note: Polarization re-use should apply only if circular polarization for terminal antenna is considered		

UEs outdoor/indoor distribution	100% outdoor distribution for UEs
UE distribution	The cell area associated to a given beam is defined as the Voronoi cell associated with the corresponding beam centers.
UE configuration	S-band: Handheld
UE orientation	Handheld: Random
UE attachment	RSRP
NOTE 1: Typical impairment values (additional frequency error, SNR loss) due to the feeder link except for delay can be considered to be negligible. When available, specific values can be considered in the evaluation and should be reported.	
NOTE 2: For the calibration purpose, the ionospheric scintillation loss shall be considered equal to zero (i.e., the UEs are located between 20 and 60 degrees of latitude).	

6.2.3.2 TN BS and UE antenna and beam forming pattern modelling

For AAS antenna, it refers to the pattern in TR 38.921 [14].

Table 6.2.3.2-1: AAS antenna parameters for 2GHz

		Rural	Macro urban
1	Base Station Antenna Characteristics		
1.1	Antenna pattern	TR 38.921[14]	
1.2	Element gain (dBi) ^(Note 2)	7.1	6.4
1.3	Horizontal/vertical 3 dB beam width of single element (degree)	90° for H 54° for V	90° for H 65° for V
1.4	Horizontal/vertical front-to-back ratio (dB)	30 for both H/V	30 for both H/V
1.5	Antenna polarization	Linear $\pm 45^\circ$	Linear $\pm 45^\circ$
1.6	Antenna array configuration (Row $\times$ Column) ^(Note 4)	8 $\times$ 8 elements	8 $\times$ 8 elements
1.7	Horizontal/Vertical radiating element spacing	0.5 of wavelength for H, 0.9 of wavelength for V	0.5 of wavelength for H, 0.7 of wavelength for V
1.8	Array Ohmic loss (dB) ^(Note 2)	2	2
1.9	Conducted power (before Ohmic loss) per antenna element (dBm) ^(Note 3)	25	25
1.10	Base station maximum coverage angle in the horizontal plane (degrees)	120	120
1.11	Base station vertical coverage range (degrees) ^(Note 1)	90-100	90-120
1.12	Mechanical downtilt (degrees)	3	10

For non-AAS antenna, the parameters in Table 6.2.3.2-1 are used for 2GHz BS antenna pattern in the NTN system simulation.

Table 6.2.3.2-1: FR1 BS Non-AAS antenna pattern for 2GHz

Parameter for BS	Values	
Antenna vertical radiation pattern (dB)	$A_{E,V}(\theta'') = -\min\left\{12\left(\frac{\theta'' - 90^\circ}{\theta_{3dB}}\right)^2, SLA_V\right\}, \theta_{3dB} = 10^\circ, SLA_V = 20\text{dB}$	
Antenna horizontal radiation pattern (dB)	$A_{E,H}(\phi'') = -\min\left\{12\left(\frac{\phi''}{\phi_{3dB}}\right)^2, A_m\right\}, \phi_{3dB} = 65^\circ, A_m = 20\text{dB}$	
Combining method for 3D antenna pattern (dB)	$A''(\theta'', \phi'') = -\min\{-[A_{E,V}(\theta'') + A_{E,H}(\phi'')], A_m\}$	
Maximum directional gain of an antenna $G_{E,max}$	17 dBi	
Conducted power	46 dBm	
Mechanical downtilt in degrees	Rural	3
	Urban	10

Only Non-AAS antenna can be used for NB-IoT.

For UE antennas, an omni-directional radiation pattern with antenna gain 0dBi is assumed.

6.2.3.3 HAPS antenna model

HAPS antenna model is based on the proposal for HIBS (HAPS as IMT base stations) study in ITU WP-5D [24]. The antenna array is composed of seven antenna panels (six side panels and one downward facing panel) as shown in Figure 6.2.3.3-1. Antenna elements on each panel are co-phased to form one beam in two crossed linear polarizations to serve one cell. There are a total of seven cells in two layers, one cell in the 1st layer and six cells in the 2nd layer. The antenna parameters are listed in Table 6.2.3.3-1. Note that a fixed beam in the direction of the panel boresight is formed by each panel to serve one cell. Figure 6.2.3.3-1 shows the antenna gain of the 1st layer cell and a 2nd layer cell projected on the ground.

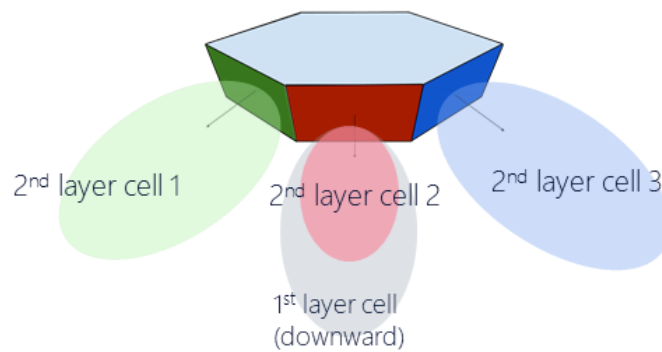
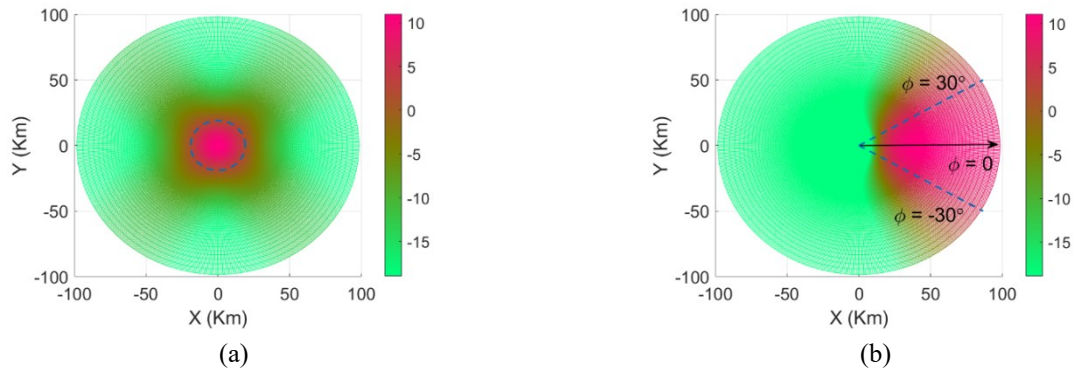
**Figure 6.2.3.3-1: HAPS antenna model**

Table 6.2.3.3-1: HAPS antenna parameters

Number of cells	7
Antenna array configuration (row x column)	2 x 2 for 1st layer cell 4 x 2 for 2nd layer cell
Antenna polarization	Linear $\pm 45^\circ$
Element gain ¹	7.8 dBi
Element HPBW horizontal/vertical	65° for both H/V
Element front-to-back ratio horizontal/vertical	30 dB for both H/V
Element spacing horizontal/vertical	0.7 wavelength for both H/V
Antenna panel tilt (from the horizon) ²	90° for 1st layer cell 23° for 2nd layer cell
EIRP/cell	56.8 dBm (1st layer cell), 59.8 dBm (2nd layer cell)
EIRP spectral density/cell	43.8 dBm/MHz (1st layer cell), 46.8 dBm/MHz (2nd layer cell)
Tx power per antenna panel	43 dBm
Conducted power (before ohmic loss) per antenna element (dBm)	34 dBm for 2 x 2 (x 2 polarizations for 1 st layer cell) 31 dBm for 4 x 2 (x 2 polarizations for 2 nd layer cell)
Antenna weight ³ $w_{m,n}$	1/2 for each of 1st layer cell element 1/2 $\sqrt{2}$ for each of 2nd layer cell element
Polarization gain for SINR	3 dB
NOTE 1: The element gain includes the Ohmic loss of 2 dB.	
NOTE 2: This is mechanical tilt only.	
NOTE 3: These antenna weights create a fixed cell beam in the antenna panel's boresight direction.	

**Figure 6.2.3.3-2: HAPS antenna gain (in dB) in a 100 km radius area on the ground. (a) Antenna gain of the 1st layer cell. (b) Antenna gain of a 2nd layer cell.**

6.2.4 ACIR model

ACLR modelling for TN and NTN co-existence study referring to clause 5.1.1.4.1 and 5.1.1.4.2 in TR 36.942[8] is used as baseline. The number of RBs refers to Table 6.2.2.1-4 and Table 6.2.2.4-1 respectively.

6.2.5 Propagation model

6.2.5.1 Propagation model between NTN SAN and UE

Propagation model between NTN SAN and UE should reference to section 6.6 in TR 38.811[5].

6.2.5.2 Propagation model between TN BS and UE

Propagation model between TN BS and UE should reference to section 7.4 in TR 38.901[10].

6.2.5.3 Propagation model between NTN SAN and TN BS

Propagation model between NTN SAN and TN BS should reference to TS 38.811[5] which is used for DL-UL cross link interference for S band.

6.2.5.4 Propagation model for HAPS

Propagation model between HAPS and UE, regardless of whether the UE is connected to HAPS or TN, should reference to section 6.6 in TR 38.811[5].

6.2.6 Transmission power control model

6.2.6.1 TN UL TPC

For uplink scenario, TPC model specified in Section 9.1 TR 36.942[8] is applied for TN with following parameters.

$$P_t = P_{\max} \times \min \left\{ 1, \max \left[R_{\min}, \left(\frac{CL}{CL_{x-ile}} \right)^\gamma \right] \right\}$$

where:

- $P_{\max} = 23\text{dBm}$,
- $R_{\min} = \text{TBD dB}$,
- CL_{x-ile} and γ are set as following:
 - $CL_{x-ile} = 88 + 10 \cdot \log_{10}(200/X) + 11 - Y$, where X is UL transmission BW (MHz) and Y is the BS noise figure
 - $\gamma = 1$ For uplink scenario,

6.2.6.2 NTN UL TPC

For the coexistence study, the same TPC model of TN for NTN UL scenarios is adopted to align with UE UL power control parameters used in TR 38.821[6].

6.2.6.3 NTN DL TPC

For downlink scenario, no power control scheme is applied.

6.2.6.4 HAPS UL TPC

HAPS uplink follows the TPC model in Section 6.2.6.1 with $X = 1.08$ (scheduled bandwidth 6 RBs) and $Y = 5$ (HAPS noise figure is 5 dB).

6.2.6.5 HAPS DL TPC

For downlink scenario, no power control scheme is applied.

6.2.7 Received power model

The received power in downlink and uplink scenarios is defined as below:

$$RX_PWR = TX_PWR - Path\ loss + G_TX + G_RX$$

where:

- RX_PWR is the received power
- TX_PWR is the transmitted power
- G_TX is the transmitter antenna gain (directional array gain)
- G_RX is the receiver antenna gain (directional array gain).

6.2.8 Performance metric

For NR, the average throughput loss and 5%-ile throughput loss should be less than 5%.

For NB-IOT, the SNR loss should refer to TR 36.802 [15].

For NTN, the average throughput loss and 5%-ile throughput loss should be less than 5%.

6.2.9 Throughput ~ SNR mapping

Adopt Section 5.2.7 of TR 38.803[20] as the SINR-Throughput performance metrics

6.3 Co-existence simulation methodology

6.3.1 Simulation procedure

Adopt following simulation steps.

- Step 1: Generate aggressor and victim networks.
 - NTN central beam is at satellite nadir, surrounded with 6 co-frequency beams. NTN FRFs higher than 1 need to be considered. Assume one NTN aggressor as default.
 - Deployment of TN network (19 cells with wraparound) refers to Table 6.2.1.1-1
- Step2: UE associations
 - TN UE are generated randomly inside the TN network, make sure enough TN UEs are associated to each TN sectors based on coupling loss.
 - Deployment of NTN UE refers to Table 6.2.1.1-1.
- Step 3: Once association is done, round robin scheduling is used. BF weights are adjusted to point to the LOS direction between BS/SAN-UE. This is done for both victim and aggressor networks.
- Step 4: Throughput is computed in the victim systems without considering ACI as below:

$$Thput_{NO\ ACI}[bpsHz] = f(SINR_{ICI}) = f\left(\frac{S}{N+I_{ICI}}\right),$$

where: I_{ICI} is the inter-cell interference.

For TN-NTN SINR calculation, the satellite receiver off angle should be considered in the satellite receiver gain calculation when calculating SINR. Note that such angle is not considered in TR 38.821[6] section 6.1.3 equations. Thus those equations should be used for SINR calculation.

- Step 5: Throughput is computed considering ACI as below:

$$Thput_{ACI}[bpsHz] = f(SINR_{ICI+ACI}) = f\left(\frac{S}{N+I_{ICI}+I_{ACI}}\right),$$

where: I_{ACI} is the adjacent channel interference.

- Step 6: RF parameters are determined based on the degradation cause by ACI as below:

$$Loss_{ACI} = 1 - \frac{Thput_{ACI}}{Thput_{SINGLE}}$$

To simplify the simulation of interference from TN to NTN UL in Case 2 and 6, following method can be used. Consider the active TN cells from central NTN beam for the ACI evaluation from TN to NTN UL. The scaling factor is to be discussed and determined if any in next meeting. There is a view that simplifying such coexistence simulation work for Case 2 may even not be required.

- Step 1: to drop NTN UE per beamprint randomly;
- Step 2: to drop N clusters consisting of 57 sectors per beamprint randomly;
- Step 3: to calculate the total ACI per beam to NTN UL by following scaling factor:

$$P_{TotalPerBeam,j} = \frac{active_TN}{N} \cdot \sum_i^N P_i$$

where: active_TN = **active_factor***round (the area per beam/the area of 57 sectors)
 active_factor = 20% (or lower, particularly for urban scenarios)

- Step 4: to calculate the total ACI from all beams (e.g. M=7) for NTN:

$$P_{total} = \sum_j^M P_{TotalPerBeam,j}$$

6.3.2 Methods and principle to process co-existence simulation results

In order to process the co-existence simulation results received for all different scenarios and assumptions, the following steps are adopted:

- Step 1: Discuss and agree on the most stringent scenario(s) for each scenario (Scenario 1, 2, 3...,6);
- Step 2: Discuss and determine the required ACIR from results of the most stringent case(s) for each scenario;
- Step 3: Use equation to derive corresponding ACLR or ACS from the agreed ACIR for each scenario

It is noted that the averaged ACIR for the most stringent case in each scenario would be derived by taking the average among the interpolated ACIR results derived from each company's results for that case.

Moreover, the following considerations are adopted to deal with major disputes for the selected cases' results in each scenario:

- If the required ACIR results, from the contributor who did not participate or their results is still not well-aligned in calibration table, has a difference larger than 10 dB with most others, this result can be not considered in the discussion.
- If the required ACIR results, from one contributor, has a difference larger than 10 dB with most others, this result can be not considered in the discussion.

The processed results by adopting above principles and methods for scenarios 1 to 7 identified in Table 6.1-2 are captured in section 6.4. It is noted that due to the space limitation, only part of the simulation results for each case are presented, and the whole results for all studied options, as listed in Table 6.1-1 and section 6.2, can be found in Annex C.

Table 6.3.2-1: Selected option for each scenario

Scenario	Aggressor system	Victim system	Environment	Contributing
1	TN DL	NTN GEO DL	Urban	NTN UE ACS
2	TN UL	NTN GEO UL	Urban	NTN SAN ACS
3	NTN LEO-600 DL	TN DL	Rural	NTN SAN LEO ACLR
	NTN GEO DL	TN DL	Rural	NTN SAN GEO ACLR
4	NTN GEO UL	TN UL	Urban	NTN UE ACLR
5	NTN GEO UL	TN DL	Rural	NTN UE ACLR
6 ¹	NR-TN DL	NTN LEO-600 UL	Rural ²	NTN SAN ACS
	NR-TN DL	NTN GEO UL	Rural ²	NTN SAN ACS
7	HAPS DL	TN DL	Rural	HAPS ACLR
8	TN UL	HAPS UL	Rural	HAPS ACS
NOTE 1: Agreed representative case for Scenario 6. NOTE 2: The initial results suggested that the NR-NTN SAN would suffer more interference in urban deployment scenario. It is agreed that a more relevant environment for case 6 is a mixture of Urban and Rural environment (e.g., urban area with a 50km diameter inside a GEO beam with a 250km diameter). Further studies based on the mixed urban environment could be considered. As compromise, rural only scenario was then selected.				

6.4 Co-existence simulation results

6.4.1 Scenario 1: 2GHz TN DL interfering NTN DL

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select the NR DL equipped with both AAS and non-AAS antenna interfering the NR-NTN GEO DL that deployed in urban environment as the most stringent case.

Table 6.4.1-1: Simulation results for average throughput loss - TN BS with AAS antenna

ACIR[dB]	6	8	10	12	14	16	18	20	22	24
Qualcomm	32.76	26.13	19.50	15.41	11.32	8.26	6.21	4.16	3.22	2.27
MTK	7.28	5.71	4.60	3.77	3.05	2.35	1.90	1.30	1.02	0.80
ZTE	31.76	24.81	18.95	14.18	10.47	7.63	5.50	3.92	2.79	1.99
Ericsson			4.2	3.0	2.1	1.5	1.1			
CATT	8.7	6.5	5.3	4.3						
Xiaomi	38.11	31.51	25.56	20.35	15.94	12.29	9.36	7.06	5.29	3.95

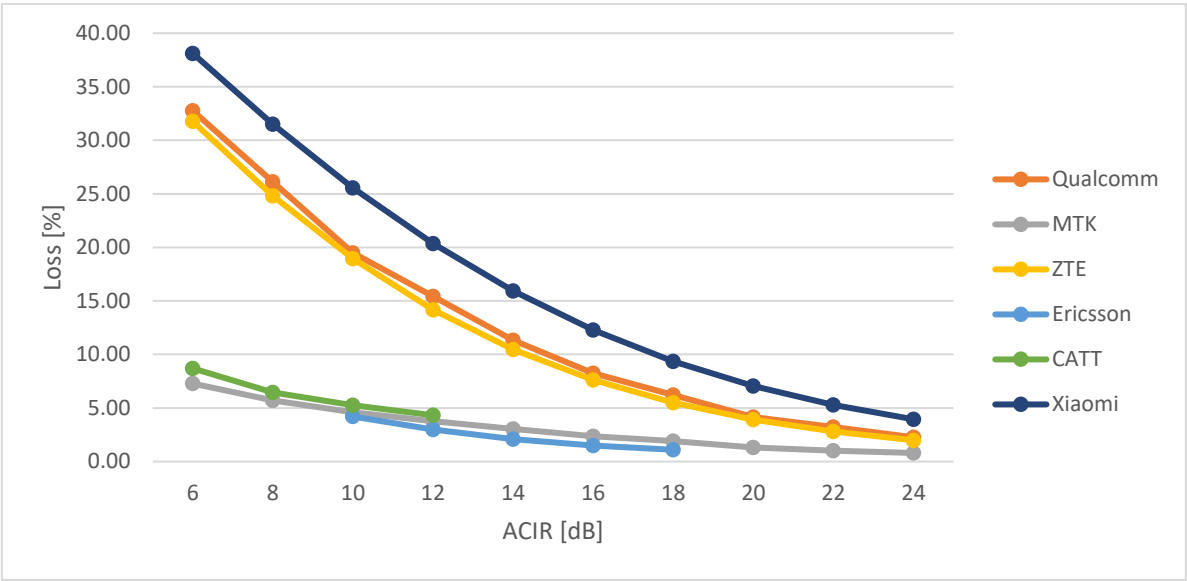


Figure 6.4.1-1: Simulation results for average throughput loss - TN BS with AAS antenna

Table 6.4.1-2: Simulation results for 5%-tile throughput loss - TN BS with AAS antenna

ACIR[dB]	14	16	18	20	22	24	26	28	30	32
Qualcomm	44.68	27.01	19.33	11.66	9.31	6.96	5.20	4.03		
MTK	7.38	4.80	3.09	1.97	1.26	0.80	0.50	0.32	0.20	0.13
ZTE	33.63	24.27	17.09	12.37	9.01	6.43	4.81	3.30	2.42	1.70
Ericsson	3.0	2.8	1.8							
CATT	13.4	12.7	11.8	9.33	8.76	8.18	7.65	5.83	4.91	
Xiaomi	65.22	54.94	44.26	33.58	24.20	16.78	11.29	7.43	4.51	

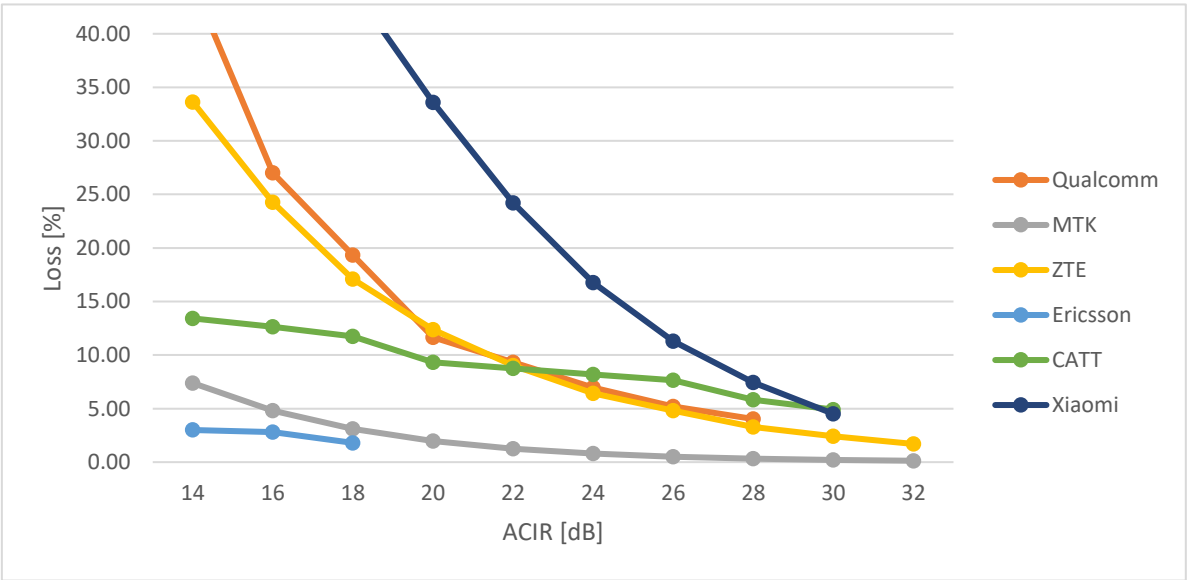


Figure 6.4.1-2: Simulation results for 5%-tile throughput loss - TN BS with AAS antenna

Table 6.4.1-3: Interpolated ACIR values for Scenario 1 to meet the 5% throughput loss criteria - TN BS with AAS antenna

Source		Interpolated ACIR[dB]
Qualcomm	Average	19.18
	5%-tile	26.34
MTK	Average	9.28
	5%-tile	15.85 ¹
ZTE	Average	18.63
	5%-tile	25.77
Ericsson	Average	
	5%-tile	12.10 ¹
CATT	Average	10.6
	5%-tile	29.80
Xiaomi	Average	22.43
	5%-tile	29.40
NOTE 1: According to the principles, these values are not treated for later process.		

Table 6.4.1-4: Average ACIR of 5%-tile values in the above worse case for Scenario 1 - TN BS with AAS antenna

	Scenario 1
ACIR value [dB]	27.83

Table 6.4.1-5: Simulation results for average throughput loss - TN BS with non-AAS antenna

ACIR[dB]	6	8	10	12	14	16	18	20	22	24
MTK	25,91	20,30	16,17	12,92	9,97	8,19	6,76	5,54	4,32	3,34
ZTE	35,27	27,40	20,83	15,57	11,46	8,34	6,04	4,32	3,06	2,14
THALES	NA	NA	NA	13,13	9,54	6,85	4,86	3,4	2,3	1,58

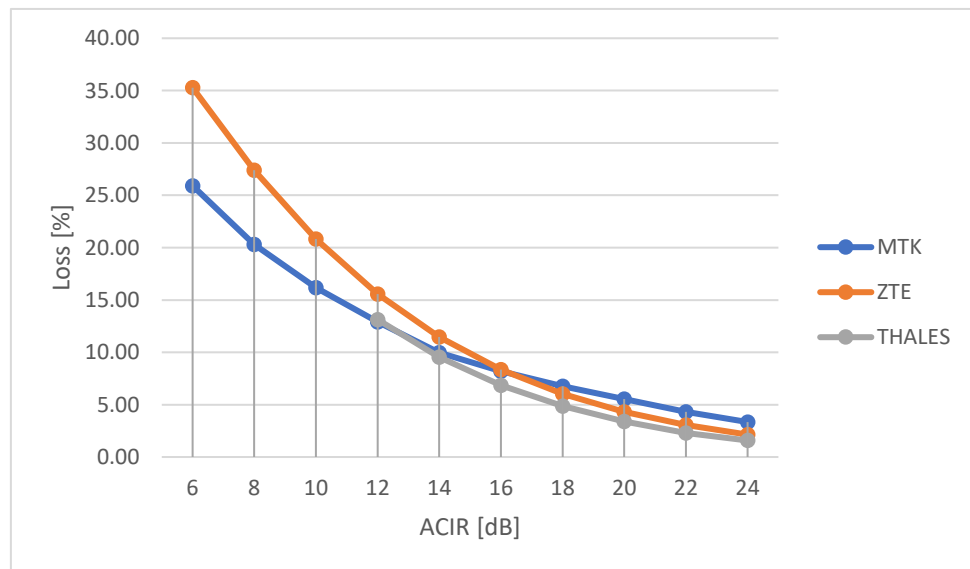
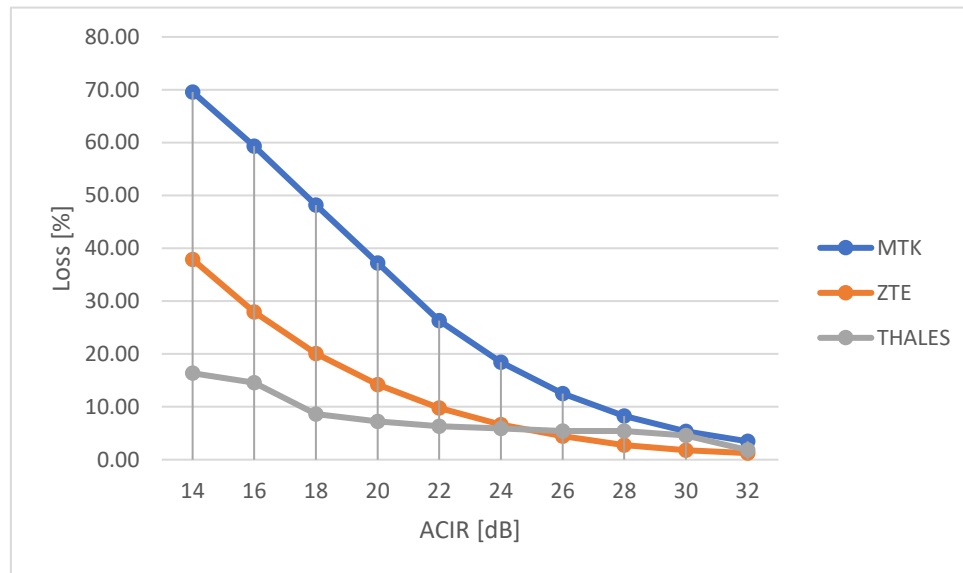


Figure 6.4.1-3: Simulation results for average throughput loss - TN BS with non-AAS antenna

Table 6.4.1-6: Simulation results for 5%-tile throughput loss - TN BS with non-AAS antenna

ACIR[dB]	14	16	18	20	22	24	26	28	30	32
MTK	69,59	59,31	48,18	37,24	26,31	18,42	12,49	8,27	5,39	3,47
ZTE	37,88	27,97	20,08	14,18	9,80	6,65	4,46	2,78	1,82	1,24
THALES	16,36	14,55	8,64	7,27	6,36	5,91	5,45	5,45	4,55	1,82

**Figure 6.4.1-4: Simulation results for 5%-tile throughput loss - TN BS with non-AAS antenna****Table 6.4.1-7: Interpolated ACIR values for Scenario 1 to meet the 5% throughput loss criteria - TN BS with non-AAS antenna**

Source		Interpolated ACIR[dB]
MTK	Average	20.89
	5%-tile	30.41
ZTE	Average	19.21
	5%-tile	25.51
Thales	Average	17.86
	5%-tile	29.00

Table 6.4.1-8: Average ACIR of 5%-tile values in the above worse case for Scenario 1 - TN BS with non-AAS antenna

	Scenario 1
ACIR value [dB]	28.31

6.4.2 Scenario 2: 2GHz TN UL interfering NTN UL

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select the NR UL interfering the NR-NTN GEO UL that deployed in urban environment as the most stringent case.

Table 6.4.2-1: Simulation results for average throughput loss - TN BS with AAS antenna

ACIR[dB]	18	20	22	24	26	28	30	32	34	36
Qualcomm	36.46	24.81	18.43	12.06	7.70	5.34	2.98			
ZTE	16.46	11.77	8.26	5.51	3.55	2.25	1.48	0.98	0.63	0.39
MTK	38.61	31.96	25.57	20.13	15.27	11.22	7.96	5.48	3.68	2.42
Ericsson		15.6	10.4	7.2	4.2	2.9	1.9			
CATT	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Xiaomi	41.11	30.58	21.75	14.92	9.96	6.53	4.22	2.70	1.72	1.09

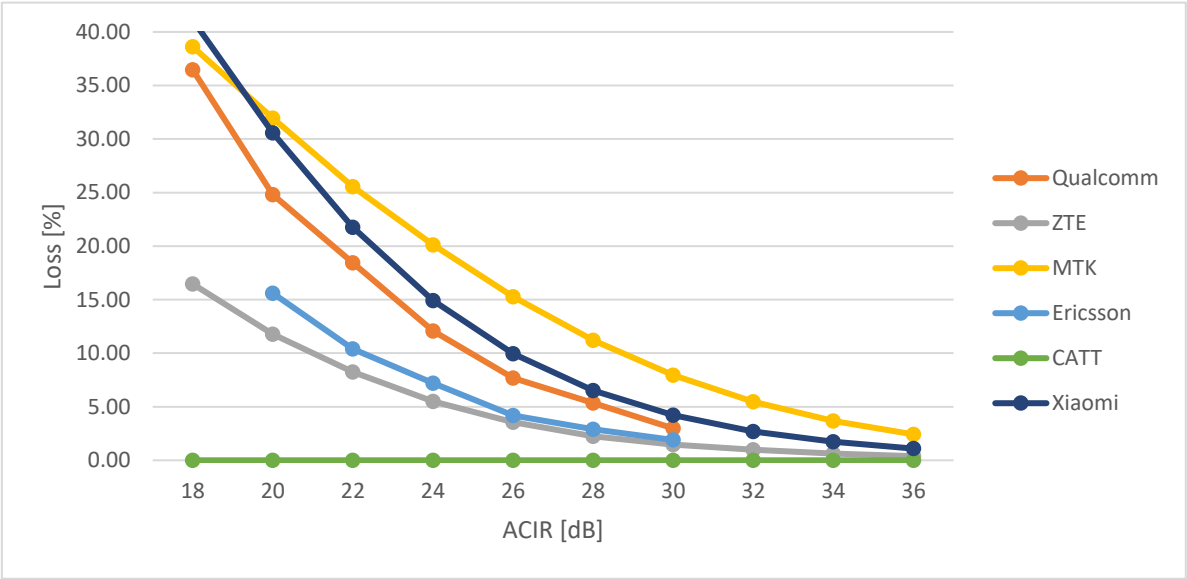


Figure 6.4.2-1: Simulation results for average throughput loss - TN BS with AAS antenna

Table 6.4.2-2: Simulation results for 5%-tile throughput loss - TN BS with AAS antenna

ACIR[dB]	22	24	26	28	30	32	34	36	38	40
Qualcomm	NA	NA	NA	NA	NA					
ZTE	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
MTK	68.80	58.62	47.26	36.43	26.97	19.13	13.06	8.68	5.67	3.67
Ericsson	NA	NA	NA	NA	NA					
CATT	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Xiaomi	NA	NA	NA	NA	NA	NA	NA	NA	NA	

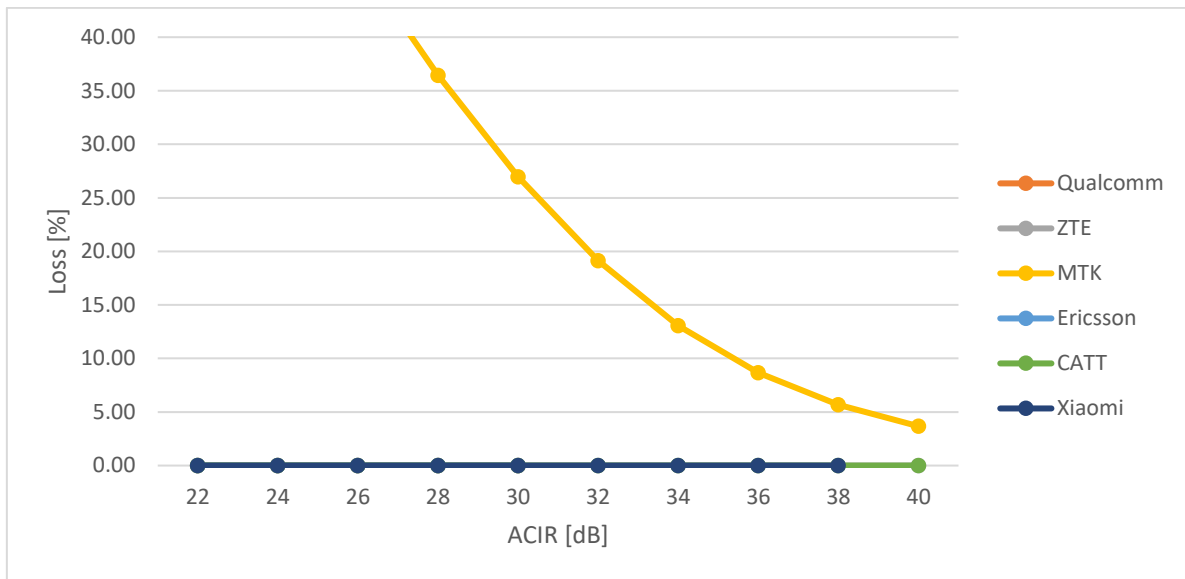


Figure 6.4.2-2: Simulation results for 5%-tile throughput loss - TN BS with AAS antenna

Table 6.4.2-3: Interpolated ACIR values for Scenario 2 to meet the 5% throughput loss criteria - TN BS with AAS antenna

Source		Interpolated ACIR[dB]
Qualcomm	Average	28.29
	5%-tile	
ZTE	Average	24.52
	5%-tile	
MTK	Average	32.53
	5%-tile	38.67 ¹
Ericsson	Average	25.47
	5%-tile	
CATT	Average	
	5%-tile	
Xiaomi	Average	29.32
	5%-tile	
NOTE 1: According to the principles, this value is not treated for later process.		

Table 6.4.2-4: Average ACIR values in the above worse case for Scenario 2 - TN BS with AAS antenna

	Scenario 2
ACIR value [dB]	28.03

Table 6.4.2-5: Simulation results for average throughput loss - TN BS with non-AAS antenna

ACIR[dB]	18	20	22	24	26	28	30	32	34	36
Ericsson	NA	NA	NA	NA	NA	10,4	6,8	4,2	2,9	NA
THALES	14,06	10,74	7,74	5,53	3,63	2,37	1,74	1,11	0,79	0,47
MTK	35,49	29,12	23,27	18,21	13,81	9,87	6,92	4,67	3,11	2,04

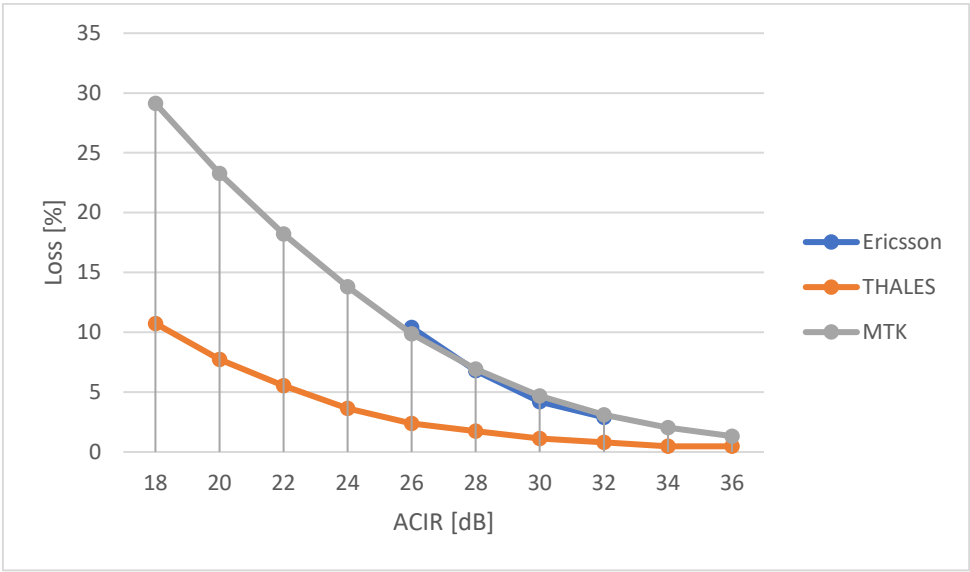


Figure 6.4.2-3: Simulation results for average throughput loss - TN BS with non-AAS antenna

Table 6.4.2-6: Simulation results for 5%-tile throughput loss - TN BS with non-AAS antenna

ACIR[dB]	22	24	26	28	30	32	34	36	38	40
Ericsson	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
THALES	0	0	0	0	0	0	0	0	0	0
MTK	57,18	45,86	34,92	25,60	18,05	12,29	8,15	5,31	3,42	2,19

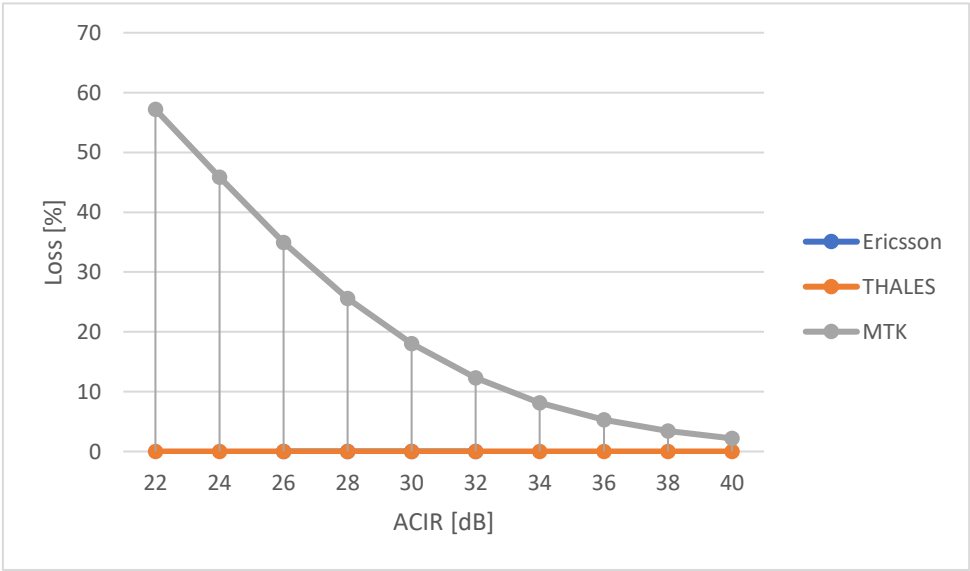


Figure 6.4.2-4: Simulation results for 5%-tile throughput loss - TN BS with non-AAS antenna

Table 6.4.2-7: Interpolated ACIR values for Scenario 2 to meet the 5% throughput loss criteria - TN BS with non-AAS antenna

Source		Interpolated ACIR[dB]
Ericsson	Average	31.38
	5%-tile	NA
Thales	Average	24.56
	5%-tile	NA
MTK	Average	31.71
	5%-tile	36.33

Table 6.4.2-8: Average ACIR values in the above worse case for Scenario 2 TN BS with non-AAS antenna

	Scenario 2
ACIR value [dB]	29.22

6.4.3 Scenario 3: 2GHz NTN DL interfering TN DL

6.4.3.1 LEO Class

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select the NR-NTN LEO-600 DL interfering the NR DL equipped with AAS antenna that deployed in rural environment as the most stringent case for LEO class.

Table 6.4.3.1-1: Simulation results for average throughput loss - TN BS with AAS antenna

ACIR[dB]	10	12	14	16	18	20	22	24	26	28
Qualcomm	21.02	16.68	12.34	8.94	6.48	4.03	2.98	1.93	1.21	0.84
Samsung	20.06	15.28	11.24	8.00	5.52	3.72	2.45	1.60	1.03	0.66
MTK	25.22	19.06	13.86	9.70	6.66	4.41	2.92	1.90	1.22	0.77
ZTE	16.65	12.34	8.84	6.15	4.17	2.77	1.81	1.17	0.75	0.48
Ericsson						3.7	2.4	1.6	1.0	
Huawei					5.94	3.97	2.52	1.64	1.24	
CATT	17.3	12.2	10.1	8.3	6.6	4.7				
Xiaomi	30.71	23.92	17.92	12.93	9.01	6.11	4.05	2.64	1.70	1.09

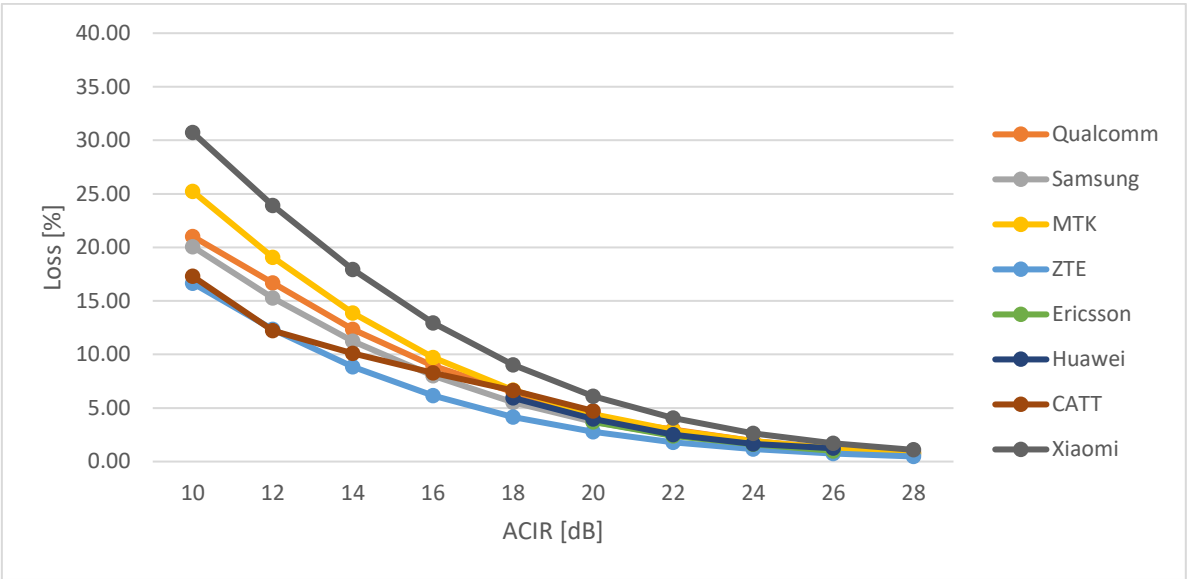


Figure 6.4.3.1-1: Simulation results for average throughput loss - TN BS with AAS antenna

Table 6.4.3.1-2: Simulation results for 5%-tile throughput loss - TN BS with AAS antenna

ACIR[dB]	14	16	18	20	22	24	26	28	30	32
Qualcomm	37.78	28.51	21.12	13.72	10.16	6.60	4.18	2.89		
Samsung	27.10	19.19	13.13	8.76	5.73	3.71	2.37	1.51	0.97	0.62
MTK	37.56	27.47	19.31	13.11	8.69	5.67	3.65	2.33	1.49	0.94
ZTE	16.27	10.85	7.06	4.59	2.82	1.91	1.31	0.88	0.60	0.39
Ericsson				9.8	6.2	4.5	2.8			
Huawei			8.61	5.01	3.12	1.94	1.32			
CATT	42.07	32.30	22.43	15.88	10.74	7.90	4.40			
Xiaomi	38.87	28.90	20.59	14.15	9.47	6.22	4.03	2.59		

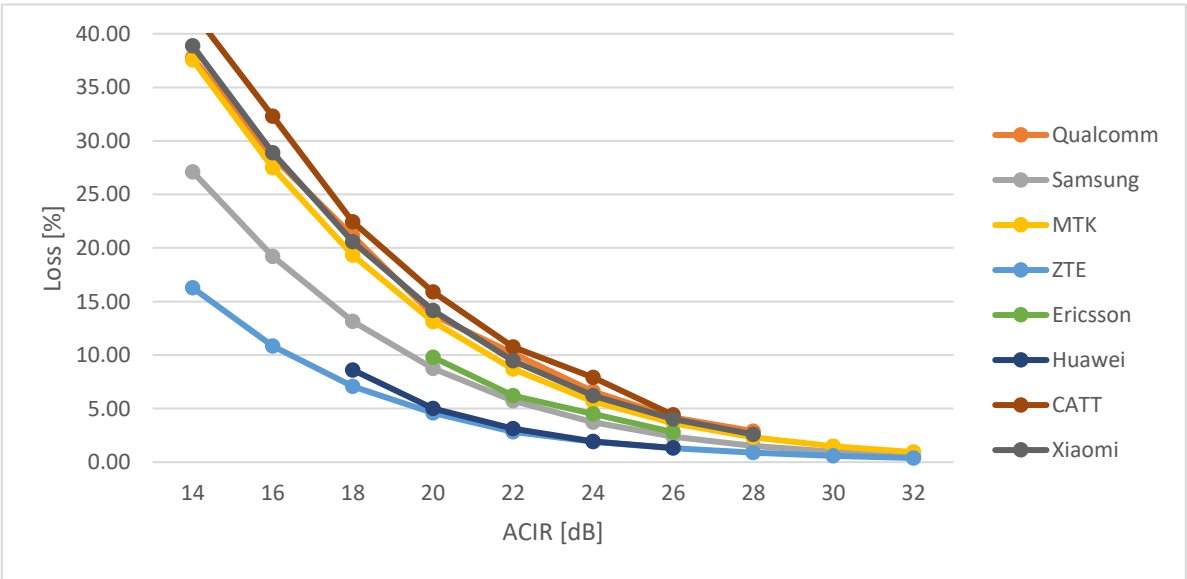


Figure 6.4.3.1-2: Simulation results for average throughput loss - TN BS with AAS antenna

Table 6.4.3.1-3: Interpolated ACIR values for Scenario 3 to meet the 5% throughput loss criteria - TN BS with AAS antenna

Source		Interpolated ACIR[dB]
Qualcomm	Average	19.21
	5%-tile	25.32
Samsung	Average	18.58
	5%-tile	22.72
MTK	Average	19.48
	5%-tile	24.66
ZTE	Average	17.16
	5%-tile	19.67
Ericsson	Average	
	5%-tile	23.41
Huawei	Average	18.95
	5%-tile	20.01
CATT	Average	19.68
	5%-tile	25.66
Xiaomi	Average	21.01
	5%-tile	25.11

Table 6.4.3.1-4: Average ACIR of 5%-tile values in the above worse case for Scenario 3 - TN BS with AAS antenna

	Scenario 3
ACIR value [dB]	23.32

Table 6.4.3.1-5: Simulation results for average throughput loss - TN BS with non-AAS antenna

ACIR[dB]	10	12	14	16	18	20	22	24	26	28
Samsung	10,42	7,20	4,86	3,21	2,09	1,35	0,86	0,55	0,35	0,22
MTK	27,33	20,75	15,22	10,59	7,32	4,81	3,29	2,16	1,45	0,93
ZTE	13,12	9,28	6,37	4,27	2,81	1,82	1,17	0,75	0,48	0,30
THALES	NA	15,84	11,43	7,99	5,43	3,61	2,35	1,52	0,97	0,62
Ericsson	NA	NA	NA	NA	NA	3,0	2,0	1,3	0,8	0,5

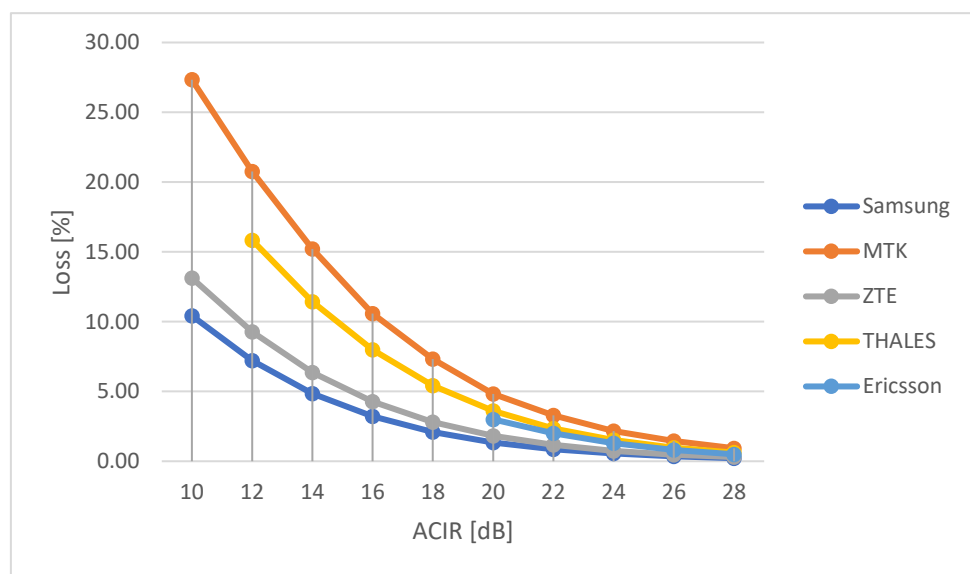
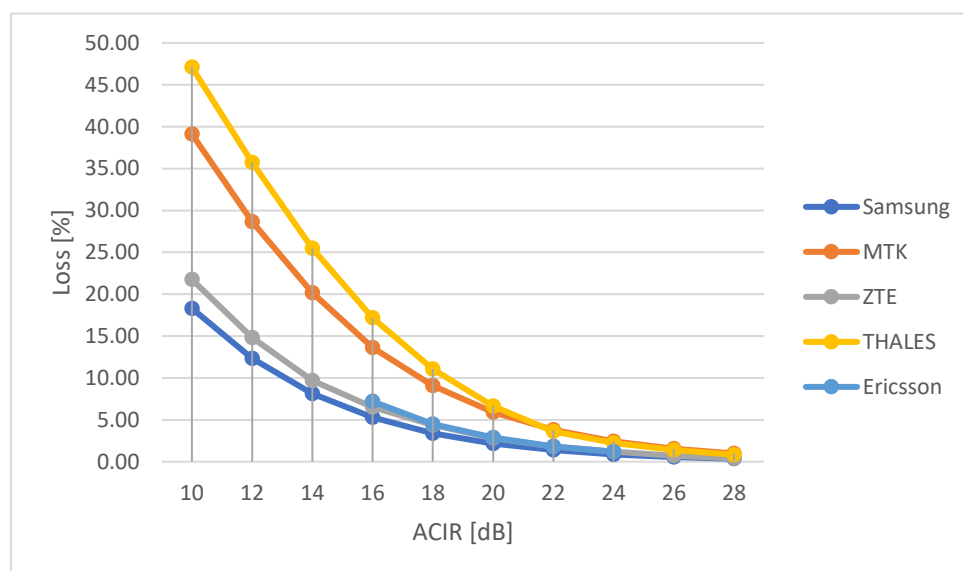


Figure 6.4.3.1-3: Simulation results for average throughput loss - TN BS with non-AAS antenna

Table 6.4.3.1-6: Simulation results for 5%-tile throughput loss - TN BS with non-AAS antenna

ACIR[dB]	14	16	18	20	22	24	26	28	30	32
Samsung	18,31	12,33	8,12	5,28	3,40	2,17	1,38	0,87	0,55	0,35
MTK	39,11	28,66	20,18	13,63	9,06	5,92	3,81	2,44	1,55	0,98
ZTE	21,75	14,80	9,68	6,52	4,36	2,83	1,82	1,20	0,71	0,39
THALES	47,09	35,73	25,48	17,17	11,08	6,65	3,6	2,22	1,39	0,83
Ericsson	NA	NA	NA	7,2	4,5	2,9	1,8	1,2	NA	NA

**Figure 6.4.3.1-4: Simulation results for average throughput loss - TN BS with non-AAS antenna****Table 6.4.3.1-7: Interpolated ACIR values for Scenario 3 LEO Class to meet the 5% throughput loss criteria - TN BS with non-AAS antenna**

Source		Interpolated ACIR[dB]
Samsung	Average	13.88
	5%-tile	20.30
MTK	Average	19.85
	5%-tile	24.87
ZTE	Average	15.3
	5%-tile	21.41
THALES	Average	18.47
	5%-tile	25.08
Ericsson	Average	
	5%-tile	21.63

Table 6.4.3.1-8: Average ACIR of 5%-tile values in the above worse case for Scenario 3 LEO Class - TN BS with non-AAS antenna

	Scenario 3
ACIR value [dB]	22.66

6.4.3.2 GEO Class

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select the NR-NTN GEO DL interfering the NR DL equipped with AAS antenna that deployed in rural environment as the most stringent case for GEO class.

Table 6.4.3.2-1: Simulation results for average throughput loss - TN BS with AAS antenna

ACIR[dB]	2	4	6	8	10	12	14	16	18	20
Qualcomm	15.65	11.48	8.25	5.96	3.67	2.70	1.74	1.09		
Samsung	15.14	11.09	7.86	5.41	3.63	2.39	1.55	1.00	0.64	0.41
MTK	20.35	15.03	10.74	7.33	4.79	3.15	2.05	1.32	0.84	0.53
ZTE	11.39	8.09	5.58	3.76	2.48	1.62	1.04	0.67	0.42	0.27
Ericsson					3.9	2.5	1.6	1.0		
CATT	9.73%	7.94%	6.42%	5.15%	4.08%	3.20%	2.48%			
Xiaomi	22.47	16.68	11.93	8.25	5.56	3.67	2.39	1.54	0.98	0.62

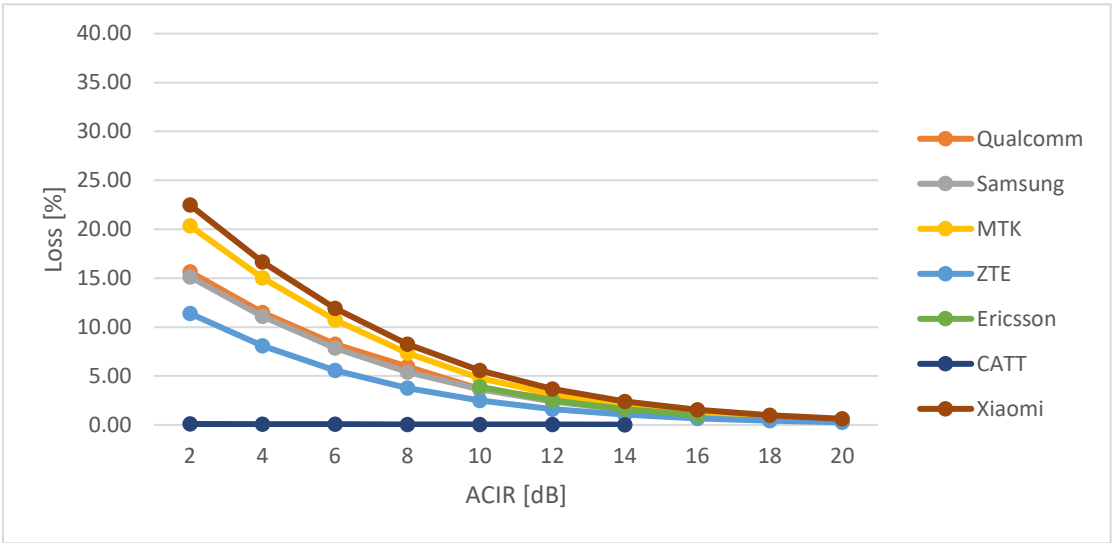


Figure 6.4.3.2-1: Simulation results for average throughput loss - TN BS with AAS antenna

Table 6.4.3.2-2: Simulation results for 5%-tile throughput loss - TN BS with AAS antenna

ACIR[dB]	4	6	8	10	12	14	16	18	20	22
Qualcomm	35.59	26.64	19.54	12.44	9.23	6.02	3.81			
Samsung	26.43	18.68	12.77	8.49	5.55	3.58	2.30	1.46	0.93	0.59
MTK	42.08	31.47	22.53	15.53	10.41	6.83	4.43	2.84	1.81	1.15
ZTE	14.19	9.38	6.19	4.19	2.58	1.62	1.00	0.57	0.38	0.25
Ericsson				8.3	5.6	3.5	2.2			
CATT	20.86	19.31	14.49	13.09	7.33	5.23	4.17			
Xiaomi	36.44	26.85	18.99	12.97	8.65	5.67	3.66	2.35	1.50	

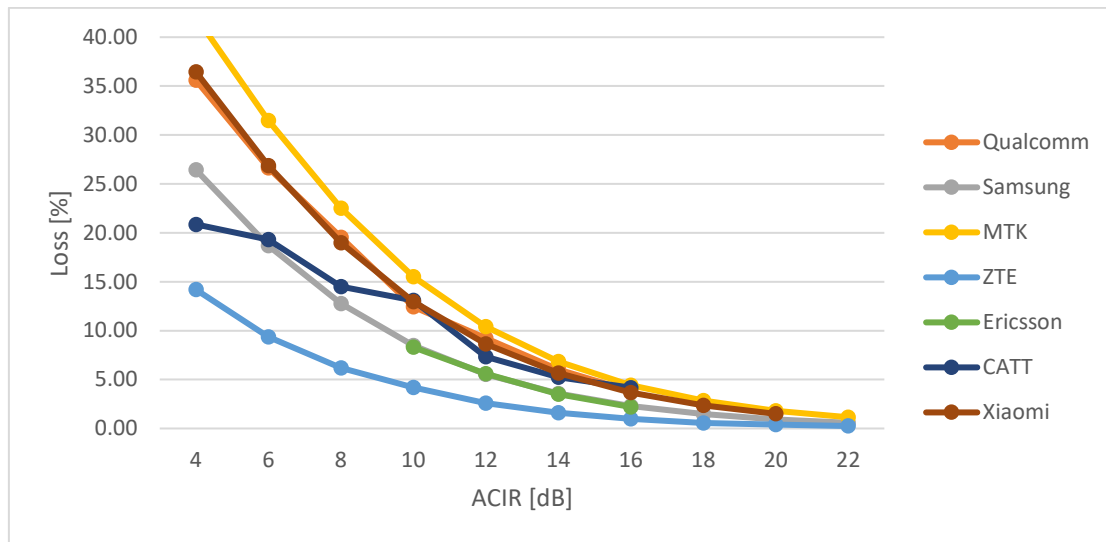


Figure 6.4.3.2-2: Simulation results for 5%-tile throughput loss - TN BS with AAS antenna

Table 6.4.3.2-3: Interpolated ACIR values for Scenario 3 GEO Class to meet the 5% throughput loss criteria - TN BS with AAS antenna

Source		Interpolated ACIR[dB]
Qualcomm	Average	8.84
	5%-tile	14.46
Samsung	Average	8.46
	5%-tile	12.56
MTK	Average	9.83
	5%-tile	15.53
ZTE	Average	6.64
	5%-tile	9.2
Ericsson	Average	
	5%-tile	12.57
CATT	Average	8.28
	5%-tile	14.43
Xiaomi	Average	10.59
	5%-tile	14.67

Table 6.4.3.2-4: Average ACIR of 5%-tile values in the above worse case for Scenario 3 GEO Class - TN BS with AAS antenna

	Scenario 3 GEO Class
ACIR value [dB]	13.35

Table 6.4.3.2-5: Simulation results for average throughput loss - TN BS with non-AAS antenna

ACIR[dB]	0	2	4	6	8	10	12	14	16	18
Samsung	10.22	7.04	4.73	3.12	2.03	1.31	0.83	0.53	0.34	0.21
MTK	28.38	21.62	15.91	11.36	7.76	5.17	3.37	2.24	1.34	0.85
ZTE	11.89	8.33	5.68	3.79	2.48	1.60	1.03	0.66	0.42	0.26
THALES							2.27	1.46	0.94	0.6
Ericsson						3.2	2.1	1.3	0.9	0.5

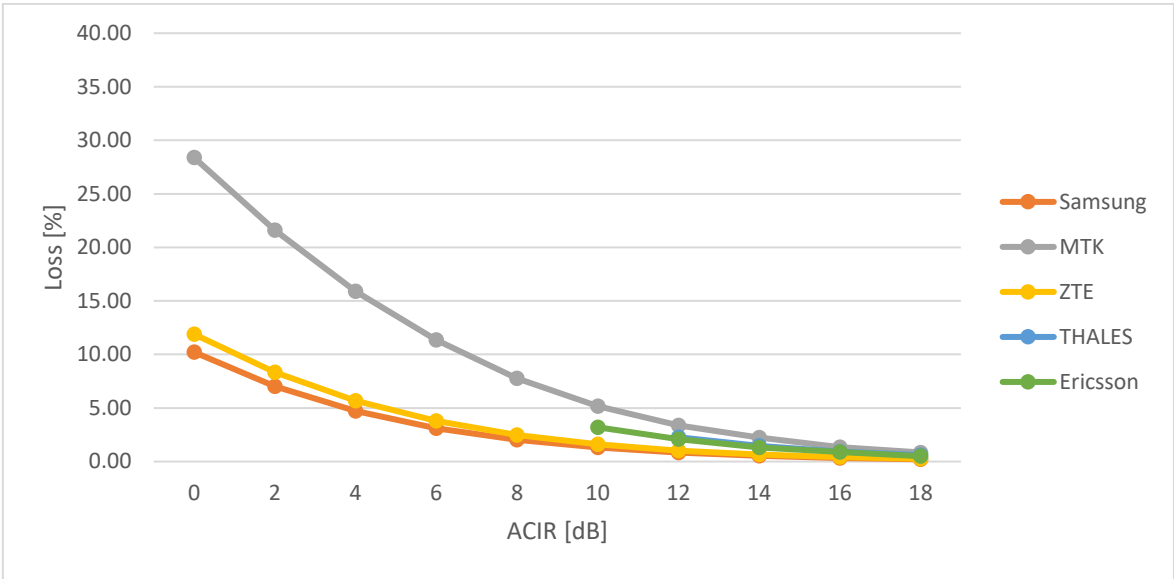


Figure 6.4.3.2-3: Simulation results for average throughput loss - TN BS with non-AAS antenna

Table 6.4.3.2-6: Simulation results for 5%-tile throughput loss - TN BS with non-AAS antenna

ACIR[dB]	6	8	10	12	14	16	18	20	22	24
Samsung	11,97	7,88	5,11	3,28	2,09	1,33	0,85	0,53	0,34	0,21
MTK	31,86	22,83	15,63	10,47	6,87	4,45	2,85	1,82	1,16	0,73
ZTE	13,72	9,33	6,08	3,88	2,38	1,50	1,13	0,75	0,55	0,30
THALES				9,14	5,54	3,6	2,22	1,39	0,83	0,55
Ericsson			9,1	5,9	4,2	2,8	1,9			

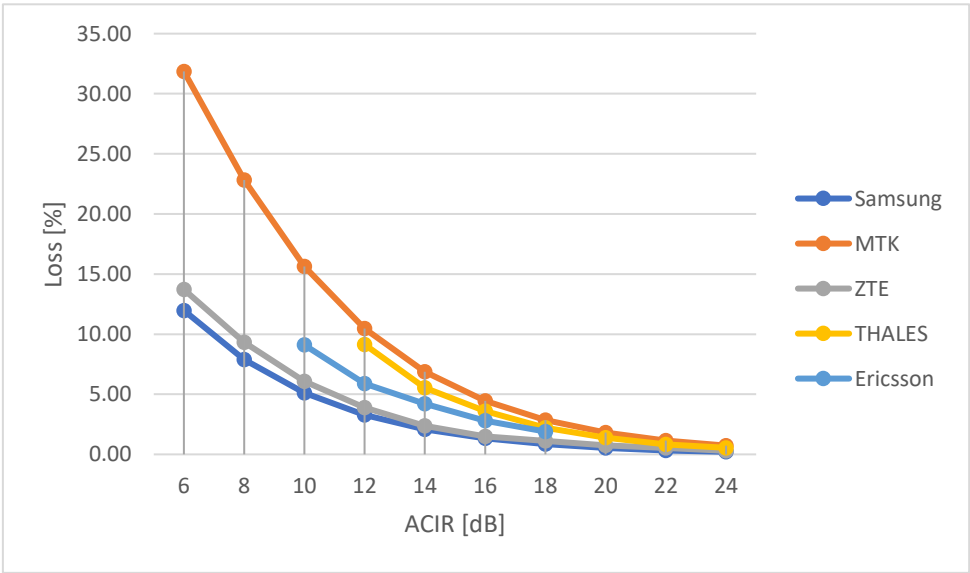


Figure 6.4.3.2-4: Simulation results for 5%-tile throughput loss - TN BS with non-AAS antenna

Table 6.4.3.2-7: Interpolated ACIR values for Scenario 3 GEO Class to meet the 5% throughput loss criteria - TN BS with non-AAS antenna

Source		Interpolated ACIR[dB]
Samsung	Average	3.77
	5%-tile	10.12
MTK	Average	10.19
	5%-tile	15.55
ZTE	Average	4.72
	5%-tile	10.98
THALES	Average	NA
	5%-tile	14.56
Ericsson	Average	NA
	5%-tile	13.06

Table 6.4.3.2-8: Average ACIR of 5%-tile values in the above worse case for Scenario 3 GEO Class - TN BS with non-AAS antenna

	Scenario 3
ACIR value [dB]	12.85

6.4.4 Scenario 4: 2GHz NTN UL interfering TN UL

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select the NR-NTN GEO UL interfering the NR UL equipped with both AAS and non-AAS antenna that deployed in urban environment as the most stringent case.

Table 6.4.4-1: Simulation results for average throughput loss - TN BS with AAS antenna

ACIR[dB]	10	12	14	16	18	20	22	24	26	28
Qualcomm	12.60	10.78	8.96	7.42	6.18	4.93	4.11	3.29	2.63	2.12
Samsung	10.83	8.69	6.92	5.47	4.30	3.35	2.60	2.00	1.53	1.17
MTK	12.16	10.25	8.40	6.66	5.09	3.75	2.67	1.84	1.24	0.82
ZTE	9.35	7.39	5.59	4.38	3.38	2.65	2.05	1.63	1.24	0.94
Ericsson ¹								1.5	1.2	1.0
CATT	2.72	2.26	1.88	1.58						
Xiaomi	10.31	8.30	6.66	5.34	4.27	3.42	2.74	2.19	1.76	1.41

NOTE 1: This result is derived by observing the NR sector having an NR-NTN transmitting UE at its sector edge.

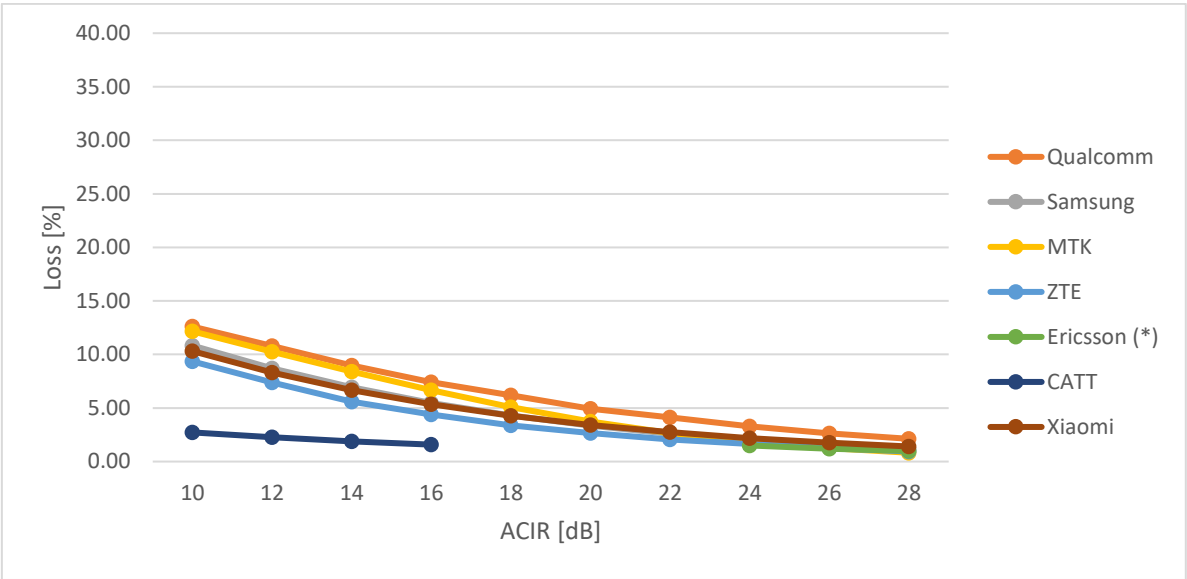


Figure 6.4.4-1: Simulation results for average throughput loss - TN BS with AAS antenna

Table 6.4.4-2: Simulation results for 5%-tile throughput loss - TN BS with AAS antenna

ACIR[dB]	20	22	24	26	28	30	32	34	36	38
Qualcomm	24.62	20.07	15.53	12.02	9.55	7.08	5.83	4.58		
Samsung	16.71	12.55	9.44	7.15	5.47	4.19	3.19	2.44	1.86	1.41
MTK	20.91	14.93	10.33	6.95	4.59	2.98	1.92	1.23	0.78	0.49
ZTE	11.43	8.86	7.23	6.23	4.71	3.34	2.66	1.47	0.87	0.75
Ericsson ¹			7.9	7.1	4.8	4.6	0.5	0.5		
CATT	8.83	7.49	5.57	4.10	3.44	2.95				
Xiaomi	16.47	11.96	8.45	5.79	3.88	2.55	1.66	1.07		

NOTE 1: This result is derived by observing the NR sector having an NR-NTN transmitting UE at its sector edge.

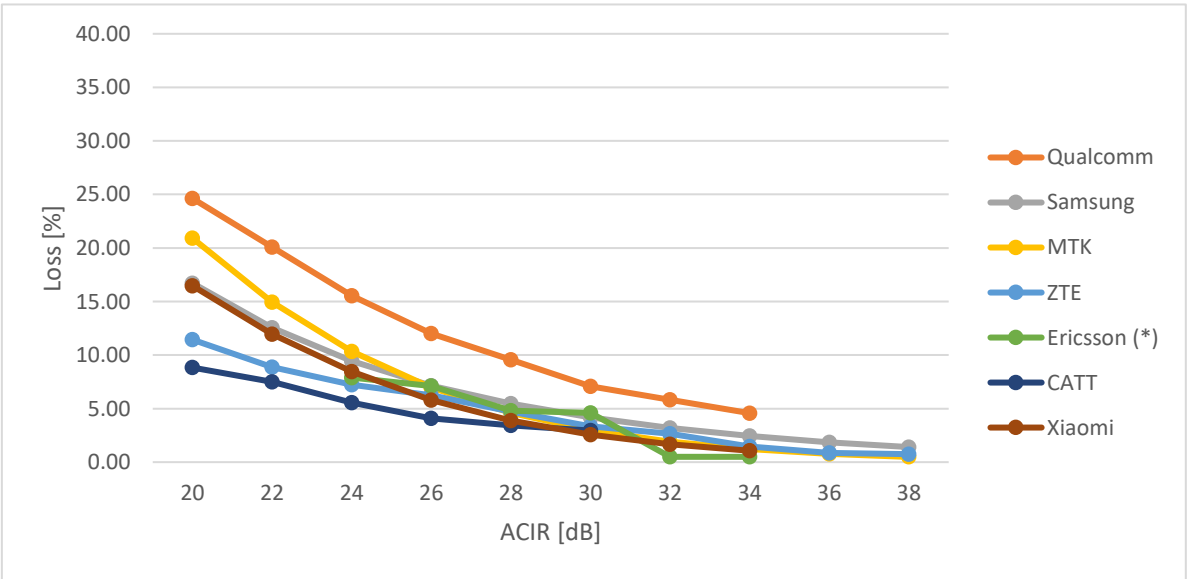


Figure 6.4.4-2: Simulation results for 5%-tile throughput loss - TN BS with AAS antenna

Table 6.4.4-3: Interpolated ACIR values for Scenario 4 to meet the 5% throughput loss criteria - TN BS with AAS antenna

Source		Interpolated ACIR[dB]
Qualcomm	Average	19.89
	5%-tile	33.33
Samsung	Average	16.80
	5%-tile	28.73
MTK	Average	18.13
	5%-tile	27.65
ZTE	Average	14.98
	5%-tile	27.62
Ericsson(*)	Average	
	5%-tile	27.83
CATT	Average	3.69
	5%-tile	24.78
Xiaomi	Average	16.64
	5%-tile	26.83

Table 6.4.4-4: Average ACIR of 5%-tile values in the above worse case for Scenario 4 - TN BS with AAS antenna

	Scenario 4
ACIR value [dB]	28.11

Table 6.4.4-5: Simulation results for average throughput loss - TN BS with non-AAS antenna

ACIR[dB]	10	12	14	16	18	20	22	24	26	28
Samsung	10,85	8,39	6,50	5,06	3,96	3,11	2,43	1,88	1,44	1,07
MTK	11,73	9,14	7,05	5,38	4,05	2,99	2,17	1,53	1,06	0,72
ZTE	6,12	5,01	4,16	3,32	2,85	2,27	1,83	1,17	0,80	0,50
THALES	NA	0,04	0,03	0,01	0,01	0	0	0	0	0
Ericsson ¹	NA	NA	NA	NA	NA	NA	NA	NA	NA	1,2

NOTE 1: This result is derived by observing the NR sector having an NR-NTN transmitting UE at its sector edge.

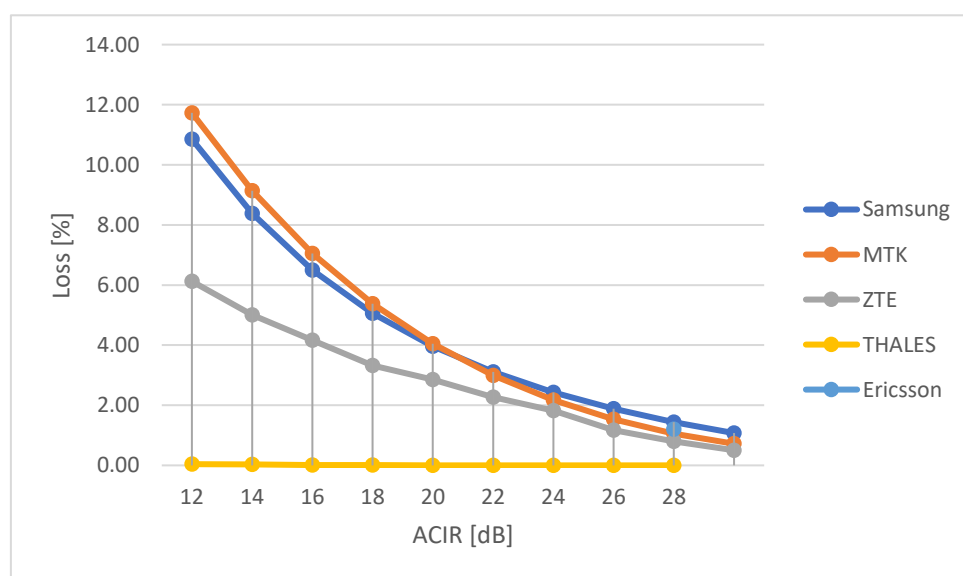
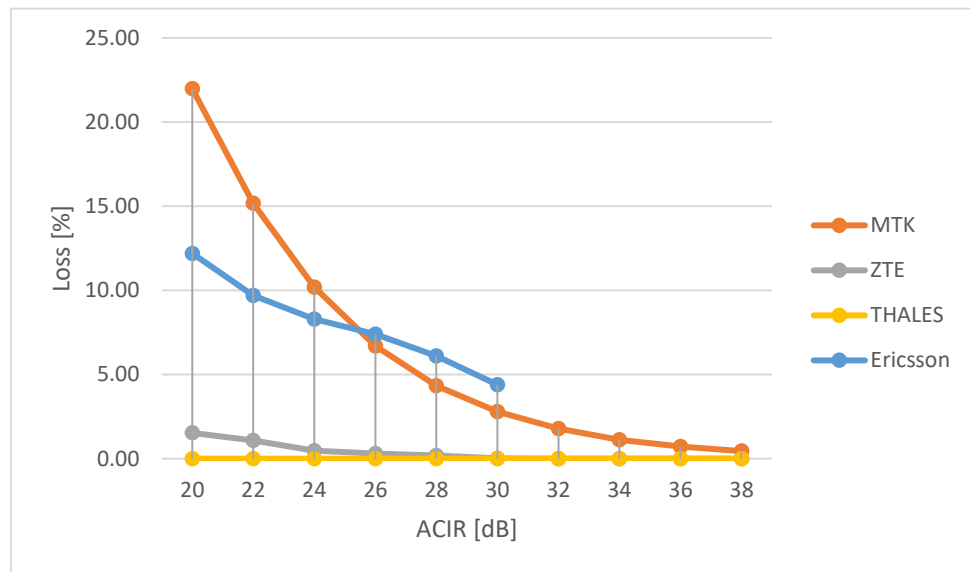


Figure 6.4.4-3: Simulation results for average throughput loss - TN BS with non-AAS antenna

Table 6.4.4-6: Simulation results for 5%-tile throughput loss - TN BS with non-AAS antenna

ACIR[dB]	20	22	24	26	28	30	32	34	36	38
Samsung	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
MTK	21,98	15,18	10,19	6,70	4,34	2,79	1,78	1,13	0,72	0,45
ZTE	1,53	1,08	0,47	0,30	0,19	0,03	0,00	0,00	0,00	0,00
THALES	0	0	0	0	0	0	0	0	0	0
Ericsson	NA	NA	NA	NA	12,2	9,7	8,3	7,4	6,1	4,4

NOTE 1: This result is derived by observing the NR sector having an NR-NTN transmitting UE at its sector edge.

**Figure 6.4.4-4: Simulation results for 5%-tile throughput loss - TN BS with non-AAS antenna****Table 6.4.4-7: Interpolated ACIR values for Scenario 4 to meet the 5% throughput loss criteria - TN BS with non-AAS antenna**

Source		Interpolated ACIR[dB]
Samsung	Average	16.11
	5%-tile	NA
MTK	Average	16.57
	5%-tile	27.44 ¹
ZTE	Average	12.02
	5%-tile	15.46
THALES	Average	NA
	5%-tile	NA
Ericsson	Average	NA
	5%-tile	37.29 ¹

NOTE 1: According to the principles, this value is not treated for later process.

Table 6.4.4-8: Average ACIR of 5%-tile values in the above worse case for Scenario 4 - TN BS with non-AAS antenna

	Scenario 4
ACIR value [dB]	15.46

6.4.5 Scenario 5: 2GHz NTN UL interfering TN DL

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select the NR-NTN GEO UL interfering the NR DL equipped with both AAS and non-AAS antenna that deployed in rural environment as the most stringent case.

Table 6.4.5-1: Simulation results for average throughput loss - TN BS with AAS antenna

ACIR[dB]	12	14	16	18	20	22	24	26	28	30
Qualcomm ⁽¹⁾	7.32	5.28	3.78	2.81	1.84	1.42	0.99	0.68		
MTK	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Xiaomi	10.31	7.75	5.70	4.11	2.91	2.03	1.39	0.94		
Samsung ⁽¹⁾	37.13	31.30	25.62	20.30	15.53	11.43	8.07	5.43	3.43	1.98
Samsung ⁽²⁾	0.16	0.13	0.11	0.09	0.07	0.06	0.05	0.03	0.02	0.02
Ericsson	0.00									

NOTE 1: These results were derived by adopting free-space path loss model for the links between NR UE and NR-NTN UE.

NOTE 2: These results were derived by adopting path loss model from TR 38.901[10] for the links between NR UE and NR-NTN UE.

NOTE3: In the meeting, views are expressed on which propagation model is more appropriate for the links between NR UE and NR-NTN UE. Due to the limited time and the fact that this scenario is not the worst case to determine the NR-NTN UE ACLR, this is not discussed nor concluded.

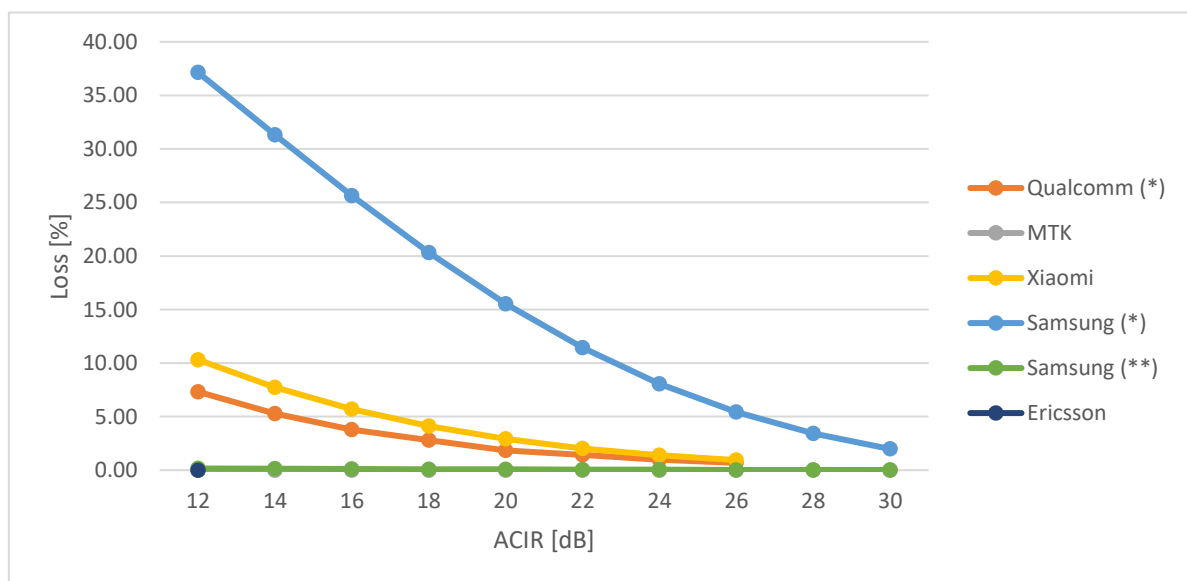


Figure 6.4.5-1: Simulation results for average throughput loss - TN BS with AAS antenna

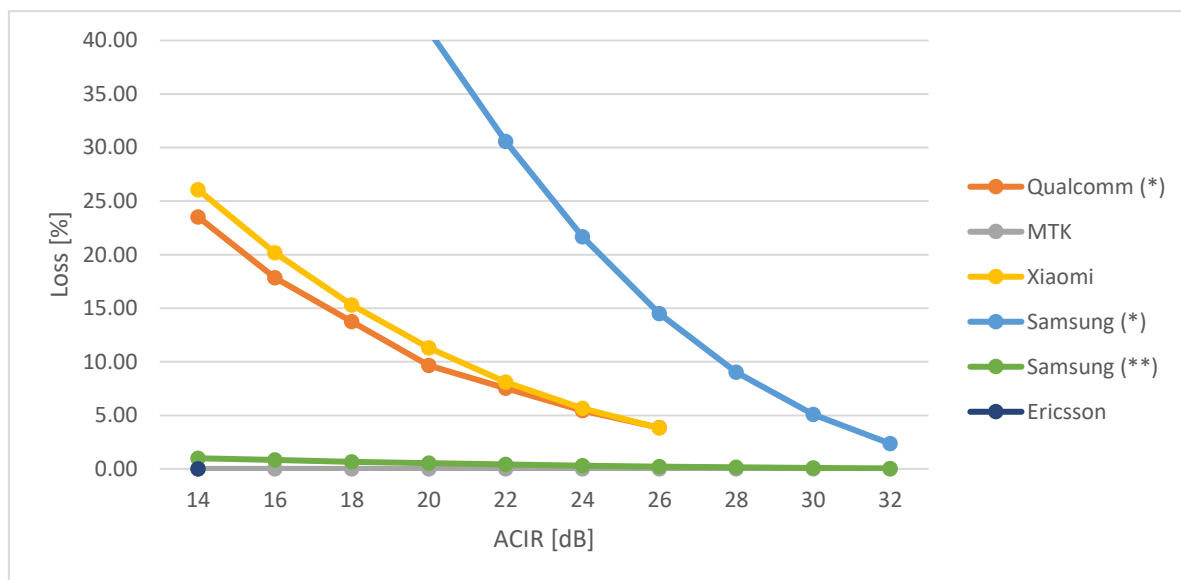
Table 6.4.5-2: Simulation results for 5%-tile throughput loss - TN BS with AAS antenna

ACIR[dB]	14	16	18	20	22	24	26	28	30	32
Qualcomm ⁽¹⁾	23.52	17.84	13.74	9.65	7.54	5.42	3.84			
MTK	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
Xiaomi	26.06	20.19	15.30	11.29	8.09	5.64	3.81			
Samsung ⁽¹⁾	72.82	63.00	52.03	40.90	30.57	21.66	14.49	9.03	5.09	2.36
Samsung ⁽²⁾	0.99	0.82	0.66	0.53	0.41	0.30	0.21	0.14	0.08	0.04
Ericsson	0.0									

NOTE 1: These results were derived by adopting free-space path loss model for the links between NR UE and NR-NTN UE.

NOTE 2: These results were derived by adopting path loss model from TR 38.901[10] for the links between NR UE and NR-NTN UE.

NOTE 3: In the meeting, views are expressed on which propagation model is more appropriate for the links between NR UE and NR-NTN UE. Due to the limited time and the fact that this scenario is not the worst case to determine the NR-NTN UE ACLR, this is not discussed nor concluded.

**Figure 6.4.5-2: Simulation results for 5%-tile throughput loss - TN BS with AAS antenna****Table 6.4.5-3: Interpolated ACIR values for Scenario 5 to meet the 5% throughput loss criteria - TN BS with AAS antenna**

Source		Interpolated ACIR[dB]
Qualcomm (*)	Average	14.37
	5%-tile	24.53
MTK	Average	
	5%-tile	
Xiaomi	Average	16.88
	5%-tile	24.70
Samsung (*)	Average	26.43
	5%-tile	30.07
Samsung (**)	Average	
	5%-tile	
Ericsson	Average	
	5%-tile	

Table 6.4.5-4: Average ACIR of 5%-tile values in the above worse case for Scenario 5 - TN BS with AAS antenna

	Scenario 5
ACIR value [dB]	26.43

Table 6.4.5-5: Simulation results for average throughput loss - TN BS with non-AAS antenna

ACIR[dB]	0	2	4	6	8	10	12	14	16	18
MTK	3,01	2,43	1,88	1,39	0,99	0,69	0,46	0,30	0,20	0,13
Samsung	0,32	0,27	0,23	0,20	0,17	0,14	0,12	0,10	0,08	0,06
THALES	NA	NA	NA	NA	NA	NA	0,07	0,06	0,05	0,04

NOTE1: In the meeting, views are expressed on which propagation model is more appropriate for the links between NR UE and NR-NTN UE. Due to the limited time and the fact that this scenario is not the worst case to determine the NR-NTN UE ACLR, this is not discussed nor concluded.

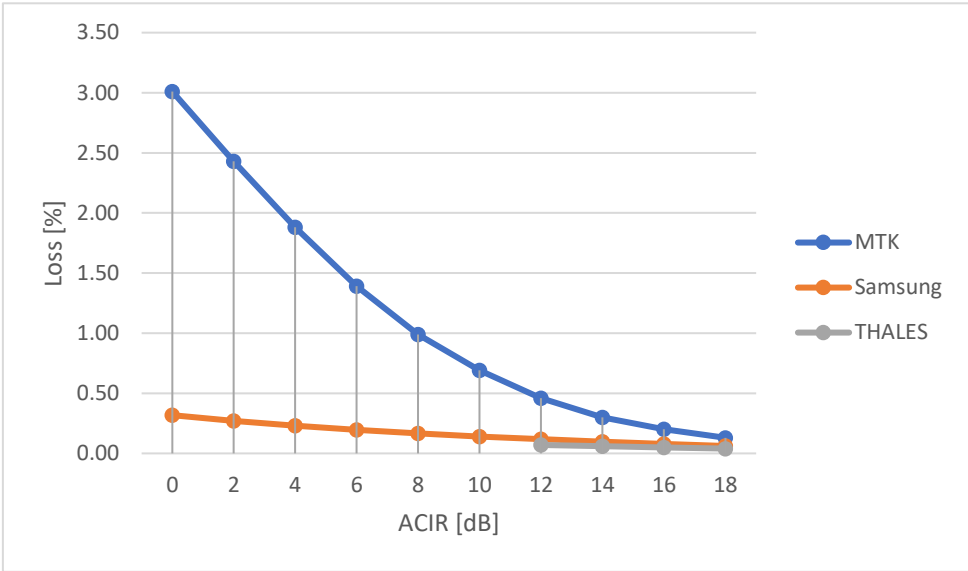


Figure 6.4.5-3: Simulation results for average throughput loss - TN BS with non-AAS antenna

Table 6.4.5-6: Simulation results for 5%-tile throughput loss - TN BS with non-AAS antenna

ACIR[dB]	6	8	10	12	14	16	18	20	22	24
MTK	15,03	10,72	7,39	4,96	3,27	2,12	1,36	0,87	0,55	0,35
Samsung	1,07	0,92	0,77	0,65	0,53	0,44	0,34	0,27	0,20	0,15
THALES				0	0	0	0	0	0	0

NOTE 1: In the meeting, views are expressed on which propagation model is more appropriate for the links between NR UE and NR-NTN UE. Due to the limited time and the fact that this scenario is not the worst case to determine the NR-NTN UE ACLR, this is not discussed nor concluded.

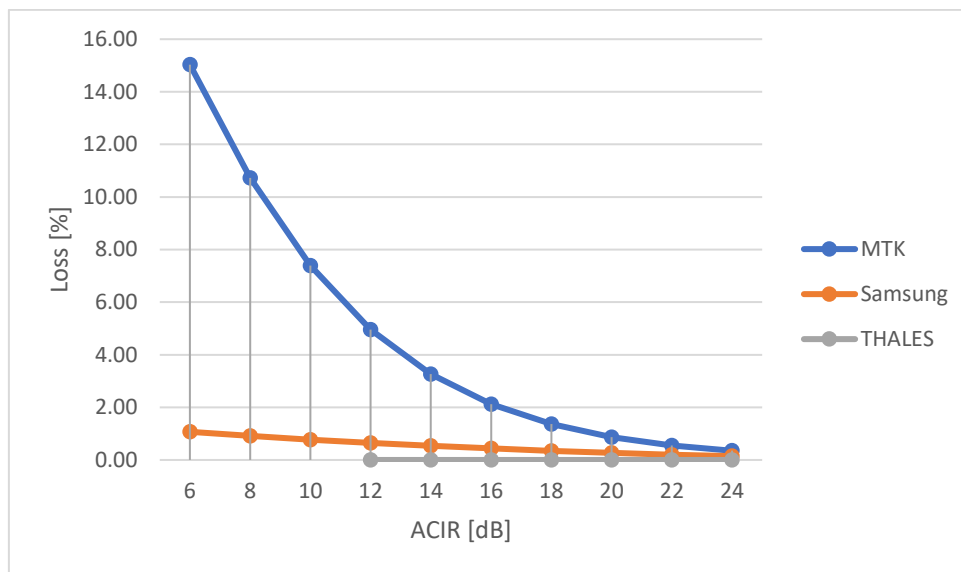


Figure 6.4.5-4: Simulation results for 5%-tile throughput loss - TN BS with non-AAS antenna

Table 6.4.5-7: Interpolated ACIR values for Scenario 5 to meet the 5% throughput loss criteria - TN BS with non-AAS antenna

Source		Interpolated ACIR[dB]
MTK	Average	0
	5%-tile	11.97 ¹
Samsung	Average	0
	5%-tile	0
THALES	Average	0
	5%-tile	0
NOTE 1: According to the principles, this value is not treated for later process.		

Table 6.4.5-8: Average ACIR of 5%-tile values in the above worse case for Scenario 5 - TN BS with non-AAS antenna

	Scenario 5
ACIR value [dB]	0

6.4.6 Scenario 6: 2GHz TN DL interfering NTN UL

The co-existence results from all concerned options in this scenario were evaluated, and the initial results suggested that the NR-NTN SAN would suffer more interference in urban deployment scenario, considering that the central beam of a GEO satellite (with the agreed minimum elevation angle of 45°) will be full of urban TNs. It was further discussed and agreed that simply using 20% active rate and assuming fully deployed NR stations with urban deployment assumptions in the whole NR-NTN beam coverage area would overestimate the density and aggregated interference from NR stations to NR-NTN in this scenario. In the end, the case of NR DL interfering the NR-NTN LEO-600 UL that deployed in a Rural environment has been selected as a representative one that could be used as basis to derive ACIR values. Further study could be considered base on a mixed Urban and Rural environment.

Table 6.4.6-1: Simulation results for average throughput loss - TN BS with non-AAS antenna

ACIR[dB]	22	24	26	28	30	32	34	36	38	40
THALES	51.17	41.28	31.94	23.87	17.01	11.87	7.91	5.26	3.4	2.22
MTK	3.05	1.98	1.28	0.82	0.52	0.33	0.21	0.13	0.08	0.05

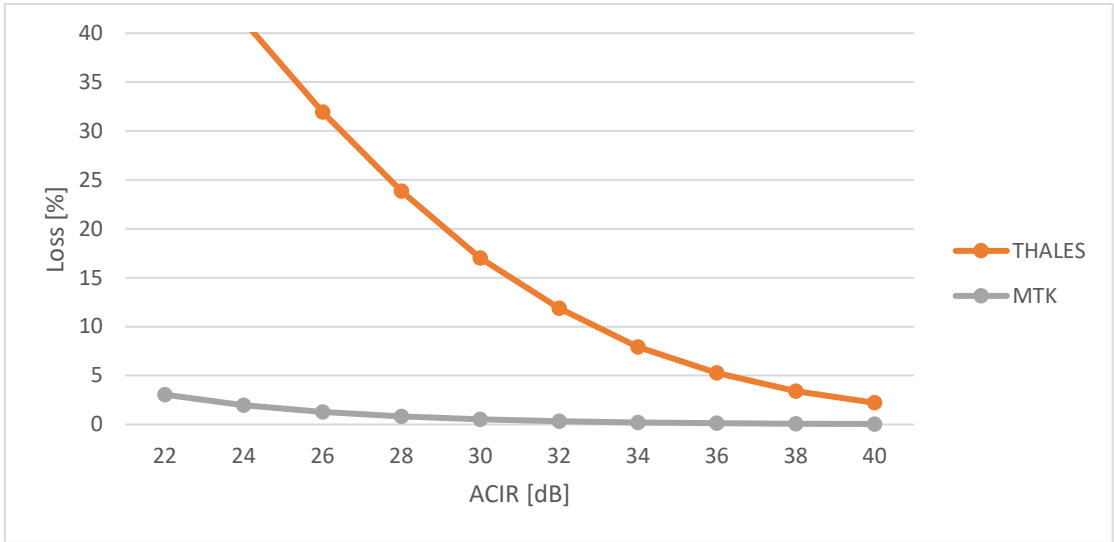


Figure 6.4.6-1: Simulation results for average throughput loss - TN BS with non-AAS antenna

Table 6.4.6-2: Simulation results for 5%-tile throughput loss - TN BS with non-AAS antenna

ACIR[dB]	22	24	26	28	30	32	34	36	38	40
THALES	53.49	43.02	33.72	25.58	17.44	12.79	8.14	5.81	3.49	2.33
MTK	8.85	5.78	3.73	2.39	1.52	0.97	0.61	0.39	0.24	0.15

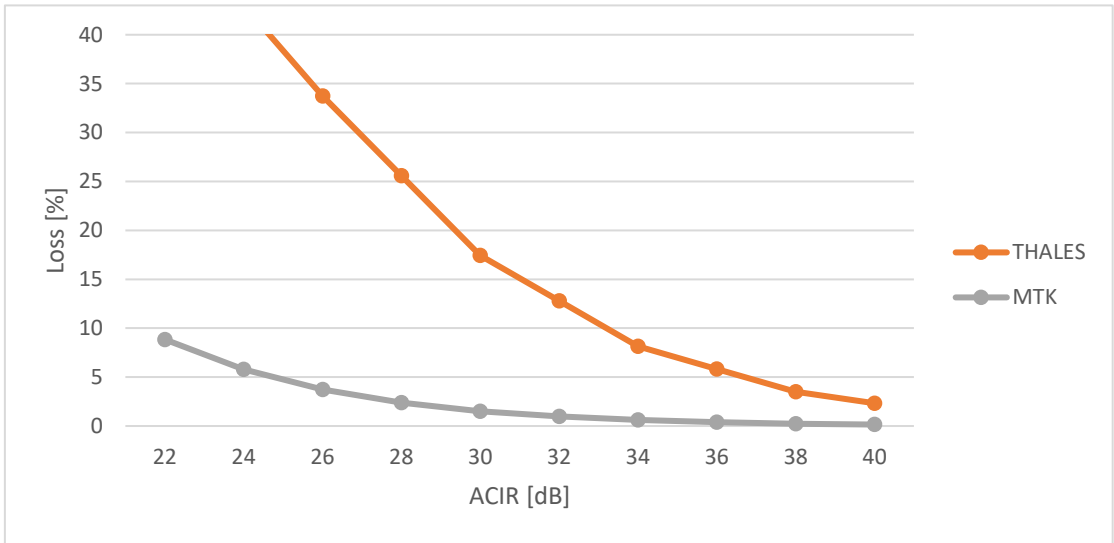


Figure 6.4.6-2: Simulation results for 5%-tile throughput loss - TN BS with non-AAS antenna

Table 6.4.6-3: Interpolated ACIR values for Scenario 5 to meet the 5% throughput loss criteria - TN BS with non-AAS antenna

Source		Interpolated ACIR[dB]
THALES	Average	36.69
	5%-tile	36.7
MTK	Average	19.66
	5%-tile	24.67 ¹
NOTE 1: According to the principles, this value is not treated for later process.		

Table 6.4.6-4: Average ACIR of 5%-tile values in the above worse case for Scenario 6 - TN BS with non-AAS antenna

	Scenario 6
ACIR value [dB]	36.7

6.4.7 Scenario 7: 2GHz HAPS DL interfering TN DL

The co-existence results contributed have been collected and compared in the following.

Table 6.4.7-1: Simulation results for average throughput loss

ACIR[dB]	5	10	15	20	25	30	35	40
Nokia	28,80	16,80	8,10	3,30	1,10	0,30	0,10	0,10
Qualcomm	38,41	23,95	12,26	5,14	1,86	0,62	0,20	0,06

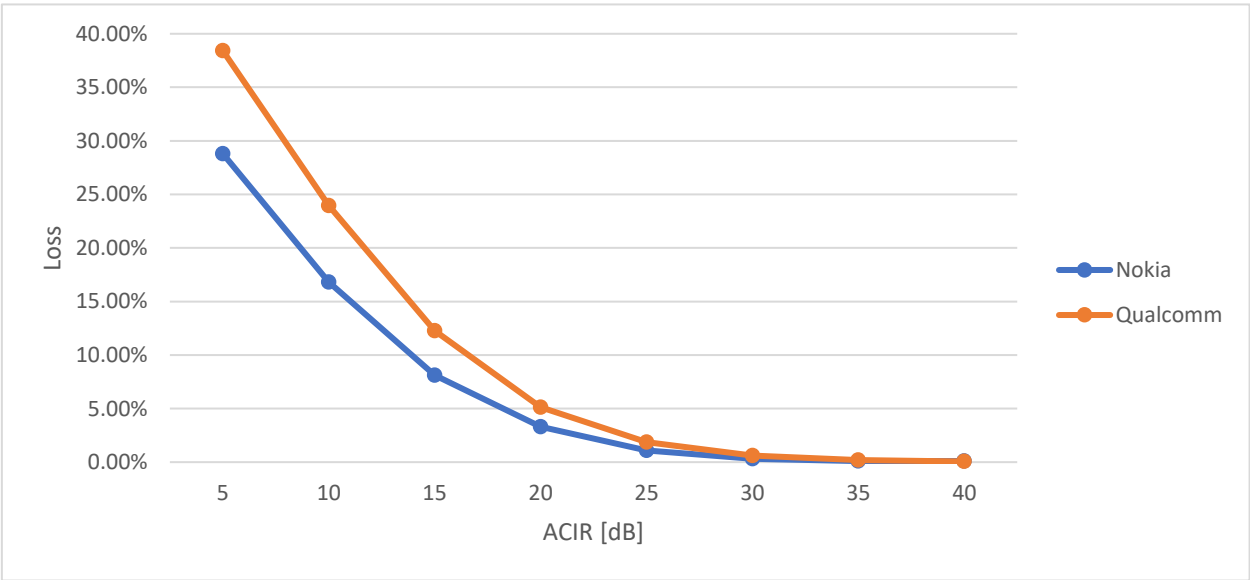


Figure 6.4.7-1: Simulation results for average throughput loss

Table 6.4.7-2: Simulation results for 5%-tile throughput loss

ACIR[dB]	5	10	15	20	25	30	35	40
Nokia	77,10	53,20	26,60	9,80	1,80	0,30	0,20	0,00
Qualcomm	100,00	68,11	39,34	17,38	6,66	2,37	0,77	0,21

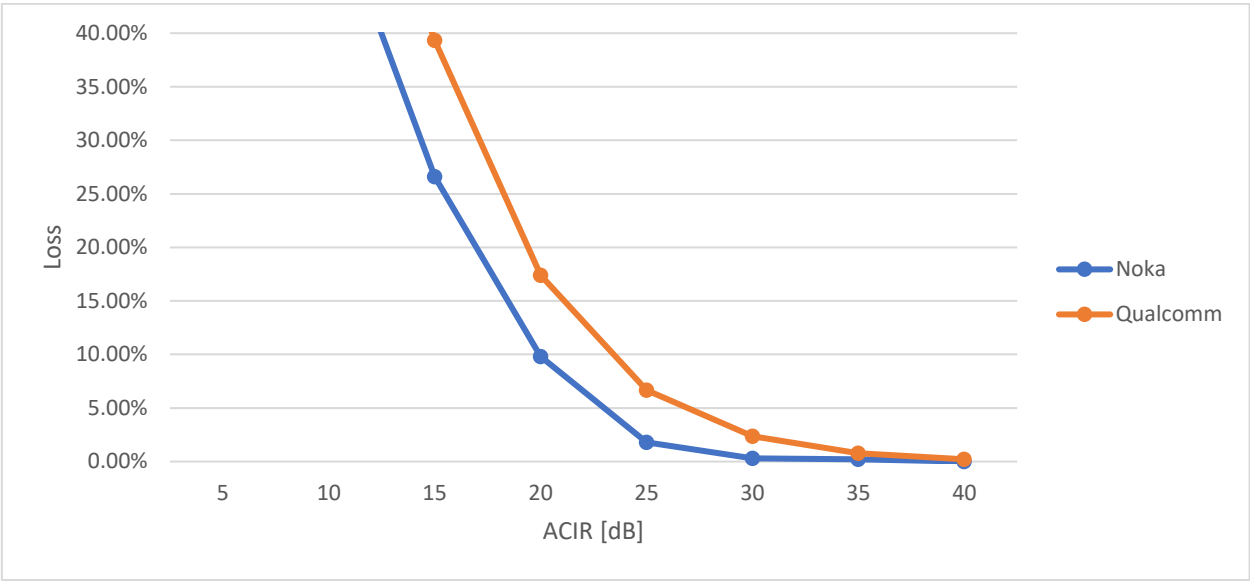


Figure 6.4.7-2: Simulation results for 5%-tile throughput loss

Table 6.4.7-3: Interpolated ACIR values for Scenario 7 to meet the 5% throughput loss criteria

Source		Interpolated ACIR[dB]
Nokia	Average	18.22
	5%-tile	23.00
Qualcomm	Average	20.10
	5%-tile	26.93

Table 6.4.7-4: Average ACIR of 5%-tile values in the above worse case for Scenario 7

	Scenario 1
ACIR value [dB]	24.97

6.4.8 Scenario 8: 2GHz TN UL interfering HAPS UL

The co-existence results contributed have been collected and compared in the following.

Table 6.4.8-1: Simulation results for average throughput loss

ACIR[dB]	5	10	15	20	25	30	35	40
Nokia	14.4%	7.2%	3.0%	1.4%	0.7%	0.6%	0.1%	0.0%

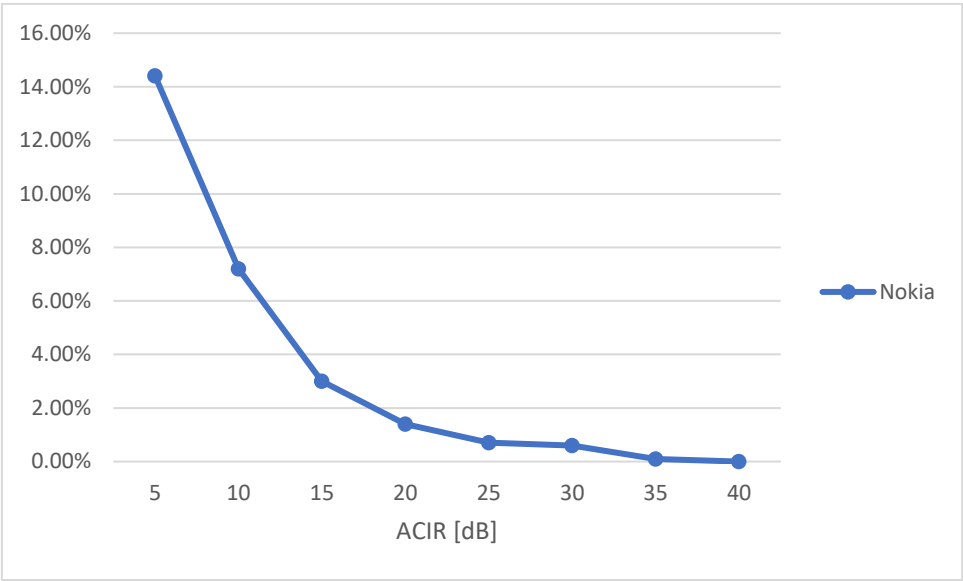


Figure 6.4.8-1: Simulation results for average throughput loss

Table 6.4.8-2: Simulation results for 5%-tile throughput loss

ACIR[dB]	5	10	15	20	25	30	35	40
Nokia	35.4%	22.6%	12.6%	11.5%	7.5%	5.0%	4.5%	1.1%

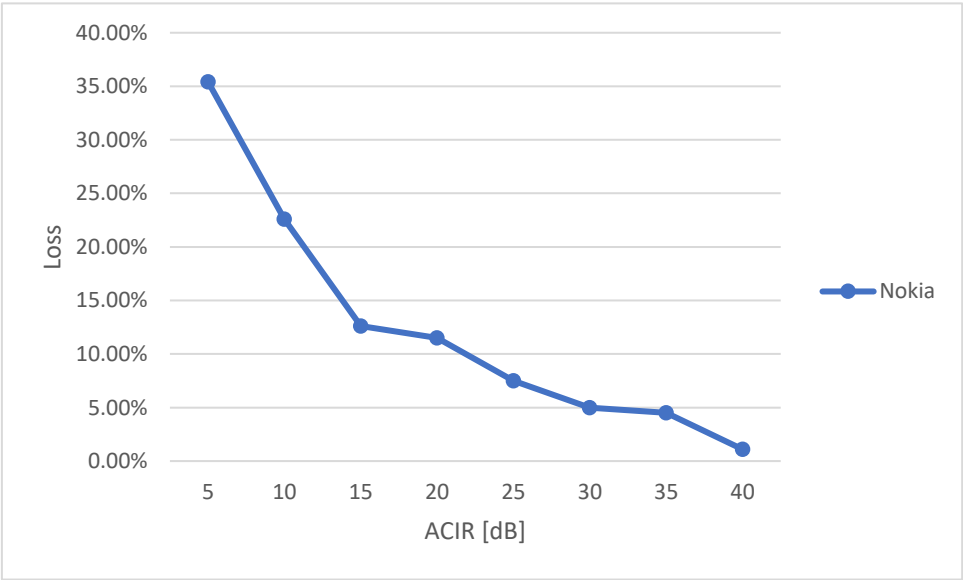


Figure 6.4.8-2: Simulation results for 5%-tile throughput loss

6.5 Summary of co-existence study

6.5.1 Coexistence at 2GHz

This sub-clause captures the summary of the co-existence studies. The averaged interpolate ACIR values for each scenario are presented in Table 6.5.1-1 below.

Table 6.5.1-1: Average ACIR values for each scenario

Scenario		1	2	3 LEO	3 GEO	4	5	6
ACIR value [dB]	TN BS with AAS	27.83	28.03	23.32	13.35	28.11	26.43	36.70
	TN BS with non-AAS	28.31	29.22	22.66	12.85	15.46	0	

Then, by considering the following ACLR and ACS of TN BS and UE in Table 6.5.1-2, the suggested ACLR and ACS of NTN SAN, HAPS and UE from each scenario are given in Table 6.5.1-3. It should be noted that the values in Table 6.5.1-3 are directly derived from the selected option of each scenario, and it is limited by the nature of assumptions and methodologies adopted in the co-existence studies.

Table 6.5.1-2: ACLR and ACS of TN

TN		Values
BS	ACLR	45 dB
	ACS	46 dB
UE	ACLR	30 dB
	ACS	33 dB

Table 6.5.1-3: Co-existence results suggested ACLR and ACS of NR-NTN

Scenario	Contributing	ACLR ACS values
1	NTN UE ACS	27.91 dB (AAS)
2	NTN SAN ACS	37.06 dB (non-AAS)
3	NTN SAN ACLR	LEO: 23.81 dB GEO: 13.39 dB
4	NTN UE ACLR	28.18 dB (AAS)
5	NTN UE ACLR	27.51 dB (AAS)
6	NTN SAN ACS	37.4 dB
7	HAPS ACLR	24.97 dB
8	HAPS ACS	30 dB

Considering the above suggested values, the agreed ACLR and ACS of NR-NTN are given in Table 6.5.1-4.

Table 6.5.1-4: ACLR and ACS of NR-NTN

NR-NTN			Values
SAN	ACLR	GEO	14 dB
		LEO	24 dB
	ACS ¹	GEO	38 dB
		LEO	38 dB
HAPS	ACLR		45 ²
	ACS		46
UE	ACLR		30 dB
	ACS		33 dB
NOTE 1: The ACS values for SAN apply to both Rural and Urban environment.			
NOTE 2: This value could be further relaxed based on co-existence studies if it is found more stringent than necessary.			

6a Co-existence study for 17/27GHz

6a.1 Co-existence simulation scenario for 17/27GHz

Scenarios for the coexistence study for 17/27GHz are listed in Table 6a.1-1

Table 6a.1-1: Scenarios for NTN-NTN/TN co-existence

FR2: 17/27GHz		NTN					
		Set 1			Set 2 ²		
		GEO ³	LEO 600km	LEO 1200km	GEO	LEO 600km	LEO 1200km
NR	Rural	N/A	N/A	N/A	N/A	N/A	N/A
	Urban macro	X	X	X	X	X	X
	Dense Urban ⁶	N/A	N/A	N/A	N/A	N/A	N/A

The aggressor and victim combinations for 17/27GHz cases are listed in Table 6a.1-2.

Table 6a.1-2: Aggressor and victim

No.	Combination	Aggressor	Victim	Frequency band	Scope of Coexistence Simulation
1a	TN with NTN	NTN UL	TN UL	27 GHz	ACLR NTN UE to be varied/defined ACS TN gNB fixed
2a	TN with NTN	TN UL	NTN UL	27 GHz	ACLR TN UE fixed ACS NTN SAN to be varied/defined
3a	TN with NTN	NTN UL	TN DL	27 GHz	ACLR NTN UE to be varied/defined ACS TN UE fixed
4a	TN with NTN	TN DL	NTN UL	27 GHz	ACLR TN gNB fixed ACS NTN SAN to be varied/defined
5a	TN with NTN	TN DL	NTN DL	17 GHz	ACLR TN gNB fixed ACS NTN UE to be varied/defined
6a	TN with NTN	NTN DL	TN DL	17 GHz	ACLR NTN SAN to be varied/defined ACS TN UE fixed
7a	TN with NTN	NTN DL	TN UL	17 GHz	ACLR NTN SAN to be varied/defined ACS TN gNB fixed
8a	TN with NTN	TN UL	NTN DL	17 GHz	ACLR TN UE fixed ACS NTN UE to be varied/defined
NOTE 1: For coexistence between Ka-Band DL and adjacent TN bands, there are no 3GPP defined/specified TN bands.					
NOTE 2: To deprioritize scenario 7a and 8a for SAN elevation angle as 90-degree cases. Scenario 7a and 8a should still be studied for SAN elevation angle as 25-degree cases (computed from the center of the central beam).					

For further visualisation of coexistence studies for 17/27GHz, the interference charts with respect to scenarios 1a to 8a are further represented with respect to NTN-TN coexistence scenarios in Figure 6a.1-1 (scenarios 1a to 4a) and Figure 6a.1-2 (scenarios 5a to 8a).

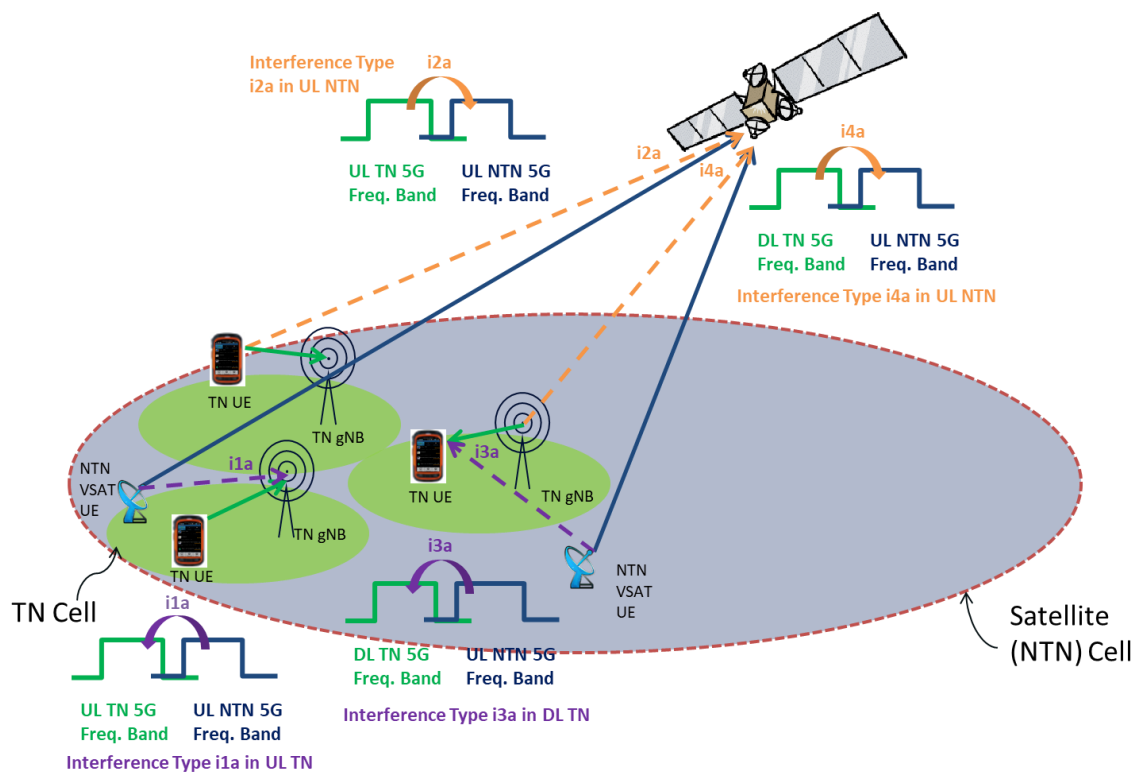


Figure 6a.1-1: NTN-TN adjacent band coexistence scenario 1a to 4a

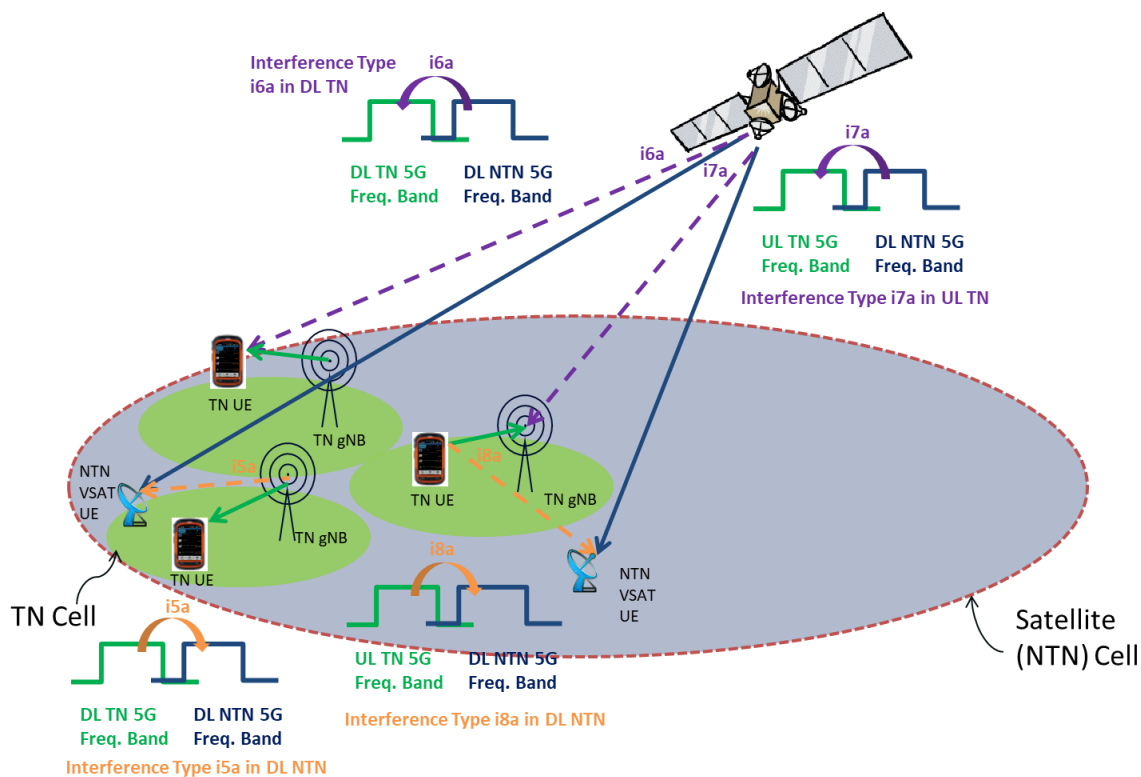


Figure 6a.1-2: NTN-TN adjacent band coexistence scenarios 5a to 8a

The frequency and bandwidth are listed in Table 6a.1-3.

Table 6a.1-3: Proposed frequency and bandwidth for co-existence study

	Frequency	Bandwidth	Duplex mode	Frequency reuse factor
TN Urban macro	17/27 GHz	200/400MHz ¹	TDD	1
GEO	17GHz (DL)/27GHz(UL)	200MHz per beam	FDD	2 ²
LEO	17GHz (DL)/27GHz(UL)	200MHz per beam	FDD	2 ²
NOTE1: Use the same channel BW for TN and NTN. TN 200MHz is the baseline for study and TN 400MHz is with second priority for study.				
NOTE2: FRF=1 with single polarization has been adopted for calibration. For co-existence simulation, assume FRF=2 with two polarizations (RHCP, LHCP). The polarization isolation is assumed to be ideal.				

6a.2 Co-existence simulation assumption for 17/27GHz

6a.2.1 Network layout model

6a.2.1.1 Co-existence between NTN satellite and TN

Cellular cell structure is considered for both NTN and TN network layout.

Referring to TR 38.811 Section 6.3 and Annex A, a 3D global coordinate system is considered (Earth-Centred Earth Fixed) for simulating NTN beams direction and location on the earth surface. It means the NTN beam location, TN randomly dropping location are generated with a set of three parameters (x,y,z)

Deployment of NTN satellite and TN cells and UEs for co-existence study is listed in Table 6a.2.1.1-1.

Table 6a.2.1.1-1: Network and UE deployment

No.	Combination	Aggressor	Victim	Which NTN cell/UE to observe?	Which TN/UE to observe?	Which TN cells in a TN to observe?
1a	TN with NTN	NTN UL	TN UL	<u>NTN cell:</u> Nadir point. <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters.	TN randomly placed in this NTN beam	Only the active TN clusters which contain NTN UE(s)
2a	TN with NTN	TN UL	NTN UL	<u>NTN cell:</u> Observe NTN central beam for SINR, 6 adjacent beams for inter-beam interference. <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters	Consider an active rate of 20% for TN	Only the active TN cells in central NTN beam
3a	TN with NTN	NTN UL	TN DL	<u>NTN cell:</u> Nadir point <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters.	TN clusters randomly placed in this NTN beam	All active TN clusters which contain NTN UE(s)
				<u>NTN cell:</u> NTN cell with satellite at low elevation (to be further discussed for GEO and LEO , interested companies can bring analysis and results for other values). <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters.	TN clusters randomly placed in this NTN beam	Only the active TN clusters which contain NTN UE(s)
4a	TN with NTN	TN DL	NTN UL	<u>NTN cell:</u> Observe NTN central beam for SINR, 6 adjacent beams for inter-beam interference. <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters	Consider the active rate of 20% for TN.	All active TN cells in central NTN beam
5a	TN with NTN	TN DL	NTN DL	<u>NTN cell:</u> Observe NTN central beam for SINR, 6 adjacent beams for inter-beam interference. <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters.	One cluster with 19 TN cells (57 sectors) randomly placed in the central NTN beam	Only the active TN clusters which contain the NTN UE(s)
6a	TN with NTN	NTN DL	TN DL	<u>NTN cell:</u> Nadir point. <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters.	TN clusters randomly placed in this NTN beam	All in central NTN beam

				<u>NTN cell:</u> NTN cell with satellite at low elevation (to be further discussed for GEO and LEO, interested companies can bring analysis and results for other values) <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters	TN clusters randomly placed in this NTN beam	
7a	TN with NTN	NTN DL	TN UL	<u>NTN cell:</u> Nadir point <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters.	TN clusters randomly placed in this NTN beam	Only the active TN clusters which contain NTN UE(s)
				<u>NTN cell:</u> NTN cell with satellite at low elevation (to be further discussed for GEO and LEO, interested companies can bring analysis and results for other values). <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters.	TN clusters randomly placed in this NTN beam	Only the active TN clusters which contain NTN UE(s)
8a	TN with NTN	TN UL	NTN DL	<u>NTN cell:</u> Observe NTN central beam for SINR, 6 adjacent beams for inter-beam interference. <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters	Consider the active rate of 20% for TN	All active TN cells in central NTN beam

6a.2.2 System parameters

6a.2.2.1 Satellite parameters

Set-1 satellite parameters are listed in Table 6a.2.2.1-2 according to TR 38.821[6].

The satellite max Tx power can be calculated by the equation as below:

$$\max Tx\ power[dBm] = EIRP\ density\left[\frac{dBW}{MHz}\right] + 30 + 10\log_{10}(N_{RB} * SCS[MHz] * 12) - Max\ Gain[dBi]$$

Table 6a.2.2.1-1: N_{RB} configuration per BandWidth size and SCS

Configuration FR2 Ka-band	N _{RB} (200MHz BW)
SCS 120 kHz	132

Table 6a.2.2.1-2: Set-1 satellite parameters for co-existence study

Satellite orbit		GEO		LEO-1200		LEO-600		
Satellite altitude		35786 km		1200 km		600 km		
Payload characteristics for DL transmissions								
Satellite EIRP density		17GHz	40 dBW/MHz		10 dBW/MHz		4 dBW/MHz	
Satellite max TX power in dBm	BW (MHz)		200	400	200	400	200	400
	SCS		120	120	120	120	120	120
	N _{RB}		132	264	132	264	132	264
Equivalent satellite antenna aperture			5.9m		0.6m		0.6m	
Satellite Tx max Gain			58.5 dBi		38.5 dBi		38.5 dBi	
3dB beamwidth or HPBW (Half-Power BandWidth) of main central beam			0.1765 deg		1.7647 deg		1.7647 deg	
ABS (Adjacent Beam Spacing) of adjacent beams from the central beam		Note 1		Note 1		Note 1		
Satellite beam diameter		110 km		40 km		20 km		
Payload characteristics for UL transmissions								
Equivalent satellite antenna aperture		27GHz	3.66m		0.36m		0.36m	
Satellite Rx max Gain			58.5 dBi		38.5 dBi		38.5 dBi	
NOTE1: ABS[rad] = sqrt(3) x sin(HPBW[degrees]/2) or ABS[rad] = sqrt(3) x sinr(HPBW[rad]/2) with ABS [degree]=180/pi x ABS[rad] and with HPBW the Half-Power BandWidth of the main lobe from the satellite antenna pattern.								

Table 6a.2.2.1-3: Other parameters for NTN satellite

Parameters	NTN satellite	Remark
Carrier frequency	27GHz(UL) / 17GHz(DL)	
The number of active UE (UL)	10 UEs and 13RBs per UE for GEO and LEO	
The number of active UE (DL)	1	Same with TN
Traffic model	Full buffer	
DL power control	NO	
UL power control	See Session 6a.2.6.2	
NTN satellite Noise Figure in dB	See Table 6a.2.2.1-4	
Handover margin	3dB	

Table 6a.2.2.1-4: NTN satellite Noise Figure in dB

Satellite	GEO	LEO 600	LEO 1200
G/T (dB K ⁻¹)	30.4	10.4	10.4
NF (dB)	3.5	3.5	3.5
NOTE: The NF of 5.9dB is used for all satellite types for calibration.			

6a.2.2.2 NTN UE parameters

NTN UE parameters are given in Table 6a.2.2.2-1.

Table 6a.2.2.2-1: NTN UE characteristics for system level simulations

Characteristics	Fixed VSAT	L-ESIM
Frequency band	27GHz UL/17GHz DL	27GHz UL/17GHz DL
Antenna type and configuration	[TBD]	[TBD]
Polarisation	Circular	Circular
Rx Antenna gain	[TBD]	[TBD]
Antenna temperature	[TBD]	[TBD]
Noise figure ¹	2.5dB for GEO [2,5dB and 6dB for LEO]	2.5dB for GEO [2,5dB and 6dB for LEO]
Tx transmit power	[TBD]	[TBD]
Tx antenna gain	[TBD]	[TBD]
Antenna elevation angle ²	25° and 90°	25° and 90°
NOTE1: The NF of 4dB is used for calibration.		
NOTE2: 30 and 90 degree are used for calibration.		

The altitudes of NTN UE for different use case are defined in Table 6a.2.2.2-2.

Table 6a.2.2.2-3: NTN UE altitude

Use case	Fixed VSAT (on roof)	M-ESIM	A-ESIM	L-ESIM
Altitude	22.5m	22.5m	3-14km	1.5m

6a.2.2.3 Void

6a.2.2.4 TN parameters

TN parameters for co-existence study are given in Table 6a.2.2.4-1 and Table 6a.2.2.4-2.

Table 6a.2.2.4-1: Simulation assumptions of TN NR

Carrier frequency in GHz	27
Size of each nominal channel BW in MHz	200
Transmission bandwidth in MHz	190.8
Environment	Urban macro
Network layout	hexagonal grid, 19 macro sites, 3 sectors per site with wrap around
Inter-site distance in meter	300m
UE number per cell	Same as the number of BS beam
System loading and activity	20%
Network location	See Table 6a.2.1.1-2
DL subcarrier spacing	120kHz
UL	OFDMA
DL power control	No
UL power control	See Section 6a.2.6.2
Frequency reuse	1
Number of scheduled UE per cell (DL)	Same as the number of BS beam
Number of scheduled UE per cell (UL)	Same as the number of BS beam
BS height	25
UE antenna height in meter	1.5
UE TX power in dBm	-40 to 23
Building penetration loss	In path loss model, TR 38.901
Cell selection margin in dB	3
BS-MS min distance in meters	35
BS noise figure in dB	10
UE noise figure in dB	10
BS-UE path-loss model	TR 38.901
Standard deviation of BS-UE log-normal shadow fading in dB	Deployment scenario related, referring to TR 38.901.
Shadowing correlation	Between cells: 1.0 Between sites: 0.5
Link-level performance model	See Section 6a.2.9
UE distribution	Uniform
Evaluation metrics	See Section 6a.2.8

Table 6a.2.2.4-2: ACLR/ACS for TN (27 GHz)

		NR
BS	ACLR	28 dB
	ACS	24 dB
UE	ACLR	17 dB
	ACS	23 dB

RAN4 did not have any consensus on the ACLR and ACS values for TN BS and TN UE at 17 GHz and therefore those values were not reported in this TR.

6a.2.3 Antenna and beam forming pattern modelling

6a.2.3.1 Satellite parameters

In Rel-18, antenna pattern in section 6.4.1 of TR38.811 is reused for NTN SAN and NTN UE. Use of phased array antenna model needs further discussion.

The following normalized antenna gain pattern, corresponding to a typical reflector antenna with a circular aperture, is considered.

$$\begin{aligned}
 &1 \quad \text{for } \theta = 0 \\
 &4 \left| \frac{J_1\left(ka \sin\left(\theta - \frac{\pi}{180}\right)\right)}{ka \sin\left(\theta - \frac{\pi}{180}\right)} \right|^2 \quad \text{for } 0 < |\theta| \leq 90^\circ
 \end{aligned}$$

$$\text{For UE, } 4 \left| \frac{J_1\left(ka \sin\left(\frac{\pi}{2}\right)\right)}{ka \sin\left(\frac{\pi}{2}\right)} \right|^2 \quad \text{for } |\theta| > 90^\circ$$

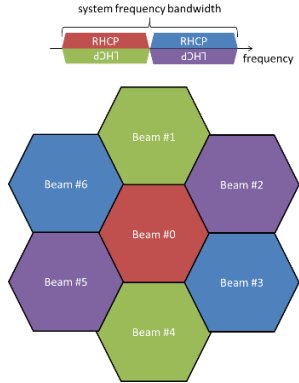
where:

- $J_1(x)$ is the Bessel function of the first kind and first order with argument 'x';
- a is the radius of the antenna's circular aperture;
- $k = 2\pi f/c$ is the wave number;
- f is the frequency of operation;
- c is the speed of light in a vacuum and θ is the angle measured from the bore sight of the antenna's main beam.

Note that ka equals to the number of wavelengths on the circumference of the aperture and is independent of the operating frequency. And the $\sin()$ function is in radian.

The beam layout definition for a single satellite simulation in Ka-Band is defined in Table 6a.2.3.1-1.

Table 6a.2.3.1-1: Beam layout definition for single satellite simulation

Beam layout definition	Same with Table 6.2.3.1-1
Number of beams	Same with Table 6.2.3.1-1
UV plane illustration (extracted from [19])	Same with Table 6.2.3.1-1
UV plane convention	Same with Table 6.2.3.1-1
Adjacent beam spacing on UV plane	Same with Table 6.2.3.1-1
Central beam bore sight direction definition	Baseline: Case 1: Central beam center is considered at nadir point Case 2: 25° for GEO and LEO
Frequency-reuse factor	FRF=2  Do not consider interference leakage from adjacent beams as starting point for co-ex study. For example for the figure above, do not consider the interference leakage from green, blue and purple beams when study SINR for the red beam.
Polarization re-use	Enable Note: Polarization re-use should apply only if circular polarization for terminal antenna is considered
UEs outdoor/indoor distribution	Same with Table 6.2.3.1-1
UE distribution	Same with Table 6.2.3.1-1
UE configuration	Fixed and mobile VSAT for GSO Fixed VSAT for NGSO
UE orientation	Random Ideal Tracking serving beam
UE attachment	RSRP

6a.2.3.2 TN BS and UE antenna and beam forming pattern modelling

The parameters in Table 6a.2.3.2-1 are used for BS antenna pattern in the coexistence study.

Table 6a.2.3.2-1: TN BS antenna modelling for Urban macro scenario

Parameters	Values
Scenario	Urban macro
Antenna pattern	TR 38.803
Element gain $G_{E,max}$ (dBi)	5.5
Horizontal 3dB /vertical 3dB beam width of single element	90° for H 90° for V
Horizontal/vertical front-to-back ratio A_m (dB)	30 for both H/V
Side lobe suppression SLA_v (dB)	30
Antenna polarization	Linear $\pm 45^\circ$
Antenna array configuration (Row $\times$ Column)	16 $\times$ 8 elements
Horizontal/Vertical radiating element spacing	$d_h = 0.5 \lambda$ $d_v = 0.5 \lambda$
Array Ohmic loss LE (dB)	2
Conducted power (before Ohmic loss) per antenna element (dBm)	22
Mechanical downtilt (degrees)	10
Antenna element vertical radiation pattern (dB)	$A_{E,V}(\theta'') = -\min \left\{ 12 \left(\frac{\theta'' - 90^\circ}{\theta_{3dB}} \right)^2, SLA_v \right\}, \theta_{3dB} = 90^\circ, SLA_v = 30dB$
Antenna element horizontal radiation pattern (dB)	$A_{E,H}(\varphi'') = -\min \left\{ 12 \left(\frac{\varphi''}{\varphi_{3dB}} \right)^2, A_m \right\}, \varphi_{3dB} = 90^\circ, A_m = 30dB$
Combining method for 3D antenna element pattern (dB)	$A''(\theta'', \varphi'') = -\min \left\{ -[A_{E,V}(\theta'') + A_{E,H}(\varphi'')], A_m \right\}$
Maximum directional gain of an antenna element $G_{E,max}$	5.5 dBi
(M_g, N_g, M, N, P) ^{note}	For 30GHz: (1, 1, 8, 16, 2)
(d_v, d_h)	$(0.5\lambda, 0.5\lambda)$
Note:	An additional 3dB gain is added to the total beamforming gain to account for the two polarization directions. Boresight direction is horizontal.

For TN UE, the antenna parameters refer to TR 38.803 and are defined in Table 6a.2.3.2-2.

Table 6a.2.3.2-2: TN UE antenna parameters

Parameter	Values
Antenna element vertical radiation pattern (dB)	$A_{E,V}(\theta'') = -\min\left\{12\left(\frac{\theta'' - 90^\circ}{\theta_{3\text{dB}}}\right)^2, SLA_V\right\}, \theta_{3\text{dB}} = 90^\circ, SLA_V = 25 \text{ dB}$
Antenna element horizontal radiation pattern (dB)	$A_{E,H}(\varphi'') = -\min\left\{12\left(\frac{\varphi''}{\varphi_{3\text{dB}}}\right)^2, A_m\right\}, \varphi_{3\text{dB}} = 90^\circ, A_m = 25 \text{ dB}$
Combining method for 3D antenna element pattern (dB)	$A''(\theta'', \varphi'') = -\min\left\{-[A_{E,V}(\theta'') + A_{E,H}(\varphi'')], A_m\right\}$
Maximum directional gain of an antenna element GE,max	5 dBi
(Mg, Ng, M, N, P)	(1, 1, 2, 2, 2)
(dv, dh)	(0.5λ, 0.5λ)
UE orientation	Random orientation in the azimuth domain: uniformly distributed between -90 and 90 degrees* Fixed elevation: 90 degrees
NOTE:	This is done to emulate two panels: the configuration is equivalent to 2 panels with 180 shift in horizontal orientation and UE orientation uniformly distributed in the azimuth domain between -180 and 180 degrees.

6a.2.4 ACIR model

ACLR modelling for TN and NTN co-existence study referring to clause 5.1.1.4.1 and 5.1.1.4.2 in TR 36.942[8] is used as baseline. The number of RBs refers to Table 6a.2.2.1-3 and Table 6a.2.2.4-1 respectively.

6a.2.5 Propagation model

Propagation models between NTN and TN components used for the coexistence study are listed in Table 6a.2.5-1.

Table 6a.2.5-1: Propagation models between NTN and TN

Link	Propagation model
TN BS to Fixed VSAT on roof	Free space path loss
TN BS to L-ESIM at 1.5 m	UMa as in 3GPP TR 38.803
TN BS to TN UE	UMa as in 3GPP TR 38.803
TN UE to Fixed VSAT on roof	UMa as in 3GPP TR 38.803 (BS is to be replaced with VSAT)
TN UE to L-ESIM	Umi
Satellite to TN BS/UE	3GPP TR 38.821
Satellite to VSAT/ESIM	3GPP TR 38.821
TN BS to Satellite	3GPP TR 38.821
Note1:	For the propagation models which use the 3GPP TR 38.821 [3], to use same assumptions as in [3] and consider the atmospheric losses and the scintillation losses.
Note2:	TN BS height is 25m

6a.2.6 Transmission power control model

6a.2.6.1 UL TPC

Following UL power control model for both NTN and TN are used as starting point.

$$P_t = P_{\max} \times \min \left\{ 1, \max \left[R_{\min}, \left(\frac{CL}{CL_{x-ile}} \right)^\gamma \right] \right\}$$

Where,

- $P_{\max}$ = maximum UE Tx power, i.e. 23 dBm for TN UE, and 33 dBm for fixed VSAT NTN UE;
- $R_{\min}$ = minimum power reduction ratio, i.e. -63 dB for TN UE, and [-63 dB by assuming NTN UE min Tx power as -30dBm as starting point] for NTN UE;
- CL_{x-ile} and γ are set as following:
 - $\gamma = 1$;
 - $CL_{x-ile} = -\text{SNR}_{\text{target}} + P_{\max} - \text{Thermal Noise} - \text{BS/SAN_NoiseFigure} - 10 \cdot \log_{10}(\text{BW})$, considering $\text{SNR}_{\text{target}}$ is 15dB and BW is actual UL BW.

6a.2.6.2 DL TPC

For downlink scenario, no power control scheme is applied.

6a.2.7 Received power model

See Section 6.2.7.

6a.2.8 Performance metric

See Section 6.2.8.

6a.2.9 Throughput ~ SNR mapping

See Section 6.2.9. [Further discuss the applicability of the mapping for NTN link.]

6a.3 Co-existence simulation methodology for 17/27GHz

6a.3.1 Simulation procedure

See Section 6.3.1, noting the FRF and network deployment for NTN and TN in above 10GHz are specified accordingly.

Following approaches for the deployment of NTN UE and satellite are used:

- The satellite should be generated in the visible area/sky of NTN UE;
- NTN UEs point to the satellite accurately;
- The position of the satellite should guarantee that NTN UE vertical angle towards the satellite;
 - For calibration, use 30 degree and 90 degree instead of the range;
- SAN elevation angle value, 25 & 90 degree (computed from the centre of the beam), for co-existence - simulation;
- Horizontal angle of NTN UEs should be calculated based on the satellite position;
- Use 35m as minimum distance assumed between VSAT and TN BS for co-ex study.

6a.3.2 Methods and principle to process co-existence simulation results

See Section 6.3.2.

6a.4 Co-existence simulation results for 17/27GHz

6a.4.1 Scenario 1a: 27GHz NTN UL interfering TN UL

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select the 5% throughput loss of NR-NTN GEO UL and NR-NTN LEO1200 UL interfering the NR UL equipped with AAS antenna 25 degree elevation angle that deployed in urban environment as the most stringent case.

Table 6a.4.1-1 Simulation results for average throughput loss for Scenario 1a - NTN GEO

ACIR(dB)	0.0	2.0	4.0	5.0	6.0	8.0	10.0	12.0	14.0	15.0	16.0	18.0	20.0
Ericsson	3.1						0.9						0.2
ZTE	6.7	5.3	4.1		3.3	2.4	1.8	1.3	0.9		0.6	0.4	0.3
Samsung	2.3	1.8	1.3		1.0	0.8	0.6	0.4	0.3		0.2	0.2	0.1
Huawei					0.0	0.0	0.0						
CATT	10.8	8.4	6.4		4.8	3.5	2.5	1.8	1.3		0.9	0.6	0.4
Qualcomm	7.60			4.6			2.5			1.3			

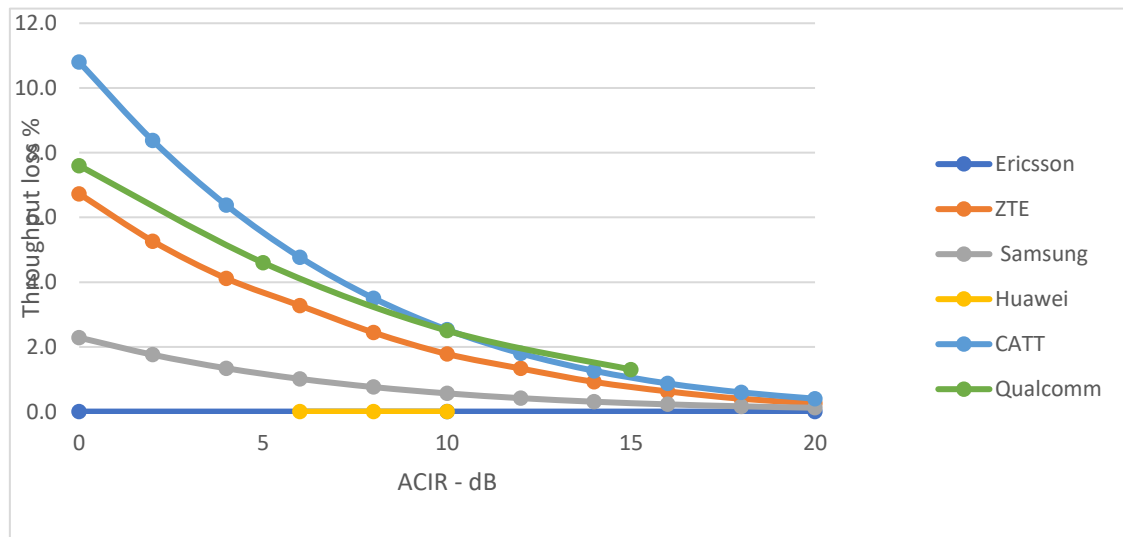


Figure 6a.4.1-1 Simulation results for average throughput loss for Scenario 1a - NTN GEO

Table 6a.4.1-2 Interpolated ACIR values for Scenario 1a to meet the 5% throughput loss criteria - NTN GEO

Company	Interpolated required ACIR
Ericsson	0.00
ZTE	2.50
Samsung	0.00
Huawei	0.00
CATT	5.80
Qualcomm	5.00
NOTE: According to the principles, these values are not treated for later process.	

Table 6a.4.1-3 Simulation results for 5%-tile throughput loss for Scenario 1a - NTN GEO

ACIR(dB)	0.0	2.0	4.0	5.0	6.0	8.0	10.0	12.0	14.0	15.0	16.0	18.0	20.0
Ericsson	11.9						4.6						1.7
ZTE	17.1	12.3	8.6		5.4	3.8	2.6	2.1	1.2		0.9	0.4	0.4
Samsung	4.3	3.2	2.4		1.7	1.2	1.0	0.7	0.5		0.3	0.2	0.2
Huawei					0.0	0.0	0.0						
CATT	35.5	26.8	19.8		13.1	8.5	5.8	4.4	2.9		1.6	0.9	0.6
Qualcomm	41.8			24.7			13.9			6.0			

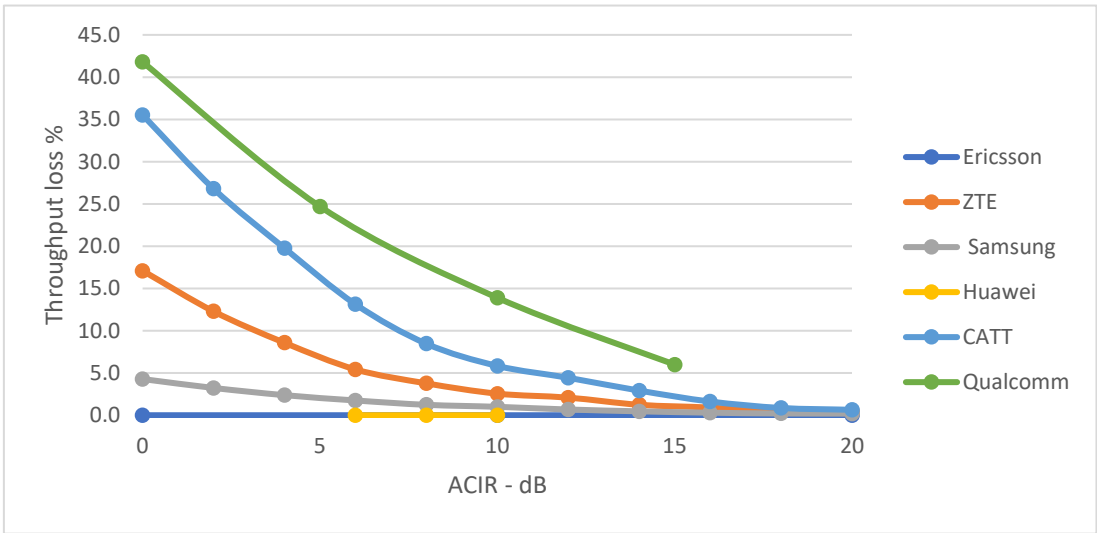


Figure 6a.4.1-2 Simulation results for 5%-tile throughput loss for Scenario 1a - NTN GEO

Table 6a.4.1-4 Interpolated ACIR values for Scenario 1a to meet the 5% throughput loss criteria - NTN GEO

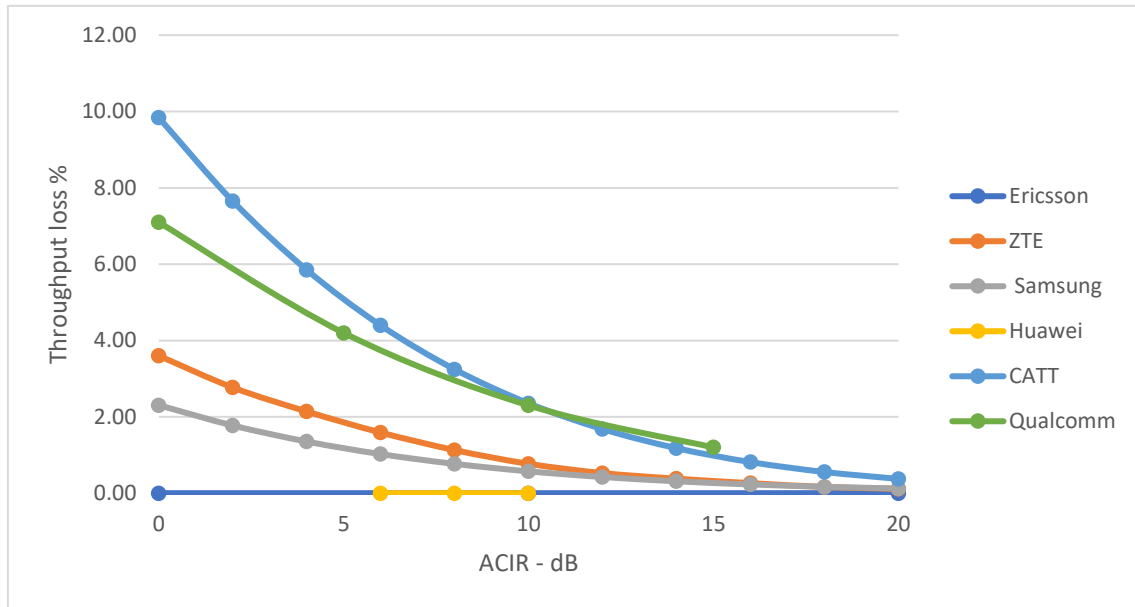
Company	Interpolated required ACIR
Ericsson	9.50
ZTE	6.50
Samsung	0.00
Huawei	0.00
CATT	11.10
Qualcomm	15.00

Table 6a.4.1-5 Averaged ACIR of 5%-tile values in the above worse case for Scenario 1a - NTN GEO

Averaged required ACIR
10.5

Table 6a.4.1-6 Simulation results for average throughput loss for Scenario 1a - NTN LEO1200

Company	0	2	4	5	6	8	10	12	14	15	16	18	20
Ericsson	6.9						2.3						0.7
ZTE	3.60	2.77	2.14		1.59	1.13	0.77	0.53	0.38		0.26	0.16	0.11
Samsung	2.30	1.77	1.35		1.02	0.77	0.57	0.42	0.31		0.23	0.17	0.12
Huawei					0.00	0.00	0.00						
CATT	9.8	7.7	5.9		4.4	3.2	2.4	1.7	1.2		0.8	0.6	0.4
Qualcomm	7.10			4.20			2.30			1.20			

**Figure 6a.4.1-3 Simulation results for average throughput loss for Scenario 1a - NTN LEO1200****Table 6a.4.1-7 Interpolated ACIR values for Scenario 1a to meet the 5% throughput loss criteria - NTN LEO1200**

Company	Interpolated required ACIR
Ericsson	4.10
ZTE	0.00
Samsung	0.00
Huawei	0.00
CATT	5.20
Qualcomm	5.00

NOTE: According to the principles, these values are not treated for later process.

Table 6a.4.1-8 Simulation results for 5%-tile throughput loss for Scenario 1a - NTN LEO1200

ACIR[dB]	0	2	4	5	6	8	10	12	14	15	16	18	20
Ericsson	35.0						13.0						5.0
ZTE	7.75	5.97	4.10		3.24	2.47	0.79	0.20	0.03		0.03	0.02	0.02
Samsung	4.56	3.30	2.63		1.80	1.32	0.89	0.60	0.46		0.35	0.21	0.13
Huawei					0.00	0.00	0.00						
CATT	35.04	25.78	18.92		13.58	8.57	5.96	4.43	2.70		1.27	0.63	0.51
Qualcomm	39.60			23.50			12.90			5.30			

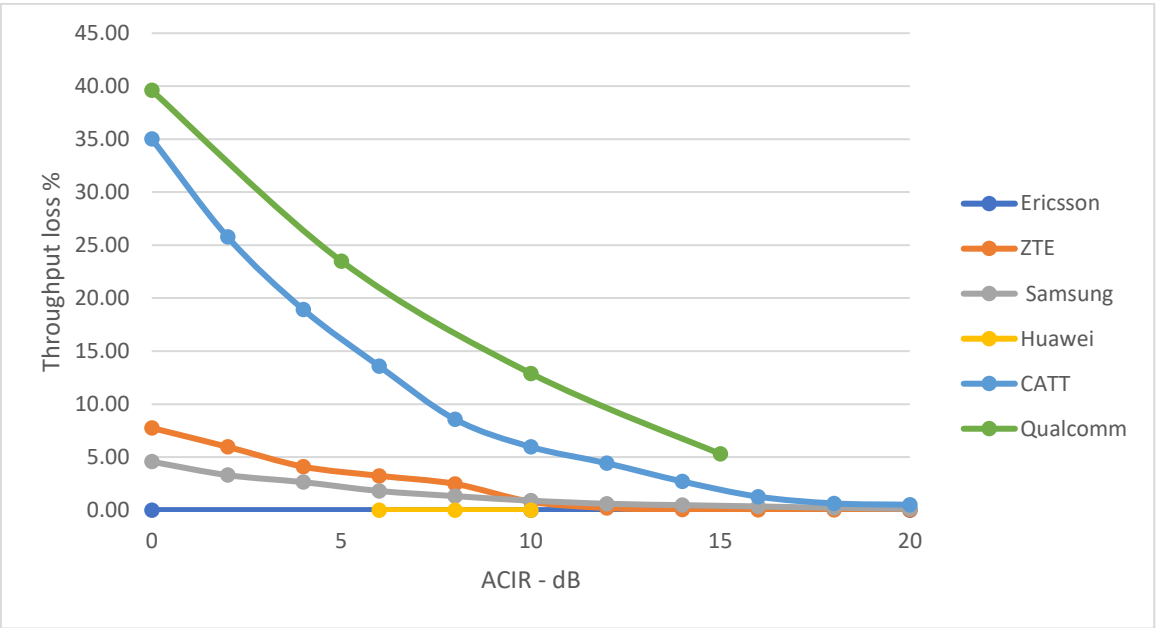


Figure 6a.4.1-4 Simulation results for 5%-tile throughput loss for Scenario 1a - NTN LEO1200

Table 6a.4.1-9 Interpolated ACIR values for Scenario 1a to meet the 5% throughput loss criteria - NTN LEO1200

Company	Interpolated required ACIR
Ericsson	20.00
ZTE	3.10
Samsung	0.00
Huawei	0.00
CATT	11.30
Qualcomm	15.00
NOTE: According to the principles, some of the values are not treated for later process.	

Table 6a.4.1-10 Averaged required ACIR of 5%-tile values in the above worse case for Scenario 1a - NTN LEO1200

Averaged required ACIR
13.15

The following values were also reported by the companies for GEO at 90° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.1-11 Simulation results for average throughput loss for Scenario 1a - NTN GEO 90° elevation angle

Required ACIR [dB]		Comp any	0	2	4	5	6	8	10	12	14	15	16	18	20	22	24	25	26	28	30	32	34	36	38	40	
Throughput Loss	Average	ZTE	0,93	0,77	0,60		0,47	0,29	0,22	0,15	0,13		0,09	0,08	0,06	0,02	0,02		0,02	0,01	0,01						
		THALES	2,81	2,41	2,02		1,62	1,23	0,83	0,71	0,58		0,45	0,33	0,20	0,17	0,14		0,10	0,07	0,04	0,03	0,02	0,02	0,01	0,00	
		Ericsson	0.5						0.2							0.0											
		Qualcomm	1,10			0,6			0,30			0,1															0,0
	5%-tile	ZTE	1,54	0,78	0,47		0,23	0,05	0,04	0,00	0,00		0,00	0,00	0,00	0,00	0,00		0,00	0,00	0,00						
		THALES	16,95	13,63	10,32		7,00	3,68	0,36	0,29	0,22		0,15	0,07	0,00	0,00	0,00		0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	
		Ericsson	3.0						1.7							0.0											
		Qualcomm	6,0			2,4			0,8				0,4														0,0

The following values were also reported by the companies for GEO at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.1-12 Simulation results for average throughput loss for Scenario 1a - NTN GEO 25° elevation angle

Required ACIR [dB]		Comp any	0	2	4	5	6	8	10	12	14	15	16	18	20	22	24	25	26	28	30	32	34	36	38	40
Throughput Loss	Average	ZTE	1,38	1,19	0,98		0,78	0,67	0,52	0,40	0,31		0,26	0,17	0,14	0,11	0,10		0,06	0,04	0,04					
		THALES	2,81	2,41	2,02		1,62	1,23	0,83	0,71	0,58		0,45	0,33	0,20	0,17	0,14		0,10	0,07	0,04	0,03	0,02	0,02	0,01	0,00
		Ericsson	1.2						0.5						0.2											
		Qualcomm	8,40			6			4,30			3			2,00			1,3			0,10					0,0
	5%-tile	ZTE	4,55	3,32	2,81		2,29	1,79	0,93	0,88	0,81		0,81	0,59	0,32	0,28	0,28		0,28	0,20	0,08					
		THALES	16,95	13,63	10,32		7,00	3,68	0,36	0,29	0,22		0,15	0,07	0,00	0,00	0,00		0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
		Ericsson	9.1						4.4						1.7											
		Qualcomm	64,4			47,3			34,0			23,3			15,5			10,3			6,5					0,0

The following values were also reported by the companies for LEO1200km at 90° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.1-13 Simulation results for average throughput loss for Scenario 1a - NTN LEO@1200km 90° elevation angle

Required ACIR [dB]		Comp any	0	2	4	5	6	8	10	12	14	15	16	18	20	22	24	25	26	28	30	32	34	36	38	40
Throughput Loss	Average	Qualcomm	0,70			0,30			0,17			0,07														
		THALES	0,63	0,53	0,43		0,32	0,22	0,12	0,10	0,08		0,06	0,03	0,01	0,01	0,01		0,01	0,00	0,00	0,00	0,00	0,00	0,00	0,00
	5%-tile	Qualcomm	3,20			1,20			0,60			0,30														
		THALES	1,29	1,03	0,77		0,52	0,26	0,00	0,00	0,00		0,00	0,00	0,00	0,00	0,00		0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00

The following values were also reported by the companies for LEO1200km at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

**Table 6a.4.1-14 Simulation results for average throughput loss for Scenario 1a - NTN LEO@1200km
25° elevation angle**

Required ACIR [dB]		Com pany	0	2	4	5	6	8	10	12	14	15	16	18	20	22	24	25	26	28	30	32	34	36	38	40
Throu ghput Loss	Aver age	Qualc omm	8,0 0			5,7 0			4,1 0			2,8 0			1,9 0			1, 3 0			0, 8 0					
		THAL ES	2,1 4	1,8 4	1,5 4		1, 2 4	0, 9 3	0,6 3	0,5 3	0,4 3		0, 3 3	0,2 3	0,1 3	0, 1 1	0,0 8		0, 0 6	0,0 4	0,0 2	0,0 2	0,0 1	0,0 1	0,0 1	0,0 0
	5%- tile	Qualc omm	61, 50			45, 00			31, 70			21, 40			15, 00			9, 9 0			6, 0 0					
		THAL ES	15, 82	13, 04	10, 26		7, 4 8	4, 7 0	1,9 1	1, 6 7	1, 4 4		1, 2 0	0, 9 6	0,7 2	0, 5 7	0,4 3		0, 2 9	0,1 4	0,0 0	0,0 0	0,0 0	0,0 0	0,0 0	0,0 0

The following values were also reported by the companies for LEO600km at 90° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

**Table 6a.4.1-15 Simulation results for average throughput loss for Scenario 1a - NTN LEO@600km
90° elevation angle**

Required ACIR [dB]		Comp any	0	2	4	5	6	8	10	12	14	15	16	18	20	22	24	25	26	28	30	32	34	36	38	40
Throu ghput Loss	Aver age	Qualc omm	0, 30			0, 10			0, 07			0, 02														
		THAL ES	0, 24	0, 20	0, 16		0, 12	0, 08	0, 03	0, 03	0, 02		0, 02	0, 01	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00
	5%- tile	Qualc omm	1, 30			0, 60			0, 26			0, 13														
		THAL ES	0, 47	0, 38	0, 28		0, 19	0, 09	0, 00	0, 00	0, 00		0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00

The following values were also reported by the companies for LEO600km at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

**Table 6a.4.1-16 Simulation results for average throughput loss for Scenario 1a - NTN LEO@600km
25° elevation angle**

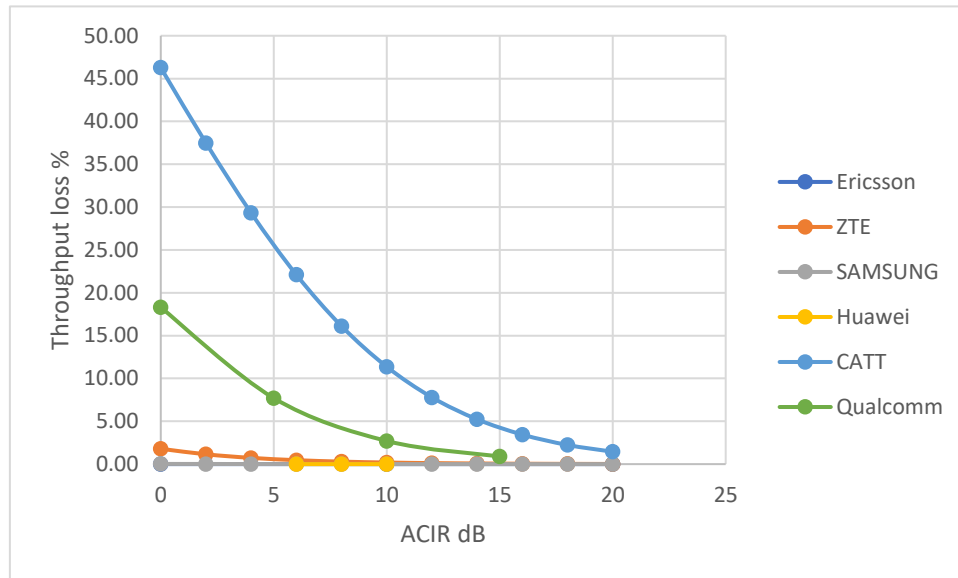
Required ACIR [dB]		Com pany	0	2	4	5	6	8	10	12	14	15	16	18	20	22	24	25	26	28	30	32	34	36	38	40
Throu ghput Loss	Aver age	Qualc omm	7,30			5,20			3,70			2,50			1,70			1,10			0,70					
		THAL ES	1,00	0,85	0,71		0,56	0,42	0,28	0,23	0,19		0,14	0,10	0,05	0,04	0,03		0,02	0,02	0,01	0,01	0,00	0,00	0,00	0,00
	5%-tile	Qualc omm	56,60			41,20			28,90			19,90			13,10			8,20			5,40					
		THAL ES	7,28	5,85	4,43		3,01	1,58	0,16	0,13	0,09		0,06	0,03	0,00	0,00		0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00

6a.4.2 Scenario 2a: 27GHz TN UL interfering NTN UL

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select the 5% throughput loss NR UL interfering the NR-NTN GEO UL and NR-NTN LEO1200 UL that 25 degrees elevation angle and deployed in urban environment as the most stringent case.

Table 6a.4.2-1 Simulation results for average throughput loss for Scenario 2a - NTN GEO

ACIR dB	0	2	4	5	6	8	10	12	14	15	16	18	20
Ericsson	1.0						0.1						0.0
ZTE	1.79	1.14	0.73		0.46	0.29	0.18	0.12	0.07		0.05	0.03	0.02
Samsung	0.02	0.02	0.01		0.01	0.00	0.00	0.00	0.00		0.00	0.00	0.00
Huawei					0.00	0.00	0.00						
CATT	46.3	37.5	29.3		22.1	16.1	11.4	7.8	5.2		3.4	2.2	1.4
Qualcomm	18.30			7.70			2.70			0.90			

**Figure 6a.4.2-1 Simulation results for average throughput loss for Scenario 2a - NTN GEO****Table 6a.4.2-2 Simulation results for 5%-tile throughput loss for Scenario 2a - NTN GEO**

ACIR dB	0	2	4	5	6	8	10	12	14	15	16	18	20
Ericsson	NA						NA						NA
ZTE	1.93	1.24	0.79		0.50	0.32	0.20	0.13	0.08		0.05	0.03	0.02
Samsung	NaN	NaN	NaN		NaN	NaN	NaN	NaN	NaN		NaN	NaN	NaN
Huawei					0.00	0.00	0.00						
CATT	58.87	49.68	40.51		31.8	24.08	17.54	12.34	8.42		5.62	3.68	2.38
Qualcomm	NaN			NaN			NaN			NaN			

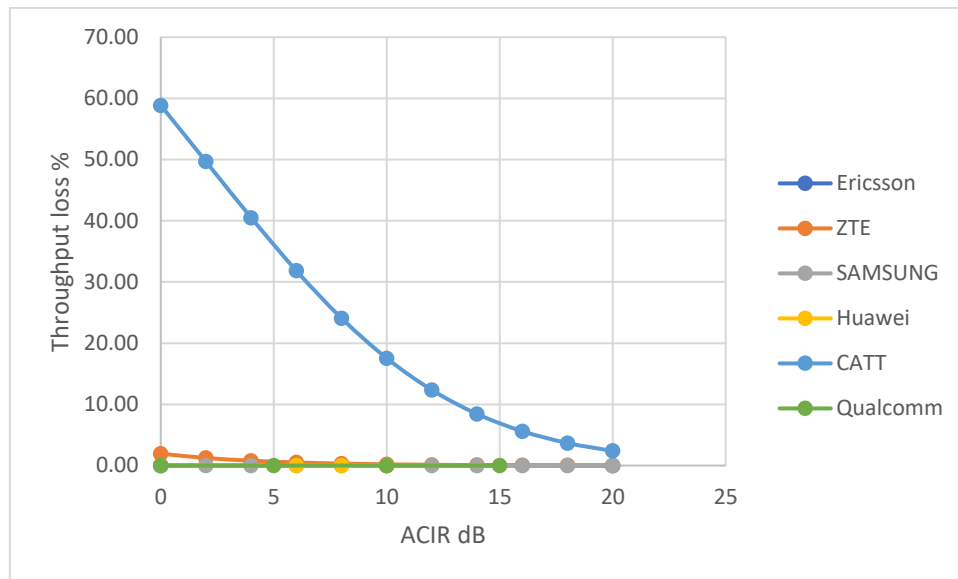


Figure 6a.4.2-2 Simulation results for 5%-tile throughput loss for Scenario 2a - NTN GEO

Table 6a.4.2-3 Interpolated ACIR values for Scenario 2a to meet the 5% throughput loss criteria - NTN GEO

Company	Interpolated required ACIR
Ericsson	0.00
ZTE	0.00
Samsung	0.00
Huawei	0.00
CATT	37.50
Qualcomm	7.7

Table 6a.4.2-4 Averaged ACIR of 5%-tile values in the above worse case for Scenario 2a - NTN GEO

Averaged required ACIR
1.54

Table 6a.4.2-5 Simulation results for average throughput loss for Scenario 2a - NTN LEO1200

ACIR dB	0	2	4	5	6	8	10	12	14	15	16	18
Ericsson	2.3						0.2					
ZTE	3.98	2.58	1.66		1.06	0.67	0.43	0.27	0.17		0.11	0.07
Samsung	0.38	0.24	0.15		0.10	0.06	0.04	0.02	0.02		0.01	0.01
Huawei					0.00	0.00	0.00					
CATT	29.5	22.3	16.3		11.5	7.9	5.3	3.5	2.3		1.5	0.9
Qualcomm	10.30			3.90			1.30			0.40		

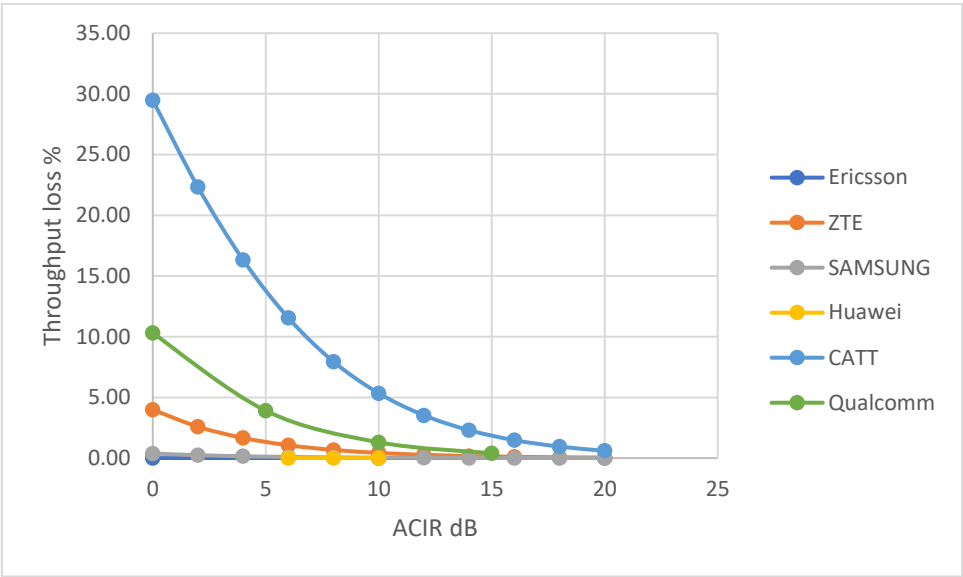


Figure 6a.4.2-3 Simulation results for average throughput loss for Scenario 2a - NTN LEO1200

Table 6a.4.2-6 Simulation results for 5%-tile throughput loss for Scenario 2a - NTN LEO1200

ACIR dB	0	2	4	5	6	8	10	12	14	15	16	18
Ericsson	NA						NA					
ZTE	3.98	2.58	1.66		1.06	0.67	0.43	0.27	0.17		0.11	0.07
Samsung	NaN	NaN	NaN		NaN	NaN	NaN	NaN	NaN		NaN	NaN
Huawei					0.00	0.00	0.00					
CATT	41.34	32.60	24.75		18.1	12.76	8.73	5.83	3.83		2.48	1.59
Qualcomm	NaN			NaN			NaN			NaN		

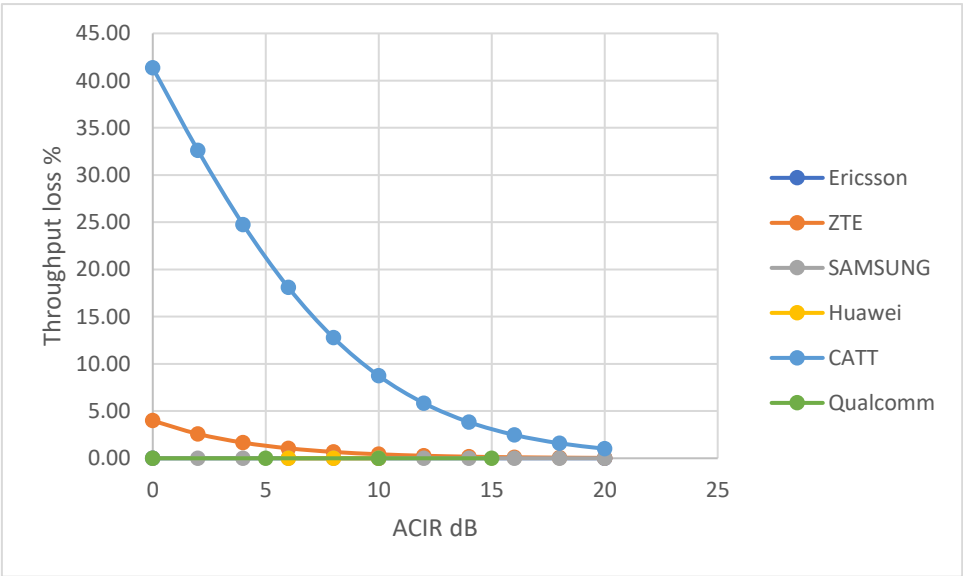


Figure 6a.4.2-4 Simulation results for 5%-tile throughput loss for Scenario 2a - NTN LEO1200

Table 6a.4.2-7 Interpolated ACIR values for Scenario 2a to meet the 5% throughput loss criteria - NTN LEO1200

Company	Interpolated required ACIR
Ericsson	0.00
ZTE	0.00
Samsung	0.00
Huawei	0.00
CATT	33.95
Qualcomm	4.10

Table 6a.4.2-8 Averaged ACIR of 5%-tile values in the above worse case for Scenario 2a - NTN LEO1200

Averaged required ACIR
0.82

The following values were also reported by the companies for GEO at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.2-9 Simulation results for average throughput loss for Scenario 2a - NTN GEO 25° elevation angle

Required ACIR [dB]	Comp any	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	ZTE	1,76	1,12	0,71	0,45	0,29	0,18	0,11	0,07	0,05	0,03	0,02	0,01	0,01	0,00	0,00	0,00				
		THALES	17,96	14,85	11,75	8,64	5,54	2,43	2,00	1,56	1,13	0,69	0,25	0,21	0,16	0,12	0,07	0,03	0,02	0,01	0,01	0,00
	5%-tile	ZTE	1,87	1,19	0,76	0,48	0,30	0,19	0,12	0,08	0,05	0,03	0,02	0,01	0,01	0,00	0,00	0,00				
		THALES	22,09	18,37	14,66	10,94	7,23	3,51	2,88	2,26	1,63	1,00	0,37	0,31	0,24	0,17	0,11	0,04	0,03	0,03	0,02	0,01

The following values were also reported by the companies for LEO1200km at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.2-10 Simulation results for average throughput loss for Scenario 2a - NTN LEO@1200km 25° elevation angle

Required ACIR [dB]	Comp any	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THALES	16,72	13,81	10,91	8,01	5,10	2,20	1,80	1,41	1,02	0,62	0,23	0,19	0,15	0,10	0,06	0,02	0,02	0,01	0,01	0,00
	5%-tile	THALES	15,16	12,51	9,86	7,21	4,56	1,91	1,57	1,22	0,88	0,54	0,20	0,16	0,13	0,09	0,06	0,02	0,02	0,01	0,01	0,00

The following values were also reported by the companies for LEO600km at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.2-11 Simulation results for average throughput loss for Scenario 2a - NTN LEO@600km 25° elevation angle

Required ACIR [dB]	Comp any	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THALES	17,35	14,34	11,33	8,32	5,31	2,30	1,89	1,47	1,06	0,65	0,24	0,20	0,15	0,11	0,07	0,02	0,02	0,02	0,01	0,00
	5%-tile	THALES	17,33	14,31	11,29	8,27	5,25	2,23	1,83	1,43	1,03	0,63	0,23	0,19	0,15	0,11	0,07	0,03	0,02	0,02	0,01	0,00

6a.4.3 Scenario 3a: 27GHz NTN UL interfering TN DL

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select the 5% throughput loss NR UL interfering the NR-NTN GEO UL and NR-NTN LEO1200 UL that 25 degrees elevation angle, and deployed in urban environment as the most stringent case.

Table 6a.4.3-1 Simulation results for average and 5%-tile throughput loss for Scenario 3a - NTN GEO

Required ACIR [dB]		Company	0	2	4	6	8	10	12	14	16	18	20
Throughput Loss	Average	ZTE	0.07	0.07	0.07	0.07	0.07	0.06	0.06	0.06	0.06	0.06	0.06
		Samsung	0.40	0.28	0.20	0.14	0.10	0.07	0.05	0.03	0.02	0.02	0.01
		Huawei				0.00	0.00	0.00					
		Qualcomm	0.00										
		CATT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	5%-tile	ZTE	0.49	0.49	0.49	0.49	0.49	0.33	0.33	0.33	0.33	0.27	0.27
		Samsung	1.16	0.76	0.54	0.33	0.22	0.16	0.13	0.09	0.05	0.04	0.01
		CATT	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Qualcomm	0.0										

Table 6a.4.3-2 Interpolated ACIR values for Scenario 3a to meet the 5% throughput loss criteria - NTN GEO

Company	Interpolated required ACIR
ZTE	0
Samsung	0
CATT	0
Qualcomm	0

Table 6a.4.3-3 Averaged ACIR of 5%-tile values in the above worse case for Scenario 3a - NTN GEO

Averaged required ACIR
0

Table 6a.4.3-4 Simulation results for average and 5%-tile throughput loss for Scenario 3a - NTN LEO1200

Required ACIR [dB]		ACIR	0	2	4	6	8	10	12	14	16	18	20
Throughput Loss	Average	ZTE	0.03	0.03	0.03	0.03	0.03	0.03	0.03	0.02	0.02	0.02	0.02
		Samsung	0.41	0.29	0.21	0.15	0.11	0.08	0.06	0.04	0.03	0.02	0.02
		Huawei				0.00	0.00	0.00					
		Qualcomm	0.00										
		CATT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	5%-tile	ZTE	0.37	0.37	0.37	0.37	0.37	0.37	0.27	0.27	0.27	0.27	0.27
		Samsung	1.35	0.98	0.75	0.59	0.40	0.27	0.17	0.14	0.10	0.09	0.08
		Huawei				0.00	0.00	0.00					
		Qualcomm	0.0										
		CATT	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table 6a.4.3-5 Interpolated ACIR values for Scenario 3a to meet the 5% throughput loss criteria - NTN LEO1200

Company	Interpolated required ACIR
ZTE	0
Samsung	0
CATT	0
Qualcomm	0

Table 6a.4.3-6 Averaged ACIR of 5%-tile values in the above worse case for Scenario 3a - NTN LEO1200

Averaged required ACIR
0

The following values were also reported by the companies for GEO at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.3-7 Simulation results for average throughput loss for Scenario 3a - NTN GEO 25° elevation angle

Required ACIR [dB]	Company	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	ZTE	0,0 5	0,0 5	0,0 5	0,0 5	0,0 5	0,0 5	0,0 5	0,0 5	0,0 5	0,0 5	0,0 5	0,0 5	0,0 5	0,0 5	0,0 5	0,0 5	0,0 5	0,0 5	0,0 5	0,0 5
		THALES	1,9 0	1,6 7	1,4 4	1,2 0	0,9 7	0,7 4	0,6 4	0,5 4	0,3 5	0,2 5	0,2 1	0,1 8	0,1 4	0,1 1	0,0 7	0,0 6	0,0 5	0,0 4	0,0 3	0,0 2
	5%-tile	ZTE	0,6 3	0,5 0	0,5 0	0,5 0	0,5 0	0,5 0	0,4 7	0,4 7	0,4 7	0,4 7	0,4 2	0,4 2	0,3 8	0,3 8	0,2 7					
		THALES	10, 39	8,9 9	7,6 0	6,2 0	4,8 0	3,4 0	2,8 3	2,2 7	1,7 1	1,1 4	0,5 8	0,4 6	0,3 5	0,2 3	0,1 2	0,0 0	0,0 0	0,0 0	0,0 0	0,0 0

The following values were also reported by the companies for LEO1200km at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.3-8 Simulation results for average throughput loss for Scenario 3a - NTN LEO@1200km 25° elevation angle

Required ACIR [dB]	Company	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THALES	1,0 9	0,9 5	0,8 0	0,6 6	0,5 1	0,3 7	0,3 2	0,2 6	0,2 1	0,1 6	0,1 1	0,1 0	0,0 8	0,0 6	0,0 5	0,0 3	0,0 3	0,0 2	0,0 2	0,0 1
	5%-tile	THALES	3,7 8	3,4 4	3,1 0	2,7 6	2,4 1	2,0 7	1,8 6	1,6 4	1,4 3	1,2 1	1,0 0	0,8 0	0,6 0	0,4 0	0,2 0	0,0 0	0,0 0	0,0 0	0,0 0	0,0 0

The following values were also reported by the companies for LEO600km at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.3-9 Simulation results for average throughput loss for Scenario 3a - NTN LEO@600km 25° elevation angle

Required ACIR [dB]	Company	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THALES	0,8 2	0,7 1	0,6 1	0,5 0	0,4 0	0,3 0	0,2 5	0,2 1	0,1 6	0,1 2	0,0 7	0,0 6	0,0 5	0,0 4	0,0 2	0,0 1	0,0 1	0,0 0	0,0 0	0,0 0
	5%-tile	THALES	5,6 3	4,8 4	4,0 5	3,2 6	2,4 7	1,6 8	1,4 1	1,1 4	0,8 6	0,5 9	0,3 2	0,3 2	0,3 1	0,3 1	0,3 1	0,2 5	0,1 8	0,1 2	0,0 6	0,0 0

6a.4.4 Scenario 4a: 27GHz TN DL interfering NTN UL

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select the 5% throughput loss NR UL interfering the NR-NTN GEO UL and NR-NTN LEO1200 UL that 25 degrees elevation angle and deployed in urban environment as the most stringent case.

Table 6a.4.4-1 Simulation results for average and 5%-tile throughput loss for Scenario 4a - NTN GEO

Required ACIR [dB]		Company	0	2	4	5	6	8	10	12	14	15	16	18	20
Throughput Loss	Average	ZTE	40.55	31.82	24.02		17.46	12.26	8.36	5.56	3.64		2.35	1.51	0.96
		Samsung	59.43	50.55	41.64		33.11	25.40	18.82	13.46	9.34		6.32	4.19	2.73
		CATT	19.4	14.1	10.0		6.8	4.6	3.0	2.0	1.3		0.8	0.5	0.3
		Qualcomm	41.40			23.20			10.70			4.20			
	5%-tile	ZTE	45.23	35.74	27.12		19.78	13.93	9.51	6.34	4.15		2.68	1.72	1.10
		Samsung	NaN	NaN	NaN		NaN	NaN	NaN	NaN	NaN		NaN	NaN	NaN
		Huawei					3.50		1.14						
		Qualcomm	NaN	NaN	NaN		NaN	NaN	NaN	NaN	NaN		NaN	NaN	NaN
		CATT	30.7	23.1	16.7		11.7	8.0	5.3	3.5	2.2		1.4	0.9	0.6

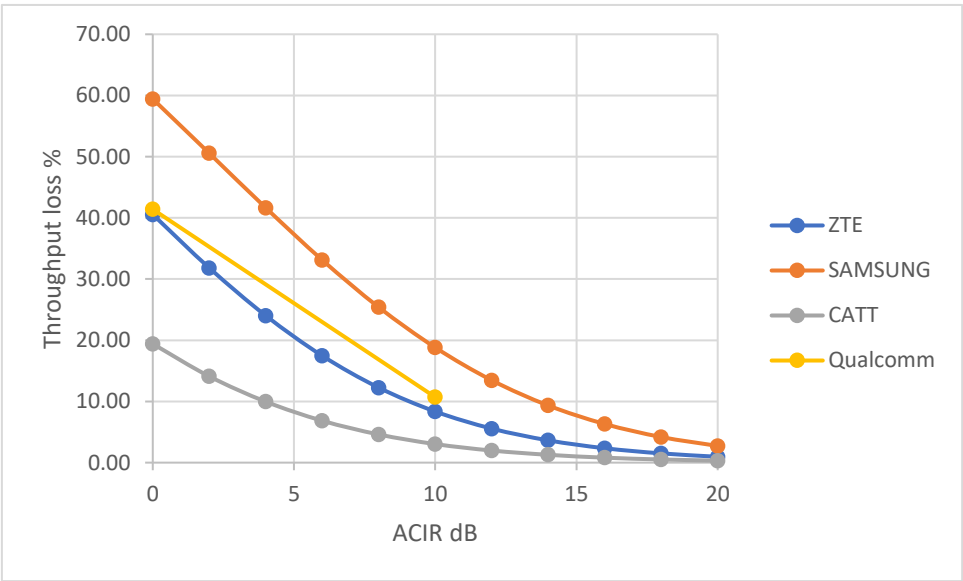


Figure 6a.4.4-1 Simulation results for average throughput loss for Scenario 4a - NTN GEO

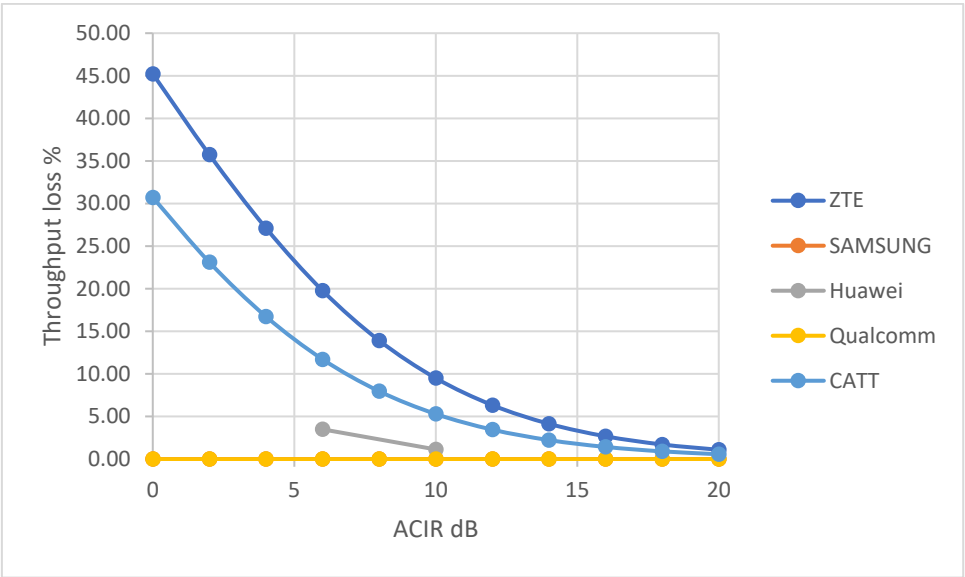


Figure 6a.4.4-2 Simulation results for 5%-tile throughput loss for Scenario 4a - NTN GEO

Table 6a.4.4-2 Interpolated ACIR values for Scenario 4a to meet the 5% throughput loss criteria - NTN GEO

Company	Interpolated required ACIR
ZTE	13.32
Samsung	17.20
CATT	12.22
Qualcomm	14.38
Huawei	<6 (3.5% @ 6)

Table 6a.4.4-3 Averaged ACIR of 5%-tile values in the above worse case for Scenario 4a - NTN GEO

Averaged required ACIR
14.3

Table 6a.4.4-4 Simulation results for average and 5%-tile throughput loss for Scenario 4a - NTN LEO1200

Required ACIR [dB]		ACIR (dB)	0	2	4	5	6	8	10	12	14	5	16	18	20	22	24	26	28	30
Throughput Loss	Average	Ericsson	81.7						36.9						6.5					0.7
		ZTE	59.80	50.62	41.41		32.65	24.78	18.11	12.77	8.74		5.84	3.83	2.48	1.59	1.01	0.64	0.41	0.26
		Samsung	45.76	37.89	30.40		23.59	17.69	12.83	9.03	6.18		4.14	2.72	1.77	1.14	0.73	0.46	0.29	0.19
		CATT	38.6	30.6	23.5		17.4	12.5	8.7	5.9	3.9		2.6	1.7	1.1	0.7	0.4	0.3	0.2	0.1
		Qualcomm	28.10			14			5.8			2.1								
	5%-tile	ZTE	59.77	50.60	41.39		32.63	24.77	18.09	12.76	8.73		5.83	3.83	2.48	1.59	1.01	0.64	0.41	0.26
		Samsung	NaN	NaN	NaN		NaN	NaN	NaN	NaN	NaN		NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
		Huawei					0.33		0.11						0.01					
		CATT	52.93	43.69	34.77		26.65	19.65	13.98	9.64	6.47		4.26	2.77	1.78	1.14	0.72	0.46	0.29	0.18
		Qualcomm	NaN	NaN	NaN		NaN	NaN	NaN	NaN	NaN		NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN

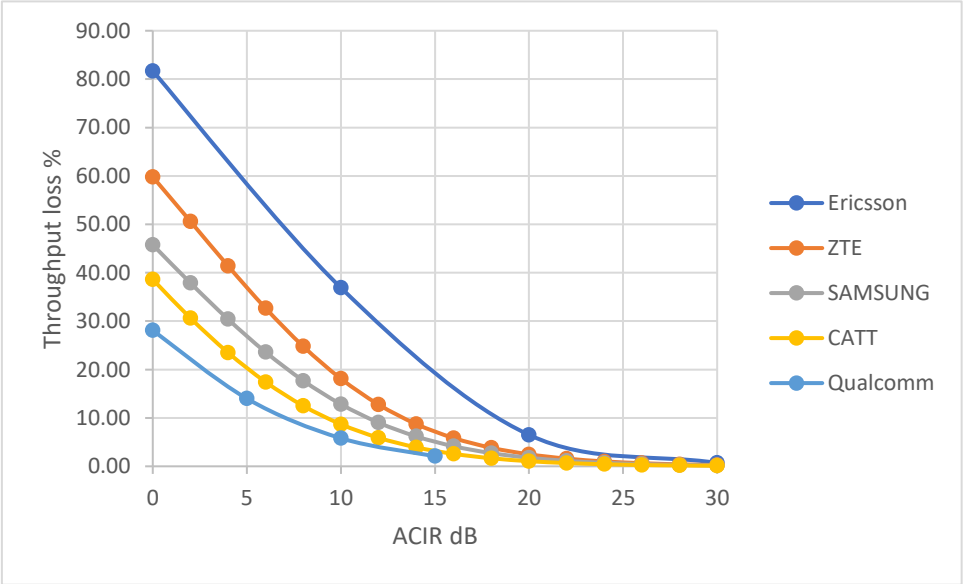


Figure 6a.4.4-3 Simulation results for average throughput loss for Scenario 4a - NTN LEO1200

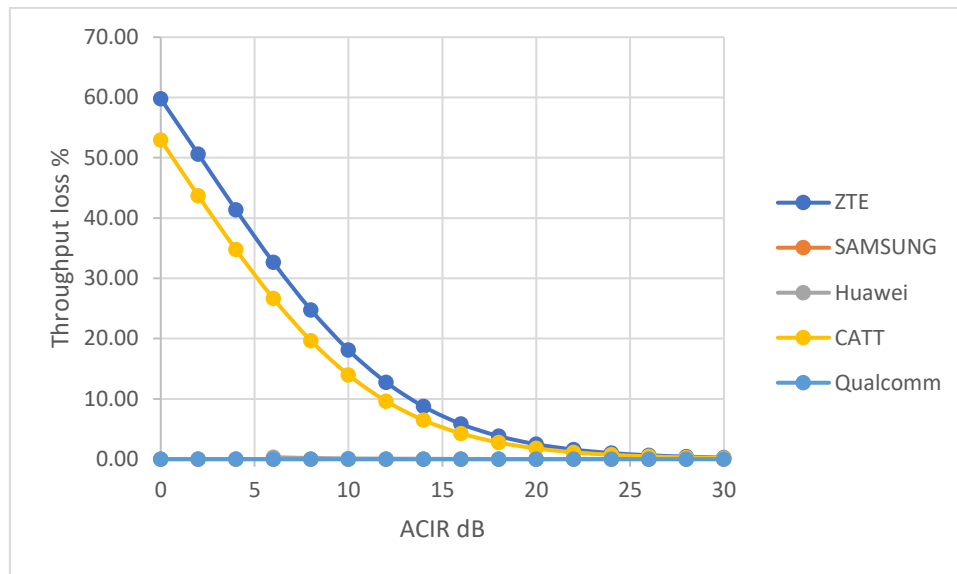


Figure 6a.4.4-4 Simulation results for 5%-tile throughput loss for Scenario 4a - NTN LEO1200

Table 6a.4.4-5 Interpolated ACIR values for Scenario 4a to meet the 5% throughput loss criteria - NTN LEO1200

Company	Interpolated required ACIR
Ericsson	22.60
ZTE	16.83
Samsung	15.10
CATT	15.33
Qualcomm	11.10
Huawei	<6 (0.33@6)

Table 6a.4.4-6 Averaged ACIR of 5%-tile values in the above worse case for Scenario 4a - NTN LEO1200

Averaged required ACIR
16.2

The following values were also reported by the companies for GEO at 90° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.4-7 Simulation results for average throughput loss for Scenario 4a - NTN GEO 90° elevation angle

Required ACIR [dB]		Company	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	ZTE	20,65	14,75	10,20	6,87	4,53	2,95	1,90	1,21	0,77	0,49	0,31	0,20	0,12	0,08	0,05	0,03	0,02	0,01	0,01	0,01	0,01
		THALES	12,57	10,36	8,15	5,94	3,73	1,52	1,25	0,97	0,70	0,43	0,16	0,13	0,10	0,07	0,04	0,02	0,01	0,01	0,01	0,01	0,01
	5%-tile	ZTE	23,33	16,71	11,58	7,81	5,16	3,35	2,16	1,38	0,88	0,56	0,35	0,22	0,14	0,09	0,06	0,04	0,02	0,01	0,01	0,01	0,01
		THALES	14,88	12,27	9,65	7,04	4,42	1,81	1,49	1,16	0,83	0,51	0,18	0,15	0,11	0,08	0,05	0,02	0,01	0,01	0,01	0,01	0,01

The following values were also reported by the companies for GEO at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.4-8 Simulation results for average throughput loss for Scenario 4a - NTN GEO 25° elevation angle

Required ACIR [dB]		Comp any	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	ZTE	41,9 5	33, 12	25, 17	18, 41	13, 00	8,9 0	5,9 5	3,9 0	2,5 3	1,6 2	1,0 3	0,6 6	0,4 2	0,2 6	0,1 7	0,1 1	0, 07	0, 04	0, 03	0, 02	0, 01
		THALES	99,6 1	97, 02	94, 44	91, 85	89, 26	86, 68	78, 85	71, 02	63, 19	55, 36	47, 53	40, 21	32, 89	25, 57	18, 25	10, 93	9, 01	7, 09	5, 17	3, 25	1, 33
	5%-tile	ZTE	45,3 2	35, 96	27, 42	20, 11	14, 22	9,7 5	6,5 2	4,2 8	2,7 7	1,7 8	1,1 3	0,7 2	0,4 6	0,2 9	0,1 8	0,1 2	0, 07	0, 05	0, 03	0, 02	0, 01
		THALES	100, 00	98, 07	96, 14	94, 21	92, 28	90, 35	82, 53	74, 71	66, 89	59, 07	51, 25	43, 14	35, 02	26, 91	18, 79	10, 68	8, 79	6, 91	5, 02	3, 14	1, 26

The following values were also reported by the companies for LEO1200km at 90° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.4-9 Simulation results for average throughput loss for Scenario 4a - NTN LEO@1200km 90° elevation angle

Required ACIR [dB]		Comp any	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THALES	10,7 8	8,8 8	6,9 8	5,0 8	3,1 7	1,2 7	1,0 4	0,8 1	0,5 9	0,3 6	0,1 3	0,1 1	0,0 8	0,0 6	0,0 4	0,0 1	0,0 1	0,0 1	0,0 1	0,0 0	0,0 0
	5%-tile	THALES	10,7 6	8,8 6	6,9 6	5,0 6	3,1 6	1,2 6	1,0 3	0,8 1	0,5 8	0,3 6	0,1 3	0,1 1	0,0 8	0,0 6	0,0 4	0,0 1	0,0 1	0,0 1	0,0 0	0,0 0	0,0 0

The following values were also reported by the companies for LEO1200km at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.4-10 Simulation results for average throughput loss for Scenario 4a - NTN LEO@1200km 25° elevation angle

Required ACIR [dB]		Comp any	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THALES	99, 45	96, 95	94, 46	91, 96	89, 46	86, 96	79, 35	71, 74	64, 13	56, 51	48, 90	41, 47	34, 03	26, 60	19, 17	11, 73	9, 68	7, 62	5, 57	3, 51	1, 45
		THALES	99, 52	97, 35	95, 18	93, 00	90, 83	88, 66	81, 22	73, 77	66, 33	58, 89	51, 45	42, 82	34, 19	25, 56	16, 93	8,3 0	6, 83	5, 36	3, 89	2, 42	0, 95
	5%-tile	THALES	99, 52	97, 35	95, 18	93, 00	90, 83	88, 66	81, 22	73, 77	66, 33	58, 89	51, 45	42, 82	34, 19	25, 56	16, 93	8,3 0	6, 83	5, 36	3, 89	2, 42	0, 95
		THALES	99, 52	97, 35	95, 18	93, 00	90, 83	88, 66	81, 22	73, 77	66, 33	58, 89	51, 45	42, 82	34, 19	25, 56	16, 93	8,3 0	6, 83	5, 36	3, 89	2, 42	0, 95

The following values were also reported by the companies for LEO600km at 90° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.4-11 Simulation results for average throughput loss for Scenario 4a - NTN LEO@600km 90° elevation angle

Required ACIR [dB]		Comp any	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THALES	10,8 7	8,9 6	7,0 4	5,1 2	3,2 0	1,2 8	1,0 5	0,8 2	0,5 9	0,3 6	0,1 3	0,1 1	0,0 8	0,0 6	0,0 4	0,0 1	0,0 1	0,0 1	0,0 1	0,0 0	0,0 0
	5%-tile	THALES	10,6 7	8,7 9	6,9 0	5,0 1	3,1 3	1,2 4	1,0 2	0,7 9	0,5 7	0,3 5	0,1 3	0,1 0	0,0 8	0,0 6	0,0 4	0,0 1	0,0 1	0,0 1	0,0 0	0,0 0	0,0 0

The following values were also reported by the companies for LEO600km at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.4-12 Simulation results for average throughput loss for Scenario 4a - NTN LEO@600km 25° elevation angle

Required ACIR [dB]		Comp any	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THALES	99,4 9	96, 95	94, 42	91, 88	89, 35	86, 82	79, 14	71, 46	63, 78	56, 10	48, 48	41, 01	33, 60	26, 20	18, 79	11, 38	9, 39	7, 39	5, 39	3, 39	1, 40
		THALES	99,4 9	96, 95	94, 42	91, 88	89, 35	86, 82	79, 14	71, 46	63, 78	56, 10	48, 48	41, 01	33, 60	26, 20	18, 79	11, 38	9, 39	7, 39	5, 39	3, 39	1, 40
	5%-tile	THALES	100, 00	97, 66	95, 33	92, 99	90, 66	88, 32	80, 99	73, 67	66, 34	59, 01	51, 69	43, 34	34, 98	26, 63	18, 28	9,9 3	8, 17	6, 42	4, 66	2, 90	1, 15
		THALES	100, 00	97, 66	95, 33	92, 99	90, 66	88, 32	80, 99	73, 67	66, 34	59, 01	51, 69	43, 34	34, 98	26, 63	18, 28	9,9 3	8, 17	6, 42	4, 66	2, 90	1, 15

6a.4.5 Scenario 5a: 17GHz TN DL interfering NTN DL

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select the 5% throughput loss NR UL interfering the NR-NTN GEO UL and NR-NTN LEO1200 UL that 25 degrees elevation angle, 2.5dB noise figure and deployed in urban environment as the most stringent case.

Table 6a.4.5-1 Simulation results for average and 5%-tile throughput loss for Scenario 5a - NTN GEO

Required ACIR [dB]	Com pany	18	20	22	24	25	26	28	30	32	34	35	36	38	40	42	44	45	46
Thro ughp ut Loss	Ave rag e	Eric son													10 .6 0	6. 70	4. 80		3. 50
		ZTE	37 .9 5	31 .0 5	24 .7 9	19 .3 2		14 .6 8	10 .8 9	7. 88	5. 57	3. 86		2. 62	1. 74	1. 15			
		Sam sung	9. 10	6. 84	5. 07	3. 72		2. 70	1. 95	1. 40	1. 01	0. 73		0. 52	0. 38	0. 28			
		CAT T	34 .9	28 .8	23 .3	18 .5		14 .3	10 .9	8. 1	5. 9	4. 2		3. 0	2. 1	1. 4	0. 9	0. 6	0. 4
		Qual com m		75 .6 0			61 .4 0			45 .5 0			30 .9 0			19 .3 0			11 .2
		Eric son													N A	N A	N A		N A
	5%- tile	ZTE	75 .4 6	66 .0 3	57 .6 4	46 .1 1		39 .0 6	28 .2 2	19 .1 5	12 .5 3	10 .1 3		4. 57	2. 31	1. 91			
		Sam sung	N a N	N a N	N a N	N a N		N a N	N a N	N a N	N a N	N a N		N a N	N a N	N a N			
		Hua wei		10 0. 0															
		CAT T	57 .2	50 .5	43 .9	37 .3		31 .0	25 .0	19 .6	14 .8	10 .8		7. 6	5. 2	3. 5	2. 29	1. 49	0. 95
		Qual com m	N a N	N a N															

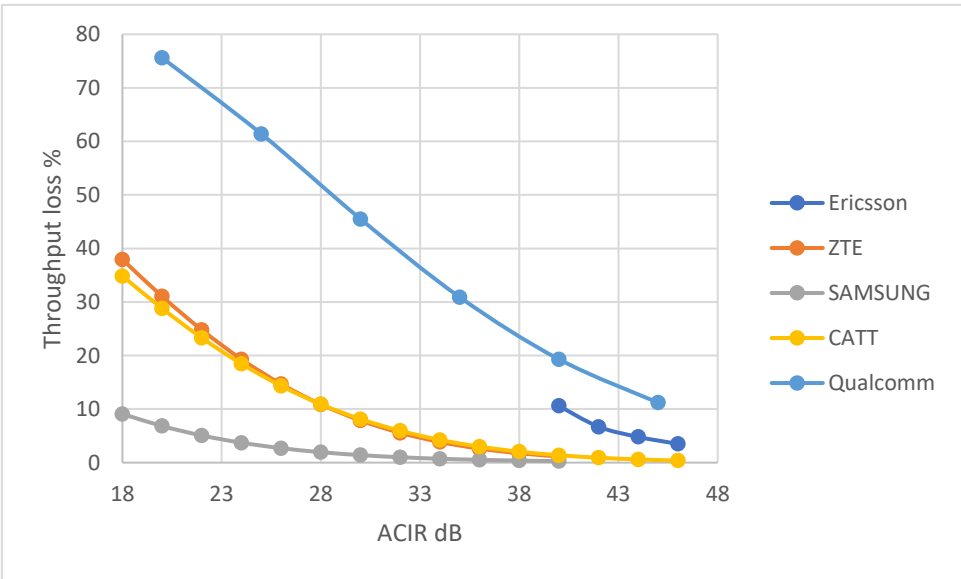


Figure 6a.4.5-1 Simulation results for average throughput loss for Scenario 5a - NTN GEO

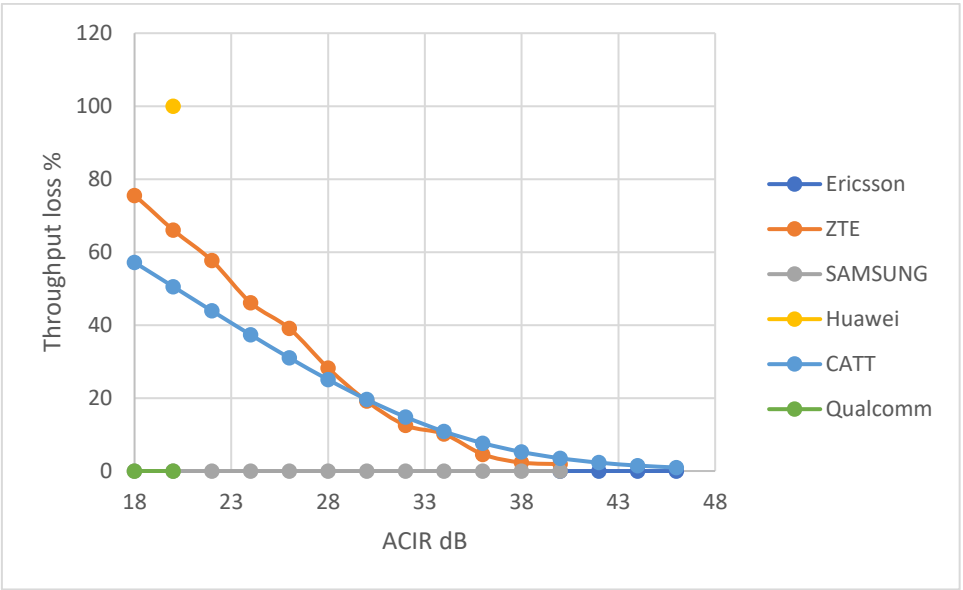


Figure 6a.4.5-2 Simulation results for 5%-tile throughput loss for Scenario 5a - NTN GEO

Table 6a.4.5-2 Interpolated ACIR values for Scenario 5a to meet the 5% throughput loss criteria - NTN GEO

Company	Interpolated required ACIR
Ericsson	43.8
ZTE	35.8
Samsung	22.1
CATT	38.2
Qualcomm	51.2
Huawei	>20 (100@20)

Table 6a.4.5-3 Averaged ACIR of 5%-tile values in the above worse case for Scenario 5a - NTN GEO

Averaged required ACIR
39.2

Table 6a.4.5-4 Simulation results for average and 5%-tile throughput loss for Scenario 5a - NTN LEO1200

Required ACIR [dB]	Company	18	20	22	24	25	26	28	30	32	34	35	36	38	40	42	44	45	46
Throughput Loss	Average	Ericsson	89.2						55.0						21.9				
		ZTE	61.19	53.45	45.65	38.06		30.97	24.63	19.15	14.54	10.80		7.84	5.57	3.88			
		Sam sung	13.72	10.66	8.20	6.21		4.66	3.45	2.53	1.84	1.32		0.94	0.66	0.47			
		CAT T	41.7	34.7	28.3	22.6		17.7	13.5	10.1	7.4	5.3		3.8	2.6	1.8	1.2	0.8	
		Qual com m		81.50			68.4			52.90			36.9			23.5			13.7
	5%-tile	Ericsson		N A					N A						N A				
		ZTE	100.00	100.00	85.54	78.45		68.55	54.72	43.78	40.13	30.75		20.65	17.19	11.77			
		Sam sung	N a N	N a N	N a N	N a N		N a N	N a N	N a N	N a N	N a N		N a N	N a N	N a N			
		Hua wei		100.0															
		CAT T	60.1	53.1	45.9	38.9		32.1	25.7	19.9	14.9	10.8		7.5	5.1	3.4	2.2	1.4	
		Qual com m	N a N	N a N															

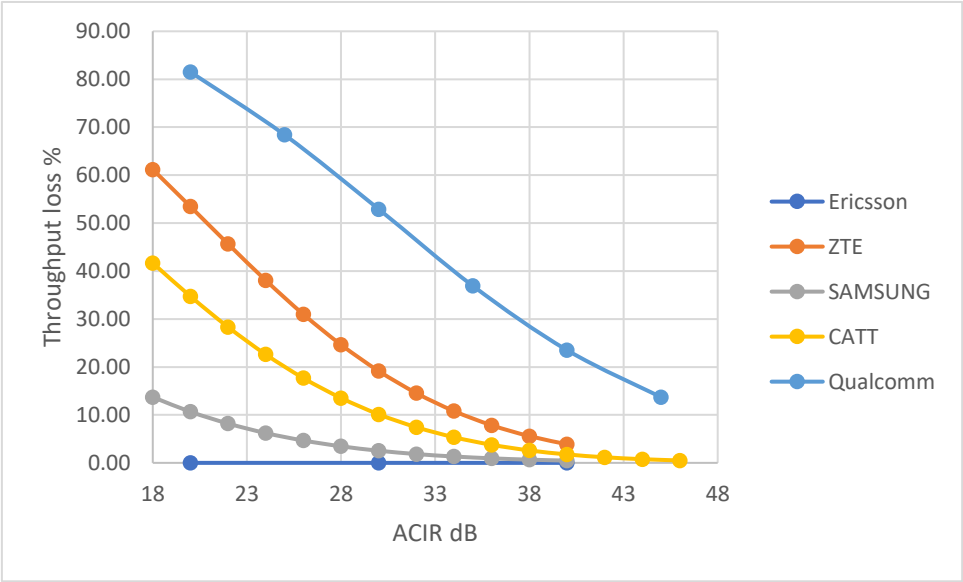


Figure 6a.4.5-3 Simulation results for average throughput loss for Scenario 5a - NTN LEO1200

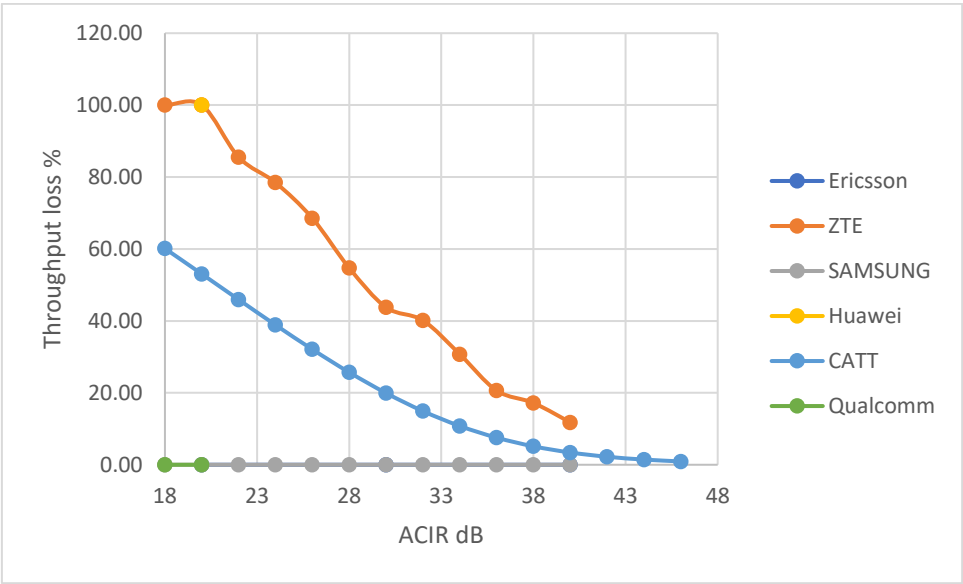


Figure 6a.4.5-4 Simulation results for 5%-tile throughput loss for Scenario 5a - NTN LEO1200

Table 6a.4.5-5 Interpolated ACIR values for Scenario 5a to meet the 5% throughput loss criteria - NTN LEO1200

Company	Interpolated required ACIR
Ericsson	>40 (118.2@40)
ZTE	38.8
SAMSUNG	25.8
CATT	41.5
Qualcomm	52.0
Huawei	>20 (100@20)

Table 6a.4.5-6 Averaged ACIR of 5%-tile values in the above worse case for Scenario 5a - NTN LEO1200

Averaged required ACIR
42.9

The following values were also reported by the companies for GEO at 90° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.5-7 Simulation results for average throughput loss for Scenario 5a - NTN GEO 90° elevation angle

Required ACIR [dB]	Company	0	2	4	5	6	8	10	12	14	15	16	18	20	22	24	25	26	28	30	32	34	35	36	38	40
Throughput Loss	Average	Ericsson																							2.9	2.2
		ZTE	36,99	31,78	26,96		22,55	18,63	15,24	12,34	9,90		7,89	6,23	4,88	3,77	2,88		2,16	1,60	1,16	0,83	0,58		0,40	0,28
		THALES	30,27	26,83	23,38		19,94	16,50	13,06	11,43	9,79		8,15	6,52	4,88	4,18	3,47		2,76	2,05	1,34	1,12	0,89		0,67	0,42
		Qualcomm	78,70			66,7			52,90			38,9			26,50			16,5			9,40			4,9		2,30
	5%-tile	Ericsson																							9.2	7.2
		ZTE	100,00	91,93	87,77		81,85	73,82	63,51	53,88	47,05		41,40	31,30	20,72	18,52	13,32		9,23	4,45	3,15	2,53	0,1		0,00	0,00
		THALES	89,17	81,08	72,99		64,90	56,82	48,73	41,14	33,56		25,97	18,39	10,81	10,19	9,57		8,95	8,33	7,11	6,43	5,15		3,88	2,63
		Qualcomm	100,0			100,0						100,0			100,0			73,5			4,77			1,76		8,7

The following values were also reported by the companies for GEO at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.5-8 Simulation results for average throughput loss for Scenario 5a - NTN GEO 25° elevation angle

Required ACIR [dB]	Company	0	2	4	5	6	8	10	12	14	15	16	18	20	22	24	25	26	28	30	32	34	35	36	38	40
Throughput Loss	Average	Ericsson																							8.8	6.5
		ZTE	48,21	43,10	38,20		33,57	29,23	25,28	21,65	18,37		15,42	12,79	10,49	8,48	6,77		5,33	4,15	3,19	2,43	1,83		1,36	0,73
		THALES	33,16	29,10	25,03		20,96	16,90	12,83	10,99	9,16		7,32	5,49	3,65	3,08	2,52		1,95	1,39	0,82	0,68	0,54		0,40	0,11
		Qualcomm	96,5			92,6			86,4			77,8			67,1			55,2			43,1			31,9		22,4
	5%-tile	Ericsson																							N/A	N/A
		ZTE	100,00	100,00	100,00		100,00	92,51	88,59	82,94	75,12		64,86	52,18	39,82	29,57	20,99		13,31	8,79	6,22	4,33	3,22		3,32	1,88
		THALES	80,08	72,16	64,24		56,32	48,41	40,49	38,89	37,30		35,70	34,10	32,51	26,32	20,12		13,93	7,74	1,55	1,25	0,94		0,64	0,32
		Qualcomm	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	

The following values were also reported by the companies for LEO1200km at 90° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

**Table 6a.4.5-9 Simulation results for average throughput loss for Scenario 5a - NTN LEO@1200km
90° elevation angle**

Required ACIR [dB]		Com pany	0	2	4	5	6	8	10	12	14	15	16	18	20	22	24	25	26	28	30	32	34	35	36	38	40
Throughput Los	Ave rage	Qual com m	80, 80			69, 10			55, 30			41, 00			28, 00			17, 50			10, 00			5, 20			2, 50
		THA LES	29, 20	25, 84	22, 48		19, 12	15, 76	12, 41	10, 81	9, 21		7, 61	6, 01	4,4 1	3, 78	3, 16		2, 53	1, 90	1, 28	1, 06	0, 85		0, 63	0, 42	0, 20
	5%- tile	Qual com m	10 0,0 0			10 0,0 0			10 0,0 0			10 0,0 0			10 0,0 0			74 1 0			43 2 0			21 6 0			8, 1 0
		THA LES	87, 48	78, 49	69, 50		60, 51	51, 52	42, 53	37, 93	33, 33		28, 73	24, 13	19, 53	16, 31	13, 09		9, 87	6, 66	3, 44	2, 75	2, 07		1, 38	0, 69	0, 01

The following values were also reported by the companies for LEO1200km at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

**Table 6a.4.5-10 Simulation results for average throughput loss for Scenario 5a - NTN LEO@1200km
25° elevation angle**

Required ACIR [dB]		Com pany	0	2	4	5	6	8	10	12	14	15	16	1 8	20	2 2	2 4	25	2 6	2 8	30	3 2	3 4	35	3 6	3 8	40
Throu ghput Loss	Ave rag e	Qual com m	97 ,7 0			94 ,6 0			89 ,1 0			80 ,9 0			70 ,6 0			58 ,8 0			46 ,6 0			34 ,9 0			25 ,1 0
		THAL ES	45 ,2 9	40 ,2 1	35 ,1 4		30 ,0 7	24 ,9 9	19 ,9 2	17 ,2 9	14 ,6 6		12 ,0 3	9, 40	6, 77	5, 73	4, 69		3, 66	2, 62	1, 58	1, 31	1, 03		0, 76	0, 48	0, 21
	5%- tile	Qual com m	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N	Na N
		THAL ES	83 ,5 9	69 ,4 8	55 ,3 8		41 ,2 8	27 ,1 8	13 ,0 7	10 ,4 7	7, 88		5, 28	2, 68	0, 08	0, 06	0, 05		0, 04	0, 03	0, 01	0, 01	0, 01		0, 01	0, 00	0, 00

The following values were also reported by the companies for LEO600km at 90° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

**Table 6a.4.5-11 Simulation results for average throughput loss for Scenario 5a - NTN LEO@600km
90° elevation angle**

Required ACIR [dB]		Com pany	0	2	4	5	6	8	10	12	14	15	16	18	20	22	24	25	26	28	30	32	34	35	36	38	40
Throughput Loss	Ave rage	Qual com m	80,90			69,10			55,40			41,00			28,00			17,50			10,00			5,20			2,50
		THALES	29,19	25,84	22,48		19,13	15,77	12,42	10,82	9,22		7,62	6,01	4,41	3,79	3,16		2,53	1,91	1,28	1,06	0,85		0,63	0,42	0,20
	5%- tile	Qual com m	10,00			10,00			10,00			10,00			10,00			74,40			42,80			21,20			8,90
		THALES	87,44	78,46	69,47		60,49	51,51	42,53	37,92	33,32		28,72	24,11	19,51	16,30	13,09		9,88	6,67	3,46	2,77	2,08		1,39	0,70	0,01

The following values were also reported by the companies for LEO600km at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

**Table 6a.4.5-12 Simulation results for average throughput loss for Scenario 5a - NTN LEO@600km
25° elevation angle**

Required ACIR [dB]		Com pany	0	2	4	5	6	8	10	12	14	15	16	18	20	22	24	25	26	28	30	32	34	35	36	38	40
Throughput Loss	Average	Qual com m	97, 80			94, 80			89, 50			81, 40			71, 20			59, 20			47, 20			35, 50			25, 30
		THA LES	39, 92	35, 61	31, 31		27, 00	22, 70	18, 40	16, 07	13, 75		11, 42	9, 09	6, 77	5, 68	4, 59		3, 50	2, 41	1, 33	1, 09	0, 86		0, 62	0, 39	0, 15
	5%- tile	Qual com m	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N	N A N
		THA LES	10 0,0 0	88, 28	76, 57		64, 85	53, 13	41, 42	34, 36	27, 30		20, 24	13, 18	6, 12	4, 90	3, 69		2, 47	1, 26	0, 04	0, 04	0, 03		0, 02	0, 01	0, 00

6a.4.6 Scenario 6a: 17GHz NTN DL interfering TN DL

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select the 5% throughput loss NR UL interfering the NR-NTN GEO UL and NR-NTN LEO1200 UL that 25 degrees elevation angle and deployed in urban environment as the most stringent case.

Table 6a.4.6-1 Simulation results for average and 5%-tile throughput loss for Scenario 6a - NTN GEO

Required ACIR [dB]	Company	0	2	4	6	8	10	12	14	16	18	20
Throughput Loss	Average	Ericsson	0.00					0.00				
		ZTE	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Samsung	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		CATT	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Qualcomm	0.00									
	5%-tile	Ericsson	0.00					0.00				
		ZTE	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Samsung	0.04	0.03	0.03	0.02	0.02	0.01	0.00	0.00	0.00	0.00
		Huawei				0.0	0.0	0.0				
		CATT	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Qualcomm	0.00									

Table 6a.4.6-2 Interpolated ACIR values for Scenario 6a to meet the 5% throughput loss criteria - NTN GEO

Company	Interpolated required ACIR
Ericsson	0
ZTE	0
Samsung	0
Huawei	0
CATT	0

Table 6a.4.6-3 Averaged ACIR of 5%-tile values in the above worse case for Scenario 6a - NTN GEO

Averaged required ACIR
0

Table 6a.4.6-4 Simulation results for average and 5%-tile throughput loss for Scenario 6a - NTN LEO1200

Required ACIR [dB]		Company	0	2	4	6	8	10	12	14	16	18	20
Throughput Loss	Average	Ericsson	0.00					0.00					0.00
		ZTE	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Samsung	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		CATT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
		Qualcomm	0.00										
	5%-tile	Ericsson	0.00					0.00					0.00
		ZTE	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Samsung	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Huawei				0.0	0.0	0.0					
		CATT	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Qualcomm	0.00										

Table 6a.4.6-5 Interpolated ACIR values for Scenario 6a to meet the 5% throughput loss criteria - NTN LEO1200

Company	Interpolated required ACIR
Ericsson	0
ZTE	0
SAMSUNG	0
Huawei	0
CATT	0

Table 6a.4.6-6 Averaged ACIR of 5%-tile values in the above worse case for Scenario 6a - NTN LEO1200

Averaged required ACIR
0

The following values were also reported by the companies for GEO at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.6-7 Simulation results for average throughput loss for Scenario 6a - NTN GEO 25° elevation angle

Required ACIR [dB]		Company	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THALES	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0
	5%-tile	THALES	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0

The following values were also reported by the companies for LEO1200km at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.6-8 Simulation results for average throughput loss for Scenario 6a - NTN LEO@1200km 25° elevation angle

Required ACIR [dB]		Company	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THALES	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0
	5%-tile	THALES	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0

The following values were also reported by the companies for LEO600km at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.6-9 Simulation results for average throughput loss for Scenario 6a - NTN LEO@600km 25° elevation angle

Required ACIR [dB]		Comp any	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THALES	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0
	5%-tile	THALES	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0	0,0

6a.4.7 Scenario 7a: 17GHz NTN DL interfering TN UL

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select the 5% throughput loss NR UL interfering the NR-NTN GEO UL and NR-NTN LEO1200 UL that 25 degrees elevation angle and deployed in urban environment as the most stringent case.

Table 6a.4.7-1 Simulation results for average and 5%-tile throughput loss for Scenario 7a - NTN GEO

Required ACIR [dB]		Company	0	2	4	6	8	10	12	14	16	18	20
Throughput Loss	Average	Ericsson	0.00					0.00					
		ZTE	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Samsung	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		CATT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	5%-tile	Ericsson	2.0					0.00					
		ZTE	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Samsung	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Huawei				0.00	0.00	0.00					
		CATT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Table 6a.4.7-2 Interpolated ACIR values for Scenario 7a to meet the 5% throughput loss criteria - NTN GEO

Company	Interpolated required ACIR
Ericsson	0
ZTE	0
Samsung	0
Huawei	0
CATT	0

Table 6a.4.7-3 Averaged ACIR of 5%-tile values in the above worse case for Scenario 7a - NTN GEO

Averaged required ACIR
0

Table 6a.4.7-4 Simulation results for average and 5%-tile throughput loss for Scenario 7a - NTN LEO1200

Required ACIR [dB]		Company	0	2	4	6	8	10	12	14	16	18	20
Throughput Loss	Average	ZTE	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Samsung	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		CATT	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	5%-tile	ZTE	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Samsung	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
		Huawei				0.00	0.00	0.00					
		CATT	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table 6a.4.7-5 Interpolated ACIR values for Scenario 7a to meet the 5% throughput loss criteria - NTN LEO1200

Company	Interpolated required ACIR
Ericsson	0
ZTE	0
Samsung	0
Huawei	0
CATT	0

Table 6a.4.7-6 Averaged ACIR of 5%-tile values in the above worse case for Scenario 7a - NTN LEO1200

Averaged required ACIR
0

The following values were also reported by the companies for GEO at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.7-7 Simulation results for average throughput loss for Scenario 7a - NTN GEO 25° elevation angle

Required ACIR [dB]		Comp any	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THALES	0,42	0,35	0,27	0,20	0,13	0,05	0,04	0,03	0,02	0,01	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
	5%-tile	THALES	0,68	0,54	0,41	0,28	0,14	0,01	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00

The following values were also reported by the companies for LEO1200km at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.7-8 Simulation results for average throughput loss for Scenario 7a - NTN LEO@1200km 25° elevation angle

Required ACIR [dB]		Comp any	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THALES	0,02	0,02	0,01	0,01	0,01	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
	5%-tile	THALES	0,11	0,10	0,08	0,06	0,04	0,02	0,02	0,01	0,01	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00

The following values were also reported by the companies for LEO600km at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.7-9 Simulation results for average throughput loss for Scenario 7a - NTN LEO@600km 25° elevation angle

Required ACIR [dB]		Company	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THALES	0,12	0,10	0,08	0,05	0,03	0,01	0,01	0,01	0,01	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00
	5%-tile	THALES	0,04	0,04	0,03	0,02	0,01	0,01	0,01	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00	0,00

6a.4.8 Scenario 8a: 17GHz TN UL interfering NTN DL

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select the 5% throughput loss NR UL interfering the NR-NTN GEO UL and NR-NTN LEO1200 UL that 25 degrees elevation angle, 2.5dB noise figure and deployed in urban environment as the most stringent case.

Table 6a.4.8-1: Simulation results for average throughput loss for Scenario 8a - NTN GEO

Required ACIR [dB]	0	2	4	6	8	10	12	14	16
Ericsson	8.20					2.60			
ZTE	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Samsung	4.98	3.61	2.57	1.81	1.25	0.86	0.58	0.39	0.26
CATT	7.5	5.7	4.3	3.2	2.3	1.6	1.1	0.8	0.5

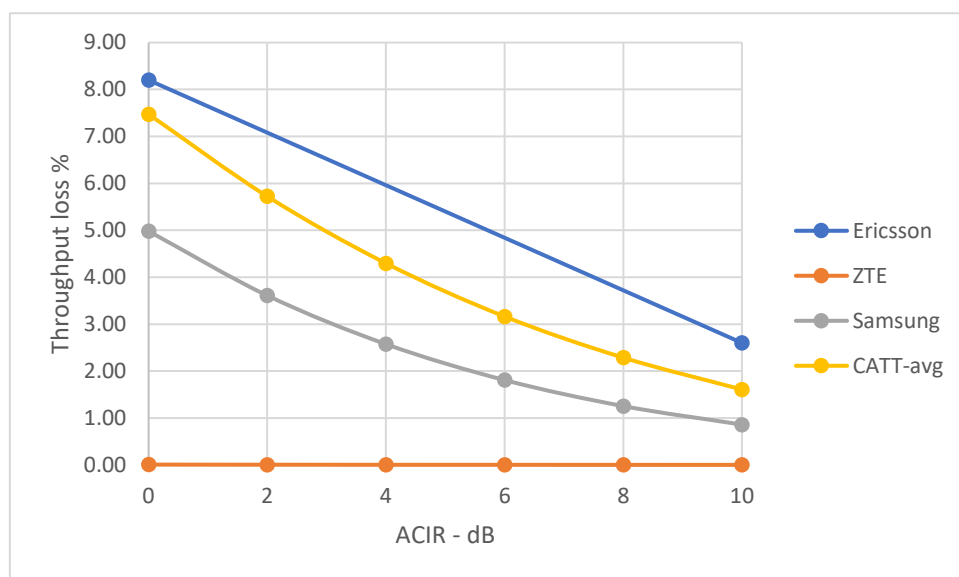


Figure 6a.4.8-1: Simulation results for average throughput loss for Scenario 8a - NTN GEO

Table 6a.4.8-2: Interpolated ACIR values for Scenario 8a to meet the 5% throughput loss criteria - NTN GEO

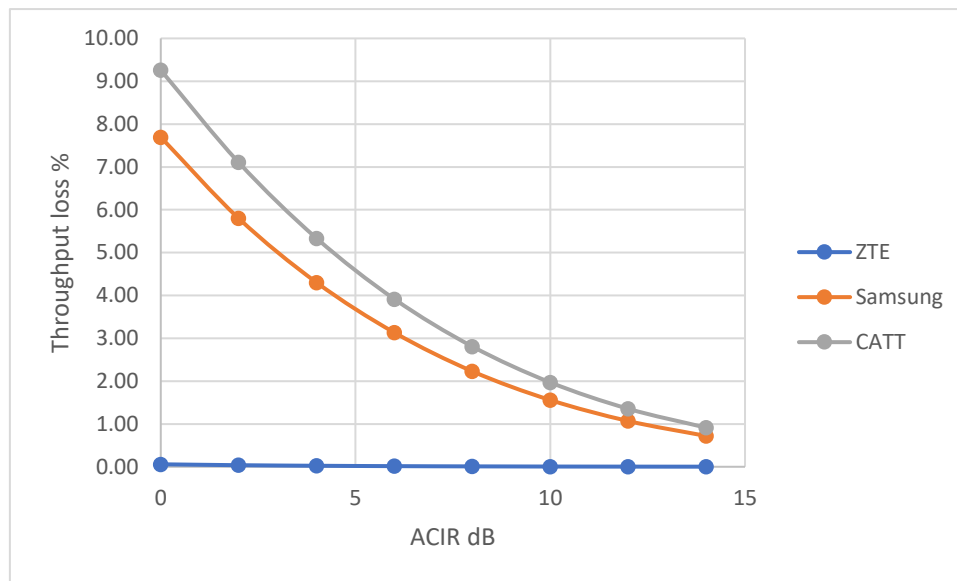
Company	Interpolated required ACIR
Ericsson	5.70
ZTE	0.00
Samsung	0.00
CATT	3.00

Table 6a.4.8-3: Averaged ACIR of 5%-tile values in the above worse case for Scenario 8a - NTN GEO

Averaged required ACIR
2.2

Table 6a.4.8-4: Simulation results for average throughput loss for Scenario 8a - NTN LEO1200

Required ACIR [dB]	0	2	4	6	8	10	12	14	16
ZTE	0.06	0.04	0.02	0.01	0.01	0.01	0.00	0.00	0.00
Samsung	7.69	5.80	4.30	3.13	2.23	1.56	1.07	0.72	0.48
CATT	9.3	7.1	5.3	3.9	2.8	2.0	1.4	0.9	0.6

**Figure 6a.4.8-2: Simulation results for average throughput loss for Scenario 8a - NTN LEO1200****Table 6a.4.8-5: Interpolated ACIR values for Scenario 8a to meet the 5% throughput loss criteria - NTN LEO1200**

Company	Interpolated required ACIR
ZTE	0.00
Samsung	3.10
CATT	4.40

Table 6a.4.8-6: Averaged ACIR of 5%-tile values in the above worse case for Scenario 8a - NTN LEO1200

Averaged required ACIR
2.5

The following values were also reported by the companies for GEO at 90° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.8-7 Simulation results for average throughput loss for Scenario 8a - NTN GEO 90° elevation angle

Required ACIR [dB]	Comp any	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	ZTE	0,0 0	0,0 0	0,0 0	0,0 0	0,0 0	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00
		THAL ES	19, 38	17, 14	14, 90	12, 67	10, 43	8, 19	7, 00	5, 80	4, 61	3, 41	2, 22	1, 86	1, 50	1, 14	0, 78	0, 42	0, 34	0, 27	0, 20	0, 12
	5%-tile	ZTE	0,0 0	0,0 0	0,0 0	0,0 0	0,0 0	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00

The following values were also reported by the companies for GEO at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.8-8 Simulation results for average throughput loss for Scenario 8a - NTN GEO 25° elevation angle

Required ACIR [dB]	Comp any	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	ZTE	0,0 2	0,0 1	0,0 1	0,0 0	0,0 0	0,0 0	0,0 0	0,0 0	0,0 0	0,0 0	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00
		THAL ES	31, 59	27, 80	24, 02	20, 23	16, 44	12, 66	11, 02	9,3 8	7,7 5	6,1 1	4,4 8	3, 92	3, 36	2, 80	2, 24	1, 69	1, 48	1, 27	1, 07	0, 86
	5%-tile	ZTE	0,0 0	0,0 0	0,0 0	0,0 0	0,0 0	0,0 0	0,0 0	0,0 0	0,0 0	0,0 0	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00	0, 00
		THAL ES	81, 99	72, 85	63, 71	54, 57	45, 43	36, 29	31, 22	26, 16	21, 09	16, 03	10, 96	9, 37	7, 78	6, 20	4, 61	3, 02	2, 81	2, 61	2, 40	1, 20

The following values were also reported by the companies for LEO1200km at 90° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.8-9 Simulation results for average throughput loss for Scenario 8a - NTN LEO@1200km 90° elevation angle

Required ACIR [dB]	Comp any	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THAL ES	27, 68	24, 49	21, 30	18, 11	14, 92	11, 74	10, 28	8,8 2	7,3 7	5,9 1	4,4 5	3,7 6	3, 07	2, 38	1, 69	1, 00	0, 83	0, 65	0, 48	0, 30
	5%-tile	THAL ES	73, 86	65, 94	58, 02	50, 10	42, 17	34, 25	30, 10	25, 95	21, 80	17, 66	13, 51	10, 83	8, 15	5, 47	2, 79	0, 12	0, 10	0, 07	0, 05	0, 03

The following values were also reported by the companies for LEO1200km at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

Table 6a.4.8-10 Simulation results for average throughput loss for Scenario 8a - NTN LEO@1200km 25° elevation angle

Required ACIR [dB]	Comp any	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THAL ES	44, 33	39, 58	34, 83	30, 07	25, 32	20, 57	18, 02	15, 47	12, 92	10, 36	7,8 1	6,8 0	5,7 9	4,7 8	3, 77	2, 76	2, 33	1, 90	1, 47	1, 03
	5%-tile	THAL ES	81, 14	75, 58	70, 01	64, 44	58, 87	53, 30	50, 35	47, 41	44, 47	41, 53	38, 58	30, 89	23, 19	15, 50	7, 80	0, 11	0, 09	0, 07	0, 05	0, 03

The following values were also reported by the companies for LEO600km at 90° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

**Table 6a.4.8-11 Simulation results for average throughput loss for Scenario 8a - NTN LEO@600km
90° elevation angle**

Required ACIR [dB]		Comp any	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THAL ES	26, 79	23, 83	20, 86	17, 90	14, 94	11, 97	10, 41	8, 84	7, 27	5, 71	4, 14	3, 49	2, 83	2, 18	1, 52	0, 87	0, 72	0, 56	0, 41	0, 26	0, 11
	5%-tile	THAL ES	60, 98	53, 53	46, 08	38, 63	31, 18	23, 74	20, 44	17, 15	13, 86	10, 57	7, 28	5, 82	4, 37	2, 92	1, 46	0, 01	0, 01	0, 00	0, 00	0, 00	0, 00

The following values were also reported by the companies for LEO600km at 25° elevation angle, with UE NF= 2.5dB and UE height 1.5m (corresponding to L-ESIM).

**Table 6a.4.8-12 Simulation results for average throughput loss for Scenario 8a - NTN LEO@600km
25° elevation angle**

Required ACIR [dB]		Com pany	0	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40
Throughput Loss	Average	THAL ES	42, 22	37, 63	33, 04	28, 45	23, 86	19, 27	16, 98	14, 69	12, 40	10, 11	7, 82	6, 78	5, 75	4, 71	3, 67	2, 63	2, 21	1, 79	1, 36	0, 94	0, 52
	5%-tile	THAL ES	92, 54	83, 52	74, 50	65, 48	56, 45	47, 43	40, 29	33, 15	26, 01	18, 87	11, 73	11, 59	11, 45	11, 31	11, 17	11, 04	8, 84	6, 64	4, 44	2, 24	0, 04

6a.5 Summary of co-existence study

6a.5.1 Coexistence at 17/27GHz

This sub-clause captures the summary of the co-existence studies. The averaged interpolate ACIR values for each scenario are presented in Table 6a.5.1-1 below.

Table 6a.5.1-1 Average ACIR values for each scenario at 17/27GHz

Scenario		1a	2a	3a	4a	5a	6a	7a	8a
ACIR value [dB]	GEO	10.5	1.54	0	14.3	-	0	0	2.2
	LEO1200	13.15	0.82	0	16.2	-	0	0	2.5

Then, by considering the following ACLR and ACS of TN BS and UE in Table 6a.5.1-2 and Clause 6a.2.2.4, the suggested ACLR and ACS of NTN SAN and VSAT from each scenario are given in Table 6a.5.1-3. It should be noted that the values in Table 6a.5.1-3 are directly derived from the selected option of each scenario, and it is limited by the nature of assumptions and methodologies adopted in the co-existence studies.

Table 6a.5.1-2: ACLR and ACS of FR2 TN (27 GHz)

TN		Values
BS	ACLR	28 dB
	ACS	24 dB
UE	ACLR	17 dB
	ACS	23 dB

Table 6a.5.1-3: Co-existence results suggested ACIR of NR-NTN at 17/27GHz

Scenario	Contributing	ACIR values
1a	NTN VSAT ACLR	GEO: 11 dBc LEO: 14 dBc
2a	SAN ACS	GEO: 2 dBc LEO: 1 dBc
3a	NTN VSAT ACLR	0 dBc
4a	SAN ACS	GEO: 14 dBc LEO: 16 dBc
5a	NTN VSAT ACS	25 dBc
6a	SAN ACLR	0 dBc
7a	SAN ACLR	0 dBc
8a	NTN VSAT ACS	GEO: 2 dBc LEO: 3 dBc

With the above suggested values, recognizing the characteristics of SAN and VSAT to the state of art and their future developments, the agreed ACLR and ACS of NR-NTN at 17/27GHz are given in Table 6a.5.1-4.

Table 6a.5.1-4: ACLR and ACS of NR-NTN at 17/27GHz

	SAN		VSAT	
	GEO	LEO	Fixed	Mobile
ACLR (dBc)	12	12	14	14
ACS (dBc)	18	24	25 ^{1, 2, 3}	25 ^{1, 2, 3}
NOTE 1: At the time of this 3GPP co-existence study, there is no TN band defined or planned near 17 GHz. The parameters are derived based on 3GPP coexistence scenarios in which a TN system is simulated to be operating in the band directly adjacent to the proposed NTN system as well as technical assumptions that may or may not be applicable in practice. The results of the study are not intended to address coexistence issues from a regulatory standpoint.				
NOTE 2: There are existing non-3GPP VSAT UEs operating in Ka-band and these will likely continue operating in the future, with a less stringent ACS value as per the ETSI EN 303 978 specification.				
NOTE 3: Additional solutions could be further considered to address coexistence issues if and when TN is deployed in 17 GHz.				

With respect to NOTE 3 at least several techniques have been identified as potential to alleviate interference from TN BS to VSAT at 17 GHz such as increasing NTN SAN transmission power, increasing TN gNB antenna gain (e.g. by increasing the number of TN gNB antenna elements while at the same time decreasing TN gNB transmission power for a constant EIRP), using a frequency guard band for a better protection from the TN gNB, increasing the antenna gain of the NTN VSAT UE.

6b Co-existence study for NTN HPUE in FR1

6b.1 Co-existence simulation scenario for NTN HPUE in FR1

Scenarios for coexistence study are listed in Table 6b.1-1.

Table 6b.1-1: Scenarios for NTN HPUE co-existence

FR1: 2GHz	NTN ^{1,4,5}		
	GEO ³	LEO 600km	LEO 1200km

NR/ NB- IoT	Rural	X	X	X
	Urban macro	X	X	X
	Dense Urban ⁶	N/A	N/A	N/A

NOTE 1:

NOTE 2:

NOTE 3:

NOTE 4:

NOTE 5:

NOTE 6:

Earth Fixed beam has been considered for co-existence studies.

Void

GEO and LEO only operate at adjacent channel.

Use GEO and LEO@600km when TN is victim.

The satellite-to-satellite coexistence scenarios are not in the scope of this study considering this is already addressed by ITU (ITU RR Article 9 etc.) and regional regulations (e.g. FCC rules).

Dense Urban is not considered as it is expected NTN UE will connect to terrestrial networks rather than satellite networks in such scenario.

The aggressor and victim combinations are listed in Table 6b.1-2.

Table 6b.1-2: Aggressor and victim

No.	Combination	Aggressor	Victim	Notes
4b	TN with NTN	NTN UL	TN UL	ACLR of NTN HPUE to be defined.

For further visualisation of coexistence studies for NTN HPUE in FR1, the interference charts with respect to scenario 4b are further represented with respect to NTN-TN coexistence scenarios in Figure 6b.1-1.

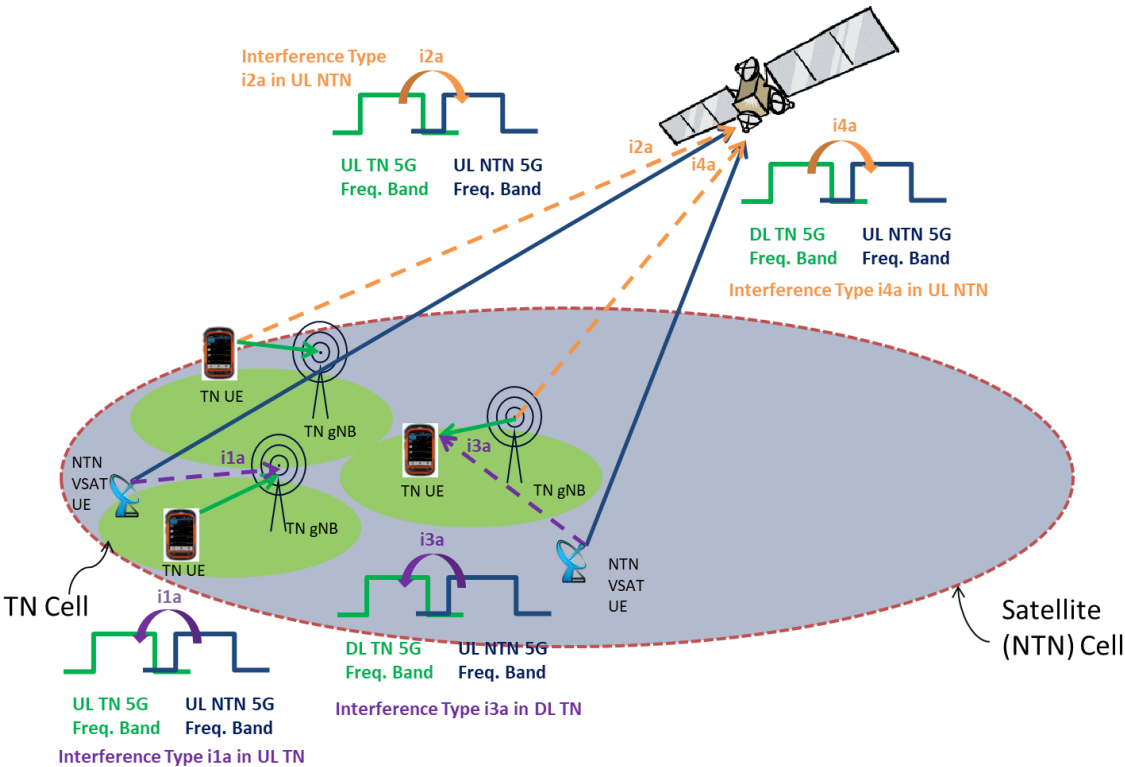


Figure 6b.1-1: NTN-TN adjacent band coexistence scenario 4b

The frequency and bandwidth are listed in Table 6b.1-3.

Table 6b.1-3: Proposed frequency and bandwidth for co-existence study

	Frequency	Bandwidth		Duplex mode	Frequency reuse factor
		NR-NTN	IoT-NTN		
TN Rural	2 GHz	20MHz		FDD, TDD	1
TN Urban macro	2 GHz	20MHz		FDD, TDD	1
GEO	2 GHz	20 MHz for FR1	0.18MHz	FDD	1, 3 ¹
LEO	2 GHz	20 MHz for FR1	0.18MHz	FDD	1, 3 ¹
NOTE: Only FRF=1 has been used in co-existence studies for simplification.					

6b.2 Co-existence simulation assumption for NTN HPUE in FR1

6b.2.1 Network layout model

6b.2.1.1 Co-existence between NTN satellite and TN

Cellular cell structure is considered for both NTN satellite and TN network layout.

Referring to TR 38.811[5] Section 2.3 and Annex A, a 3D global coordinate system is considered (Earth-Centred Earth Fixed) for simulating NTN satellite beams direction and location on the earth surface. It means the NTN satellite beam location, TN randomly dropping location are generated with a set of three parameters (x,y,z).

Deployment of NTN satellite and TN cells and UEs for co-existence study is listed in Table 6b.2.1.1-1.

As defined in Figure 6b.2.1.1-1, isolation distance is the distance between the blue-dotted line which represents TN cell boarder and the red line. No UEs deployed in the isolation region is assumed to reduce the calculation complexity. The baseline is to use 2 x ISD (ISD = 750 meters) as the isolation distance

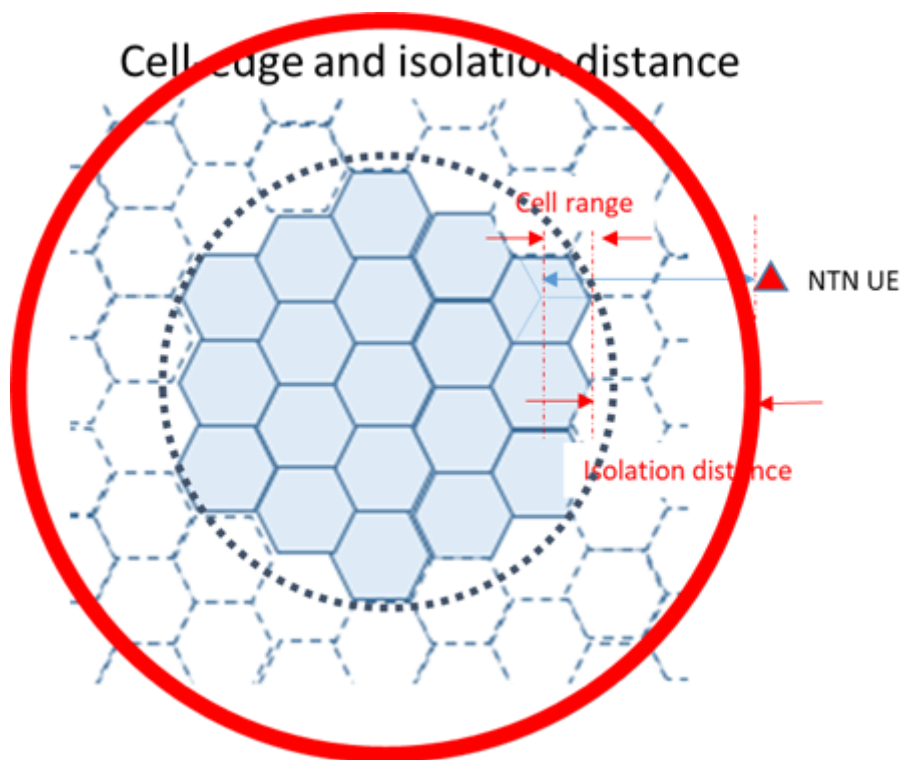
**Figure 6b.2.1.1-1: Isolation distance**

Table 6b.2.1.1-1: Network and UE deployment

No.	Combination	Aggressor	Victim	Which NTN cell/UE to observe?	Which TN/UE to observe?	Which TN cells in a TN to observe?
4b	TN with NTN satellite	NTN UL	TN UL	NTN cell: Nadir point. NTN UE: NTN UEs dropped at the edge of TN clusters	TN randomly placed in this NTN beam	Option 1: All active TN clusters which has the NTN UE(s) at its edge. Option 2: Only the TN sectors which have NTN UE(s) at their edges.

6b.2.2 System parameters

6b.2.2.1 Satellite parameters

Set-1 satellite parameters are listed in Table 6b.2.2.1-2 and Table 6b.2.2.1-3 according to TR 38.821[6].

The satellite max Tx power can be calculated by the equation as below:

$$\max Tx\ power[dBm] = EIRP\ density\left[\frac{dBW}{MHz}\right] + 30 + 10\log_{10}(N_{RB} * SCS[MHz] * 12) - Max\ Gain[dBi]$$

Table 6b.2.2.1-1: N_{RB} configuration per BandWidth size and SCS

Configuration FR1 S-band	For IoT-NTN	For NR-NTN			
	N _{RB} (Subcarriers)	N _{RB} (5MHz BW)	N _{RB} (10MHz BW)	N _{RB} (15MHz BW)	N _{RB} (20MHz BW)
For IoT-NTN : SCS 3.75kHz	1 (12)	-	-	-	-
For NR-NTN and IoT-NTN : SCS 15 kHz	1 (48)	25	52	79	106
For NR-NTN : SCS 30 kHz	-	11	24	38	51

Table 6b.2.2.1-2a: Satellite parameters for co-existence study

(Note: For NR-NTN)

Satellite orbit		GEO				LEO-1200				LEO-600				
Satellite altitude		35786 km				1200 km				600 km				
Payload characteristics for DL transmissions														
Satellite EIRP density		2GHz	59 dBW/MHz				40 dBW/MHz				34 dBW/MHz			
Satellite max TX power in dBm	BW (MHz)		5	10	15	20	5	10	15	20	5	10	15	20
	SCS 15kHz		44.53	47.71	49.53	50.81	46.53	49.71	51.53	52.81	40.53	43.71	45.53	46.81
	SCS 30kHz		43.98	47.37	49.36	50.64	45.98	49.37	51.36	52.64	39.98	43.37	45.36	46.64
Satellite Tx max Gain			51 dBi				30 dBi				30 dBi			
Channel bandwidth			5/10/15/20MHz				5/10/15/20MHz				5/10/15/20MHz			
3dB beamwidth or HPBW (Half-Power BandWidth) of main central beam			0.4011 deg				4.4127 deg				4.4127 deg			
ABS (Adjacent Beam Spacing) of adjacent beams from the central beam			0.3474 deg				3.8206 deg				3.8206 deg			
Satellite beam diameter		250 km				90 km				50 km				
Payload characteristics for UL transmissions														
G/T		2 GHz	19 dB K ⁻¹				1.1 dB K ⁻¹				1.1 dB K ⁻¹			
Satellite Rx max Gain			51 dBi				30 dBi				30 dBi			

Table 6b.2.2.1-2b Satellite parameters for co-existence study

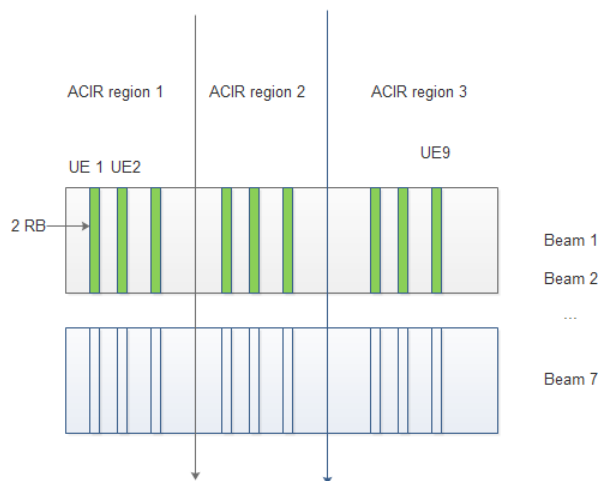
(Note: For IoT-NTN)

Satellite orbit		GEO	LEO-1200	LEO-600	
Satellite altitude		35786 km	1200km	600 km	
		Payload characteristics for DL transmissions			
Satellite EIRP density		2GHz	59 dBW/MHz	40 dBW/MHz	34 dBW/MHz
Satellite max TX power in dBm	BW (MHz)		0.18 (NB-IoT)	0.18 (NB-IoT)	0.18 (NB-IoT)
	SCS 15kHz		30.55	32.55	26.55
	SCS 3.75kHz		30.55	32.55	26.55
Satellite Tx max Gain			51 dBi	30 dBi	30 dBi
3dB beamwidth or HPBW (Half-Power BandWidth) of main central beam			0.4011 deg	4.4127 deg	4.4127 deg
ABS (Adjacent Beam Spacing) of adjacent beams from the central beam			0.3474 deg	3.8206 deg	3.8206 deg
Satellite beam diameter			250 km	90 km	50 km
		Payload characteristics for UL transmissions			
G/T		2 GHz	19 dB K ⁻¹	1.1 dB K ⁻¹	1.1 dB K ⁻¹
Satellite Rx max Gain			51 dBi	30 dBi	30 dBi

Table 6b.2.2.1-3: Other parameters for NTN satellite

Parameters	NR-NTN	IoT NTN
Carrier frequency	2GHz	2GHz
The number of active UE (UL)	9 UEs and 2RBs per UE for GEO and LEO ¹ NOTE 2	9 for 15kHz SCS 18 and 36 for 3.75kHz SCS (1 subcarrier per UE)
The number of active UE (DL)	1	1
Traffic model	Full buffer	Full buffer
DL power control	NO	NO
UL power control	See Session 2.2.6.2 NOTE 2	36.942 section 5.1.1.6 (set 1) by bandwidth scale, target SNR at BS is 15 dB
UE Tx power in dBm (PC3)	-40 to 23	-40 to 23
UE Tx power in dBm (PC2)	-40 to 26	-40 to 26
UE Tx power in dBm (PC1.5)	-40 to 29	-
UE Tx power in dBm (PC1)	-40 to 31	-40 to 31
NTN satellite Noise Figure in dB	See Table 2.3-3-1	See Table 2.3-3-1
Handover margin	3dB	3dB

NOTE 1: UEs are equally splitted inside the channel bandwidth into ACIR 3 regions. Scheduled PRB position for UE1 per satellite beam should be also fully aligned to simulate the worst case for co-channel interference and this is also aligned with full buffer case.



NOTE 2: For NR-NTN HPUE: Option 1: 3 dB SNR target and 5MHz UL BW (One company proposed for LEO only); Option 2: 15 dB SNR target and 2RB UL BW as baseline assumption

NOTE 3: Assuming all NR-NTN/IoT-NTN UEs are HPUE, i.e. 100% HPUE ratio as baseline.

Table 6b.2.2.1-5: NTN satellite Noise Figure in dB

Satellite	GEO	LEO 600	LEO 1200
G/T (dB K ⁻¹)	19	1.1	1.1
G Rx (dBi)	51	30	30
NF (dB)	7.4	4.3	4.3

6b.2.2.2 NTN UE parameters

NTN UE parameters are given in Table 6b.2.2.2-1

Table 6b.2.2.2-1: NTN UE characteristics for system level simulations

Characteristics	For NR-NTN and IoT-NTN
Frequency band	S band (i.e. 2 GHz)
Antenna type and configuration	(1, 1, 2) with omni-directional antenna element
Polarisation	Linear: +/-45°X-pol
Rx Antenna gain	0 dBi per element
Antenna temperature	290 K
Noise figure	9 dB
Tx antenna gain	0 dBi per element

6b.2.2.3 TN parameters

TN parameters for co-existence study are given in Table 6b.2.2.3-1, 6b.2.2.3-2 and 6b.2.2.3-3.

Table 6b.2.2.3-1: Simulation assumptions of TN respectively based on NR

	NR
Carrier frequency in GHz	2
Size of each nominal channel BW in MHz	20
Transmission bandwidth in MHz	N/A
Environment	Deployment scenario related, see Table 6b.2.2.3-2.
Network layout	19-sites 57 sectors with wrap-around
Inter-site distance in meter	Deployment scenario related, see Table 6b.2.2.3-2
System loading and activity	See Table 6b.2.1.1-1
Network location	See Table 6b.2.1.1-1
DL subcarrier spacing	15kHz
UL	OFDMA
DL power control	No
UL power control	TR 36.942[8]
Frequency reuse	1
Number of scheduled UE per cell (DL)	1
Number of scheduled UE per cell (UL)	3
UE antenna height in meter	1.5m
UE TX power in dBm	-40 to 23
UE antenna gain in dBi	0
Building penetration loss	In pathloss model, TR 38.901[10]
Cell selection margin in dB	3
BS-MS min distance in meters	35
BS noise figure in dB	5
UE noise figure in dB	9
BS-UE path-loss model	TR 38.901[10]
Standard deviation of BS-UE log-normal shadow fading in dB	Deployment scenario related, referring to TR 38.901[10]
Shadowing correlation	Inter-cell 0.5 Intra-cell 1
Link-level performance model	See Section 6b.2.8 Throughput-SINR mapping
UE distribution	Uniform
Evaluation metrics	See Section 6b.2.8 Throughput or SNR loss criteria

Table 6b.2.2.3-2: Deployment-related parameters of TN (2 GHz)

	Urban Macro	Rural Macro	Remarks
ISD in meters	750	7500	ITU-R Report M.2292[11]
BS Antenna height in meters	25	30	
UE Outdoor/indoor	100% Outdoor		
UE height in meter	1.5	1.5	RP-200559 [12] 3GPP LS to ITU-R WP5D and ITU-R WP5D [IMT_Parameters] [13]

Table 6b.2.2.3-3: ACLR/ACS for TN (2GHz)

		NR
BS	ACLR	45 dB
	ACS	46 dB
UE	ACLR	30dB (ACLR1) 43dB (ACLR2)
	ACS	33

6b.2.3 Antenna and beam forming pattern modelling

6b.2.3.1 NTN SAN and NTN UE

Satellite and UE Antenna and beam forming pattern modelling of satellite could be referred to section 6.4.1 in TR 38.811[5].

The following normalized antenna gain pattern, corresponding to a typical reflector antenna with a circular aperture, is considered.

$$\begin{aligned}
 &1 && \text{for } \theta = 0 \\
 &4 \left| \frac{J_1(ka \sin \theta)}{ka \sin \theta} \right|^2 && \text{for } 0 < |\theta| \leq 90^\circ
 \end{aligned}$$

where:

- $J_1(x)$ is the Bessel function of the first kind and first order with argument 'x', where x is $ka \sin \theta$;
- a , is the radius of the antenna's circular aperture;
- $k = 2\pi f/c$ is the wave number;
- f is the frequency of operation;
- c is the speed of light in a vacuum and θ is the angle measured from the bore sight of the antenna's main beam.

Note that ka equals to the number of wavelengths on the circumference of the aperture and is independent of the operating frequency.

The antenna patterns for LEO 600km, 1200km and GEO are shown in Figure 6b.2.3.1-1 and 6b.2.3.1-2.

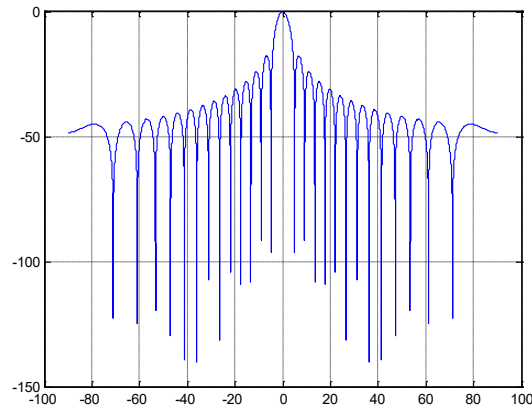


Figure 6b.2.3.1-1: Antenna pattern for LEO 600KM and 1200KM (4.4127 deg for 3dB beamwidth)

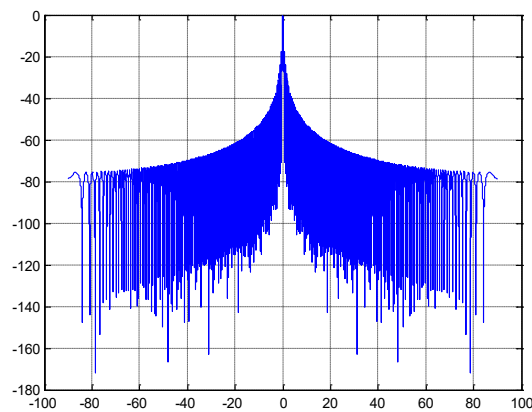


Figure 6b.2.3.1-2: Antenna pattern for antenna aperture of GEO (0.4011 deg for 3dB beamwidth)

The beam layout definition for a single satellite simulation in S-Band is defined in Table 6b.2.3.1-1.

Table 6b.2.3.1-1: Beam layout definition for single satellite simulation

Beam layout definition	Baseline: Hexagonal mapping of the beam bore sight directions on UV plane defined in the satellite reference frame. Only the 3dB beam width parameters should be used. The beam diameter and beam spacing values can be computed directly from the 3 dB beam width assumptions and should be considered as informative.
Number of beams	Baseline: 7-beam layout (i.e. 6 co-frequency beams surrounding the central beam)
UV plane illustration (extracted from [19])	Spherical Coordinate System Projection of Coordinate System On the X-Y Plane (view from above Z-axis) Direction Cosines $\begin{cases} u = \sin \theta \cos \phi \\ v = \sin \theta \sin \phi \end{cases}$
UV plane convention	U axis is defined as the perpendicular line to the satellite-earth line on the orbital plane as illustrated here after: The straight line being orthogonal to UV plane is pointing towards the Earth centre. UV coordinates of the nadir of the reference satellite is (0,0)
Adjacent beam spacing on UV plane	Baseline: Adjacent beam spacing computation based on 3dB beam width of the satellite antenna pattern: $ABS[\text{rad}] = \sqrt{3} \times \sin(\text{HPBW}[\text{degrees}]/2)$ or $ABS[\text{rad}] = \sqrt{3} \times \sin(\text{HPBW}[\text{rad}]/2)$ with $ABS[\text{degree}] = 180/\pi \times ABS[\text{rad}]$ and with HPBW the Half-Power BandWidth of the main lobe from the satellite antenna pattern.
Central beam bore sight direction definition	Baseline: Case 1: Central beam center is considered at nadir point Case 2: 45° for GEO and LEO
	Option 1: FRF=1
Polarization re-use	Option 1: Disable Option 2: Enable Note: Polarization re-use should apply only if circular polarization for terminal antenna is considered

UEs outdoor/indoor distribution	100% outdoor distribution for UEs
UE distribution	The cell area associated to a given beam is defined as the Voronoi cell associated with the corresponding beam centers.
UE orientation	Omnidirectional
UE attachment	RSRP
NOTE 1: Typical impairment values (additional frequency error, SNR loss) due to the feeder link except for delay can be considered to be negligible. When available, specific values can be considered in the evaluation and should be reported.	
NOTE 2: For the calibration purpose, the ionospheric scintillation loss shall be considered equal to zero (i.e., the UEs are located between 20 and 60 degrees of latitude).	

6b.2.3.2 TN BS and TN UE

For AAS antenna, it refers to the pattern in TR 38.921 [14].

Table 6b.2.3.2-1: AAS antenna parameters for 2GHz

		Rural	Macro urban
1	Base Station Antenna Characteristics		
1.1	Antenna pattern	TR 38.921[14]	
1.2	Element gain (dBi) ^(Note 2)	7.1	6.4
1.3	Horizontal/vertical 3 dB beam width of single element (degree)	90° for H 54° for V	90° for H 65° for V
1.4	Horizontal/vertical front-to-back ratio (dB)	30 for both H/V	30 for both H/V
1.5	Antenna polarization	Linear $\pm 45^\circ$	Linear $\pm 45^\circ$
1.6	Antenna array configuration (Row $\times$ Column) ^(Note 4)	8 $\times$ 8 elements	8 $\times$ 8 elements
1.7	Horizontal/Vertical radiating element spacing	0.5 of wavelength for H, 0.9 of wavelength for V	0.5 of wavelength for H, 0.7 of wavelength for V
1.8	Array Ohmic loss (dB) ^(Note 2)	2	2
1.9	Conducted power (before Ohmic loss) per antenna element (dBm) ^(Note 3)	25	25
1.10	Base station maximum coverage angle in the horizontal plane (degrees)	120	120
1.11	Base station vertical coverage range (degrees) ^(Note 1)	90-100	90-120
1.12	Mechanical downtilt (degrees)	3	10

For non-AAS antenna, the parameters in Table 6b.2.3.2-1 are used for 2GHz BS antenna pattern in the NTN system simulation.

Table 6b.2.3.2-1: FR1 BS Non-AAS antenna pattern for 2GHz

Parameter for BS	Values	
Antenna vertical radiation pattern (dB)	$A_{E,V}(\theta'') = -\min\left\{12\left(\frac{\theta'' - 90^\circ}{\theta_{3\text{dB}}}\right)^2, SLA_V\right\}, \theta_{3\text{dB}} = 10^\circ, SLA_V = 20\text{dB}$	
Antenna horizontal radiation pattern (dB)	$A_{E,H}(\phi'') = -\min\left\{12\left(\frac{\phi''}{\phi_{3\text{dB}}}\right)^2, A_m\right\}, \phi_{3\text{dB}} = 65^\circ, A_m = 20\text{dB}$	
Combining method for 3D antenna pattern (dB)	$A''(\theta'', \varphi'') = -\min\{-[A_{E,V}(\theta'') + A_{E,H}(\varphi'')], A_m\}$	
Maximum directional gain of an antenna $G_{E,max}$	17 dBi	
Conducted power	46 dBm	
Mechanical downtilt in degrees	Rural	3
	Urban	10

For UE antennas, an omni-directional radiation pattern with antenna gain 0dBi is assumed.

6b.2.4 ACIR model

6b.2.4.1 NR-NTN

ACLR modelling for TN and NTN co-existence study referring to clause 5.1.1.4.1 and 5.1.1.4.2 in TR 36.942[8] is used as baseline. The number of RBs refers to Table 6b.2.2.1-4 and Table 6b.2.2.3-1 respectively.

6b.2.4.2 IoT-NTN

For scenario 4b, symmetric bandwidth IoT-NTN ACLR model is assumed according to TR 36.942 section 5.1.1.4.2 with flat NTN ACLR model and FACLR = 0 dB, assuming flat ACS.

6b.2.5 Propagation model

6b.2.5.1 Propagation model between NTN SAN and UE

Propagation model between NTN SAN and UE should reference to section 6.6 in TR 38.811[5].

6b.2.5.2 Propagation model between TN BS and UE

Propagation model between TN BS and UE should reference to section 7.4 in TR 38.901[10].

6b.2.5.3 Propagation model between NTN SAN and TB BS

Propagation model between NTN SAN and TN BS should reference to TS 38.811[5] which is used for DL-UL cross link interference for S band.

6b.2.6 Transmission power control model

6b.2.6.1 TN uplink transmission power control

For uplink scenario, transmission power control model specified in Section 9.1 TR 36.942[8] is applied for TN with following parameters.

$$P_t = P_{\max} \times \min \left\{ 1, \max \left[R_{\min}, \left(\frac{CL}{CL_{x-ile}} \right)^\gamma \right] \right\}$$

where:

- $P_{\max} = 23\text{dBm}$,
- $R_{\min} = -63\text{ dB}$,
- CL_{x-ile} and γ are set as following:
 - $CL_{x-ile} = 88 + 10 \cdot \log_{10}(200/X) + 11 - Y$, where X is UL transmission BW (MHz) and Y is the BS noise figure
 - $\gamma = 1$ For uplink scenario,

6b.2.6.2 NTN uplink transmission power control

For NTN HPUE uplink transmission power control model, the following parameters are assumed:

$$P_t = P_{\max} \times \min \left\{ 1, \max \left[R_{\min}, \left(\frac{CL}{CL_{x-ile}} \right)^\gamma \right] \right\}$$

where:

- $P_{\max} = 26\text{dBm}$ for PC2, 29 dBm for PC1.5, 31 dBm for PC1.
- $R_{\min} = -66\text{ dB}$ for PC2, -69 dB for PC1.5, -71 dB, assuming -40dBm minimum output power,
- CL_{x-ile} and γ are set as following:
 - $CL_{x-ile} = 10 \cdot \log_{10}(P_{\max}) - (\text{SNR}_{\text{target}} + 10 \cdot \log_{10}(kTB) + \text{NF})$,
 - o Where:
 - o $\text{SNR}_{\text{target}}$ and BW

For NR-NTN HPUE:

- Option 1: 3 dB SNR target and 5MHz UL BW (One company proposed for LEO only)
- Option 2: 15 dB SNR target and 2RB UL BW as baseline assumption
- o NF is the SAN noise figure, i.e. 4.3dB.
- o $10 \log_{10}(kT) = -174\text{dBm/Hz}$.
- o $\gamma = 1$ For uplink scenario.

6b.2.6.3 Downlink transmission power control

For downlink scenario, no power control scheme is applied to both NTN and TN scenarios.

6b.2.7 Received power model

See Section 6.2.7.

6b.2.8 Performance metric

See Section 6.2.8.

6b.2.9 Throughput ~ SNR mapping

See Section 6.2.9.

6b.3 Co-existence simulation methodology for NTN HPUE in FR1

6b.3.1 Simulation procedure

See section 6.3.1.

6b.3.2 Methods and principle to process co-existence simulation results

See section 6.3.2, noting that the following scenario is selected, in principle, to draw conclusion for NTN HPUE co-existence study.

Table 6b.3.2-1: Selected option for each scenario

Scenario	Aggressor system	Victim system	Environment	Contributing
4b	NTN UL	TN UL	Urban	NTN UE ACLR

6b.4 Co-existence simulation results for NTN HPUE in FR1

6b.4.1 NR-NTN Power Class 2: Scenario 4b

The co-existence results from all concerned options for NR-NTN power class 2 in this scenario were evaluated, and the detailed results can be found in Annex F to this TR. The numbers in below table have taken into account all the submitted results from these contributing companies for both average throughput loss and cell edge loss, and picked from the worst results from each source.

Table 6b.4.1-1 Interpolated required ACLR values for Scenario 4b with isolation distance 1.5km assumed

Company	Interpolated required ACLR
Samsung	<20
Vivo	<20
Ericsson	<20
ZTE	30.6
Mediatek	<20
Thales	<20
Xiaomi	<20
Qualcomm	<20
Average	<20
Note: Some company's results meet performance metrics at certain starting ACIR/ACLR level they used, e.g. 20dBc. These results are listed as smaller than those starting level.	

Table 6b.4.1-2 Interpolated required ACLR values for Scenario 4b without isolation distance assumed

Company	Interpolated required ACLR
Samsung	25.9
Vivo	27.1
Ericsson	30.1
Mediatek	33
Huawei	23.7
Xiaomi	24.7
Qualcomm	20
Average	26.4

6b.4.2 NR-NTN Power Class 1.5: Scenario 4b

The co-existence results from all concerned options for NR-NTN power class 1.5 in this scenario were evaluated, and the detailed results can be found in Annex F to this TR. The numbers in below table have taken into account all the submitted results from these contributing companies for both average throughput loss and cell edge loss, and picked from the worst results from each source.

Table 6b.4.2-1 Interpolated required ACLR values for Scenario 4b with isolation distance 1.5km assumed

Company	Interpolated required ACLR
Samsung	<20
Vivo	<20
Ericsson	14.4
ZTE	31.4
Mediatek	22.6
Thales	13
Xiaomi	23.2
Qualcomm	<20
Average	<20.6
Note: Some company's results meet performance metrics at certain starting ACIR/ACLR level they used, e.g. 20dBc. These results are listed as smaller than those starting level.	

Table 6b.4.2-2 Interpolated required ACLR values for Scenario 4b without isolation distance assumed

Company	Interpolated required ACLR
Samsung	29.3
Vivo	31.2
Ericsson	37.9
Mediatek	37.4
Huawei	26.2
Xiaomi	43.9
Qualcomm	20
Average	32.3

6b.4.3 NR-NTN Power Class 1: Scenario 4b

The co-existence results from all concerned options for NR-NTN power class 1 in this scenario were evaluated, and the detailed results can be found in Annex F to this TR. The numbers in below table have taken into account all the submitted results from these contributing companies for both average throughput loss and cell edge loss, and picked from the worst results from each source.

Table 6b.4.3-1 Interpolated required ACLR values for Scenario 4b with isolation distance 1.5km assumed

Company	Interpolated required ACLR
Samsung	<20
Vivo	20.5
Ericsson	14.4
ZTE	33.7
Mediatek	24.6
Thales	<13
Xiaomi	25.1
Average	<21.6
Note: Some company's results meet performance metrics at certain starting ACIR/ACLR level they used, e.g. 20dBc. These results are listed as smaller than those starting level.	

Table 6b.4.3-2 Interpolated required ACLR values for Scenario 4b without isolation distance assumed

Company	Interpolated required ACLR
Samsung	30.9
Vivo	34.5
Ericsson	40.3
Mediatek	39.4
Huawei	27.3
Xiaomi	38.6
Qualcomm	20
Average	35.2

6b.4.4 IoT-NTN Power Class 2: Scenario 4b

The co-existence results from all concerned options for IoT-NTN power class 2 in this scenario were evaluated, and the detailed results can be found in Annex F to this TR. The numbers in below table have taken into account all the submitted results from these contributing companies for both average throughput loss and cell edge loss, and picked from the worst results from each source.

Table 6b.4.4-1 Interpolated required ACLR values for Scenario 4b with isolation distance 1.5km assumed

Company	Interpolated required ACLR
Samsung	<20
Vivo	<20
ZTE	39.2
Mediatek	9.2
Xiaomi	34.3
Qualcomm	<20
Ericsson	13.4
Average	<22.8
Note: Some company's results meet performance metrics at certain starting ACIR/ACLR level they used, e.g. 20dBc. These results are listed as smaller than those starting level.	

Table 6b.4.4-2 Interpolated required ACLR values for Scenario 4b without isolation distance assumed

Company	Interpolated required ACLR
Samsung	35.3
Vivo	<20
Mediatek	33.6
Qualcomm	<20
Average	<27.2
Note: Some company's results meet performance metrics at certain starting ACIR/ACLR level they used, e.g. 20dBc. These results are listed as smaller than those starting level.	

6b.4.5 IoT-NTN Power Class 1: Scenario 4b

The co-existence results from all concerned options for IoT-NTN power class 1 in this scenario were evaluated, and the detailed results can be found in Annex F to this TR. The numbers in below table have taken into account all the submitted results from these contributing companies for both average throughput loss and cell edge loss, and picked from the worst results from each source.

Table 6b.4.5-1 Interpolated required ACLR values for Scenario 4b with isolation distance 1.5km assumed

Company	Interpolated required ACLR
Samsung	<20
Vivo	<20
ZTE	41.8
Mediatek	9.2
Xiaomi	42.4
Qualcomm	<20
Ericsson	19.0
Average	<24.7
Note: Some company's results meet performance metrics at certain starting ACIR/ACLR level they used, e.g. 20dBc. These results are listed as smaller than those starting level.	

Table 6b.4.5-2 Interpolated required ACLR values for Scenario 4b without isolation distance assumed

Company	Interpolated required ACLR
Samsung	39.8
Vivo	<20
Mediatek	33.6
Qualcomm	32
Average	<31.4
Note: Some company's results meet performance metrics at certain starting ACIR/ACLR level they used, e.g. 20dBc. These results are listed as smaller than those starting level.	

6b.5 Summary of co-existence study for NTN HPUE in FR1

The averaged interpolate ACLR values for each power class of NR-NTN and IoT-NTN in Scenario 4b are presented in Table 6b.5-1 below. It should be noted that only co-ex study results assuming 1.5km isolation distance are taken into account in this summary.

Table 6b.5-1: Average ACLR values for NTN HPUE in FR1

Power classes		PC 2	PC 1.5	PC 1
ACLR values (dBc)	NR-NTN	<20	<20.6	<21.6
	IoT-NTN	<22.8	Note 2	<24.7
Note 1: Some company's results meet performance metrics at certain starting ACIR/ACLR level they used, e.g. 20dBc. These results are listed as smaller than those starting level.				
Note 2: The power class 1.5 for IoT-NTN is not studied in this version.				

With the above averaged values from co-existence study and recognizing the characteristics of NTN HPUE in FR1 to the state of art and their future developments; and the protection to the operators in adjacent channel, the agreed ACLR of NR-NTN and IoT NTN HPUE are given in Table 6b.5.1-2.

Table 6b.5.1-2: ACLR and ACS of NTN HPUE in FR1

Power classes		PC 2	PC 1.5	PC 1
ACLR (dBc)	NR-NTN	31	31	34 ³
	NB-IoT based IoT-NTN	37 ¹	Note 2	37 ¹
	eMTC based IoT-NTN	31	Note 2	[34] ³
Note 1: NB-IoT based IoT-NTN PC2 and PC1 ACLR reused from NB-IoT based IoT-NTN PC3 UTRA_ACLR.				
Note 2: The power class 1.5 for IoT-NTN is not studied in this version.				
Note 3: It is observed in some company co-existence results that the 5%-ile throughput degradation is slightly higher than 5% with zero km isolation distance assumption. It was however agreed to progress with this ACLR as it facilitates UE RF implementation.				

6c Co-existence study for 11/14GHz

6c.1 Co-existence simulation scenario for 11/14GHz

Scenarios for the coexistence study for 11/14GHz are listed in Table 6c.1-1

Table 6c.1-1: Scenarios for NTN-NTN/TN co-existence

11/14GHz		NTN		
		GEO	LEO 600km	LEO 1200km
NR	Rural	N/A	N/A	N/A
	Urban macro	X	X	N/A
	Dense Urban	N/A	N/A	N/A
Note: It was agreed to focus on Urban Macro for GEO and LEO600 as they represent the worst-case.				

The aggressor and victim combinations for 11/14GHz cases are listed in Table 6c.1-2.

Table 6c.1-2: Aggressor and victim

No	Combination	Aggressor	Victim	Frequency Band	Notes on the Ku Band	Scope of Coexistence Simulation
1	TN with NTN	VSAT UL	UE UL	14 GHz	Ku Band VSAT uplink interfering with gNodeB receiver.	The ACLR of the VSAT to be varied/defined. The ACS of the gNodeB is fixed.
2	TN with NTN	UE UL	VSAT UL	14 GHz	UE uplink interfering with Ku Band Satellite receiver.	The ACLR of the UE is fixed. The ACS of the satellite to be varied/defined.
3	TN with NTN	VSAT UL	gNodeB DL	14 GHz	Ku Band VSAT uplink interfering with UE receiver.	The ACLR of the VSAT to be varied/defined. The ACS of the UE is fixed.

4	TN with NTN	gNodeB DL	VSAT UL	14 GHz	gNodeB downlink interfering with Ku Band Satellite receiver.	The ACLR of the gNodeB is fixed. The ACS of the Satellite to be varied/defined.
5	TN with NTN	gNodeB DL	Satellite DL	11 GHz	gNodeB downlink interfering with Ku Band VSAT receiver.	The ACLR of the gNodeB is fixed. The ACS of the VSAT to be varied/defined.
6	TN with NTN	Satellite DL	gNodeB DL	11 GHz	Ku Band Satellite downlink interfering with UE receiver.	The ACLR of the Satellite to be varied/defined. The ACS of the UE is fixed.
7	TN with NTN	Satellite DL	UE UL	11 GHz	Ku Band Satellite downlink interfering with gNodeB receiver.	The ACLR of the Satellite to be varied/defined. The ACS of the gNodeB is fixed.
8	TN with NTN	UE UL	Satellite DL	11 GHz	UE uplink interfering with Ku Band VSAT receiver.	The ACLR of the UE is fixed. The ACS of the VSAT to be varied/defined.

The frequency and bandwidth are listed in Table 6c.1-3.

Table 6c.1-3: Proposed frequency and bandwidth for co-existence study

	Frequency	Bandwidth	Duplex mode	Frequency reuse factor
TN Urban macro	11/14 GHz	100MHz	TDD	1
GEO	11GHz (DL)/14GHz(UL)	100MHz per beam	FDD	2
LEO	11GHz (DL)/14GHz(UL)	100MHz per beam	FDD	2
Note: FRF=1 with single polarization has been adopted for calibration. For co-existence simulation, assume FRF=2 with two polarizations (RHCP, LHCP).				

6c.2 Co-existence simulation assumption for 11/14GHz

6c.2.1 Network layout model

6c.2.1.1 Co-existence between NTN satellite and TN

Cellular cell structure is considered for both NTN and TN network layout.

Referring to TR 38.811 Section 6.3 and Annex A, a 3D global coordinate system is considered (Earth-Centred Earth Fixed) for simulating NTN beams direction and location on the earth surface. It means the NTN beam location, TN randomly dropping location are generated with a set of three parameters (x,y,z).

Deployment of NTN satellite and TN cells and UEs for co-existence study is listed in Table 6c.2.1.1-1.

Table 6c.2.1.1-1: Network and UE deployment

No.	Combination	Aggressor	Victim	Which NTN cell/UE to observe?	Which TN/UE to observe?	Which TN cells in a TN to observe?
1x	TN with NTN	NTN UL	TN UL	<u>NTN cell:</u> Nadir point. <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters.	TN randomly placed in this NTN beam	Only the active TN clusters which contain NTN UE(s)
2x	TN with NTN	TN UL	NTN UL	<u>NTN cell:</u> Observe NTN central beam for SINR, 6 adjacent beams for inter-beam interference. <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters	Consider an active rate of 20% for TN	Only the active TN cells in central NTN beam
3x	TN with NTN	NTN UL	TN DL	<u>NTN cell:</u> Nadir point <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters.	TN clusters randomly placed in this NTN beam	All active TN clusters which contain NTN UE(s)
				<u>NTN cell:</u> NTN cell with satellite at low elevation (to be further discussed for GEO and LEO , interested companies can bring analysis and results for other values). <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters.	TN clusters randomly placed in this NTN beam	Only the active TN clusters which contain NTN UE(s)
4x	TN with NTN	TN DL	NTN UL	<u>NTN cell:</u> Observe NTN central beam for SINR, 6 adjacent beams for inter-beam interference. <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters	Consider the active rate of 20% for TN.	All active TN cells in central NTN beam
5x	TN with NTN	TN DL	NTN DL	<u>NTN cell:</u> Observe NTN central beam for SINR, 6 adjacent beams for inter-beam interference. <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters.	One cluster with 19 TN cells (57 sectors) randomly placed in the central NTN beam	Only the active TN clusters which contain the NTN UE(s)
6c	TN with NTN	NTN DL	TN DL	<u>NTN cell:</u> Nadir point. <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters.	TN clusters randomly placed in this NTN beam	All in central NTN beam

				<u>NTN cell:</u> NTN cell with satellite at low elevation (to be further discussed for GEO and LEO, interested companies can bring analysis and results for other values) <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters	TN clusters randomly placed in this NTN beam	
7x	TN with NTN	NTN DL	TN UL	<u>NTN cell:</u> Nadir point <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters.	TN clusters randomly placed in this NTN beam	Only the active TN clusters which contain NTN UE(s)
				<u>NTN cell:</u> NTN cell with satellite at low elevation (to be further discussed for GEO and LEO, interested companies can bring analysis and results for other values). <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters.	TN clusters randomly placed in this NTN beam	Only the active TN clusters which contain NTN UE(s)
8x	TN with NTN	TN UL	NTN DL	<u>NTN cell:</u> Observe NTN central beam for SINR, 6 adjacent beams for inter-beam interference. <u>NTN UE:</u> NTN UEs randomly dropped in TN clusters	Consider the active rate of 20% for TN	All active TN cells in central NTN beam

6c.2.2 System parameters

6c.2.2.1 Satellite parameters

Set-1 satellite parameters are listed in Table 6c.2.2.1-2 according to TR 38.821 [6].

The satellite max Tx power can be calculated by the equation as below:

$$\max Tx\ power[dBm] = EIRP\ density\left[\frac{dBW}{MHz}\right] + 30 + 10\log_{10}(N_{RB} * SCS[MHz] * 12) - Max\ Gain[dBi]$$

Table 6c.2.2.1-1: N_{RB} configuration per BandWidth size and SCS

Configuration Ku-band	N _{RB} (100MHz BW)
SCS 30 kHz	273

Table 6c.2.2.1-2: Satellite parameters for co-existence study

Satellite orbit		GEO	LEO-600
Satellite altitude		35786 km	600 km
Satellite antenna pattern		Section 6.4.1 in [2]	Section 6.4.1 in [2]
Payload characteristics for DL transmissions			
Equivalent satellite antenna aperture	Ku-band (i.e. 11 GHz for DL)	4 m	0.91 m
Satellite EIRP density		43 dBW/MHz	7 dBW/MHz
Satellite Tx max Gain		43 dBi	38.5 dBi
3dB beamwidth		0.400 deg	1.768 deg
Satellite beam diameter		250 km	19 km
Payload characteristics for UL transmissions			
Equivalent satellite antenna aperture	Ku-band (i.e. 14 GHz for UL)	4 m	0.714 m
G/T		16.5 dB K ⁻¹	10.9 dB K ⁻¹
Satellite RX max Gain		44 dBi	38.5 dBi

Table 6c.2.2.1-3: Other parameters for NTN satellite

Parameters	NTN satellite	Remark
Carrier frequency	14GHz(UL) / 11GHz(DL)	
The number of active UE (UL)	10 UEs and 27RBs per UE for GEO and LEO	
The number of active UE (DL)	1	
Traffic model	Full buffer	
DL power control	NO	
UL power control	See Session 6c.2.6.2	
NTN satellite Noise Figure in dB	See Table 6c.2.2.1-4	
Handover margin	3dB	

Table 6c.2.2.1-4: NTN satellite Noise Figure in dB

Satellite	GEO	LEO 600
G/T (dB K ⁻¹)	16.5	10.9
NF (dB)	3	3

6c.2.2.2 NTN UE parameters

The NTN UE antenna characteristics are given in Table 6c.2.2.2-1

Table 6c.2.2.2-1: NTN UE antenna characteristics for system level simulations

VSAT Antenna Characteristics				
1		VSAT phased array – 45x45	VSAT phased array – 78x78	VSAT Parabolic
1.1	Antenna pattern	TR 38.922	TR 38.922	Circular Aperture TR 38.863 Clause 6.2.3.1
1.2	Element gain (dBi) (Note 1)	3.5	3.5	N/A
1.3	Y axis / Z axis element 3 dB beam width of single element (degree)	90° for Y axis 90° for Z axis	90° for Y axis 90° for Z axis	N/A
1.4	Y axis / Z axis element front-to-back ratio (dB)	30 dB	30 dB	N/A
1.5	Antenna polarization (Note 5)	Circular (RHCP or LHCP)	Circular (RHCP or LHCP)	Circular (RHCP or LHCP)
1.6	Antenna array configuration (Y axis × Z axis)(Note 3)	45x45 elements	78x78 elements	60 cm diameter aperture with Rx antenna gain 34.3 dBi for 11GHz (DL) and Tx antenna gain 36.6 dBi for 14GHz (UL)
1.7	Number of supported polarizations, <i>P</i>	1	1	1
1.8	Y axis / Z axis radiating element spacing	0.5 lambda	0.5 lambda	N/A
1.9	Array Ohmic loss (dB) (Note 1)	2	2	N/A
1.10	Conducted power (before Ohmic loss) per antenna element (dBm) (Note 2)	6.9	9.1	N/A
1.11	Tx Transmit power	10 Watts (Both GEO and LEO)	50 Watts (Only GEO)	10 Watts (Both GEO and LEO)
1.12	Maximum coverage angle in the horizontal plane (degrees)	0~360 degrees	0~360 degrees	0~360 degrees
1.13	Vertical coverage range (degrees) between beam direction and normal direction	0~60 degrees	0~60 degrees	0~60 degrees
1.14	Normal direction	Toward X+ axis	Toward X+ axis	Toward X+ axis
1.15	Max EIRP Value	76.6 dBm	88.2 dBm	76.6 dBm
1.16	Noise Figure	6 dB	6.0 dB	2.5 dB

Note 1: The element gain in row 1.2 includes the loss given in row 1.9.

Note 2: The conducted power per element assumes Y axis × Z axis × Number of supported polarizations elements (i.e. power per Y axis / Z axis polarized element).

Note 3: All of the Y axis × Z axis elements are assumed as the horizontal radiating elements in the horizontal plane.

Note 4: Antenna model for NTN VSAT uses circular polarization, but since many GEO satellites use linear polarization evaluations are conducted by converting to linear polarization

The altitudes of NTN UE for different use case are defined in Table 6c.2.2.2-2.

Table 6c.2.2.2-2: NTN UE altitude

Use case	M-ESIM	A-ESIM	L-ESIM
Altitude	22.5m	3-14km	1.5m

Note 1: Co-existence evaluation is only limited to L-ESIM.

6c.2.2.3 TN parameters

TN parameters for co-existence study are given in Table 6c.2.2.3-1.

Table 6c.2.2.3-1: Simulation assumptions of TN NR

Carrier frequency in GHz	11 and 14
Size of each nominal channel BW in MHz	100
Transmission bandwidth in MHz	98.28
Environment	Urban macro
Network layout	hexagonal grid, 19 macro sites, 3 sectors per site with wrap around
Inter-site distance in meter	450m
UE number per cell	3
System loading and activity	20%
Network location	See Table 6c.2.1.1-2
DL subcarrier spacing	30kHz
UL	OFDMA
DL power control	No
UL power control	See Section 6c.2.6.2
Frequency reuse	1
Number of scheduled UE per cell (DL)	Same as the number of BS beam
Number of scheduled UE per cell (UL)	Same as the number of BS beam
BS height	25
UE antenna height in meter	1.5
UE TX power in dBm	-40 to 23
Building penetration loss	3GPP TR 38.803
Cell selection margin in dB	3
BS-MS min distance in meters	35
BS noise figure in dB	10
UE noise figure in dB	10
BS-UE path-loss model	TR 38.901
Standard deviation of BS-UE log-normal shadow fading in dB	Deployment scenario related, referring to TR 38.901.
Shadowing correlation	Between cells: 1.0 Between sites: 0.5
Link-level performance model	See Section 6c.2.9
UE distribution	Uniform
Evaluation metrics	See Section 6c.2.8

6c.2.3 Antenna and beam forming pattern modelling

6c.2.3.1 Satellite parameters

In Rel-18, antenna pattern in section 6.4.1 of TR 38.811 is reused for NTN SAN and NTN UE.

The following normalized antenna gain pattern, corresponding to a typical reflector antenna with a circular aperture, is considered.

$$\begin{aligned}
 &1 \quad \text{for } \theta = 0 \\
 &4 \left| \frac{J_1\left(ka \sin\left(\theta - \frac{\pi}{180}\right)\right)}{ka \sin\left(\theta - \frac{\pi}{180}\right)} \right|^2 \quad \text{for } 0 < |\theta| \leq 90^\circ
 \end{aligned}$$

$$\text{For UE, } 4 \left| \frac{J_1\left(ka \sin\left(\frac{\theta}{2}\right)\right)}{ka \sin\left(\frac{\theta}{2}\right)} \right|^2 \quad \text{for } |\theta| > 90^\circ$$

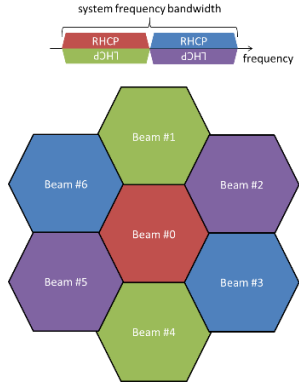
where:

- $J_1(x)$ is the Bessel function of the first kind and first order with argument 'x';
- a is the radius of the antenna's circular aperture;
- $k = 2\pi f/c$ is the wave number;
- f is the frequency of operation;
- c is the speed of light in a vacuum and θ is the angle measured from the bore sight of the antenna's main beam.

Note that ka equals to the number of wavelengths on the circumference of the aperture and is independent of the operating frequency. And the $\sin()$ function is in radian.

The beam layout definition for a single satellite simulation in Ku-Band is defined in Table 6c.2.3.1-1.

Table 6c.2.3.1-1: Beam layout definition for single satellite simulation

Beam layout definition	Same with Table 6.2.3.1-1
Number of beams	Same with Table 6.2.3.1-1
UV plane illustration (extracted from [19])	Same with Table 6.2.3.1-1
UV plane convention	Same with Table 6.2.3.1-1
Adjacent beam spacing on UV plane	Same with Table 6.2.3.1-1
Central beam bore sight direction definition	Baseline: Case 1: Central beam center is considered at nadir point Case 2: 35° for GEO and LEO
Frequency-reuse factor	FRF=2  Do not consider interference leakage from adjacent beams for co-ex study. For example for the figure above, do not consider the interference leakage from green, blue and purple beams when study SINR for the red beam.
Polarization re-use	Enable Note: Polarization re-use should apply only if circular polarization for terminal antenna is considered
UEs outdoor/indoor distribution	100% outdoor
UE distribution	Same with Table 6.2.3.1-1
UE configuration	Mobile VSAT for GSO and NGSO
UE orientation	Random Ideal Tracking serving beam
UE attachment	RSRP

6c.2.3.2TN BS antenna and beam forming pattern modelling

The parameters in Table 6c.2.3.2-1, Table 6c.2.3.2-2, Table 6c.2.3.2-3 and Table 6c.2.3.2-4 are used for BS antenna pattern in the coexistence study.

Table 6c.2.3.2-1: TN BS antenna modelling for Urban macro scenario for 11 GHz

Parameters	Values for 11 GHz
Scenario	Urban macro
Antenna pattern	AAS model see below (from TR 38.922)
Element gain $G_{E,max}$ (dBi)	6.4
Horizontal 3dB /vertical 3dB beam width of single element	90° for H 65° for V
Horizontal/vertical front-to-back ratio (dB)	30 for both H/V
Antenna polarization	Linear $\pm 45^\circ$
Antenna array configuration (Row $\times$ Column) (Note 4)	8x16
Horizontal/Vertical radiating sub-array or element spacing (Note 5)	0.5 of wavelength for H, 2.1 of wavelength for V
Number of element rows in sub-array	3
Vertical element separation in sub-array ($d_{v,sub}$)	0.7 of wavelength for V
Pre-set sub-array down-tilt (degrees) (Note 6)	3
Array Ohmic loss LE (dB)	2
Conducted power (before Ohmic loss) per antenna element (dBm)	22
Mechanical downtilt (degrees)	6

Table 6c.2.3.2-2: Array antenna model details

Description	Equation	Unit
Peak normalized element radiation pattern	$A(\theta, \varphi) = -\min \left[-\min \left[12 \left(\frac{\varphi}{\varphi_{3dB}} \right)^2, A_m \right] - \min \left[12 \left(\frac{\theta - 90}{\theta_{3dB}} \right)^2, SLA_v \right], A_m \right]$	dB
Peak gain normalized element radiation pattern	$A_E(\theta, \varphi) = G_{E,max} + A(\theta, \varphi)$	dBi
Composite array radiation pattern	$A_A(\theta, \varphi) = A_E(\theta, \varphi) + 10 \log_{10} \left(\left \sum_{m=1}^M \sum_{n=1}^N w_{m,n} v_{m,n} \right ^2 \right)$, where $v_{m,n} = \exp \left(j2\pi \left((m-1) \frac{d_v}{\lambda} \cos(\theta) + (n-1) \frac{d_h}{\lambda} \sin(\theta) \sin(\varphi) \right) \right)$ $w_{m,n} = \frac{1}{\sqrt{MN}} \exp \left(j2\pi \left((m-1) \frac{d_v}{\lambda} \sin(\theta_{etilt}) - (n-1) \frac{d_h}{\lambda} \cos(\theta_{etilt}) \sin(\varphi_{escan}) \right) \right)$	dBi

Table 6c.2.3.2-3: TN BS antenna modelling for Urban macro scenario for 14 GHz

Parameters	Values for 14GHz
Scenario	Urban macro
Antenna pattern	Extended AAS model see below (from TR 38.922)
Element gain $G_{E,max}$ (dBi)	6.4
Horizontal 3dB /vertical 3dB beam width of single element	90° for H 65° for V
Horizontal/vertical front-to-back ratio A_m (dB)	30 for both H/V
Side lobe suppression SLA_v (dB)	30
Antenna polarization	Linear $\pm 45^\circ$
Antenna array configuration (Row $\times$ Column)	16 $\times$ 24 elements
Horizontal/Vertical radiating element spacing	H = 0.5 λ V = 2.8 λ
Number of element rows in sub-array	4
Vertical element separation in sub-array ($d_{v,sub}$)	0.7 of wavelength of V
Pre-set sub-array down-tilt (degrees)	3
Array Ohmic loss LE (dB)	2
Conducted power (before Ohmic loss) per antenna element (dBm)	17.14
Base station horizontal coverage range (degrees)	+/-60
Base station vertical coverage range (degrees) ^(Note 1)	90-100
Mechanical down-tilt (degrees)	6

Table 6c.2.3.2-4: Extended AAS model

Description	Equation
Peak normalized element radiation pattern	$A(\theta, \varphi) = -\min \left[-\left(-\min \left[12 \left(\frac{\varphi}{\varphi_{3dB}} \right)^2, A_m \right] - \min \left[12 \left(\frac{\theta - 90}{\theta_{3dB}} \right)^2, SLA_v \right] \right), A_m \right]$
Peak gain normalized element radiation pattern	$A_E(\theta, \varphi) = G_{E,max} + A(\theta, \varphi)$
Sub-array excitation	$w_m = \frac{1}{\sqrt{M_{sub}}} \exp \left(j2\pi(m-1) \frac{d_{v,sub}}{\lambda} \sin(\theta_{subtilt}) \right)$
Sub-array radiation pattern	$A_{sub}(\theta, \varphi) = A_E(\theta, \varphi) + 10 \log_{10} \left(\left \sum_{m=1}^{M_{sub}} w_m v_m \right ^2 \right)$, where $v_m = \exp \left(j2\pi(m-1) \frac{d_{v,sub}}{\lambda} \cos(\theta) \right)$
Array excitation	$w_{m,n} = \frac{1}{\sqrt{MN}} \exp \left(j2\pi \left((m-1) \frac{d_v}{\lambda} \sin(\theta_{etilt}) - (n-1) \frac{d_h}{\lambda} \cos(\theta_{etilt}) \sin(\varphi_{escan}) \right) \right)$
Composite array radiation pattern	$A_A(\theta, \varphi) = A_{sub}(\theta, \varphi) + 10 \log_{10} \left(\left \sum_{m=1}^M \sum_{n=1}^N w_{m,n} v_{m,n} \right ^2 \right)$, where $v_{m,n} = \exp \left(j2\pi \left((m-1) \frac{d_v}{\lambda} \cos(\theta) + (n-1) \frac{d_h}{\lambda} \sin(\theta) \sin(\varphi) \right) \right)$

6c.2.3.3 TN UE antenna model

The UE is expected to have a conducted interface assuming an isotropic radiation pattern antenna and without beamforming.

6c.2.4 ACIR model

ACLR modelling for TN and NTN co-existence study referring to clause 5.1.1.4.1 and 5.1.1.4.2 in TR 36.942 [8] is used as baseline. The number of RBs refers to Table 6c.2.2.1-3 and Table 6c.2.2.4-1 respectively.

6c.2.5 Propagation model

Propagation models between NTN and TN components used for the coexistence study are listed in Table 6c.2.5-1.

Table 6a.2.5-1: Propagation models between NTN and TN

Link	Propagation model
TN BS to Mobile VSAT on roof	Free space path loss
TN BS to L-ESIM at 1.5 m	UMa as in 3GPP TR 38.803
TN BS to TN UE	UMa as in 3GPP TR 38.803
TN UE to Mobile VSAT on roof	UMa as in 3GPP TR 38.803 (BS is to be replaced with VSAT)
TN UE to L-ESIM	Umi
Satellite to TN BS/UE	3GPP TR 38.821, Refer to Table 6.6.2-2 TR 38.811 for NLOS Clutter Loss assumptions
Satellite to VSAT/ESIM	3GPP TR 38.821, Refer to Table 6.6.2-2 TR 38.811 for NLOS Clutter Loss assumptions
TN BS to Satellite	3GPP TR 38.821, Refer to Table 6.6.2-2 TR 38.811 for NLOS Clutter Loss assumptions
Note1:	For the propagation models which use the 3GPP TR 38.821 [3], to use same assumptions as in [3] and consider the atmospheric losses and the scintillation losses.
Note2:	TN BS height is 25m

6c.2.6 Transmission power control model

6c.2.6.1 UL TPC

Following UL power control model for both NTN and TN are used as starting point

$$P_t = P_{\max} \times \min \left\{ 1, \max \left[R_{\min}, \left(\frac{CL}{CL_{x-ile}} \right)^\gamma \right] \right\}$$

Where,

- $P_{\max}$ = maximum UE Tx power, i.e. 23 dBm for TN UE;
- $R_{\min}$ = minimum power reduction ratio, i.e. -63 dB for TN UE, and [-63 dB by assuming NTN UE min Tx power as -30dBm as starting point] for NTN UE;
- CL_{x-ile} and γ are set as following:
 - $\gamma = 1$;
 - $CL_{x-ile} = -\text{SNR_target} + P_{\max} - \text{Thermal Noise} - \text{BS/SAN_NoiseFigure} - 10 \cdot \log_{10}(\text{BW})$, considering SNR_target is 15dB and BW is actual UL BW.

6c.2.6.2 DL TPC

For downlink scenario, no power control scheme is applied.

6c.2.7 Received power model

The received power in downlink and uplink scenarios is defined as below:

$$RX_PWR = TX_PWR - Path\ loss + G_TX + G_RX$$

where:

- RX_PWR is the received power
- TX_PWR is the transmitted power
- G_TX is the transmitter antenna gain (directional array gain)
- G_RX is the receiver antenna gain (directional array gain).

6c.2.8 Performance metric

For NR, the average throughput loss and 5%-ile throughput loss should be less than 5%.

For NTN, the average throughput loss and 5%-ile throughput loss should be less than 5%.

6c.2.9 Throughput ~ SNR mapping

Adopt Section 5.2.7 of TR 38.803 [20] as the SINR-Throughput performance metrics

6c.3 Co-existence simulation methodology for 11/14GHz

6c.3.1 Simulation procedure

See Section 6.3.1 of TR 38.863, noting the FRF and network deployment for NTN and TN in above 10GHz are specified accordingly.

Following approaches for the deployment of NTN UE and satellite are used:

- The satellite should be generated in the visible area/sky of NTN UE;
- NTN UEs point to the satellite accurately;
- The position of the satellite should guarantee that NTN UE vertical angle towards the satellite;
 - For calibration, use 35 degree instead of the range;
 - SAN elevation angle value, 35, & 90 degrees (computed from the centre of the beam), for co-existence - simulation.
- Horizontal angle of NTN UEs should be calculated based on the satellite position;
- Use 35m as minimum distance assumed between VSAT and TN BS for co-ex study.

6c.3.2 Methods and principle to process co-existence simulation results

In order to process the co-existence simulation results received for all different scenarios and assumptions, the following steps are adopted:

- Step 1: Discuss and agree on the most stringent scenario(s) for each scenario (Scenario 1, 2, 3...,8);
- Step 2: Discuss and determine the required ACIR from results of the most stringent case(s) for each scenario;

- Step 3: Use equation to derive corresponding ACLR or ACS from the agreed ACIR for each scenario

It is noted that the averaged ACIR for the most stringent case in each scenario would be derived by taking the average among the interpolated ACIR results derived from each company's results for that case.

Moreover, the following considerations are adopted to deal with major disputes for the selected cases' results in each scenario:

- If the required ACIR results, from the contributor who did not participate or their results is still not well-aligned in calibration table, has a difference larger than 10 dB with most others, this result can be not considered in the discussion.
- If the required ACIR results, from one contributor, has a difference larger than 10 dB with most others, this result can be not considered in the discussion.

6c.4 Co-existence simulation results for 11/14GHz

It was agreed to simulate only Scenarios 1x, 4x, 5x, and 6c among the possible combinations, as these adequately cover the relevant ACLR/ACS conditions for both SAN and VSAT.

6c.4.1 Scenario 1x: 14GHz NTN UL interfering TN UL

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select worst case between 35 and 90 degrees elevation angle of GEO and LEO-600. The antenna assumptions for VSAT UE include Parabolic, Phase array 45x45 and Phase array 78x78 (only applicable for GEO). The required ACIR for throughput loss of 5% is selected as the most stringent case. In cases where 5th percentile cannot be achieved, 50th percentile is considered to determine the worst case.

Table 6c.4.1-1 Simulation results for Scenario 1x - GEO

VSAT	Xiaomi	Ericsson	Qualcomm	Mediatek	CATT	ZTE
Parabolic	21.2	3	20	21.53	18.67	34.03
Phased array - 45x45	17.3	16	25	21.16	31.55	22.57
Phased array - 78x78	18.1		30	21.45		20.71

Table 6c.4.1-2 Simulation results for Scenario 1x – LEO600

ACIR (dB)	Xiaomi	Ericsson	Qualcomm	Mediatek	CATT	ZTE
Parabolic	13.7	0	20	0	14.52	30.15
Phased array - 45x45	12.7	6	25	0	27.77	18.91

The average ACIR values from above simulation is calculated based on the methods and principle to process co-existence simulation results defined in clause 6.3x.2.

Table 6c.4.1-3 Averaged ACIR for Scenario 1x - GEO

	Average required ACIR (dB)
Phase array 45x45	20.06
Phase array 78x78	21.18
Parabolic	21.08

Table 6c.4.1-4 Averaged ACIR for Scenario 1x – LEO600

	Average required ACIR (dB)
Parabolic	17.26
Phased array - 45x45	12.45

6c.4.2 Scenario 4x: 14GHz TN DL interfering NTN UL

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select worst case between 35 and 90 degrees elevation angle of GEO and LEO-600. The antenna assumptions for VSAT UE include Parabolic, Phase array 45x45 and Phase array 78x78 (only applicable for GEO). The required ACIR for throughput loss of 5% is selected as the most stringent case. In cases where 5th percentile cannot be achieved, 50th percentile is considered to determine the worst case.

Table 6c.4.2-1 Simulation results for Scenario 4x - GEO

VSAT	Ericsson	Qualcomm	Mediatek	CATT	ZTE
Parabolic	10	20	0	38.43	33.36
Phased array - 45x45	14	20	0	37.77	33.04
Phased array - 78x78		15	0		34.25

Table 6c.4.2-2 Simulation results for Scenario 4x – LEO600

ACIR (dB)	Ericsson	Qualcomm	Mediatek	CATT	ZTE
Parabolic	13	30	14.51	40	30
Phased array - 45x45	12	25	14.51	40	26

The average ACIR values from above simulation is calculated based on the methods and principle to process co-existence simulation results defined in clause 6.3x.2.

Table 6c.4.2-3 Averaged ACIR for Scenario 4x - GEO

	Average required ACIR (dB)
Phase array 45x45	15
Phase array 78x78	17
Parabolic	15

Table 6c.4.2-4 Averaged ACIR for Scenario 4x – LEO600

	Average required ACIR (dB)
Parabolic	30
Phased array - 45x45	21.83

6c.4.3 Scenario 5x: 11GHz TN DL interfering NTN DL

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select worst case between 35 and 90 degrees elevation angle of GEO and LEO-600. The antenna assumptions for VSAT UE include Parabolic, Phase array 45x45 and Phase array 78x78 (only applicable for GEO). The required ACIR for throughput loss of 5% is selected as the most stringent case. In cases where 5th percentile cannot be achieved, 50th percentile is considered to determine the worst case.

Table 6c.4.2-1 Simulation results for Scenario 5x - GEO

VSAT	Xiaomi	Ericsson	Qualcomm	Mediatek	CATT	ZTE
Parabolic	41.47	32	40	35.53	33.58	35.8
Phased array - 45x45	40.09	42	45	35.92	40	30.04
Phased array - 78x78	35.23		45	31.01		20.57

Table 6c.4.2-2 Simulation results for Scenario 5x – LEO600

ACIR (dB)	Xiaomi	Ericsson	Qualcomm	Mediatek	CATT	ZTE
Parabolic	39.84	29	40	35.88	38.41	46.37
Phased array - 45x45	40.34	38	45	36.26	40	16.77

The average ACIR values from above simulation is calculated based on the methods and principle to process co-existence simulation results defined in clause 6.3b.2.

Table 6c.4.2-3 Averaged ACIR for Scenario 5x - GEO

	Average required ACIR (dB)
Phase array 45x45	36.39
Phase array 78x78	38.84
Parabolic	33.12

Table 6c.4.2-4 Averaged ACIR for Scenario 5b – LEO600

	Average required ACIR (dB)
Parabolic	38.25
Phased array - 45x45	39.92

6c.4.4 Scenario 6c: 11GHz NTN DL interfering TN DL

The co-existence results from all concerned options in this scenario were evaluated, and it has been agreed to select worst case between 35 and 90 degrees elevation angle of GEO and LEO-600. The antenna assumptions for VSAT UE include Parabolic, Phase array 45x45 and Phase array 78x78 (only applicable for GEO). The required ACIR for throughput loss of 5% is selected as the most stringent case. In cases where 5th percentile cannot be achieved, 50th percentile is considered to determine the worst case.

Table 6c.4.2-1 Simulation results for Scenario 6c - GEO

VSAT	Xiaomi	Ericsson	Qualcomm	Mediatek	CATT	ZTE
Parabolic	0	0	0	0	0	0
Phased array - 45x45	0	0	0	0	0	0
Phased array - 78x78	0		0	0		0

Table 6c.4.2-2 Simulation results for Scenario 6c – LEO600

ACIR (dB)	Xiaomi	Ericsson	Qualcomm	Mediatek	CATT	ZTE
Parabolic	0	0	0	0	0	0
Phased array - 45x45	0	0	0	0	0	0

The average ACIR values from above simulation is calculated based on the methods and principle to process co-existence simulation results defined in clause 6.3b.2.

Table 6c.4.2-3 Averaged ACIR for Scenario 6c - GEO

	Average required ACIR (dB)
Phase array 45x45	0
Phase array 78x78	0
Parabolic	0

Table 6c.4.2-4 Averaged ACIR for Scenario 6c – LEO600

	Average required ACIR (dB)
Parabolic	0
Phased array - 45x45	0

6c.5 Summary of co-existence study

6c.5.1 Coexistence at 11/14GHz

This sub-clause captures the summary of the co-existence studies. The averaged ACIR values for each scenario are presented in Table 6c.5.1-1 below.

Table 6c.5.1-1 Average ACIR values for each scenario at 17/27GHz

Co-existence results suggested ACIR values					
			ACIR values		
Scenario	Contributing	NTN	Parabolic	Phased array - 45x45	Phased array - 78x78
1x	VSAT ACLR	GEO	20,06	21,18	21,08
		LEO600	17,26	12,45	
4x	SAN ACS	GEO	15	17	15
		LEO600	30	21,83	
5x	VSAT ACS	GEO	36,39	38,84	33,12
		LEO600	38,25	39,92	
6c	SAN ACLR	GEO	0	0	0
		LEO600	0	0	

By considering the ACLR and ACS of TN BS and UE in Table 6c.5.1-2, and the average ACIR values for each scenario at 11/14GHz in Table 6c.5.1-1, the suggested ACLR and ACS of NTN SAN and VSAT from each scenario are given in Table 6c.5.1-3. It should be noted that the values in Table 6c.5.1-3 are directly derived from the selected option of each scenario, and it is limited by the nature of assumptions and methodologies adopted in the co-existence studies.

Table 6c.5.1-2: ACLR and ACS of TN for 11/14GHz

	TN		Values
11 GHz (IMT studies for 10 GHz from TR 38.921)	BS	ACLR	37 dB
		ACS	40 dB
	UE	ACLR	24 dB
		ACS	31 dB
14 GHz (IMT studies for 15 GHz from TR 38.922)	BS	ACLR	30 dB
		ACS	30 dB
	UE	ACLR	24 dB
		ACS	24 dB

Table 6c.5.1-3: Co-existence results suggested ACIR of NR-NTN at 11/14GHz

	SAN		VSAT		
	GEO	LEO 600	Parabolic	Phased array 45x45	Phased array 78x78
ACLR (dBc)	Note 1	Note 1	GEO: 20.52 LEO 17.49	GEO: 21.79 LEO: 12.52	GEO: 21.67
ACS (dBc)	17.22	22.54	GEO: 45.21 LEO: Note 2	Note 2	GEO: 35.4
Note 1: NTN/TN coexistence simulations indicate that there is no interference condition under current assumptions.					
Note 2: The current simulation assumptions do not allow the required ACIR obtained from simulation.					

With the above suggested values, recognizing the characteristics of SAN and VSAT to the state of art and their future developments, the agreed ACLR and ACS of NR-NTN at 11/14 GHz are given in Table 6c.5.1-4.

Table 6c.5.1-4: ACLR and ACS of NR-NTN at 11/14GHz

	SAN		VSAT
	GEO	LEO 600	
ACLR (dBc)	16	16	18.5 ¹
ACS (dBc)	20	26	26 ²
Note 1: Co-existence results suggested stricter ACLR values. However, a relaxed requirement was selected to balance feasibility and implementation practicality			
Note 2: The VSAT ACS is derived from co-existence results and taking several factors into considerations including the likelihood of encountering similar geometrical conditions (i.e., a very weak NTN link and a very strong adjacent TN link) in realistic VSAT operations.			

7 RF requirements

7.1 SAN Reference points for RF requirements

7.1.1 General

This section describes the SAN reference points for RF requirements with respect to different SAN types.

7.1.2 SAN type 1-C

This requirement is not applicable in Release 17.

7.1.3 SAN type 1-H

For SAN type 1-H, the requirements are defined for two points of reference, signified by radiated requirements and conducted requirements.

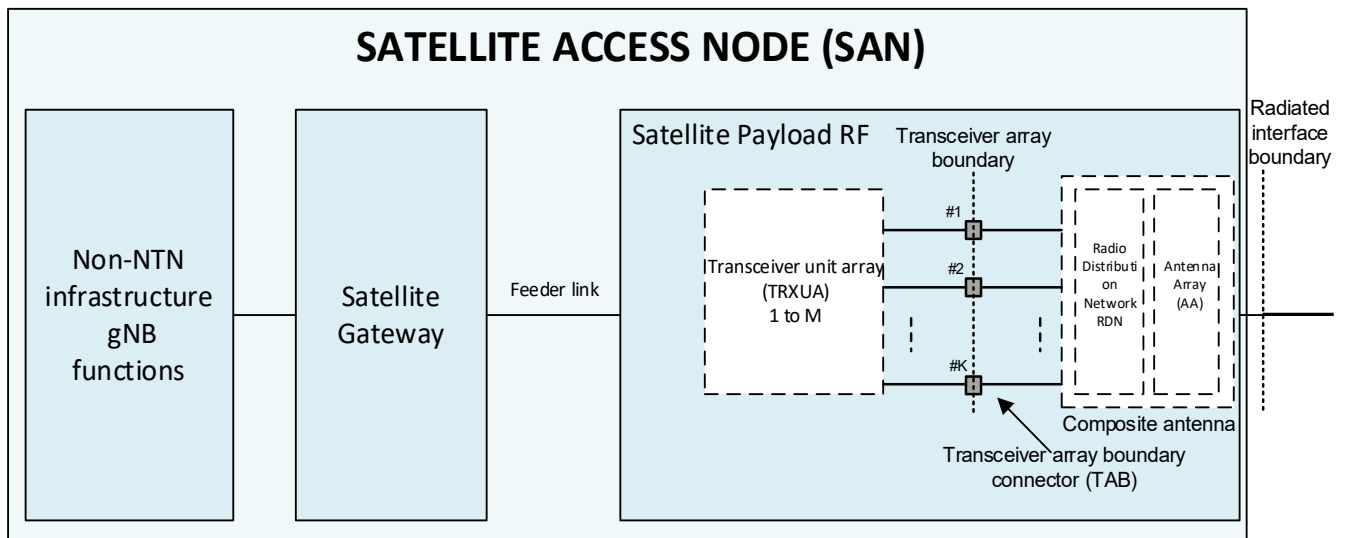


Figure 7.1.3-1: Radiated and conducted reference points for SAN type 1-H

Radiated characteristics are defined over the air (OTA), where the *operating band* specific radiated interface is referred to as the *Radiated Interface Boundary* (RIB). Radiated requirements are also referred to as OTA requirements. The (spatial) characteristics in which the OTA requirements apply are detailed for each requirement.

Conducted characteristics are defined at individual or groups of *TAB connectors* at the *transceiver array boundary*, which is the conducted interface between the transceiver unit array and the composite antenna.

The transceiver unit array is part of the composite transceiver functionality receiving and transmitting modulated signals to ensure radio links with ground base stations and users.

The Non-NTN infrastructure gNB functions implement the modulation and demodulation functionalities of the NTN gNB.

The Satellite NTN Gateway is part of the NTN ground infrastructure and is responsible for up- and down-converting, as well as transmitting and receiving modulated signals between the Non-NTN infrastructure gNB functions and the Satellite payload via the Feeder link.

The satellite payload is composed by a transceiver unit array and a composite antenna array. The transceiver unit array contains an implementation of specific number of transmitter units and an implementation of specific number of receiver units.

The composite antenna contains a radio distribution network (RDN) and an antenna array. The RDN is a linear passive network which distributes the RF power generated by the transceiver unit array to the transmit antenna array.

How a conducted requirement is applied to the *transceiver array boundary* is detailed in the respective requirement clause.

7.1.4 SAN type 1-O

For SAN type 1-O the radiated characteristics are defined over the air (OTA), where the *operating band* specific radiated interface is referred to as the *Radiated Interface Boundary* (RIB). Radiated requirements are also referred to as OTA requirements. The (spatial) characteristics in which the OTA requirements apply are detailed for each requirement.

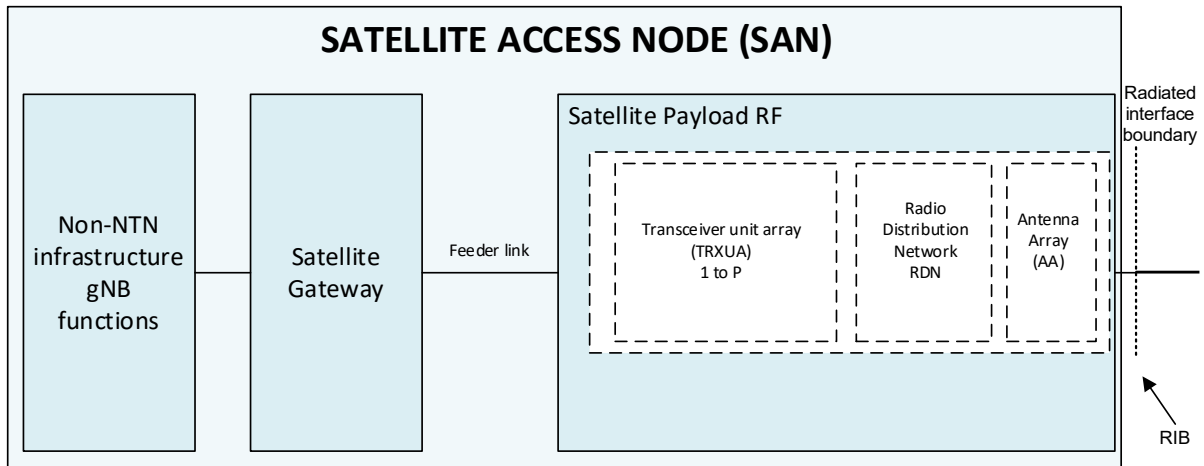


Figure 7.1.4-1: Radiated reference points for SAN type 1-O

The SAN type 1-O payload includes an active antenna with performances characterized in radiated mode (OTA).

7.2 Common issues for satellite access node and NTN UE

7.2.1 General

This section aims to present the agreed definitions for the newly introduced NTN operating bands. The possible configurations in terms of channel bandwidth, SCS, channel and synchronization raster for NTN operations are introduced as well.

7.2.2 Operating bands

The following bands in Table 7.2.2-1 are agreed NTN satellite bands. Regarding the band numbering for NTN satellite bands, it is agreed to start from the largest band number in FR1 range for NTN satellite bands which fully within FR1 frequency ranges to better differentiate NTN satellite band and TN bands and ensure contiguous band number allocation for NTN satellite bands, the number can be taken in a decreased order with first come, first service.

Table 7.2.2-1: NTN satellite bands in FR1

NTN satellite band #	Uplink (UL) operating band Satellite Access Node receive / UE transmit $F_{UL,low} - F_{UL,high}$	Downlink (DL) operating band Satellite Access Node transmit / UE receive $F_{DL,low} - F_{DL,high}$	Duplex mode
n256	1980MHz – 2010 MHz	2170 MHz – 2200 MHz	FDD
n255	1626.5 MHz – 1660.5 MHz	1525 MHz – 1559 MHz	FDD
n253	1668 MHz – 1675 MHz	1518 MHz – 1525 MHz	FDD
n252	2000 MHz – 2020 MHz	2180 MHz – 2200 MHz	FDD
n251	1626.5 MHz – 1660.5 MHz	1518 MHz – 1559 MHz	FDD
n250	1668 MHz – 1675 MHz	1518 MHz – 1559 MHz	FDD

NOTE: NTN bands are numbered in descending order from n256.

7.2.3 Channel bandwidth, SCS and spectral utilization

The common definition for channel bandwidth, transmission bandwidth configuration, minimum guard band, and RB alignment in TS 38.104[16] and TS 38.101-1[17] can be reused for NTN satellite system.

The supported channel bandwidths per operating band should be defined based on NTN operator input. The details are given in Table 7.2.3-1.

Table 7.2.3-1: Channel bandwidths and SCS per NTN satellite operating band in FR1

NTN satellite band #	SCS kHz	3MHz	5 MHz	10 MHz	15 MHz	20 MHz	25 MHz ¹	30 MHz ¹	35 MHz ¹	40 MHz ¹
n256	15	Yes	Yes	Yes	Yes	Yes				
	30			Yes	Yes	Yes				
	60			Yes	Yes	Yes				
n255	15	Yes	Yes	Yes	Yes	Yes				
	30			Yes	Yes	Yes				
	60			Yes	Yes	Yes				
n253	15		Yes							
	30									
	60									
n252	15		Yes	Yes	Yes	Yes				
	30			Yes	Yes	Yes				
	60			Yes	Yes	Yes				
n251	15		Yes	Yes	Yes	Yes	[Yes]	[Yes]	[Yes ²]	[Yes ²]
	30			Yes	Yes	Yes	[Yes]	[Yes]	[Yes ²]	[Yes ²]
	60			Yes	Yes	Yes	[Yes ²]	[Yes]	[Yes ²]	[Yes ²]
n250	15		Yes	Yes ²	Yes ²	Yes ²	[Yes ²]	[Yes ²]	[Yes ²]	[Yes ²]
	30			Yes ²	Yes ²	Yes ²	[Yes ²]	[Yes ²]	[Yes ²]	[Yes ²]
	60			Yes ²	Yes ²	Yes ²	[Yes ²]	[Yes ²]	[Yes ²]	[Yes ²]

NOTE 1: Deployment of 30, 35 and 40 MHz channel bandwidths for NTN needs to be preceded by introduction of all applicable Tx RF, Rx RF, and demodulation requirements.
NOTE 2: This channel bandwidth is applicable only to downlink.

7.2.3.1 Asymmetric channel bandwidth

The UE channel bandwidth can be asymmetric in downlink and uplink. In asymmetric channel bandwidth operation, the narrower carrier shall be confined within the frequency range of the wider channel bandwidth.

In FDD, the confinement is defined as a maximum deviation to the Tx-Rx carrier center frequency separation (defined in table 7.4.1.1-1) as following:

$$\Delta F_{TX-RX} = |(BW_{DL} - BW_{UL})/2|$$

The operating bands and supported asymmetric channel bandwidth combinations are defined in table 7.2.3-1.

Table 7.2.3-1: FDD asymmetric UL and DL channel bandwidth combinations

NR Band	Channel bandwidths for UL (MHz)	Channel bandwidths for DL (MHz)	Asymmetric channel bandwidth combination set
n252	5	10, 15, 20	0
	10	15, 20	
	15	20	
n251	5	10, 15, 20	0
	10	15, 20	0
	15	20	0
n250	5	10, 15, 20	0

NOTE 1: The assignment of the paired UL and DL channels are subject to a TX-RX separation as specified in clause 5.4.4.

NOTE 2: As indicated in TS38.306 [11], it is mandatory for UEs to support asymmetric channel BCS0 if there is an asymmetric BCS0 defined for the band.

7.2.4 Channel raster and sync raster

7.2.4.1 Channel raster

In order to be compatible with current design of sync raster below 3 GHz, the ARFCN parameters below 3000 MHz will be reused and 100 kHz was chosen as the channel raster for NR NTN satellite bands below 3 GHz.

The applicable ARFCNs per operating and for satellite access node are defined in table 7.2.4.1-1.

Table 7.2.4.1-1: ARFCNs per operating band satellite access node

NTN satellite band #	ΔF_{Raster} (kHz)	Uplink range of N_{REF} (First – <Step size> – Last)	Downlink range of N_{REF} (First – <Step size> – Last)
n256	100	396000 – <20> – 402000	434000 – <20> – 440000
n255	100	325300 – <20> – 332100	305000 – <20> – 311800
n253	100	333600 – <20> – 335000	303600 – <20> – 305000
n252	100	400000 – <20> – 404000	436000 – <20> – 440000
n251	100	325300 – <20> – 332100	303600 – <20> – 311800
n250	100	333600 – <20> – 335000	303600 – <20> – 311800

In addition to the 100 kHz channel raster, Enhanced channel raster is defined with $\Delta F_{\text{Raster}} = 2 \times \Delta F_{\text{Global}}$. In this case every 2th NR-ARFCN within the operating band are applicable for the channel raster within the operating band and the step size for the channel raster in Table 7.2.4.1-2 is given as <2>.

Table 7.2.4.1-2: ARFCNs per operating band satellite access node

NTN satellite band #	ΔF_{Raster} (kHz)	Uplink Range of N_{REF} (First – <Step size> – Last)	Downlink Range of N_{REF} (First – <Step size> – Last)
n256	10	396000 – <2> – 402000	434000 – <2> – 440000
n255	10	325300 – <2> – 332100	305000 – <2> – 311800
n253	10	333600 – <2> – 335000	303600 – <2> – 305000
n252	10	400000 – <2> – 404000	436000 – <2> – 440000
n251	10	325300 – <2> – 332100	303600 – <2> – 311800
n250	10	333600 – <2> – 335000	303600 – <2> – 311800

7.2.4.2 Sync raster

The applicable SS raster entries per operating band are defined in Table 7.2.4.2-1 and 7.2.4.2-2.

Table 7.2.4.2-1: Applicable SS raster entries per operating band (for above 3MHz channel)

NTN satellite band #	SS Block SCS	SS Block pattern	Range of GSCN (First – <Step size> – Last)
n256	15 kHz	Case A	5429 – <1> – 5494
n255	15 kHz	Case A	3818 – <1> – 3892
	30 kHz	Case B	3824 – <1> – 3886
n253	15 kHz	Case A	3800 – <1> – 3807
n252	15 kHz	Case A	5456 – <1> – 5494
n251	15 kHz	Case A	3800 – <1> – 3892
	30 kHz	Case B	3806 – <1> – 3886
n250	15 kHz	Case A	3800 – <1> – 3892
	30 kHz	Case B	3806 – <1> – 3886

Table 7.2.4.2-2: Applicable SS raster entries per operating band (for the 3MHz channel)

NTN satellite band #	SS Block SCS	SS Block pattern	Range of GSCN (First – <Step size> – Last)
n256	15 kHz	Case A	37492 – <1> – 37629
n255	15 kHz	Case A	34267 – <1> – 34424

7.3 Satellite Access Node requirements

7.3.1 General

This section aims to provide some recommendations for the Satellite Access Node RF requirements definition in TS 38.181 [25] and TS 38.108 [26]. The technical justifications which support the requirements definition are also presented in the following sub-clauses.

NTN products are assumed to use AAS BS architecture, with both 1-H and 1-O being included in Release 17 WI. Therefore, both conducted, as well as radiated requirements are required to be considered.

7.3.1.1 SAN class

The requirements in TS 38.108 [26] apply to Satellite Access Node otherwise stated. The associated deployment scenarios are exactly the same for SAN with and without connectors.

For SAN type 1-O and SAN type 1-H, two SAN classes (LEO and GEO) are currently defined in Table 7.3.1.1-1.

Table 7.3.1.1-1: SAN classes

SAN Class	Satellite Constellation
GEO class	GEO satellite
LEO class	LEO 600 km satellite LEO 1200 km satellite

The classification is based on the requirements associated with SAN classes in case-by-case manner including [IoT level/ICS, NF, ACLR/ACS, Emission] requirements. The satellite altitude or orbit type were not identified as part of the class differentiation, and for this reason LEO at 600 km and LEO at 1200 km were grouped together in the same class. At least NF, ACLR and ACS values are the same for LEO 600 km and LEO 1200 km and therefore are not requiring class differentiation. On the other hand, as resulted from the coexistence studies, ACLR requirement values are different between GEO and LEO constellations, requiring a class differentiation. Differences exist as well in terms of emission

requirements (e.g. Out-of-band emissions and spurious emissions) between GEO and LEO. At the same time, ACS requirement values from the coexistence studies are the same between GEO and LEO constellations.

7.3.2 Transmission characteristics

7.3.2.1 General

Radiated characteristics are defined over the air (OTA), where the radiated interface is referred to as the *Radiated Interface Boundary* (RIB) as shown in Figure 7.3.2.1-1.

Conducted characteristics are defined at individual or groups of *TAB connectors* at the *transceiver array boundary*, which is the conducted interface between the transceiver unit array and the composite antenna. The TAB connector is shown in Figure 7.3.2.1-1.

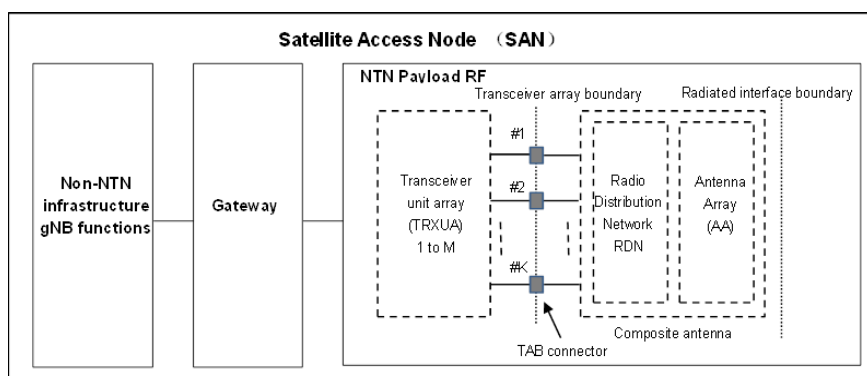


Figure 7.3.2.1-1: Reference point for SAN type 1-H

7.3.2.2 Conducted transmitter requirements

7.3.2.2.1 SAN output power

The SAN conducted output power requirement applies at *TAB connector* for SAN type 1-H.

The *rated carrier output power* of the *SAN type 1-H* shall be as specified in Table 7.3.2.2.1-1.

Table 7.3.2.2.1-1: SAN type 1-H rated output power limits for SAN classes

SAN class	$P_{\text{rated,c,sys}}$	$P_{\text{rated,c,TABC}}$
SAN GEO class	(NOTE)	(NOTE)
SAN LEO class	(NOTE)	(NOTE)
NOTE: $P_{\text{rated,c,sys}}$ or $P_{\text{rated,c,TABC}}$ of SAN shall be based on manufacture declaration and comply with regulation requirement.		

In normal conditions, $P_{\text{max,c,TABC}}$ shall remain within +2 dB and -2 dB of the *rated carrier output power* $P_{\text{rated,c,TABC}}$ for each *TAB connector* as declared by the manufacturer.

In extreme conditions, $P_{\text{max,c,TABC}}$ shall remain within +2.5 dB and -2.5 dB of the *rated carrier output power* $P_{\text{rated,c,TABC}}$ for each *TAB connector* as declared by the manufacturer.

7.3.2.2.2 Output power dynamics

7.3.2.2.2.1 RE power dynamic range

The RE power control dynamic range is the difference between the power of an RE and the average RE power for a SAN at maximum output power for a specified reference condition.

RE power control dynamic range requirement is define in Table 7.3.2.2.2.1-1.

Table 7.3.2.2.2.1-1: RE power control dynamic range

Modulation scheme used on the RE	RE power control dynamic range (dB)	
	(down)	(up)
QPSK (PDCCH)	-6	+4
QPSK (PDSCH)	-6	+3
16QAM (PDSCH)	-3	+3
NOTE: The output power per carrier shall always be less or equal to the maximum output power of the satellite access node.		

7.3.2.2.2.2 Total power dynamic range

Total power dynamic range for SAN will reuse the requirement from TN BS for the same channel bandwidth.

The downlink (DL) total power dynamic range for each carrier shall be larger than or equal to the level in table 7.3.2.2.2.2-1.

Table 7.3.2.2.2.2-1: Total power dynamic range

SAN channel bandwidth (MHz)	Total power dynamic range (dB)		
	15 kHz SCS	30 kHz SCS	60 kHz SCS
5	13.9	10.4	N/A
10	17.1	13.8	10.4
15	18.9	15.7	12.5
20	20.2	17	13.8

7.3.2.2.3 Transmitted signal quality

7.3.2.2.3.1 Frequency error

For SAN type 1-H, the modulated carrier frequency of each carrier configured by the satellite access node shall be accurate to within 0.05ppm observed over 1 ms.

7.3.2.2.3.2 Modulation quality (EVM)

It is agreed to support QPSK and 16QAM for SAN. 64QAM will be optionally supported based on manufacture declaration.

For SAN type 1-H, the EVM levels of each carrier for different modulation schemes on PDSCH outlined in table 7.3.2.2.3.2-1 shall be met.

Table 7.3.2.2.3.2-1: EVM requirements for *satellite access node type 1-H*

Modulation scheme for PDSCH	Required EVM
QPSK	17.5 %
16QAM	12.5 %
64QAM ¹	8 %
NOTE: Support of 64QAM is based on manufacture declaration	

7.3.2.2.4 Unwanted emissions

7.3.2.2.4.1 ACLR

Adjacent Channel Leakage power Ratio (ACLR) is the ratio of the filtered mean power centred on the assigned channel frequency to the filtered mean power centred on an adjacent channel frequency.

From *Section 6.5 Summary of co-existence study* (see TR 38.863 related section), it can be observed in *Table 6.5.1-3 Co-existence results suggested ACLR and ACS of NR-NTN* the resulted study value of NTN SAN ACLR:

Table 7.3.2.2.4.1-1: NTN SAN ACLR

NTN LEO class ACLR	23.81 dB
NTN GEO class ACLR	13.39 dB

Therefore, the value of 24 dB is the ACLR LEO class limit for the conducted transmitter requirements of SAN as shown in Table 7.3.2.2.4.1-2 and the value 14 dB is the ACLR GEO class limit for the conducted transmitter requirements of SAN as shown in Table 7.3.2.2.4.1-3:

Table 7.3.2.2.4.1-2: Satellite Access Node LEO class ACLR limit

SAN channel bandwidth of lowest/highest carrier transmitted BW_{Channel} (MHz)	SAN adjacent channel centre frequency offset below the lowest or above the highest carrier centre frequency transmitted	Assumed adjacent channel carrier (informative)	Filter on the adjacent channel frequency and corresponding filter bandwidth	ACLR limit
5, 10, 15, 20	BW_{Channel}	NR of same BW^2	Square $(BW_{\text{Config}})^1$	24 dB
	$2 \times BW_{\text{Channel}}$	NR of same BW^2	Square $(BW_{\text{Config}})^1$	24 dB
NOTE 1: BW_{Channel} and BW_{Config} are the SAN channel bandwidth and transmission bandwidth configuration of the lowest/highest carrier transmitted on the assigned channel frequency.				
NOTE 2: With SCS that provides largest transmission bandwidth configuration (BW_{Config}).				

Table 7.3.2.2.4.1-3: Satellite Access Node GEO class ACLR limit

SAN channel bandwidth of lowest/highest carrier transmitted BW_{Channel} (MHz)	SAN adjacent channel centre frequency offset below the lowest or above the highest carrier centre frequency transmitted	Assumed adjacent channel carrier (informative)	Filter on the adjacent channel frequency and corresponding filter bandwidth	ACLR limit
5, 10, 15, 20	BW_{Channel}	NR of same BW^2	Square $(BW_{\text{Config}})^1$	14 dB
	$2 \times BW_{\text{Channel}}$	NR of same BW^2	Square $(BW_{\text{Config}})^1$	14 dB
NOTE 1: BW_{Channel} and BW_{Config} are the SAN channel bandwidth and transmission bandwidth configuration of the lowest/highest carrier transmitted on the assigned channel frequency.				
NOTE 2: With SCS that provides largest transmission bandwidth configuration (BW_{Config}).				

7.3.2.2.4.2 Out-of-band emissions

The SAN out-of-band emissions requirements for GEO and LEO classes are therefore defined as described in Table 7.3.2.2.4.2-1.

Table 7.3.2.2.4.2-1: **SAN** out-of-band emissions basic limits

Frequency offset of measurement filter -3dB point, Δf	Frequency offset of measurement filter centre frequency, f_{offset}	Basic limits dBm	Measurement bandwidth
$0 \text{ MHz} \leq \Delta f < 2 \times \text{BW}_{\text{Channel}}$	$0.002 \text{ MHz} \leq f_{\text{offset}} < 2 \times \text{BW}_{\text{Channel}} + 0.002 \text{ MHz}$	$\max \left(SE \text{ limit}, PSD_{\text{channel}} - \Delta_{\text{Sat_Type}} [\text{dB}] \right. \\ \left. - 40 \times \log_{10} \left(\frac{f_{\text{offset}} - 0.002}{\text{BW}_{\text{Channel}}} * 2 + 1 \right) \right) \text{ dBm}$	4kHz
NOTE 1: $PSD_{\text{channel}} = P_{\text{rated,c,sys}} - 10 \log_{10}(\text{BW}_{\text{Channel}}) - 24$, unit dBm/4kHz. NOTE 2: SE limit is spurious emission limit specified in spurious emission clause. NOTE 3: PSD attenuation as in Recommendation ITU-R SM.1541-6[29], Annex 5 OoB domain emission limits for space services. NOTE 4: $\Delta_{\text{Sat_Type}} = 0 \text{ dB}$ for GEO class and $\Delta_{\text{Sat_Type}} = 3 \text{ dB}$ for LEO class.			

Recommendation ITU-R SM.1541-6 [29] specifies the OoB mask requirements. The OoB domain emission as attached Annex 5 should be attenuated below the maximum psd by the following and the attenuation can't be greater than the spurious emission limit.

The OoB domain emissions of a station operating in the bands allocated to the FSS and MSS should be attenuated below the maximum psd, in a reference bandwidth of 4 kHz (for systems operating above 15 GHz a reference bandwidth of 1 MHz may be used in place of 4 kHz) within the necessary bandwidth, by the following:

$$40 \log \left(\frac{F}{50} + 1 \right) \quad \text{dBsd}$$

where F is the frequency offset from the edge of the total assigned band, expressed as a percentage of necessary bandwidth. It is noted that the OoB emission domain starts at the edges of the total assigned band.

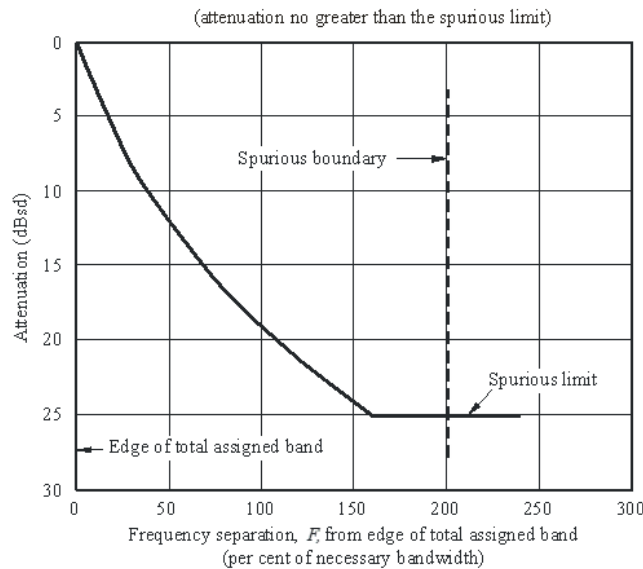


Figure 7.3.2.2.4.2-1: Example of OoB mask assuming the spurious limit is equivalent to 25 dBsd

The “F” = Frequency offset / channel bandwidth $\times 100$ can be get as above. Thus, the OoB mask requirements can be specified as below based on the ITU regulation when we use f_{offset} and $\text{BW}_{\text{channel}}$ to replace “F”.

7.3.2.2.5 Transmitter spurious emission

Transmitter basic limits for spurious emissions are presented in Table 7.3.2.2.5-1.

Table 7.3.2.2.5-1: SAN transmitter spurious emission limits

Spurious frequency range	Basic limit GEO class (dBm)	Basic limit LEO class (dBm)	Measurement bandwidth	Notes
30 MHz – 5 th harmonic (NOTE 2, 4, 5)	When $P_{\text{rated,c,sys}} \leq 47\text{dBm}$: -13 When $P_{\text{rated,c,sys}} > 47\text{dBm}$: $P_{\text{rated,c,sys}} - 60$ (NOTE 3)		4 kHz (NOTE 1)	See NOTES 1, 2, 3, 4, 5
NOTE 1: Measurement bandwidths as in Recommendation ITU-R SM.329 [30], s4.1. NOTE 2: Lower and Upper frequency as in Recommendation ITU-R SM.329 [30], s2.5, Table 1. Applies only for the fundamental frequency range 600 MHz - 5.2 GHz (e.g. for n255 and n256 frequency bands). NOTE 3: Values as in Recommendation ITU-R SM.329 [30], Table 10. NOTE 4: Lower frequency limit is replaced by 0.7 times the waveguide cut-off frequency, according to ITU-R SM.329-12[30], for systems having an integral antenna incorporating a waveguide section, or with an antenna connection in such form, and of unperturbed length equal to at least twice the cut-off NOTE 5: This spurious frequency range applies only to SAN type 1-H.				

Recommendation ITU-R SM.329 [30] defines Unwanted emissions in the spurious domain. Measurement bandwidth is listed in section 4.1, lower and upper frequencies are specified in the Table 1 of section 2.5 and the values are taken from the table 10.

7.3.2.3 Radiated transmitter requirements

7.3.2.3.1 Reserved

7.3.2.3.2 Reserved

7.3.2.3.3 Reserved

7.3.2.3.4 Reserved

7.3.2.3.5 Reserved

7.3.2.3.6 Reserved

7.3.2.3.7 Reserved

7.3.2.3.7.1 Reserved

7.3.2.3.7.2 Reserved

7.3.2.3.7.3 OTA Adjacent Channel Leakage Power Ratio (ACLR)

7.3.2.3.7.3.1 General

OTA Adjacent Channel Leakage power Ratio (ACLR) is the ratio of the filtered mean power centred on the assigned channel frequency to the filtered mean power centred on an adjacent channel frequency. The measured power is TRP.

7.3.2.3.7.3.2 Minimum requirement for SAN type 1-O

The OTA ACLR limit requirement for SAN type 1-O is the same as the ACLR limit defined in *Section 7.3.2.2.4.1 ACLR* of TR 38.863. For the OTA SAN ACLR limit refer to Table 7.3.2.2.4.1-2: Satellite Access Node ACLR limit.

The OTA SAN ACLR absolute basic limit from Table 7.3.2.2.4.1-3: SAN ACLR absolute basic limit does not apply to SAN type 1-O.

7.3.2.3.7.3.3 Minimum requirement for SAN type 2-O

This requirement is not applicable in Release 17.

7.3.3 Receiver characteristics

7.3.3.1 General

For Rx requirements, the TAB connector is located in the satellite payload, while the throughput measurement is done in the “non-NTN infrastructure gNB”, as shown in the Figure 7.3.3.1-1 for reference sensitivity requirement.[18][19]

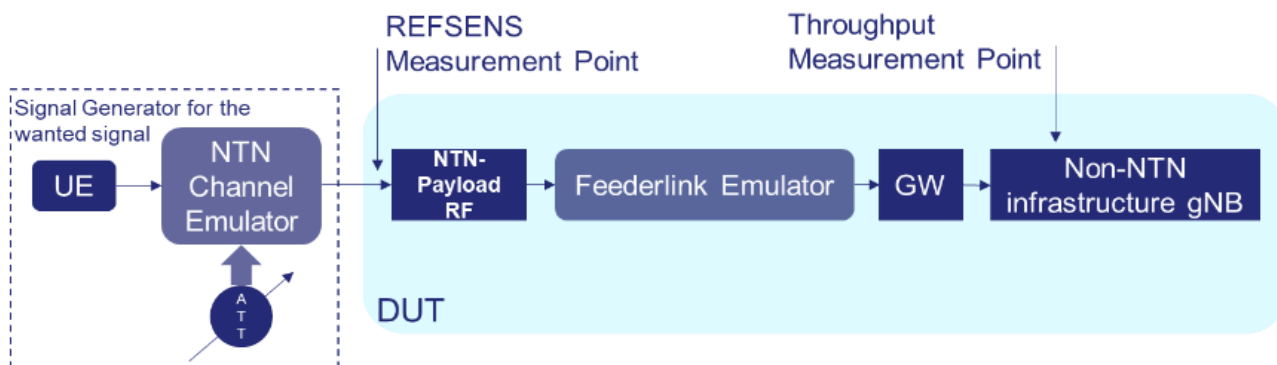


Figure 7.3.3.1-1: Reference points for measurement of Rx requirements

The FRCs specified in TS 38.104[16] shall be re-used to specify the satellite access node Rx requirements. Further revisit is not precluded if the spectrum utilization (SU) would be different.

For derivation of the Rx requirements, the following assumptions were made:

- The FRCs specified in TS 38.104 [16] for NR BS were re-used to specify the NTN SAN Rx requirements.
- The Implementation Margin (IM) value was assumed to be 2 dB for NTN SAN.
- The following Noise Figure (NF) values were used for the derivation of Rx requirements for NTN SAN:

Table 7.3.3.1-1: NF values for NTN SAN requirements derivation

	GEO	LEO 600	LEO 1200
G/T (dB K ⁻¹)	19	1.1	1.1
G _{Rx} (dBi)	51	30	30
NF (dB)	7.4	4.3	4.3

7.3.3.2 Conducted receiver characteristics

7.3.3.2.1 Reference sensitivity level

The following equation was used to derive the reference sensitivity for NTN SAN requirement development, reusing the same methodology for the reference sensitivity derivation as done for the NR BS, as captured in TR 38.817-02 [31]:

$$P_{\text{REFSENS}} = -174 \text{ dBm/Hz} + 10 * \log_{10} \text{ BW} + \text{NF} + \text{IM} + \text{SNR} [\text{dBm}]$$

Where:

- BW is the maximum transmission bandwidth,
- NF is noise figure of NTN SAN receiver in dB,
- IM is implementation margin reserved for NTN SAN manufacturer in dB;
- SNR is the value to satisfy the 95% throughput of the FRC channel.

7.3.3.2.2 Dynamic range

The Rx dynamic range is specified as a measure of the capability of the receiver to receive a wanted signal in the presence of an interfering signal at the *TAB connector* for *SAN type 1-H* inside the received SAN channel bandwidth. In this condition, a throughput requirement shall be met for a specified reference measurement channel. As the Rx dynamic range requirement measures the effects of NTN SAN receiver's impairments, the uncertainty due to the receiver's own thermal noise floor should be minimized. Therefore, the interference level should be increased by a certain amount of margin to mask the receiver's own noise floor, represented by Interference over Thermal (IoT).

In order to define the dynamic range requirement, both wanted and interfering signal power levels need to be defined. The mean power levels of interfering and the wanted signals were defined in the following method:

$$P_{\text{Intf}} = -174 \text{ dBm/Hz} + 10 * \log_{10}(N_{\text{RB}} * \text{SCS} * 12) + \text{NF} + \text{IoT}$$

$$P_{\text{wanted}} = -174 \text{ dBm/Hz} + 10 * \log_{10}(N_{\text{RB}} * \text{SCS} * 12) + \text{NF} + \text{IoT} + \text{SNR} + \text{IM}$$

Where:

- N_{RB} is the transmission bandwidth configuration of the lowest SCS defined for SAN for each *BS channel bandwidth*;
- SCS is the lowest sub-carrier spacing defined for SAN for each *BS channel bandwidth*;
- NF is noise figure of SAN receiver in dB as defined in table 6.2.2.1-5,
- IoT: during the discussion various values of the IoT were considered for the definition of the dynamic range requirement. In case of LEO600, IoT values in range of 15-18 dB were collected based on the simulation results. In case of LEO1200, IoT values in range of 10-15 dB were collected based on the simulation results, as captured in annex D.
- SNR is the value to satisfy the 95% throughput of measurement channel. As the derivation methodology and the FRC's from the NR BS was reused, more details on the SNR simulation results can be found in TR 38.817-02 [31];
- IM is implementation margin reserved for SAN manufacturer in dB. The IM values was reused from the NR BS as 2 dB;

For SAN it was agreed to define a single dynamic range requirement for LEO SAN (both 600 and 1200) based on the IoT value of 15 dB. Based on the collected simulation results for NTN scenarios, it was decided not to define dynamic range requirement for SAN GEO class.

The interfering signal for the dynamic range requirement was defined as AWGN signal, same as for NR BS requirements.

7.3.3.2.3 In-band selectivity and blocking

7.3.3.2.3.1 Adjacent Channel Selectivity (ACS)

7.3.3.2.3.1.1 General

Adjacent channel selectivity (ACS) is a measure of the receiver's ability to receive a wanted signal at its assigned channel frequency at the *antenna connector* for *NTN Satellite Access Node (SAN)* in the presence of an adjacent channel signal with a specified centre frequency offset of the interfering signal to the band edge of a victim system.

The ACS value is coming from coexistence analysis. A worst case value of 38 dB ACS is found in GEO case and the interfering signal mean power (dBm) is computed given a NF 7.4 dB for a 5 MHz band.

It is worth noting that OFDM is considered as interfering waveform (rather than DFT-s-OFDM) corresponding to a coexistence scenario 6 for the MSS S-band where TN BS DL is interfering with SAN UL.

7.3.3.2.3.1.2 Minimum requirements for Satellite Access Node

The throughput shall be $\geq 95\%$ of the maximum throughput of the reference measurement channel.

For SAN, the wanted and the interfering signal coupled to the *antenna connector* are specified in table 7.3.3.2.3.1.2-1 and table 7.3.3.2.3.1.2-2 and the frequency offset between the wanted and interfering signal in table 7.3.3.2.3.1.2-3 for ACS.

The ACS requirement is applicable outside the *SAN RF Bandwidth* or *Radio Bandwidth*. The interfering signal offset is defined relative to the *SAN RF Bandwidth* edges or *Radio Bandwidth* edges.

Minimum conducted requirement is defined at the *antenna connector*.

Table 7.3.3.2.3.1.2-1: SAN GEO class ACS requirement

<i>SAN channel bandwidth of the lowest/highest carrier received (MHz)</i>	<i>Wanted signal mean power (dBm)</i>	<i>Interfering signal mean power (dBm)</i>
5, 10, 15, 20 (NOTE 1)	$P_{\text{REFSENS}} + 6 \text{ dB}$	-57
NOTE 1: The SCS for the lowest/highest carrier received is the lowest SCS supported by the SAN for that bandwidth.		
NOTE 2: For SAN, P_{REFSENS} depends on the <i>SAN channel bandwidth</i> .		

Table 7.3.3.2.3.1.2-2: SAN LEO class ACS requirement

<i>SAN channel bandwidth of the lowest/highest carrier received (MHz)</i>	<i>Wanted signal mean power (dBm)</i>	<i>Interfering signal mean power (dBm)</i>
5, 10, 15, 20 (NOTE 1)	$P_{\text{REFSENS}} + 6 \text{ dB}$	-60
NOTE 1: The SCS for the lowest/highest carrier received is the lowest SCS supported by the SAN for that bandwidth.		
NOTE 2: For SAN, P_{REFSENS} depends on the <i>SAN channel bandwidth</i> .		

The wanted signal and interfering signal power level are calculated in the following way in FR1:

$$\text{Wanted signal power level} = \text{REFSENS} + 6 \text{ dB}$$

$$\text{Interfering signal power level} = \text{SAN noise floor} + \text{ACS} + 4.7 \text{ dB} = -174 \text{ dBm/Hz} + 10 \cdot \log_{10}(\text{BW}) + \text{NF} + \text{ACS} + 4.7 \text{ dB}$$

Where:

- BW is wanted signal bandwidth in Hz, e.g. 25 PRB for 5 MHz SCS 15 kHz;
- NF is noise figure which is agreed as 7.4 dB for GEO, 4.3 dB for LEO@600, 4.3 dB for LEO@1200;
- SAN ACS is agreed as 38 dBc (more precisely according to coexistence Scenario 6, it has been noticed from the coexistence simulations that the required SAN ACS is in the order of 38 dBc ACS);
- 4.7 dB is calculated from $10 \log_{10}(10^{(6/10)} - 1)$.

If the equation considers the worst case of 38 dB ACS, with NF of 7.4 dB for GEO class and NF of 4.3 dB for LEO class, for 5 MHz signal bandwidth (as for TS 38.104[16]), the interfering signal mean power level is:

$-174 + 10 \cdot \log_{10}(5 \cdot 10^6) + 7.4 + 38 + 4.7 = -57 \text{ dBm}$ for GEO class, according to the interfering signal mean power value from Table 7.4.1.2-1.

$-174 + 10 \cdot \log_{10}(5 \cdot 10^6) + 4.3 + 38 + 4.7 = -60 \text{ dBm}$ for LEO class, according to the interfering signal mean power value from Table 7.4.1.2-2.

Table 7.3.3.2.3.1.2-3: SAN ACS interferer frequency offset values

SAN channel bandwidth of the lowest/highest carrier received (MHz)	Interfering signal centre frequency offset from the lower/upper SAN RF Bandwidth edge or sub-block edge inside a sub-block gap (MHz)	Type of interfering signal
5	±2.5025	5 MHz OFDM NR signal 15 kHz SCS, 25 RBs
10	±2.5075	
15	±2.5125	
20	±2.5025	

7.3.3.2.3.2 In-band blocking

No In-band blocker is considered for SAN.

7.3.3.2.4 Out-of-band blocking

7.3.3.2.4.1 General

The out-of-band blocking characteristics is a measure of the receiver ability to receive a wanted signal at its assigned channel at the *TAB connector* for *SAN type 1-H* in the presence of an unwanted interferer out of the *operating band*, which is a CW signal for out-of-band blocking.

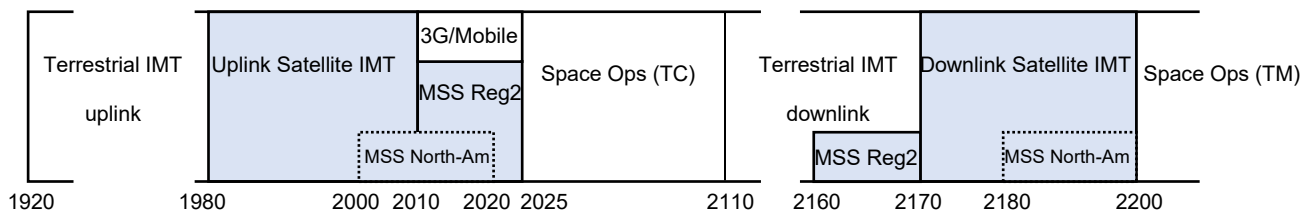


Figure 7.3.3.2.4.1-1: Band of concern to determine OOB blocking

The identified worst case is for the band n256 with the SpaceOps TC (TeleCommand) band between 2025 and 2110 MHz. This band is used to send commands to the satellite platform, taking into account the fact that a very robust radio link is established. To do so, the maximum EIRP of SpaceOps (Space Operations) TC (TeleCommand) stations is 80 dBW.

For a LEO satellite at 600 km (LEO@600km), in a worst case scenario (i.e. SpaceOps TC station at satellite Nadir), the free space losses are 154 dB.

Therefore, the signal level at satellite level is -44 dBm.

7.3.3.2.4.2 Minimum requirements for Satellite Access Node

The throughput shall be $\geq 95\%$ of the maximum throughput of the reference measurement channel, with a wanted and an interfering signal coupled to *SAN type 1-H TAB connector* using the parameters in table 7.3.3.2.4.2-1.

The out-of-band blocking requirement apply from 1 MHz to $F_{UL,low} - \Delta f_{OOB}$ and from $F_{UL,high} + \Delta f_{OOB}$ up to 12750 MHz, including the downlink frequency range of the FDD *operating band*.

Minimum conducted requirement is defined at the *antenna connector* at the *TAB connector* for *SAN type 1-H*.

Table 7.3.3.2.4.2-1: Out-of-band blocking performance requirement for NR

Wanted Signal mean power (dBm)	Interfering Signal mean power (dBm)	Type of Interfering Signal
$P_{\text{REFSENS}} + 6 \text{ dB}$ (NOTE)	-44	CW carrier
NOTE: P_{REFSENS} depends on the RAT. For NR, P_{REFSENS} depends also on the <i>SAN channel bandwidth</i> .		

7.3.3.2.5 Receiver spurious emissions

The requirement is not applicable in this version of the specification.

7.3.3.2.6 Receiver intermodulation

The requirement is not applicable in this version of the specification.

7.3.3.2.7 In-channel selectivity

The requirement framework from terrestrial network (TN) NR specification is agreed to be reused.

7.3.3.3 Radiated receiver characteristics

7.3.3.3.1 OTA sensitivity

On the same principle as for the NR TN BS Specification TS 38.104[16], the SAN OTA sensitivity testing procedure should be based on SAN manufacturer declarations of sensitivity direction and EIS level.

7.3.3.3.2 OTA reference sensitivity level

OTA reference sensitivity level is a directional requirement specified as an EIS level. On the same principle as for the NR TN BS specification TS38.104[16], the SAN OTA reference sensitivity EIS_{REFSENS} requirements should be derived from:

- the conducted reference sensitivity P_{REFSENS} which can be calculated based on the formula presented in Section 7.3.3.2.1;
- a hypothetical antenna gain of a passive satellite antenna;
- some implementation margin reserved for the NTN SAN manufacturer.

The last two contributions are both taken into account under $\Delta_{\text{OTAREFSSENS}}$ parameter.

As OTA reference sensitivity is used as a reference for many of the interference requirements where it is important that the interferer is polarization matched to the receiver antenna it is also important that the wanted signal is polarization matched. Hence OTA reference sensitivity is specifically declared per polarization and the EIS values are under the assumption of *polarization match*.

For SAN type 1-O, OTA reference sensitivity is intended to provide equivalent protection and performance as the conducted reference sensitivity level. The OTA reference sensitivity level is calculated based on the conducted reference sensitivity level for SAN type 1-H and antenna gain of a passive antenna which would be used with a non-AAS SAN in the same deployment scenario.

7.3.3.3.3 OTA dynamic range

The requirement assumes that the wanted signal and interfering signal come from the same direction. And since both wanted signal and AWGN will be transmitted from the test equipment in the same direction, the same combining gain will be achieved for both wanted signal and the AWGN. Hence, the OTA dynamic range for FR1 can be defined based on conductive requirements plus an offset, i.e., $\Delta_{\text{OTAREFSSENS}}$.

7.3.3.3.4 OTA in-band selectivity and blocking

7.3.3.3.4.1 OTA adjacent channel selectivity

OTA Adjacent channel selectivity (ACS) is a measure of the receiver's ability to receive an OTA wanted signal at its assigned channel frequency in the presence of an OTA adjacent channel signal with a specified centre frequency offset of the interfering signal to the band edge of a victim system. For SAN type 1-O, interfering signal mean power should be offset by the value $\Delta_{\min\text{SENS}}$, other parameters could be reused from SAN type 1-H and respective ACS clause.

7.3.3.3.5 OTA out-of-band blocking

7.3.3.3.5.1 General

The OTA out-of-band blocking characteristics are a measure of the receiver unit ability to receive a wanted signal at the RIB at its assigned channel in the presence of an unwanted interferer.

7.3.3.3.5.2 Minimum requirement for SAN type 1-O

7.3.3.3.5.2.1 General minimum requirement

The requirement shall apply at the RIB when the AoA of the incident wave of the received signal and the interfering signal are from the same direction and are within the *minSENS RoAoA*.

The wanted signal applies to each supported polarization, under the assumption of *polarization match*. The interferer shall be *polarization matched* in-band and the polarization maintained for out-of-band frequencies.

The values for the minimum requirements have been obtained with the hypothesis from section 7.3.3.2.4 Out-of-band blocking, considering the maximum EIRP of SpaceOps (Space Operations) TC (TeleCommand) stations of 80 dBW, and the interfering signal mean power of -44 dBm, for a LEO satellite at 600 km of altitude (LEO@600km).

For OTA wanted and OTA interfering signals provided at the RIB using the parameters in table 7.3.5.6.2.1-1, the following requirements shall be met:

- The throughput shall be $\geq 95\%$ of the maximum throughput of the reference measurement channel.

For *SAN type 1-O* the OTA out-of-band blocking requirement apply from 30 MHz to $F_{\text{UL},\text{low}} - \Delta f_{\text{OOB}}$ and from $F_{\text{UL},\text{high}} + \Delta f_{\text{OOB}}$ up to 12750 MHz, including the downlink frequency range of the FDD operating band.

Table 7.3.3.3.5.2.1-1: OTA out-of-band blocking performance requirement

Wanted signal mean power (dBm)	Interfering signal RMS field-strength (V/m)	Type of interfering Signal
$E_{\text{IS}_{\min\text{SENS}}} + 6 \text{ dB}$ (NOTE 1)	0.091	CW carrier
NOTE 1: $E_{\text{IS}_{\min\text{SENS}}}$ depends on the <i>channel bandwidth</i> . NOTE 2: The RMS field-strength level in V/m is related to the interferer EIRP level at a distance described as $E = \frac{\sqrt{30\text{EIRP}}}{r}$, where EIRP is in W and r is in m.		

7.3.3.3.5.2.2 Co-location minimum requirement

This requirement is not applicable in Release 17.

7.3.3.3.6 OTA receiver spurious emissions

The requirement is not applicable in this version of the specification.

7.3.3.3.7 OTA receiver intermodulation

This requirement is not applicable in Release 17.

7.3.3.3.8 OTA in-channel selectivity

For SAN type 1-O, power level of wanted signal and interfering signal should be offset by the value Δ_{minSENS} . Other parameters could be reused from SAN type 1-H and respective in-channel selectivity (ICS) clause.

7.3.4 Others

7.3.4.1 SAN RF requirements for 3MHz channel bandwidth

7.3.4.1.1 Conducted Tx/Rx requirements

The agreed changes to SAN RF conducted Tx/Rx requirements for 3MHz channel bandwidth are provided in Table 7.3.4.1.1-1.

Table 7.3.4.1.1-1: Conducted Tx/Rx requirements for 3MHz channel bandwidth

NR SAN RF conducted Tx/Rx requirement in TS 38.108 [26]	Proposal for 3MHz channel bandwidth in FR1-NTN bands
6.2 Satellite Access Node output power	No specification impact.
6.3.2 RE power control dynamic range	No specification impact.
6.3.3 Total power dynamic range	Minimum requirement calculated according to transmission bandwidth configuration in Table 6.3.3.2-1.
6.4 Transmit ON/OFF power	No specification impact.
6.5 Transmitted signal quality	No specification impact.
6.6.2 Occupied bandwidth	No specification impact.
6.6.3 Adjacent Channel Leakage Power Ratio	Requirement limits as other channel bandwidth, but assume different adjacent channel carriers in Table 6.6.3.2-1 and Table 6.6.3.2-2.
6.6.4 Out-of-band emissions	No specification impact.
6.6.5 Transmitter spurious emissions	No specification impact
6.7 Transmitter intermodulation	No specification impact.
7.2 Reference sensitivity level	Add minimum requirements for 3 MHz in Table 7.2.2-1 and Table 7.2.2-2. Scaling from 5 MHz can be used. Use FRC G-FR1-A1-7, same as in TN.
7.3 Dynamic range	Add minimum requirements for 3 MHz in Table 7.3.2-1. Scaling from 5 MHz can be used. Reuse TN FRC G-FR1-A2-15 (to be introduced to A.2).

7.4.1 Adjacent Channel Selectivity (ACS)	Define ACS requirement similar to TN for 3MHz in Table 7.4.1.2-1, and assume same interferer frequency offset values as TN 3 MHz in Table 7.4.1.2-2.
7.4.2 In-band blocking	No specification impact.
7.5 Out-of-band blocking	No specification impact.
7.6 Receiver spurious emissions	No specification impact.
7.7 Receiver intermodulation	No specification impact.
7.8 In-channel selectivity	Add minimum requirements for 3 MHz in Table 7.8.2-1 and Table 7.8.2-2. Reuse TN FRC G-FR1-A1-20 (to be introduced to A.1).
A.1 Fixed Reference Channels for RF Rx requirements (QPSK, R=1/3)	FRC for 3MHz in-channel selectivity need to be added in Table A.1-1. Reuse TN FRC G-FR1-A1-20. Rename the FRC to indicate “NTN” can be discussed as Rel-18 maintenance.
A.2 Fixed Reference Channels for dynamic range (16QAM, R=2/3)	FRC for 3MHz needs to be added in Table A.2-1. Reuse TN FRC G-FR1-A2-15. Rename the FRC to indicate “NTN” can be discussed as Rel-18 maintenance.

7.3.4.1.2 Radiated Tx/Rx requirements

The agreed changes to SAN RF radiated Tx/Rx requirements for 3MHz channel bandwidth are provided in Table 7.3.4.x.2-1.

Table 7.3.4.1.2-1: Radiated Tx/Rx requirements for 3MHz channel bandwidth

NR SAN RF conformance requirement in TS 38.181 [25]	Expected updates for 3MHz channel bandwidth in FR1-NTN bands
4.1.2 Acceptable uncertainty of Test System	Add maximum test system uncertainty for 3 MHz in Table 4.1.2.2-1 and Table 4.1.2.2-2.
4.7.2 Test signal used to build Test Configurations	Add a note in Table 4.7.2-1 to use 3 MHz channel bandwidth if supported.
6.2 Satellite Access Node output power	No specification impact.
6.3.2 RE power control dynamic range	No specification impact.
6.3.3 Total power dynamic range	Add minimum requirements for 3 MHz in Table 6.3.3.5-1.
6.4 Transmit ON/OFF power	No specification impact.
6.5 Transmitted signal quality	Add minimum requirements for 3 MHz in Table 6.5.3.5-2.
6.6.2 Occupied bandwidth	Add minimum requirements for 3 MHz in Table 6.6.2.4.2-1.

6.6.3 Adjacent Channel Leakage Power Ratio	Requirement limits as other channel bandwidth, but assume different adjacent channel carriers in Table 6.6.3.5.2-1 and Table 6.6.3.5.2-2.
6.6.4 Out-of-band emissions	No specification impact.
6.6.5 Transmitter spurious emissions	No specification impact
6.7 Transmitter intermodulation	No specification impact.
7.2 Reference sensitivity level	Add minimum requirements for 3 MHz in Table 7.2.5-1 and Table 7.2.5-2. Scaling from 5 MHz can be used. Use FRC G-FR1-A1-7, same as in TN.
7.3 Dynamic range	Add minimum requirements for 3 MHz in Table 7.3.5-1. Scaling from 5 MHz can be used. Reuse TN FRC G-FR1-A2-15 (to be introduced to A.2).
7.4.1 Adjacent Channel Selectivity (ACS)	Define ACS requirement similar to TN for 3MHz in Table 7.4.1.5-1, and assume same interferer frequency offset values as TN 3 MHz in Table 7.4.1.5-2.
7.4.2 In-band blocking	No specification impact.
7.5 Out-of-band blocking	No specification impact.
7.6 Receiver spurious emissions	No specification impact.
7.7 Receiver intermodulation	No specification impact.
7.8 In-channel selectivity	Add minimum requirements for 3 MHz in Table 7.8.5-1 and Table 7.8.5-2. Reuse TN FRC G-FR1-A1-20 (to be introduced to A.1).
9.2 Radiated transmit power	No specification impact.
9.3 OTA SAN Output power dynamics	No specification impact.
9.4 OTA output power dynamic range	Similar update as conductive requirements to add 3MHz channel bandwidth.
9.5 OTA transmit ON/OFF Power	No specification impact.
9.6 OTA transmitted signal quality	Similar update as conductive requirements to add 3MHz channel bandwidth.
9.7 OTA unwanted emissions	Similar update as conductive requirements to add 3MHz channel bandwidth.
9.8 OTA Transmitter intermodulation	No specification impact.
10.2 OTA sensitivity	Similar update as conductive requirements to add 3MHz channel bandwidth.
10.3 OTA reference sensitivity level	Similar update as conductive requirements to add 3MHz channel bandwidth.
10.4 OTA dynamic range	Similar update as conductive requirements to add 3MHz channel bandwidth.
10.5 OTA In-band selectivity and blocking	Similar update as conductive requirements to add 3MHz channel bandwidth.
10.6 OTA Out-of-band blocking	No specification impact.

10.7 OTA receiver spurious emissions	No specification impact.
10.8 OTA receiver intermodulation	No specification impact.
10.9 OTA In-channel selectivity	Similar update as conductive requirements to add 3MHz channel bandwidth.
A.1 Fixed Reference Channels for RF Rx requirements (QPSK, R=1/3)	FRC for 3MHz in-channel selectivity need to be added in Table A.1-1. Reuse TN FRC G-FR1-A1-20. Rename the FRC to indicate “NTN” can be discussed as Rel-18 maintenance.
A.2 Fixed Reference Channels for dynamic range (16QAM, R=2/3)	FRC for 3MHz needs to be added in Table A.2-1. Reuse TN FRC G-FR1-A2-15. Rename the FRC to indicate “NTN” can be discussed as Rel-18 maintenance.

7.4 NTN UE requirements

7.4.1 General

In general, NTN UE products are assumed to meet TN UE requirements as specified in TS 38.101-1[17]. The structure and content of the NTN UE specification TS 38.101-5 should be based on TS 38.101-1[17]. The development of TS 38.101-5 should follow the set of conditions that were agreed at the RAN#94e meeting in RP-213618[27].

7.4.1.1 TX-RX frequency separation

The default TX channel (carrier centre frequency) to RX channel (carrier centre frequency) separation for operating bands is specified in Table 7.4.1.1-1 for FR1-NTN.

Table 7.4.1.1-1: UE TX-RX frequency separation (FR1-NTN)

NTN Satellite Operating Band	TX – RX carrier centre frequency separation
n253	-150 MHz ¹ -148 to -152 MHz ²
n252	180 MHz ¹ 165 – 195 MHz ²
n251	-108.5 MHz ¹ -72.5 to -137.5 MHz ²
n250	-150 MHz ¹ -114 to -152 MHz ²
NOTE 1: Default Tx-Rx separation	
NOTE 2: The verification of flexible Tx-Rx frequency separation within this range is limited to reference sensitivity.	

7.4.2 UE Transmission characteristics for satellite access

7.4.2.1 General

Unless otherwise stated, the transmitter characteristics are specified at the antenna connector of the UE with a single or multiple transmit antenna(s). For UE with integral antenna only, a reference antenna with a gain of 0dBi is assumed.

7.4.2.2 Conducted transmitter characteristics

7.4.2.2.1 Maximum output power

The following UE Power Classes define the maximum output power for any transmission bandwidth within the channel bandwidth of NR carrier unless otherwise stated. The period of measurement shall be at least one sub frame (1ms).

Table 7.4.2.2.1-1: UE Power Class

NTN satellite band #	Class 1 (dBm)	Tolerance (dB)	Class 2 (dBm)	Tolerance (dB)	Class 3 (dBm)	Tolerance (dB)
n256	31	+2/-3	26	+2/-3	23	±2
n255	31	+2/-3	26	+2/-3	23	±2
n253					23	±2
n252					23	+2
n251					23	±2
n250					23	±2

NOTE 1: $P_{PowerClass}$ is the maximum UE power specified without taking into account the tolerance
 NOTE 2: Power class 3 is default power class unless otherwise stated
 NOTE 3: Generally, PC1 UE are not targeted for smartphone form factor in this version of the specification.

7.4.2.2.2 MPR/AMPR

7.4.2.2.2.1 MPR

UE is allowed to reduce the maximum output power (MPR) due to higher order modulations and transmit bandwidth configurations to meet the general transmission requirements which includes in-band emission, ACLR, out-of-band emission and general spurious emission. As it was agreed those general transmission requirements for PC3 NTN UE can be reused from TN, the same requirements specified in Table 6.2.2-1 in TS 38.101-1 [17] can be reused for NTN satellite access UEs.

For PC2 UE with 1Tx and 2Tx, the allowed maximum power reduction (MPR) is defined as Table 6.2.2-2 in 3GPP TS 38.101-1[17] clause 6.2.2 and Table 6.2D.2-1 in 3GPP TS 38.101-1[17] clause 6.2D.2 respectively, except for 256QAM. For UE power class 1 with 1Tx, the allowed maximum power reduction (MPR) is defined in Table 7.4.2.2.2.1-1. And for UE power class 1 with 4Tx, the allowed maximum power reduction (MPR) is defined as Table 7.4.2.2.2.1-1a.

Table 7.4.2.2.2.1-1 Maximum power reduction (MPR) for power class 1 with 1Tx

Modulation		MPR (dB)		
		Edge RB allocations	Outer RB allocations	Inner RB allocations
DFT-s-OFDM	Pi/2 BPSK	≤ 10.0	≤ 0.5	≤ 0
	QPSK	≤ 10.5	≤ 1.0	≤ 0
	16 QAM	≤ 10.5	≤ 2.0	≤ 1.0
	64 QAM	≤ 10.5	≤ 2.5	
CP-OFDM	QPSK	≤ 11.0	≤ 3	≤ 1.5
	16 QAM	≤ 11.0	≤ 3	≤ 2.0
	64 QAM	≤ 11.0	≤ 3.5	

Table 7.4.2.2.2.1-1a Maximum power reduction (MPR) for power class 1 with 4Tx

Modulation		MPR (dB)		
		Edge RB allocations	Outer RB allocations	Inner RB allocations
DFT-s-OFDM	Pi/2 BPSK	≤ 10.0	≤ 2.0	≤ 0.5
	QPSK	≤ 10.5	≤ 2.5	≤ 0.5
	16 QAM	≤ 10.5	≤ 3.5	≤ 1.5
	64 QAM	≤ 10.5	≤ 4.0	≤ 3.5
CP-OFDM	QPSK	≤ 11.0	≤ 4.5	≤ 2.0
	16 QAM	≤ 11.0	≤ 4.5	≤ 2.5
	64 QAM	≤ 11.0	≤ 5.0	≤ 5.0

NOTE 1: This table is targeted to devices with 15 dB or above antenna isolation.

The definitions for Edge, Outer and Inner RB allocations defined in clause 6.2.2 of 3GPP TS 38.101-1 [17] shall apply to NTN UE, except for PC1 NTN UE.

For PC1 NTN UE, RB allocation is an Edge RB allocation if

$$L_{CRB} \leq L_{CRB,edge} \text{ AND } (RB_{start} \leq RB_{start,edge} \text{ OR } RB_{start} \geq N_{RB} - RB_{start,edge} - L_{CRB}),$$

where

$$L_{CRB,edge} = 8 \text{ and } RB_{start,edge} = 0.$$

7.4.2.2.2.2 A-MPR

A-MPR is band specific requirement which is used to meet the additional requirements indicated by network signalling (NS) value.

For n256, based on the regulation conclusion in section 5.2, the A-MPR requirements for n65 could be as starting point. From TS 38.101-1, since the maximum carrier centre frequency of NS_05 and NS_05U is lower than 1960 MHz and the maximum carrier centre frequency of NS_51 is lower than 1980 MHz, it is proposed NS_05, NS_05U and NS_51 don't need to be considered for n256. The requirements of NS_01, NS_24 and NS_100 specified in Table 6.2.3.1-1 in TS 38.101-1 [17] for band n65 can be reused for n256. For the band n256 3MHz channel bandwidth, additional power back-off simulation results are provided in Annex X, based on which the corresponding A-MPR values are adopted.

For PC2 and PC1 NTN UE for band n256, the A-MPR analysis have been performed as documented in Annex F, and the concluded A-MPR requirements can be found in sub-clause 7.4.2.2.2.2.6.

For n255, based on regulatory requirements, a new NS is required. The following observations were made from the measurement results using an LTE modem and transceiver supporting 1626.5 – 1656.5 MHz:

- For 5 MHz channel, PA emissions for DFT-s-OFDM are below the spurious emission limits between 1559 and 1606 MHz.
- For 10 MHz channel, PA emissions for DFT-s-OFDM are exceeding the spurious emission limits for 1 MHz MBW by 5 dB between 1604 and 1606 MHz.
- For 15 MHz channel, PA emissions for DFT-s-OFDM are exceeding the spurious emission limits for 1 MHz MBW by 9 dB between 1599 and 1606 MHz.
- For 20 MHz channel, PA emissions for DFT-s-OFDM are exceeding the spurious emission limits by 9 dB between 1584 and 1606 MHz.
- For 700 Hz MBW, PA emissions for all channel bandwidths are below the spurious emission limits between 1559 and 1606 MHz.

A n255 duplexer with rejection of more than 10 dB will be a sufficient to satisfy the spurious emission limits specified in Table 7.4.2.2.8-1.

Based on this measurement observations, the expected filter attenuation of more than 10 dB below 1610 MHz is going to be adequate to meet the additional emissions requirements between 1559 – 1610 MHz, and therefore, no A-MPR needs to be specified for n255.

In view of the above, the additional requirements with their associated network signalling values and the allowed A-MPR and applicable operating band(s) for each NS value shall be specified in Table 7.4.2.2.2.2-1 and the mapping of NR satellite band numbers and values of the *additionalSpectrumEmission* to network signalling labels is specified in Table 7.4.2.2.2.2-1A.

For PC2 and PC1 NTN UE for band n255, the A-MPR analysis have been performed as documented in Annex G, and the concluded A-MPR requirements can be found in sub-clause 7.4.2.2.2.2.5. These PC2 and PC1 A-MPR values for NS_02N are specified with duplexer filter achieving at least 10 dB attenuation at frequencies 1610 MHz and below. Previous evaluation of duplexer data in R4-2016896 during the development of specifications of b24/n24 band showed that it is possible to achieve a higher attenuation than 10 dB which will likely to reduce the A-MPR being needed to meet the additional out of band emissions requirements related to NS_02N.

For n252, the following three NSs are defined:

- NS_06N: for protection of aeronautical radionavigation-satellite service. This requirement is based on FCC rule 47 C.F.R § 25.216 (e)
- NS_07N and NS_08N: for protection of n2/n25. This requirement applies based on coordination between operators and subject to regional/national regulation. See section 7.4.2.2.8 for detail on UE-to-UE coexistence additional spurious requirements. From UE perspective, UE applies this requirement based on the received NS value. For implementation flexibility, the requirement for aeronautical radionavigation-satellite service is also included in these NSs.

NS_09N and NS_10N are defined to capture additional spectrum emission requirements based on ETSI 301 681, while NS_13N and NS_14N are defined to capture additional spectrum emission requirements based on the new initial draft ETSI requirements for “Harmonised Standard for NTN capable UE” under discussion in ETSI TC SES SCN as referenced by the ETSI LS received at RAN4#116 (R4-2509023_SES(25)000071).

Table 7.4.2.2.2-1: Additional maximum power reduction (A-MPR)

Network signalling label	Requirements (clause)	NR satellite Band	Channel bandwidth (MHz)	Resources blocks (N_{RB})	A-MPR (dB)
NS_01		Table 5.2.1-1	3, 5, 10, 15, 20	Table 5.3.2-1 in TS 38.101-1	N/A
NS_02N	6.5.3.3.1	n255	3, 5, 10, 15, 20	Table 7.4.2.2.2.5-1, Table 7.4.2.2.2.5-3	Table 7.4.2.2.2.5-2, Table 7.4.2.2.2.5-4, Table 7.4.2.2.2.5-4a
NS_24N	6.5.3.3.5 in TS 38.101-5	n256	3, 5, 10, 15, 20	Table 6.2.3.15-1 in TS 38.101-1, Table 7.4.2.2.2.6-1, Table 7.4.2.2.2.6-3	Clause 6.2.3.5 in TS 38.101-5 ² , Table 7.4.2.2.2.6-2, Table 7.4.2.2.2.6-4
NS_100	6.5.2.4.2 in TS 38.101-1	n256 ¹			Table 6.2.3.1-2 in TS 38.101-1
NS_06N	7.4.2.2.8	n252	5, 10, 15, 20		N/A
NS_07N	7.4.2.2.8	n252	5, 10, 15, 20		Table 7.4.2.2.2.1-2
NS_08N	7.4.2.2.8	n252	5, 10, 15, 20		Table 7.4.2.2.2.2-2
NS_09N	6.5.2.3.3 6.5.3.3.8	n250, n253	5		Table 7.4.2.2.2.5-1
NS_10N	6.5.2.3.3 6.5.3.3.9	n251	5, 10, 15, 20		Table 7.4.2.2.2.6-1
NS_13N ⁵	6.5.3.3.10		5		
NS_14N ⁵	6.5.3.3.10		5, 10, 15, 20		Table 7.4.2.2.2.6-3

NOTE 1: This NS can be signalled for NTN satellite bands that have UTRA services deployed.

NOTE 2: A-MPR for the upper 5 MHz is not specified, and therefore shall be used as a guard band.

NOTE 3: The NS_07N and NS_08N is signalled based on coordination between operators and subject to regional/national regulation

NOTE 5: NS_09N (n253, n250), NS_10N (n251) are currently in force. NS_13N (n253, n250), NS_14N (n251) may supersede NS_09N, NS_10N when the new ETSI requirements get ratified.

Table 7.4.2.2.2-1A: Mapping of network signalling label

NTN satellite band	Value of additionalSpectrumEmission							
	0	1	2	3	4	5	6	7
n256	NS_01	NS_24	NS_100					
n255	NS_01	NS_02N						
n253	NS_01	NS_02N	NS_09N					
n252	NS_01	NS_06N	NS_07N	NS_08N				
n251	NS_01	NS_02N	NS_10N					
n250	NS_01	NS_02N	NS_09N					
NOTE: <i>additionalSpectrumEmission</i> corresponds to an information element of the same name defined in clause 6.3.2 of TS 38.331.								

For n256, FFS for how to protect n2, n25, n70 and whether A-MPR required.

7.4.2.2.2.1 AMPR for NS_07N (UE-to-UE coexistence requirement of -40dBm/MHz)

Table 7.4.2.2.2.1-1: A-MPR regions for NS_07N

Channel BW	Carrier Frequency, Fc, MHz	RB _{Start} *12*SCS (MHz)	L _{CRB} *12*SCS (MHz)	A-MPR
5MHz	< 2007.5	≥ 0	≥ 2.16	A1
10MHz	< 2010	≤ 1.44	≥ 0	A2
		> 1.44	≥ 3.6	A1
		> 7.74	≤ 0.72	A1
		≥ 0	≥ 4.5	A1
15MHz	< 2012.5	≤ 3.24	≥ 0	A3
		> 3.24	$\geq 4.5, \leq 7.2$	A1
		> 3.24	> 7.2	A2
		> 10.8	≤ 0.72	A1
	2012.5	≤ 1.44	≥ 0	A2
		> 1.44	≥ 6.48	A1
		> 12.6	≤ 0.72	A1
		≤ 4.68	≥ 0	A3
20 MHz	2010	> 4.68	$\geq 4.5, \leq 7.2$	A1
		> 4.68	> 7.2	A2
		> 12.6	≤ 0.72	A1

Table 7.4.2.2.2.1-2: A-MPR for NS_07N

Modulation/Waveform	A1	A2	A3
	Outer/Inner	Outer/Inner	Outer/Inner
DFT-s-OFDM PI/2 BPSK	≤ 4	≤ 7	≤ 11
DFT-s-OFDM QPSK	≤ 4.5	≤ 7.5	≤ 11.5
DFT-s-OFDM 16 QAM	≤ 5	≤ 8	≤ 12
DFT-s-OFDM 64 QAM	≤ 5.5	≤ 8	≤ 12
CP-OFDM QPSK	≤ 7	≤ 9	≤ 13
CP-OFDM 16 QAM	≤ 7	≤ 9	≤ 13
CP-OFDM 64 QAM	≤ 7	≤ 9	≤ 13

7.4.2.2.2.2.2 AMPR for NS_08N (UE-to-UE coexistence requirement of -30dBm/MHz)

Table 7.4.2.2.2.2-1: A-MPR regions for NS_08N

Channel BW	Carrier Frequency, F_c , MHz	$RB_{start} \cdot 12 \cdot SCS$ (MHz)	$LCRB \cdot 12 \cdot SCS$ (MHz)	A-MPR
5 MHz	-	-	-	-
10MHz	< 2010	≤ 1.44	≤ 2.7	A1
		≤ 1.44	$> 2.7, < 4.5$	A2
		≤ 1.44	≥ 4.5	A1
		≥ 1.44	≥ 5.4	A2
15MHz	< 2012.5	≤ 3.24	≤ 2.7	A1
		≤ 3.24	$> 2.7, < 4.5$	A2
		≤ 3.24	≥ 4.5	A1
		≥ 3.24	≥ 5.4	A2
	2012.5	≤ 1.44	≤ 2.7	A1
		≤ 1.44	$> 2.7, < 5.4$	A2
		≤ 1.44	≥ 5.4	A1
		≥ 1.44	≥ 7.2	A2
20 MHz	2010	≤ 4.68	≤ 3.6	A1
		≤ 4.68	$> 3.6, \leq 5.4$	A2
		≤ 4.68	> 5.4	A1
		> 4.68	≥ 7.2	A2

Table 7.4.2.2.2.2-2: A-MPR for NS_08N

Modulation/Waveform	A1	A2
	Outer/Inner	Outer/Inner
DFT-s-OFDM PI/2 BPSK	≤ 4	≤ 2.5
DFT-s-OFDM QPSK	≤ 4.5	≤ 3
DFT-s-OFDM 16 QAM	≤ 5	≤ 3.5
DFT-s-OFDM 64 QAM	≤ 5.5	≤ 4
CP-OFDM QPSK	≤ 7	≤ 4.5
CP-OFDM 16 QAM	≤ 7	≤ 4.5
CP-OFDM 64 QAM	≤ 7	≤ 4.5

7.4.2.2.2.2.3 A-MPR for NS_09N and NS_13N

Table 7.4.2.2.2.3-1: A-MPR regions for NS_09N

Channel Bandwidth (MHz)	Carrier Centre Frequency, F_c (MHz)	Region		
		$RB_{start} \cdot 12 \cdot SCS$	$LCRB \cdot 12 \cdot SCS$	A-MPR
5MHz	$1670.5 \leq F_c \leq 1672.5$	≥ 3.6		A1

Table 7.4.2.2.2.3-2: A-MPR values for NS_09N

	Modulation	A1
DFT-s-OFDM	Pi/2 BPSK	1
	QPSK	1.5
	16QAM	1.5
	64QAM	2
CP-OFDM	QPSK	3.0
	16QAM	3
	64QAM	3

7.4.2.2.2.4

A-MPR for NS_10N and NS_14N

Table 7.4.2.2.2.4-1: A-MPR regions for NS_10N

Channel Bandwidth (MHz)	Carrier Centre Frequency, F_c (MHz)	Region		
		RBstart*12*SCS	LCRB*12*SCS	A-MPR
5MHz	1629 $\leq F_c < 1632.5$	≤ 0.9		A1
	1654 $< F_c \leq 1658$	≥ 3.24		A1
			≥ 3.24	A1
10MHz	1631.5 $\leq F_c < 1636$	≤ 2.7	< 7.2	A1
			≥ 7.2	A2
	1650 $< F_c \leq 1655.5$	≥ 6.84		A3
15MHz	1634 $\leq F_c < 1641$	≤ 4.32		A2
		≤ 6.12	≥ 7.2	A2
	1641 $\leq F_c \leq 1646$	$\leq 1.8, \geq 12.02$	≤ 1.08	A3
	1646 $< F_c \leq 1653$	≥ 10.08		A4
		< 10.8	≥ 3.6	A3
		≤ 3.6	≤ 3.6	A5
20MHz	1636.5 $\leq F_c < 1644$	$> 3.6, \leq 5.4$	≤ 0.72	A1
		≤ 3.6	> 3.6	A4
		≥ 18.0		A1
	1644 $< F_c \leq 1650.5$	< 13.32	≥ 3.6	A3
		≥ 13.32		A4
		≤ 1.08		A1

Table 7.4.2.2.2.4-2: A-MPR values for NS_10N

	Modulation	A1	A2	A3	A4	A5
DFT-s-OFDM	Pi/2 BPSK	1.5	2	2	3	3.5
	QPSK	2	2.5	2.5	3.5	4.5
	16QAM	2	2.5	3	4	5
	64QAM	2.5	3	3	4	5
CP-OFDM	QPSK	3.5	4	4.0	5.5	6
	16QAM	3.5	4	4	5.5	6
	64QAM	3.5	4	4	5.5	6

Table 7.4.2.2.2.4-3: A-MPR regions for NS_14N

Channel Bandwidth (MHz)	Carrier Centre Frequency, F_c (MHz)	Region		
		RBstart*12*SCS	LCRB*12*SCS	A-MPR
5MHz	1629 $\leq F_c \leq 1632.5$	≤ 0.9		A1
10MHz	1631.5 $\leq F_c \leq 1636$	≤ 2.7	< 5.4	A1
			≥ 5.4	A1
15MHz	1634 $\leq F_c < 1641$	≤ 4.32		A2
20MHz	1636.5 $\leq F_c < 1644$	≤ 3.6	≤ 3.6	A5
		$> 3.6, \leq 5.4$	≤ 0.72	A1
		≤ 3.6	> 3.6	A4
		≥ 18		A1

Table 7.4.2.2.2.4-4: A-MPR values for NS_14N

	Modulation	A1	A2	A3	A4	A5
DFT-s-OFDM	Pi/2 BPSK	1.5	2	2	3	3.5
	QPSK	2	2.5	2.5	3.5	4.5
	16QAM	2	2.5	3	4	5
	64QAM	2.5	3	3	4	5
CP-OFDM	QPSK	3.5	4	4.0	5.5	6
	16QAM	3.5	4	4	5.5	6
	64QAM	3.5	4	4	5.5	6

7.4.2.2.2.5 A-MPR for NS_24N

Table 7.4.2.2.2.5-1: A-MPR regions and values for NS_24N

Channel Bandwidth (MHz)	Carrier Centre Frequency, Fc (MHz)	Region		
		RBend*12*SCS	RBend*12*SCS	RBend*12*SCS
3MHz	2000.5 < fc <= 2003.5		> 1.08	A8
			< 1.08	A9
	1997.5 < fc <= 2000.5		>= 1.8	A11
			< 1.8	A10

	Modulation	A8	A9	A10	A11
DFT-s-OFDM	Pi/2 BPSK	6.0	3	2	2.0
	QPSK	7.0	3	2	3.5
	16QAM	8.0	3.5	2.5	3.5
	64QAM	8.5	3.5	2.5	3.5
CP-OFDM	QPSK	9.0	3.5	2.5	5.0
	16QAM	9.0	3.5	2.5	5.0
	64QAM	9.0	3.5	2.5	5.0

7.4.2.2.2.6 A-MPR for NS_24

Table 7.4.2.2.2.6-1: A-MPR regions for NS_24 (Power Class 2)

Channel Bandwidth, MHz	Carrier Centre Frequency, Fc, MHz	Region A			Region B			Region C			Region D		
		RBend* 12*SCS MHz	LCRB*1 2*SCS MHz	A-M P R	RBend* 12*SCS MHz	LCRB*1 2*SCS MHz	A-M P R	RBend* 12*SCS MHz	LCRB*1 2*SCS MHz	A-M P R	RBend* 12*SCS MHz	LCRB*1 2*SCS MHz	A-M P R
5MHz	1987.5 < Fc ≤ 1992.5		>3.24	A7									
5MHz	1992.5 < Fc ≤ 1997.5		>3.24	A4		≤ 3.24	A8						
5MHz	1997.5 < Fc ≤ 2002.5		>1.98	A1	>3.6	>1.08 ≤ 1.98	A2	≤ 3.6	≤ 1.98	A3			
						≤ 1.08	A6						
10MHz	Fc = 1985	>5.4		A4		≥ 4.5	A8						
10MHz	1985 < Fc ≤ 1995		>4.32	A1	≥ 7.2	>1.08 ≤ 4.32	A2	< 7.2	≤ 4.32	A3			
						≤ 1.08	A6						
10MHz	1995 < Fc ≤ 2000	≥ 5.76		A5	< 3.06		A5	≥ 3.06 < 5.76	> 1.44	A6	≥ 3.06 < 5.76	≤ 1.44	A8
15MHz	Fc = 1987.5		>6.84	A1	≥ 10.8	>1.08 ≤ 6.84	A2	< 10.8	≤ 6.84	A3			
						≤ 1.08	A6						
15MHz	1987.5 < Fc ≤ 1997.5	≥ 8.64		A5	< 4.04		A5	≥ 4.04 < 8.64	> 1.44	A6	≥ 4.04 < 8.64	≤ 1.44	A8
20MHz	Fc = 1990	≥ 12.96		A5	< 4.68		A5	≥ 4.68 < 12.96	> 2.16	A6	≥ 4.68 < 12.96	≤ 2.16	A8
20MHz	1990 < Fc ≤ 1995	≥ 11.52		A5	< 5.58		A5	≥ 5.58 < 11.52	> 1.44	A6	≥ 5.58 < 11.52	≤ 1.44	A8

Table 7.4.2.2.2.6-2: A-MPR for NS_24 (Power Class 2)

Modulation/ Waveform		A1	A2	A3	A4	A5	A6	A7	A8
		Outer/Inner	Outer/Inner	Outer/Inner	Outer/Inner	Outer/Inner	Outer/Inner	Outer/Inner	Outer/Inner
DFT-s-OFDM	π/2 BPSK	≤ 14	≤ 8	≤ 6.5	≤ 8.5	≤ 21	≤ 13	≤ 5.0	≤ 3
	QPSK	≤ 14	≤ 8	≤ 6.5	≤ 8.5	≤ 21	≤ 13	≤ 5.0	≤ 3
	16 QAM	≤ 14	≤ 8	≤ 6.5	≤ 8.5	≤ 21	≤ 13	≤ 5.0	≤ 4
	64 QAM	≤ 14	≤ 8	≤ 7	≤ 8.5	≤ 22	≤ 13	≤ 5.0	≤ 4.5
CP-OFDM	QPSK	≤ 15	≤ 9	≤ 7	≤ 8.5	≤ 22	≤ 15	≤ 6.5	≤ 4.5
	16 QAM	≤ 15	≤ 9	≤ 7	≤ 8.5	≤ 22	≤ 15	≤ 6.5	≤ 5
	64 QAM	≤ 15	≤ 9	≤ 7	≤ 8.5	≤ 22	≤ 15	≤ 6.5	≤ 5

Table 7.4.2.2.2.6-3: A-MPR regions for NS_24 (Power Class 1)

Channel Bandwidth, MHz	Carrier Centre Frequency, Fc, MHz	Region A			Region B			Region C			Region D		
		RBend*12*SCS MHz	LCRB*12*SCS MHz	A-M P R	RBend*12*SCS MHz	LCRB*12*SCS MHz	A-M P R	RBend*12*SCS MHz	LCRB*12*SCS MHz	A-M P R	RBend*12*SCS MHz	LCRB*12*SCS MHz	A-M P R
5MHz	1982.5 < Fc ≤ 1992.5		>3.24	A7		≤ 3.24	A9						
5MHz	1992.5 < Fc ≤ 1997.5		>3.24	A4		≤ 3.24	A8						
5MHz	1997.5 < Fc ≤ 2002.5		>1.98	A1	>3.6	>1.08 ≤ 1.98	A2	≤ 3.6	≤ 1.98	A3			
						≤ 1.08	A6						
10MHz	Fc = 1985	>5.4		A4		≤ 5.4 ≥ 4.5	A8		< 4.5	A9			
10MHz	1985 < Fc ≤ 1995		>4.32	A1	≥ 7.2	>1.08 ≤ 4.32	A2	< 7.2	≤ 4.32	A3			
						≤ 1.08	A6						
10MHz	1995 < Fc ≤ 2000	≥ 5.76		A5	< 3.06		A5	≥ 3.06 < 5.76	> 1.44	A6	≥ 3.06 < 5.76	≤ 1.44	A8
15MHz	Fc = 1987.5		>6.84	A1	≥ 10.8	>1.08 ≤ 6.84	A2	< 10.8	≤ 6.84	A3			
						≤ 1.08	A6						
15MHz	1987.5 < Fc ≤ 1997.5	≥ 8.64		A5	< 4.04		A5	≥ 4.04 < 8.64	> 1.44	A6	≥ 4.04 < 8.64	≤ 1.44	A8
20MHz	Fc = 1990	≥ 12.96		A5	< 4.68		A5	≥ 4.68 < 12.96	> 2.16	A6	≥ 4.68 < 12.96	≤ 2.16	A8
20MHz	1990 < Fc ≤ 1995	≥ 11.52		A5	< 5.58		A5	≥ 5.58 < 11.52	> 1.44	A6	≥ 5.58 < 11.52	≤ 1.44	A8

Table 7.4.2.2.2.6-2: A-MPR for NS_24 (Power Class 1)

Modulation/Waveform		A1	A2	A3	A4	A5	A6	A7	A8	A9
		Outer/Inner	Outer/Inner	Outer/Inner	Outer/Inner	Outer/Inner	Outer/Inner	Outer/Inner	Outer/Inner	Outer/Inner
DFT-s-OFDM	Pi/2 BPSK	≤ 19	≤ 13	≤ 10.5	≤ 11.5	≤ 26	≤ 18	≤ 7	≤ 8	≤ 4.5
	QPSK	≤ 19	≤ 13	≤ 10.5	≤ 11.5	≤ 26	≤ 18	≤ 7	≤ 8	≤ 4.5
	16 QAM	≤ 19	≤ 13	≤ 10.5	≤ 11.5	≤ 26	≤ 18	≤ 7	≤ 9	≤ 4.5
	64 QAM	≤ 19	≤ 13	≤ 11	≤ 11.5	≤ 27	≤ 18	≤ 7	≤ 9.5	≤ 5
CP-OFDM	QPSK	≤ 20	≤ 14	≤ 11	≤ 11.5	≤ 27	≤ 20	≤ 8	≤ 9.5	≤ 5.5
	16 QAM	≤ 20	≤ 14	≤ 11	≤ 11.5	≤ 27	≤ 20	≤ 8	≤ 10	≤ 5.5
	64 QAM	≤ 20	≤ 14	≤ 11	≤ 11.5	≤ 27	≤ 20	≤ 8	≤ 10	≤ 5.5

7.4.2.2.2.7 A-MPR for NS_02N

Table 7.4.2.2.2.7-1: A-MPR regions for NS_02N (Power Class 2)

Channel BW	Carrier Frequency, Fc, MHz	RB _{Start} *12*SCS (MHz)	L _{CRB} *12*SCS (MHz)	A-MPR
15MHz	1634 < Fc ≤ 1645		> 10.8	A2
20MHz	1636.5 ≤ Fc ≤ 1650.5	≤ 0.54	< 1.8	A1
			> 10.8	A1

Table 7.4.2.2.2.7-2: A-MPR for NS_02N (Power Class 2)

Modulation/Waveform		A1 (dB)	A2 (dB)
		Outer/Inner	Outer/Inner
DFT-s-OFDM	Pi/2 BPSK	≤ 2.5	≤ 1.0
	QPSK	≤ 3.0	≤ 2.0
	16 QAM	≤ 3.5	≤ 2.5
	64 QAM	≤ 4.0	≤ 3.0
CP-OFDM	QPSK	≤ 4.5	≤ 3.5
	16 QAM	≤ 4.5	≤ 3.5
	64 QAM	≤ 4.5	≤ 4

Table 7.4.2.2.2.7-3: A-MPR regions for NS_02N (Power Class 1 with 1Tx or 4Tx)

Channel BW	Carrier Frequency, Fc, MHz	RB _{Start} *12*SCS (MHz)	L _{CRB} *12*SCS (MHz)	A-MPR
15MHz	1634 < Fc ≤ 1645		> 10.8	A2
20MHz	1636.5 ≤ Fc ≤ 1650.5	≤ 0.54	< 1.8	A1
			> 14.4	A1
			>10.8, ≤ 14.4	A2

Table 7.4.2.2.2.7-4: A-MPR for NS_02N (Power Class 1 with 1Tx)

Modulation/Waveform		A1 (dB)	A2 (dB)
		Outer/Inner	Outer/Inner
DFT-s-OFDM	Pi/2 BPSK	≤ 5.0	≤ 3.0
	QPSK	≤ 5.0	≤ 3.0
	16 QAM	≤ 5.0	≤ 3.5
	64 QAM	≤ 5.5	≤ 4.0
CP-OFDM	QPSK	≤ 6.0	≤ 4.0
	16 QAM	≤ 6.0	≤ 4.0
	64 QAM	≤ 6.0	≤ 4.0

Table 7.4.2.2.2.7-4a: A-MPR for NS_02N (Power Class 1 with 4Tx)

Modulation/Waveform		A1 (dB)	A2 (dB)
		Outer/Inner	Outer/Inner
DFT-s-OFDM	Pi/2 BPSK	≤ 6.0	≤ 4.0
	QPSK	≤ 6.0	≤ 4.0
	16 QAM	≤ 6.0	≤ 4.5
	64 QAM	≤ 6.5	≤ 5.0
CP-OFDM	QPSK	≤ 7.0	≤ 5.0
	16 QAM	≤ 7.0	≤ 5.0
	64 QAM	≤ 7.0	≤ 5.0

7.4.2.2.3 Output power dynamics

For Transmit OFF power and Power control, the framework and requirements for existing TN UE can be reused for satellite UE. The transmit OFF power is -50 dBm for 20MHz.

Since handheld UE always transmits the maximum output power for satellite access scenario based on the NR NTN calibration summary in R4-2115628, FFS whether relaxed value needed or not based on existing TN UE requirements

7.4.2.2.4 Frequency error

The UE basic measurement interval of modulated carrier frequency is 1 UL slot. The mean value of basic measurements of UE modulated carrier frequency shall be accurate to within ± 0.1 PPM observed over a period of 1 ms of cumulated measurement intervals compared to the carrier frequency received from the Satellite Access Node.

NOTE: The requirements is applicable only when PVT ephemeris updated at least once [10] seconds.

7.4.2.2.5 Transmit modulation quality

For the EVM (Error Vector Magnitude), the same requirements specified in TS 38.101-1 [17] sub-clause 6.4.2.1 for QPSK, 16QAM, and 64QAM can be reused for NTN satellite access UEs. NTN satellite access UEs support UL/DL 64QAM for NTN operation as optional with per band.

For the carrier leakage, the same requirements specified in TS 3.101-1 [17] sub-clause 6.4.2.2 can be reused for NTN satellite access UEs.

For the in-band emission, the same requirements specified in TS 3.101-1 [17] sub-clause 6.4.2.3 can be reused for NTN satellite access UEs.

7.4.2.2.6 Spectrum emission mask

For SEM of NTN satellite access UEs, the same requirements of TN UE specified in TS 38.101-1 [17] sub-clause 6.5.2.2 can be reused.

7.4.2.2.7 ACLR

For ACLR, for PC3 and PC2 the same ACLR for TN NR UE which is specified in TS 38.101-1 [17] sub-clause 6.5.2.4.1 can be applied for NTN satellite access UEs. For PC1, a relaxed ACLR requirement of 34dB compared to that

in TN NR has been specified as a trade-off between the required protection of adjacent bands, as determined by coexistence studies, and the challenges of UE implementation.

The requirement of ALCR is specified in Table 7.4.2.2.7-1.

Table 7.4.2.2.7-1: NTN satellite access UE ACLR requirement

	Power class 1	Power class 2	Power class 3
NTN UE ACLR	34 dB	31 dB	30 dB

7.4.2.2.8 Spurious emissions

The spurious emissions limits specified in TS 38.101-1[17] sub-clause 6.5.3 would also be applicable to NTN satellite access UEs referring to the ERC Recommendation 74-01[28] Annex 3 item 3.1.4 for Mobile Earth Stations (MES) (see Note 1bis) transmitting in the Mobile Satellite Service between 1 GHz and 3 GHz.

Considering FCC rule 47 C.F.R § 25.216 (h) [23], TN UE's general spurious emission requirements will be inadequate for L-band NTN satellite UE operation in the US. A new NS value and a subclause with Table 7.4.2.2.8-1 needs to be specified for n255 for NTN satellite operations in the US.

Table 7.4.2.2.8-1: Additional requirements for NS_02N

Frequency band (MHz)	Channel bandwidth / Spectrum emission limit ¹ (dBW)	Measurement bandwidth	NOTE
	3 MHz, 5 MHz, 10 MHz, 15 MHz, 20 MHz		
1559≤ f ≤ 1605	-80	700 Hz	Averaged over any 2 millisecond active transmission interval
1605≤ f ≤ 1610	-80 + 24/5 (f-1605)	700Hz	
1559 ≤ f ≤ 1605	-70	1MHz	Averaged over any 2 millisecond active transmission interval
1605≤ f ≤ 1610	-70 + 24/5 (f-1605)	1MHz	
NOTE 1: The EIRP requirement in regulation is converted to conducted requirement using a 0 dBi antenna.			

For n252, to protect n2 and n25, the following two additional spurious emission requirements are defined, each is associated to an NS value, and a corresponding A-MPR (See section 7.4.2.2.2).

- -40 dBm/MHz (NS_07N)
- -30 dBm/MHz (NS_08N)

The above requirements apply based on coordination between operators and subject to regional/national regulation. From UE perspective, UE applies this requirement based on the received NS value.

For n252, as described in section 5.2.4, FCC rule 47 C.F.R § 25.216 (e) shall be applied to protect aeronautical radionavigation-satellite service. This requirement is included in all three NSs (NS_06N, NS_07N, NS_08N) for n252 for implementation flexibility purpose.

Table 7.4.2.2.8-2: Additional out-of-band requirements for "NS_06N"

Frequency range (MHz)	Channel bandwidth / Spectrum emission limit (dBm)	Measurement bandwidth	NOTE
	5 MHz, 10 MHz, 15 MHz, 20MHz		
$1559 \leq f < 1610$	-50	700 Hz	Averaged over any 2 millisecond active transmission interval
$1559 \leq f < 1610$	-40	1MHz	

Table 7.4.2.2.8-3: Additional out-of-band requirements for "NS_07N"

Frequency range (MHz)	Channel bandwidth / Spectrum emission limit (dBm)	Measurement bandwidth	NOTE
	5 MHz, 10 MHz, 15 MHz, 20MHz		
$1559 \leq f < 1610$	-50	700 Hz	Averaged over any 2 millisecond active transmission interval
$1559 \leq f < 1610$	-40	1MHz	
$1930 \leq f \leq 1995$	-40	1MHz	

Table 7.4.2.2.8-4: Additional out-of-band requirements for "NS_08N"

Frequency range (MHz)	Channel bandwidth / Spectrum emission limit (dBm)	Measurement bandwidth	NOTE
	5 MHz, 10 MHz, 15 MHz, 20MHz		
$1559 \leq f < 1610$	-50	700 Hz	Averaged over any 2 millisecond active transmission interval
$1559 \leq f < 1610$	-40	1MHz	
$1930 \leq f \leq 1995$	-30	1MHz	

7.4.2.2.9 Transmit intermodulation

The transmit intermodulation limits specified in TS 38.101-1[17] sub-clause 6.5.4 (Table 7.4.2.2.9-1) would also be applicable to NTN satellite access UEs.

The requirement of transmit intermodulation is specified in Table 7.4.2.2.9-1.

Table 7.4.2.2.9-1: Transmit Intermodulation

Wanted signal channel bandwidth	BW_{Channel}	
Interference signal frequency offset from channel center	BW_{Channel}	$2 \cdot BW_{\text{Channel}}$
Interference CW signal level	-40 dBc	
Intermodulation product	< -29 dBc	< -35 dBc
Measurement bandwidth	The maximum transmission bandwidth configuration among the different SCS's for the channel BW as defined in Table 6.5.2.4.1-1 from TS 38.101-1[17]	
Measurement offset from channel center	BW_{Channel} and $2 \cdot BW_{\text{Channel}}$	$2 \cdot BW_{\text{Channel}}$ and $4 \cdot BW_{\text{Channel}}$

7.4.2.2.10 UE to UE coexistence issues between NTN and TN

7.4.2.2.10.1 UE-to-UE coexistence for n256

In recent RAN4#102 and RAN4#103 meetings, the spurious emissions requirements (-50dBm/MHz) for protected bands UE co-existence (especially for band n256 UL: 1980~2010MHz DL: 2170~2200MHz) were widely discussed. The adjacent/overlapping TN bands to NTN band n256 are shown in Figure 5.2.1-1.

1) Coexistence protection for band n34 DL

The NS_24 including associated A-MPR value as specified in TS 38.101-1[17] is reused to solve this issue in Release 17. However, other companies pointed out the n256 UL frequency range can't be leveraged due to 5MHz guard band at the upper of n256 UL frequency range to protect band n34. If NS_24 is used, n256 will be reduced due to the 5 MHz guard band that has been considered with NS_24, frequency range 2005-2010 MHz can not be used anymore. Besides, when NS_24 is indicated to protect the DL frequency range of band n34, satellite UE has to reduce the out power in band n256. It will lead to UL throughput loss and even UL radio link failure for some UL RB configurations.

2) Coexistence protection for band n39 DL

-50dBm/MHz co-existence requirement can be met considering 20MHz channel bandwidth for band n256 without A-MPR requirement, even if a band n65 duplexer is reused for band n256.

Based on extrapolating the SEM mask to the ACLR4 region which overlaps n39 for the lowest n256 20MHz channel it is feasible to specify -50dBm/MHz protection level for n39 by n256 even when using n65 duplexer (no filter help assumed). This assumes the maximum n256 bandwidth is 20MHz.

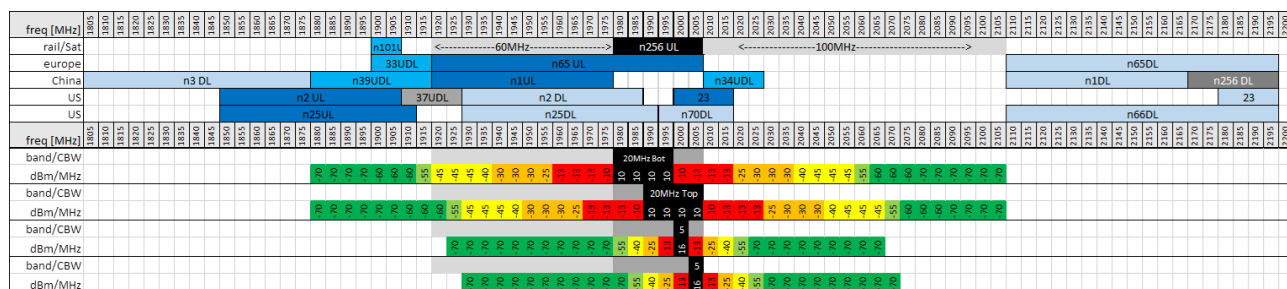


Figure 7.4.2.2.10.1-1: Coexistence analysis based on SEM mask extension for band n256

3) Coexistence protection for band n2, n25, n70 DL

From RAN4 perspective, UE co-existence requirements can't be fulfilled (-50dBm/MHz) for the TN bands (i.e., n2, n25 and n70) which are overlapping n256. In Release 17, it is RAN4 understanding that the co-existence issue between n256 and those TN bands shall rely on regional or national regulation.

Other proposals:

Other proposals were submitted to solve the UE to UE coexistence issues between NTN and TN especially for the protected DL frequency range which is overlapping with UL frequency range of band n256 or next to the edge of band n256 UL filter.

For example:

- Isolation regions could be defined between NTN and TN coverage. And a corresponding signaling could be designed by RAN2.
- Before transmitting UL band n256 signals, satellite UE should determine (or be informed) whether the IMT services exist for the protected DL frequency range in the vicinity.
- Other potential mechanisms/solutions are not precluded.

Considering these proposals will have impacts on RAN1/RAN2 specifications, these proposals were not further discussed in Release 17.

7.4.2.2.10.2 UE-to-UE coexistence for n252

The adjacent/overlapping TN bands to NTN band n252 are shown in Figure 5.2.4-1. Band n252 UL has 5MHz frequency separation from TN band n25, and 10 MHz separation from TN band n2. Furthermore, band n252 UL is overlapped with TN band n70 DL, and band 252 DL is overlapped with TN band n66 DL. The following are RAN4 agreements regarding UE-to-UE coexistence between band n252 and the adjacent/overlapping TN bands.

1. Coexistence protection for band n2, n25 DL

It is very challenging to meet the traditional -50 dBm/MHz UE-to-UE coexistence spurious emission requirement for n2, n25 without significant amount of power reduction (A-MPR) to n252. RAN4 agrees to define a more relaxed UE-to-UE coexistence spurious emission requirement compared to -50 dBm/MHz, that allows the NTN service to work effectively, across the full n252 bands, whilst still provide protection to n2/n25 DL UE from n252 UL UE.

The following two (2) different additional spurious emission requirements, each with the associated NS value and A-MPR, are agreed.

- -30 dBm/MHz
- -40 dBm/MHz

The above requirement applies based on coordination between operators and subject to regional/national regulation

To allow operation flexibility, RAN4 also agreed to define asymmetric channel bandwidth with 5, 10 and 15 MHz UL channel bandwidth for band n252, with BCS0 including all permutation based on UL channel bandwidth. See Table 7.2.3-1 for asymmetric channel bandwidth definition.

2. Coexistence protection for band n70 DL

From RAN4 perspective, there is no 3GPP solution to address coexistence between overlapping bands. Coexistence between n252 and band n70 is subject to regional/national regulation.

3. Coexistence protection for band n66 DL

UE-to-UE coexistence requirements of -50dBm/MHz between n252 UL and TN band n66 DL can be met without additional power reduction (A-MPR).

7.4.2.3 Radiated transmitter characteristics for Ka band VSAT

7.4.2.3.1 VSAT maximum output power

There are many kinds of implementations for VSAT in Ka band. Based on the ITU regulation, VSAT could be fixed or mobile. The antenna beam steering types could be mechanical or electronic. Thus, the following VSAT types are considered.

Table 7.4.2.3.1-1: Assumptions of UE Types

UE class	UE type	Type description
Fixed VSAT	1	Fixed VSAT communicating with GSO and LEO with mechanical steering antenna.
	2	Fixed VSAT communicating with GSO and LEO with electronic steering antenna.
	3	Fixed VSAT communicating with LEO only with electronic steering antenna.
Mobile VSAT	4	Mobile VSAT communicating with GSO with mechanical steering antenna.
	5	Mobile VSAT communicating with GSO with electronic steering antenna.
Note 1: The UE types are assuming UE has only one antenna beam towards one satellite at a given time in this release.		
Note 2: The Mobile VSAT communicating with non-GSO is not considered in this release.		

As the 76.2dBm maximum peak EIRP is assumed for the adjacent coexistence studies, the 76.2dBm maximum peak EIRP was specified in Rel-18.

For type 3 in LEO600 scenario, at least 61dBm EIRP is assumed with the following UL budget.

Table 7.4.2.3.1-2: The calculation of UL budget for LEO600 scenario

	LEO600
NTN VSAT EIRP (dBm)	61
Elevation angle (degree)	45
Slant range (km)	814.83
center frequency (GHz)	27
freePathloss	179.29
SAN UL Received SINR (dB)	> 0.7

The 61dBm minimum peak EIRP can guarantee the uplink budget without failure.

For other types in GSO scenario, at least 70dBm EIRP is assumed with the following UL budget.

Table 7.4.2.3.1-3: The calculation of UL budget for GSO scenario

	GSO
NTN VSAT EIRP (dBm)	70
Elevation angle (degree)	30
Slant range (km)	38611
center frequency (GHz)	27
freePathloss	212.80
SAN UL Received SINR (dB)	> -1

The 70dBm minimum peak EIRP can guarantee the uplink budget without failure.

7.4.3 UE Receiver characteristics for satellite access

7.4.3.1 General

Unless otherwise stated the receiver characteristics are specified at the antenna connector(s) of the UE. For UE(s) with an integral antenna only, a reference antenna(s) with a gain of 0 dBi is assumed for each antenna port(s). UE with an integral antenna(s) may be taken into account by converting these power levels into field strength requirements, assuming a 0 dBi gain antenna. For UEs with more than one receiver antenna connector, identical interfering signals shall be applied to each receiver antenna port if more than one of these is used (diversity).

7.4.3.2 Conducted receiver characteristics

7.4.3.2.1 General and diversity characteristics

Unless otherwise stated the receiver characteristics are specified at the antenna connector(s) of the UE for satellite access.

The handheld UE for satellite access is required to be equipped with a minimum of two Rx antenna ports in all NTN satellite operating bands in FR1.

7.4.3.2.2 Reference sensitivity

The reference sensitivity power level REFSENS is the minimum mean power applied to each one of the UE antenna ports for all UE categories, at which the throughput shall meet or exceed the requirements for the specified reference measurement channel. For below 6GHz, the REFSENS level can be calculated by the equation below in TR 38.817-01[32]:

$$\text{Sensitivity} = -174\text{dBm(kT)} + 10 \cdot \log(\text{RX BW}) + \text{NF} + \text{SNR} + \text{IM} - \text{diversity gain}$$

It is noted that the Rx BW is identical to the transmission bandwidth configuration, which is determined by the spectrum utilization. The RB values in the analysis of this contribution are based on the agreed SU for NR.

SNR in Nagoya meeting was agreed as -1dB for NR.

The REFSENS for E-UTRA assumes that the receiver is equipped with two Rx port as a baseline. If 2Rx is considered, the diversity gain is 3dB and the Implementation Margin (IM) uses 2.5dB. In the following calculation, same assumptions are used for NR NTN satellite bands.

For band n255, 9dB noise figure is assumed which is aligned with band n24.

Both dedicate 30MHz duplexer and n65 duplexer are allowed to be used for band n256. Finally, 9dB noise figure is assumed for band n256 as well.

For n252, RAN4 agreed that as a starting point the UE REFSENS value for n252 is aligned with the values for n256.

The throughput shall be $\geq 95\%$ of the maximum throughput of the reference measurement channels as specified in Annexes A.2.2.2 and A.3.2 from TS 38.101-1 [17] (with one sided dynamic OCNG Pattern OP.1 FDD for the DL-signal as described in Annex A.5.1.1 from TS 38.101-1 [17]) with parameters specified in Table 7.4.3.2.2-1 and Table 7.4.3.2.2-2.

Table 7.4.3.2.2-1: Two antenna port reference sensitivity QPSK REFSENS

NTN satellite band #	Operating band / SCS / Channel bandwidth / Duplex-mode						Duplex Mode
	SCS kHz	3 MHz (dBm)	5 MHz (dBm)	10 MHz (dBm)	15 MHz (dBm)	20 MHz (dBm)	
n255	15	-101.7	-100.0	-96.8	-95.0	-93.8	FDD
	30			-97.1	-95.1	-94.0	
	60			-97.5	-95.4	-94.2	
n256	15	-101.7	-99.5	-96.3	-94.5	-93.3	FDD
	30			-96.6	-94.6	-93.5	
	60			-97	-94.9	-93.7	
n253	15	-100.0				15	FDD
	30					30	
	60					60	
n252	15		-99.5	-96.3	-94.5	-93.3	FDD
	30			-96.6	-94.6	-93.5	
	60			-97.0	-94.9	-93.7	
n251	15	-100.0	-96.8	-95.0	-93.8	15	FDD
	30		-97.1	-95.1	-94.0	30	
	60		-97.5	-95.4	-94.2	60	
n250	15	-100.0	-96.8	-95.0	-93.8	15	FDD
	30		-97.1	-95.1	-94.0	30	
	60		-97.5	-95.4	-94.2	60	

For power class 2 and power class 1 UEs, certain degradation of the reference sensitivity in Table 7.4.3.2.2-1 is allowed. For a power class 2 UE the maximum amount of degradation is specified in Table 7.4.3.2.2-1a, and in Table 7.4.3.2.2-1b for a UE that indicates *txDiversity-r16* or *txDiversity2Tx-r18*. For a power class 1 UE the maximum amount of degradation is specified in Table 7.4.3.2.2-1c, and in Table 7.4.3.2.2-1d for a UE that indicates *txDiversity-4Tx-r18*.

Table 7.3.2-1a Reference Sensitivity Degradation from PC3 to PC2 for FDD bands for UE with 1Tx

Operating Band	5 MHz (dB)	10 MHz (dB)	15 MHz (dB)	20 MHz (dB)	25 MHz (dB)	30 MHz (dB)	35 MHz (dB)	40 MHz (dB)	45 MHz (dB)	50 MHz (dB)
n256	0	0	0	0						
n255	0.8	0.8	0.9	0.9						
NOTE 1: The transmitter shall be set to P_{UMAX} as defined in clause 6.2.4 of 3GPP TS 38.101-1 [5].										

Table 7.3.2-1b Reference Sensitivity Degradation from PC3 to PC2 for FDD bands for UE with 2Tx

Operating Band	5 MHz (dB)	10 MHz (dB)	15 MHz (dB)	20 MHz (dB)	25 MHz (dB)	30 MHz (dB)	35 MHz (dB)	40 MHz (dB)	45 MHz (dB)	50 MHz (dB)
n256	0	0	0	0						
n255	1.6	1.6	1.7	1.7						
NOTE 1: The transmitter shall be set to P_{UMAX} as defined in clause 6.2.4 of 3GPP TS 38.101-1 [5].										

Table 7.3.2-1c Reference Sensitivity Degradation from PC3 to PC1 for FDD bands for UE with 1Tx

Operating Band	5 MHz (dB)	10 MHz (dB)	15 MHz (dB)	20 MHz (dB)	25 MHz (dB)	30 MHz (dB)	35 MHz (dB)	40 MHz (dB)	45 MHz (dB)	50 MHz (dB)
n256	1.5	1.5	1.5	1.5						
n255	2	2	2	2						
NOTE 1: The transmitter shall be set to P_{UMAX} as defined in clause 6.2.4 of 3GPP TS 38.101-1 [5].										

Table 7.3.2-1d Reference Sensitivity Degradation from PC3 to PC1 for FDD bands for UE with 4Tx

Operating Band	5 MHz (dB)	10 MHz (dB)	15 MHz (dB)	20 MHz (dB)	25 MHz (dB)	30 MHz (dB)	35 MHz (dB)	40 MHz (dB)	45 MHz (dB)	50 MHz (dB)
n256	2.1	2.1	2.1	2.1						
n255	3.3	3.3	3.3	3.3						
NOTE 1: The transmitter shall be set to P_{UMAX} as defined in clause 6.2.4 of 3GPP TS 38.101-1 [5].										

Table 7.4.3.2.2-2: Uplink configuration for reference sensitivity

NTN satellite band #	Operating band / SCS / Channel bandwidth / Duplex-mode						Duplex Mode
	SCS kHz	3 MHz (dBm)	5 MHz (dBm)	10 MHz (dBm)	15 MHz (dBm)	20 MHz (dBm)	
n255	15	15	25	50	75	100	FDD
	30			24	36	50	
	60			10	18	24	
n256	15	15	25	50	75	100	FDD
	30			24	36	50	
	60			10	18	24	
n253	15		25				FDD
	30						
	60						
n252	15		25	50	75	100	FDD
	30			24	36	50	
	60			10	18	24	
n251	15		25	50	75	75 ² 50 ³	FDD
	30			24	36	36 ² 24 ³	
	60			10	18	18 ² 10 ³	
n250	15		25			75 ² 50 ³	FDD

Table 7.4.3.2.2-3: TX – RX carrier centre frequency separation for REFSENS verification

Operating Band	Channel bandwidth	TX – RX carrier centre frequency separation for REFSENS verification
n253	5 MHz	-148MHz, -152MHz
n251	5 MHz	-72.5 MHz, -137.5 MHz
	20 MHz	-87.5 MHz, -122.5MHz
n250	5 MHz	-114 MHz, -152 MHz
	20 MHz	-129 MHz, -137 MHz

7.4.3.2.3 Maximum input level

Maximum input level is defined as the maximum mean power received at the UE antenna port, at which the specified relative throughput shall meet or exceed the minimum requirements for the specified reference measurement channel. In defining requirements for maximum input level, there are two effects should be considered. One is the dynamic range for Rx link since the maximum input level would determine the high limit of dynamic range. The other is the maximum received input power at UE in real deployment. If the maximum received input signal level is increased, then pre-LNA attenuation is required, which adds complexity to UE RX chain. Similar as the method in TN system, in order to evaluate the level of maximum received power, the Min distance for consideration is given in Table 7.4.3.2.3-1.

Table 7.4.3.2.3-1: Min distance for conderation

	Parameters	Satellite Type		
		GEO	LEO 1200	LEO 600
	Carrier frequency [GHz]	2	2	2
	Min Distance from Satellite D_Min [Km]	35786	1200	600
	The free space path loss (FSPL) (dB)	189.5	160.1	154.0
	Satellite EIRP density (dBW/MHz)	59	40	34
SET 1	Bandwidth (15kHz SCS) (MHz)	20		
	Rx signal at min Distance (dBm)	-87.7	-77.2	-77.2
	Satellite EIRP density (dBW/MHz)	53.3	34	28
SET 2	Bandwidth (15kHz SCS) (MHz)	20		
	Rx signal at min Distance (dBm)	-93.4	-83.2	-83.2

From above table, the maximum received power is -77.2 dBm for 20 MHz which is very lower than -25dBm in TN. It is therefore the maximum input level can be relaxed by 15 dB compared with TN requirement.

7.4.3.2.4 Adjacent channel selectivity

For ACS, the coexistence studies outcome shows that the same ACS values as for NR TN UE are appropriate, and therefore reused.

ACS test parameters in case 1 is applied. ACS requirements in case 2 can be specified based on the maximum input level of -40dBm.

7.4.3.2.5 Blocking characteristics

For blocking characteristics, it is agreed, the same requirements as for NR TN UEs are applicable, and therefore reused. For NR satellite band n255, OOB requirements of n24 are reused. For n256, it is assumed to keep UE implementation freedom with different options (e.g., reusing n65 duplexer or dedicated 30MHz duplexer).

For n252, the agreed Out-of-band blocking requirement does not preclude any implementation of UE duplexer architecture.

Table 7.4.3.2.5-1 describes the agreed out-of-band blocking requirement for n252.

Table 7.4.3.2.5-1: Out-of-band blocking for n252

Operating Band	Parameter	Unit	Range 1	Range 2	Range 3
	$P_{\text{interferer}}$	dBm	-44	-30	-15
n252	$F_{\text{interferer}}$ (CW)	MHz	$-110 < f - F_{\text{DL_low}} < -15$ or $15 < f - F_{\text{DL_high}} < 60$	$-155 < f - F_{\text{DL_low}} \leq -110$ or $60 \leq f - F_{\text{DL_high}} < 85$	$1 \leq f \leq F_{\text{DL_low}} - 155$ or $F_{\text{DL_high}} + 85 \leq f \leq 12750$

7.4.3.2.6 Spurious response

Spurious response is a measure of the ability of the receiver to receive a wanted signal on its assigned channel frequency without exceeding a given degradation due to the presence of an unwanted CW interfering signal at any other frequency for which a response is obtained, i.e. for which the out-of-band blocking limit is not met. To assume the same interfering signals for guaranteeing TN and NTN UE performance, TN spurious response requirements in section 7.7 of TS 38.101-1[17] could be applicable to NTN satellite access UEs.

7.4.3.2.7 Intermodulation characteristics

Intermodulation response rejection is a measure of the capability of the receiver to receive a wanted signal on its assigned channel frequency in the presence of two or more interfering signals which have a specific frequency relationship to the wanted signal. To assume the same interfering signals for guaranteeing TN and NTN UE performance, TN wide-band intermodulation parameters in Table 7.8.2-1 of TS 38.101-1[17] could be applicable to NTN satellite access UEs.

7.4.3.2.8 Rx Spurious emissions

The spurious emissions limits specified in TS 38.101-1[17] sub-clause 7.9 would also be applicable to NTN satellite access UEs referring to the ERC Recommendation 74-01[28] Annex 3 item 3.1.5 for Mobile Earth Stations (MES) (see Note 1 bis) operating below 3 GHz: receivers, receive only MES and transmitters in the carrier-off state.

7.4.3.2.9 Void

7.4.3.3 Radiated receiver characteristics for Ka band VSAT

7.4.3.3.1 Polarization characteristics

Due to the faraday rotation effect in the atomosphere, the circular polarization is usually used for satellite communication. The minimum requirements on the receiver characteristics apply under RHCP (right hand circular polarization) or LHCP (left hand circular polarization).

7.4.3.3.2 OTA reference sensitivity level

If 62dBm EIRP can be assumed for LEO600 SAN with 30 degree elevation angle scenario, the DL budget calculation is shown below to get the DL received signal at OTA (effective isotropic received signal).

Table 7.4.3.3.2-1: The calculation of DL budget for LEO600 scenario

	LEO600
NTN SAN EIRP (dBm)	62
Elevation angle (degree)	30
Slant range (km)	1075
center frequency (GHz)	17
freePathloss	177.68
VSAT DL Received signal level at OTA (dBm)	-115.68

If -5dB G/T can be assumed to calculate the EIS requirements, the following calculation can be used to derived it.

$$EIS = 10 \cdot \log_{10}(k) + 30 + 10 \cdot \log_{10}(BW) + SNR + IM - G/T$$

BW is channel bandwidth. (candidate value: 50MHz)

IM is implementation margin. 2dB can be assumed.

SNR is the required SNR for demodulation. -1dB can be assumed.

G/T: -5dB.

K: is Boltzmann constant, $1.38 \cdot 10^{-23}$ J/K

Based on the calculation, the 50MHz minimum EIS for fixed VSAT communicating with LEO only with electronical steering antenna is -115.6dBm, which is valid for DL link budget.

If 82dBm EIRP can be assumed for GSO SAN with 30 degree elevation angle scenario, the DL budget calculation is shown below to get the DL received signal at OTA (effective isotropic received signal).

Table 7.4.3.3.2-1: The calculation of DL budget for GSO scenario

	GSO
NTN SAN EIRP (dBm)	87
Elevation angle (degree)	30
Slant range (km)	38611
center frequency (GHz)	17
freePathloss	208.78
VSAT DL Received signal level at OTA (dBm)	-121.78

If 6dB G/T can be assumed to calculate the EIS requirements, the following calculation can be used to derived it.

$$\text{EIS} = 10 \cdot \log_{10}(k) + 30 + 10 \cdot \log_{10}(\text{BW}) + \text{SNR} + \text{IM} - \text{G/T}$$

BW is channel bandwidth. (candidate value: 50MHz)

IM is implementation margin. 2dB can be assumed.

SNR is the required SNR for demodulation. -1dB can be assumed.

G/T: 6 dB.

K: is Boltzmann constant, $1.38 \cdot 10^{-23}$ J/K

Based on the calculation, the 50MHz minimum EIS for other types is -122 dBm, which is valid for DL link budget.

7.4.4 UE antenna characteristics for satellite access

7.4.4.1 Handheld UE antenna characteristics for satellite access

In Release 17, a reference antenna with a gain of 0dBi is assumed for handheld UE. However, the actual antenna gain can't meet 0dBi in each direction considering the antenna efficiency which is not 100%. Since all the Tx and Rx requirements are specified at conduct connector for handheld UE, the actual antenna characteristic will have no impact on the RF specification for Release 17 handheld UE.

8 Conclusion

It is recommended to take due consideration of contents in Chapter 6 and Chapter 7 when developing Technical Specifications of RF requirements for SAN, HAPS and UE using NR to support non-terrestrial networks (NTN).

Annex A: Calibration results of NTN components

A.1 Calibration assumptions at 2GHz

Assumptions in Section 6.2 are adopted as the baseline for calibration. It should be noted there are different parts which are listed in Table A.1-1.

Table A.1-1: NTN Assumptions for calibration

Propagation model 38.811[5] considerations	Basic path loss	Yes
	Atmospheric loss	0
	Ionospheric or scintillation loss	0
	O2I / building-entry loss	N/A
NTN SINR	SINR statistics target	Central beam (UL/DL)
	Interference	Co-channel interference from 6 adjacent beams
	BW / #UE	20MHz / 1 DL, 3UL
	Polarization gain with 3dB	not considered
	Elevation angle	90 degrees for GEO and LEO
HAPS SINR	SINR statistics target	7 cells for DL and UL, HAPS UE is uniformly distributed in 7 cells
	Interference	Co- channel interference from other 6 cells
	BW / #UE	20MHz/1DL, 3UL and each UE BW is 0.36MHz
	Polarization gain with 3dB	considered
Propagation model 38.811[5] considerations for HAPS	Basic path loss	yes
	Atmospheric loss	0
	Ionospheric or scintillation loss	0
	O2I / building-entry loss	0
HAPS	power control parameter	$\gamma = 1$, CL-ile = 121.45
	rural vs. urban difference	only reflect on the propagation model. Other assumptions are the same.

A.1a Calibration assumptions at 11/14GHz

Assumptions in Section 6c.2 are adopted as the baseline for calibration. It should be noted there are different parts which are listed in Table A.1a-1.

Table A.1a-1: NTN Assumptions for calibration

Propagation model 38.811[5] considerations	Basic path loss	Yes (including Clutter Loss from Table 6.6.2-1 TR 38.811)
	Atmospheric loss	0
	Ionospheric or scintillation loss	0
	O2I / building-entry loss	N/A
NTN SINR	SINR statistics target	Central beam (UL/DL)
	Interference	Co-channel interference from 6 adjacent beams
	BW / #UE	100MHz / 1 DL, 10UL
	Polarization gain with 3dB	not considered
	Elevation angle	35 degrees for GEO and LEO
	Scenario	Urban
VSAT phased array	Antenna characteristics	45x45 antenna elements - Refer to Table 6a.2.2.2-1
	VSAT NF	6 dB

A.2 Calibration results at 2GHz

The calibration results include SINR and coupling loss distributions in DL, and SINR, coupling loss and transmit power distributions in UL. Both rural and urban propagation models defined in TR 38.811[5] are considered. For NTN simulation, HAPS, LEO-600, LEO-1200 and GEO are calibrated and results are summarized in Table A.2-1, 2-2, 2-3 and 2-4.

NOTE: The “-” means the data was not provided by its corresponding contributor.

Table A.2-1: Calibration summary for NTN DL Rural cases

Calibration metrics		DL Coupling Loss			DL Geometry SINR		
CDF percentile		@5%	@50%	@95%	@5%	@50%	@95%
LEO-600	Samsung	123.60	125.25	126.98	-2.49	-0.41	2.23
	Qualcomm	123.69	125.38	127.13	-2.72	-0.72	2.16
	CATT	123.59	125.21	127.10	-3.07	-0.84	2.22
	THALES	123.75	125.84	127.90	-3.86	-1.40	2.17
	Huawei	123.54	125.63	129.16	7.44	10.91	12.97
	Xiaomi	123.61	125.15	126.93	-3.01	-0.59	2.02
	ZTE	123.66	125.57	127.64	-3.05	-0.22	3.65
	Nokia	123.68	125.33	127.05	-3.10	-1.16	0.94
	Ericsson	-	-	-	-2.58	6.97	9.95
	FhG	123.61	125.37	127.07	-3.48	-1.19	1.36
	Variance	0.00	0.04	0.46	10.03	16.13	14.92
	Mean	123.64	125.41	127.44	-1.99	1.14	3.97
LEO-1200	Samsung	129.61	131.27	132.99	-2.47	-0.40	2.23
	Qualcomm	129.60	131.21	132.91	-2.35	-0.42	2.13
	CATT	129.56	131.20	132.86	-2.21	-0.54	2.08
	THALES	129.68	131.51	133.28	-2.91	-0.79	2.27
	Huawei	129.48	131.28	133.83	1.47	7.62	12.76
	Xiaomi	129.68	131.21	132.94	-2.55	-0.70	1.65
	ZTE	129.71	131.53	133.31	-2.42	-0.18	3.66
	Nokia	129.70	131.36	133.10	-3.09	-1.20	0.94
	Ericsson	-	-	-	-3.04	6.14	10.22
	FhG	129.75	131.30	132.97	-3.36	-1.11	1.20
	Variance	0.01	0.01	0.08	1.70	9.31	15.16
	Mean	129.64	131.32	133.13	-2.29	0.84	3.91
GEO	Samsung	138.15	139.85	141.61	-3.77	-1.79	0.50
	Qualcomm	138.11	139.76	141.49	-3.75	-1.97	0.29
	CATT	137.97	139.80	141.51	-3.74	-2.08	0.36
	THALES	138.23	140.07	141.90	-3.86	-1.85	1.05
	Huawei	139.45	140.61	142.00	-1.04	0.26	1.27
	Xiaomi	138.54	140.07	141.77	-3.76	-1.94	0.23
	ZTE	137.81	140.36	142.92	-3.81	-2.24	-0.02
	Nokia	137.95	140.21	142.49	-3.91	-2.06	-0.08
	Ericsson	-	-	-	-6.62	-0.18	3.60
	FhG	138.43	139.99	141.80	-5.18	-2.75	-0.59
	Variance	0.21	0.07	0.20	1.72	0.79	1.22
	Mean	138.29	140.08	141.94	-3.94	-1.66	0.66
HAPS	Nokia	112.15	118.78	142.92	-7.82	10.20	17.05
	Qualcomm	114.14	118.70	141.92	-12.54	10.19	18.46
	Variance	0.99	0.00	0.25	5.58	0.00	0.50
	Mean	113.14	118.74	142.42	-10.18	10.20	17.75

Table A.2-2: Calibration summary for NTN DL Urban cases

Calibration metrics		DL Coupling Loss			DL Geometry SINR		
CDF percentile		@5%	@50%	@95%	@5%	@50%	@95%
LEO-600	Samsung	118.57	125.31	132.31	-2.67	-0.50	2.17
	Qualcomm	118.56	125.21	132.93	-2.94	-0.84	2.12
	CATT	118.94	125.50	132.25	-2.51	-0.35	2.25
	THALES	119.00	125.67	132.50	-3.93	-1.50	2.13
	Huawei	120.16	125.74	131.63	1.80	9.01	15.14
	Xiaomi	119.19	125.45	132.56	-2.69	-0.79	1.57
	ZTE	118.90	125.58	132.40	-3.05	-0.61	3.37
	Ericsson	-	-	-	-3.66	4.92	13.77
	Nokia	118.71	125.16	131.68	-3.53	-1.15	2.42
	FhG	119.50	125.89	133.01	-11.24	-3.00	3.96
	Variance	0.23	0.05	0.20	9.11	11.73	23.37
	Mean	119.06	125.50	132.36	-3.44	0.52	4.89
LEO-1200	Samsung	124.61	131.33	138.34	-2.64	-0.50	2.17
	Qualcomm	124.35	131.34	138.33	-2.49	-0.50	2.12
	CATT	125.02	131.78	137.85	-2.63	-0.37	2.23
	THALES	124.60	131.40	138.00	-3.00	-0.88	2.23
	Huawei	123.64	131.62	139.59	-4.73	6.35	14.64
	Xiaomi	125.02	131.28	138.73	-2.62	-0.77	1.57
	ZTE	124.71	131.52	138.22	-2.40	-0.32	3.45
	Ericsson	-	-	-	-3.67	4.81	14.30
	Nokia	123.97	130.42	136.92	-3.58	-1.27	2.28
	FhG	125.31	131.45	138.08	-10.40	-2.51	4.21
	Variance	0.25	0.13	0.45	5.29	7.18	23.32
	Mean	124.58	131.35	138.23	-3.82	0.40	4.92
GEO	Samsung	133.18	139.90	146.93	-5.72	-2.09	1.10
	Qualcomm	132.93	139.89	146.94	-5.83	-2.21	0.92
	CATT	132.50	139.69	146.40	-5.39	-2.16	1.03
	THALES	133.30	139.97	146.60	-4.97	-2.16	1.34
	Huawei	132.43	138.44	145.29	-5.93	-0.54	6.53
	Xiaomi	133.23	139.68	146.97	-5.24	-2.63	-0.56
	ZTE	133.72	140.50	146.99	-5.23	-2.51	-0.11
	Ericsson	-	-	-	-7.49	-0.59	7.41
	FhG	132.74	139.69	146.97	-11.58	-3.57	4.35
	Variance	0.17	0.30	0.30	3.86	0.81	7.54
	Mean	133.00	139.72	146.64	-6.38	-2.05	2.45
HAPS	Nokia	111.39	145.66	164.30	-29.10	-10.47	16.29
	Qualcomm	112.00	143.12	159.76	-30.86	-10.70	20.84
	Variance	0.09	1.61	5.14	0.77	0.01	5.17
	Mean	111.69	144.39	162.03	-29.98	-10.58	18.57

Table A.2-3: Calibration summary for NTN UL Rural cases

Calibration metrics		UL Coupling Loss			UL Geometry SINR			UL UE Tx Power		
CDF percentile		@5%	@50%	@95%	@5%	@50%	@95%	@5%	@50%	@95%
LEO-600	Samsung	123.60	125.25	126.98	-5.51	-3.07	-0.70	23.00	23.00	23.00
	Qualcomm	123.64	125.37	127.14	-5.43	-2.92	-0.42	23.00	23.00	23.00
	CATT	123.68	125.47	127.29	-7.22	-1.78	0.42	-	-	-
	THALES	123.75	125.84	127.90	-6.16	-3.76	-1.31	23.00	23.00	23.00
	Huawei	123.54	125.63	129.16	-5.12	-1.31	0.60	23.00	23.00	23.00
	Xiaomi	123.66	125.22	126.92	-5.14	-2.60	0.18	23.00	23.00	23.00
	ZTE	123.83	125.65	127.46	-5.45	-3.04	-0.55	23.00	23.00	23.00
	Ericsson	-	-	-	-14.64	-8.63	-3.87	-	-	-
	FhG	123.85	125.27	127.12	-5.98	-3.47	-1.38	23.00	23.00	23.00
	Variance	0.01	0.04	0.48	8.19	3.96	1.63	0.00	0.00	0.00
	Mean	123.69	125.46	127.50	-6.74	-3.40	-0.78	23.00	23.00	23.00
LEO-1200	Samsung	129.62	131.27	132.99	-9.44	-7.60	-5.83	23.00	23.00	23.00
	Qualcomm	129.56	131.18	132.89	-9.40	-7.61	-5.78	23.00	23.00	23.00
	CATT	129.72	131.31	132.85	-8.84	-6.97	-5.28	-	-	-
	THALES	129.68	131.51	133.28	-9.46	-7.68	-5.75	23.00	23.00	23.00
	Huawei	129.48	131.28	133.83	-9.48	-6.85	-5.18	23.00	23.00	23.00
	Xiaomi	129.61	131.23	132.93	-7.97	-6.09	-4.20	23.00	23.00	23.00
	ZTE	129.60	131.38	133.25	-9.27	-7.32	-5.63	23.00	23.00	23.00
	Ericsson	-	-	-	-19.05	-13.42	-9.84	-	-	-
	FhG	129.68	131.28	133.06	-9.51	-7.69	-5.98	23.00	23.00	23.00
	Variance	0.01	0.01	0.09	9.87	4.03	2.16	0.00	0.00	0.00
	Mean	129.62	131.31	133.14	-10.27	-7.91	-5.94	23.00	23.00	23.00
GEO	Samsung	138.15	139.85	141.59	-20.31	-18.54	-16.85	23.00	23.00	23.00
	Qualcomm	138.06	139.73	141.47	-20.21	-18.47	-16.78	23.00	23.00	23.00
	CATT	138.07	139.61	141.26	-19.86	-18.24	-16.70	-	-	-
	THALES	138.23	140.07	141.90	-20.37	-18.55	-16.68	23.00	23.00	23.00
	Huawei	139.45	140.61	142.00	-17.89	-16.73	-15.61	23.00	23.00	23.00
	Xiaomi	138.10	139.59	141.16	-18.98	-17.45	-15.92	23.00	23.00	23.00
	ZTE	137.76	140.33	142.91	-18.35	-15.72	-13.37	23.00	23.00	23.00
	Ericsson	-	-	-	-30.59	-25.37	-21.63	-	-	-
	FhG	138.33	139.76	141.56	-20.26	-18.49	-17.06	23.00	23.00	23.00
	Variance	0.22	0.12	0.27	12.84	6.56	4.15	0.00	0.00	0.00
	Mean	138.27	139.94	141.73	-20.76	-18.62	-16.73	23.00	23.00	23.00
HAPS	Nokia	112.20	118.96	143.51	-11.47	10.01	14.49	13.75	20.51	23.00
	Qualcomm	114.14	118.70	141.92	-9.93	10.59	14.85	15.69	20.25	23.00
	Variance	0.94	0.02	0.64	0.60	0.09	0.03	0.93	0.02	0.00
	Mean	113.17	118.83	142.72	-10.70	10.30	14.67	14.72	20.38	23.00

Table A.2-4: Calibration summary for NTN UL Urban cases

Calibration metrics		UL Coupling Loss			UL Geometry SINR			UL UE Tx Power		
CDF percentile		@5%	@50%	@95%	@5%	@50%	@95%	@5%	@50%	@95%
LEO-600	Samsung	118.58	125.30	132.31	-11.38	-3.64	3.57	23.00	23.00	23.00
	Qualcomm	118.71	125.16	132.66	-11.83	-3.91	2.64	23.00	23.00	23.00
	CATT	118.41	125.45	132.54	-11.13	-3.89	2.97	-	-	-
	THALES	119.00	125.67	132.50	-11.15	-3.92	2.70	23.00	23.00	23.00
	Huawei	120.16	125.74	131.63	-11.52	-4.25	1.70	23.00	23.00	23.00
	Xiaomi	118.63	125.29	132.62	-11.20	-3.19	4.57	23.00	23.00	23.00
	ZTE	118.92	125.71	132.43	-11.19	-3.75	3.56	23.00	23.00	23.00
	Ericsson	-	-	-	-17.17	-8.77	0.15	-	-	-
	FhG	117.88	125.14	132.15	-11.34	-4.04	3.87	23.00	23.00	23.00
	Variance	0.37	0.05	0.10	3.40	2.49	1.52	0.00	0.00	0.00
	Mean	118.79	125.43	132.35	-11.99	-4.37	2.86	23.00	23.00	23.00
LEO-1200	Samsung	124.60	131.32	138.32	-15.10	-7.94	-1.13	23.00	23.00	23.00
	Qualcomm	124.40	131.30	138.36	-15.23	-8.01	-0.95	23.00	23.00	23.00
	CATT	123.95	131.27	137.95	-13.61	-6.83	-0.26	-	-	-
	THALES	124.60	131.40	138.00	-14.36	-7.79	-1.27	23.00	23.00	23.00
	Huawei	123.64	131.62	139.59	-16.30	-7.25	-0.41	23.00	23.00	23.00
	Xiaomi	124.84	131.32	138.61	-13.72	-6.53	0.40	23.00	23.00	23.00
	ZTE	124.73	131.47	138.22	-14.30	-7.40	-0.51	23.00	23.00	23.00
	Ericsson	-	-	-	-21.09	-13.52	-5.47	-	-	-
	FhG	124.30	130.66	137.02	-14.02	-7.35	-0.89	23.00	23.00	23.00
	Variance	0.15	0.07	0.45	4.83	3.93	2.55	0.00	0.00	0.00
	Mean	124.38	131.30	138.26	-15.30	-8.07	-1.17	23.00	23.00	23.00
GEO	Samsung	133.17	139.89	146.92	-25.64	-18.62	-11.90	23.00	23.00	23.00
	Qualcomm	132.96	139.86	146.90	-25.69	-18.61	-11.73	23.00	23.00	23.00
	CATT	133.28	139.84	147.04	-24.99	-18.36	-11.24	-	-	-
	THALES	133.30	139.97	146.60	-25.25	-18.47	-11.80	23.00	23.00	23.00
	Huawei	132.43	138.44	145.29	-24.06	-17.38	-11.26	23.00	23.00	23.00
	Xiaomi	133.03	139.55	146.75	-24.19	-17.39	-10.63	23.00	23.00	23.00
	ZTE	133.58	140.24	146.91	-22.79	-15.32	-8.86	23.00	23.00	23.00
	Ericsson	-	-	-	-32.17	-25.29	-17.44	-	-	-
	FhG	133.46	140.15	147.08	-25.79	-18.93	-12.26	23.00	23.00	23.00
	Variance	0.11	0.28	0.30	6.22	6.50	4.73	0.00	0.00	0.00
	Mean	133.15	139.74	146.69	-25.62	-18.71	-11.90	23.00	23.00	23.00
HAPS	Nokia	111.39	145.66	164.30	-28.44	-10.17	14.88	12.94	23.00	23.00
	Qualcomm	111.89	143.11	158.93	-26.02	-8.42	14.90	13.44	23.00	23.00
	Variance	0.06	1.62	7.21	1.46	0.76	0.00	0.06	0.00	0.00
	Mean	111.64	144.38	161.61	-27.23	-9.29	14.89	13.19	23.00	23.00

Annex B: Calibration results of TN components

B.1 Calibration assumptions at 2GHz

Assumptions in Section 6.2 are adopted as the baseline for calibration. It should be noted there are different parts which are listed in Table B.1-1.

Table B.1-1: TN Assumptions for calibration

Calibration assumptions		
TN AAS	Rural	Element gain: 7.1 dBi
		3dB: H 90 / V 54
		Front-back: 30 H/V
		Array: 8x8
		Element spacing: H 0.5/V 0.9
		Conducted Tx: 25 dBm
		Ohmic loss: 2 dB
		Mechanical downtilt: 3 deg
		Polarization gain 3 dB
		#UE: 1 DL/ 3 UL
		100% Outdoor
		ISD 7.5 KM
	Urban	Element gain: 6.4 dBi
		3dB: H 90 / V 65
		Front-back: 30 H/V
		Array: 8x8
		Element spacing: H 0.5/V 0.7
		Conducted Tx: 25 dBm
		Ohmic loss: 2 dB
		Mechanical downtilt: 10 deg
		Polarization gain 3 dB
		#UE: 1 DL/ 3 UL
		100% Outdoor
		ISD 0.75 KM
TN non-AAS	Antenna gain	17 dBi
	Conducted Tx	46 dBm
	3dB	R4-2108645 Section 2.4.2
	Front-back	
	Mechanical downtilt	Rural 3 / Urban 10

B.1a Calibration assumptions at 11/14GHz

Assumptions in Section 6c.2 are adopted as the baseline for calibration. It should be noted there are different parts which are listed in Table B.1a-1.

Table B.1a-1: TN Assumptions for calibration

Calibration assumptions		
TN AAS BS	Antenna characteristics	Refer to Table 6a.2.3.2-1 and Table 6a.2.3.2-2 for 11 GHz Refer to Table 6a.2.3.2-3 and Table 6a.2.3.2-4 for 14 GHz
	Scenario	Urban
		#UE: 1 DL/ 3 UL
		100% Outdoor
		ISD 0.45 KM
TN UE	Antenna characteristics	Omni-directional radiation pattern with 0 dBi gain

B.2 Calibration results at 2GHz

The calibration results include SINR and coupling loss distributions in DL, and SINR, coupling loss and transmit power distributions in UL. Both rural macro and urban macro propagation models defined in TR 38.901 [10] are considered. NR BS with AAS and non-AAS antennas are calibrated separately and results are summarized in Table B.2-1, 2-2 and 2-3. NB-IoT scenarios are not calibrated.

NOTE: The “-” means the data was not provided by its corresponding contributor.

Table B.2-1: Calibration summary for TN NR AAS DL cases

Calibration metrics		DL Coupling Loss			DL Geometry SINR		
CDF percentile		@5%	@50%	@95%	@5%	@50%	@95%
Urban Macro	Samsung	60.05	88.58	98.98	8.47	24.01	45.77
	Huawei	52.84	85.79	103.53	5.75	22.23	43.40
	Qualcomm	57.89	87.11	100.61	5.08	19.51	43.65
	Nokia	59.04	87.59	99.32	12.42	20.15	36.50
	Ericsson	-	-	-	6.66	19.58	39.57
	ZTE	58.76	86.90	99.74	5.63	20.56	45.14
	Variance	6.42	0.83	2.69	6.35	2.62	10.70
	Mean	57.72	87.19	100.44	7.33	21.01	42.34
Rural Macro	Samsung	78.09	116.25	127.96	4.52	16.53	37.71
	Huawei	77.53	114.99	128.13	3.02	15.15	31.35
	Qualcomm	78.27	116.81	131.58	0.17	12.60	35.83
	Nokia	78.55	116.18	127.72	5.60	14.93	28.23
	Ericsson	-	-	-	10.03	20.17	37.72
	Variance	0.14	0.44	2.51	10.51	6.21	14.21
	Mean	78.11	116.06	128.85	4.67	15.87	34.17

Table B.2-2: Calibration summary for TN NR non-AAS DL cases

Calibration metrics		DL Coupling Loss			DL Geometry SINR		
CDF percentile		@5%	@50%	@95%	@5%	@50%	@95%
Urban Macro	Samsung	73.23	99.36	110.33	-1.76	6.99	14.85
	Huawei	70.73	99.50	113.81	-5.02	5.47	14.44
	Qualcomm	71.85	99.44	113.47	-4.03	4.52	14.81
	ZTE	71.00	99.00	112.09	-2.89	5.24	14.99
	THALES	71.84	100.22	112.26	-3.03	4.25	14.40
	Variance	0.95	0.04	1.88	1.50	0.81	0.04
	Mean	71.73	99.50	112.39	-3.35	5.29	14.70
Rural Macro	Samsung	87.82	125.94	138.22	-4.25	5.77	15.42
	Huawei	93.87	126.75	141.60	-8.93	1.10	15.33
	Qualcomm	89.63	127.43	142.65	-7.37	2.54	15.47
	THALES	88.83	127.81	139.57	-5.31	2.57	15.24
	Variance	6.43	0.37	3.58	3.78	3.81	0.00
	Mean	90.04	126.98	140.51	-6.47	3.00	15.37

Table B.2-3: Calibration summary for TN NR AAS UL cases

Calibration metrics		UL Coupling Loss			UL Geometry SINR			UL UE Tx Power		
CDF percentile		@5%	@50%	@95%	@5%	@50%	@95%	@5%	@50%	@95%
Urban Macro	Samsung	60.00	88.58	99.02	3.42	14.01	14.96	-25.77	2.81	13.25
	Huawei	52.84	85.79	103.53	7.53	13.89	14.89	-31.46	4.14	16.51
	Qualcomm	62.00	91.09	104.80	1.33	10.10	13.87	-23.77	5.32	19.03
	Nokia	59.04	87.59	99.32	10.98	14.55	15.10	-27.23	1.32	13.05
	Ericsson	-	-	-	6.83	12.94	15.08	-28.52	-1.70	13.98
	ZTE	58.43	87.22	98.46	5.47	13.42	14.81	-27.59	1.20	12.44
	Variance	9.37	3.11	6.81	9.41	2.11	0.18	5.61	5.14	5.41
	Mean	58.46	88.05	101.03	5.93	13.15	14.78	-27.39	2.18	14.71
Rural Macro	Samsung	78.00	116.25	127.95	-4.86	7.03	14.96	-7.77	23.00	23.00
	Huawei	77.53	114.99	128.13	-10.84	4.42	14.86	-3.46	23.00	23.00
	Qualcomm	82.05	120.47	135.46	-13.41	2.07	14.69	-3.72	23.00	23.00
	Nokia	78.42	116.20	127.85	-4.15	7.69	15.17	-7.85	22.99	22.99
	Ericsson	-	-	-	3.30	13.55	15.14	-8.00	21.55	23.00
	Variance	3.20	4.32	10.50	33.91	14.88	0.03	4.41	0.34	0.00
	Mean	79.00	116.98	129.85	-5.99	6.95	14.96	-6.16	22.71	23.00

Table B.2-4: Calibration summary for TN NR non-AAS UL cases

Calibration metrics		UL Coupling Loss			UL Geometry SINR			UL UE Tx Power		
CDF percentile		@5%	@50%	@95%	@5%	@50%	@95%	@5%	@50%	@95%
Urban Macro	Samsung	73.18	99.31	110.33	-10.99	4.31	9.49	-12.59	13.54	23.00
	Huawei	70.73	99.50	113.81	-6.46	3.08	8.88	-10.26	18.51	23.00
	Qualcomm	71.53	99.51	113.45	-3.52	3.36	8.00	-14.24	13.74	23.00
	ZTE	70.11	98.57	111.77	-1.39	3.87	8.01	-16.23	12.57	23.00
	THALES	71.84	100.22	112.26	-0.01	4.13	6.80	-14.05	14.90	22.76
	Variance	1.32	0.15	1.94	12.98	0.22	0.39	4.79	5.32	0.00
	Mean	71.48	99.42	112.32	-4.48	3.75	8.24	-13.47	14.65	22.95
Rural Macro	Samsung	87.78	125.93	138.22	-17.23	-3.82	14.65	2.01	23.00	23.00
	Huawei	93.87	126.75	141.60	-24.83	-9.72	14.75	12.88	23.00	23.00
	Qualcomm	89.53	127.64	142.62	-21.39	-5.65	14.33	3.76	23.00	23.00
	THALES	88.83	127.81	139.57	-17.29	-5.16	14.26	3.55	22.42	22.94
	Variance	6.55	0.49	3.54	9.66	6.07	0.03	22.71	0.00	0.00
	Mean	90.00	127.03	140.50	-20.18	-6.09	14.50	5.55	22.86	22.99

Annex C:

Summary of NR-NTN co-existence study

All NR-NTN co-existence study results have been captured in the list. Please see attachment Summary of NR-NTN co-existence study.

[Editor note: see attachment Summary of NR-NTN coexistence study_r1]

C1: Summary of n252 UE-to-UE co-existence study/simulations

C.1.1 UE-to-UE separation distance and UE-to-UE protection limit (R4-2416459)

The origins of the -50 dBm/MHz limit can be tracked back to 2008 [R4-080710]. The analysis back then was largely based on approximately 1m separation between UEs and assumptions on human body and antenna losses. For terrestrial networks, this scenario can realize e.g. in indoor environments or in a bus where multiple people are using data intensive applications simultaneously. Given the penetration loss of walls, it is also possible that indoors UEs both would need to use high transmit power to reach the gNB outdoors, as well as operate near sensitivity.

For NTN, outdoor environment is the typical condition. One could also expect outdoor rural conditions to be more appropriate, as one would expect urban environments to have TN coverage always available. As rural environment is very different from indoor environment, we did evaluations on how different UE-to-UE distances and what would need to be the emission level at UE antenna to provide same protection as -50 dBm/MHz requirement does for 1m separation. We looked at different distances and two different propagation (path loss) models, i.e. free space and urban micro. Urban micro was included as TR 38.863 has documented it to be used for UE-to-UE interference in Ka-band studies.

The summary of the results is provided in Table C.1.1.1-1.

Table C.1.1.1-1: impact of UE-to-UE separation distance to required UE-to-UE protection limit

sensitivity of n25 for 5 MHz	dBm	-96.5							
Allowed noise increase	dB	2.5							
Total allowed interference	dBm	-97.6							
Total antenna loss (Tx and Rx)	dB	-10							
Human body loss (Tx and Rx)	dB	-6							
distance between UEs	m	1	5	10	15	20	30	40	50
Propagation loss (free space)	dB	-38.5	-52.5	-58.5	-62.0	-64.5	-68.0	-70.5	-72.5
Propogation loss (Umi NLOS)	dB	-38.4	-60.7	-70.3	-75.9	-79.9	-85.5	-89.5	-92.6
Emission power at antenna connector (free space)	dBm	-43.1	-29.1	-23.1	-19.6	-17.1	-13.6	-11.1	-9.1
	dBm/MHz	-49.7	-35.7	-29.7	-26.1	-23.6	-20.1	-17.6	-15.7
Emission power at antenna connector (Umi NLOS)	dBm	-43.2	-20.9	-11.3	-5.7	-1.7	4.0	7.9	11.0
	dBm/MHz	-49.7	-27.4	-17.8	-12.2	-8.2	-2.6	1.4	4.5

2GHz center frequency was used in the path loss calculation. It can be seen that with the assumed parameters 1m separation distance sets the emission power to -49.7 dBm/MHz. At 10 m distance there is 20 dB more power allowed assuming free space propagation. Then every doubling of the distance allows 6 dB more emissions. For example, if 20 m would be used in deriving the coex-limit, -24 dBm/MHz could be used as the limit.

C.1.2: AMPR study for n252

C.1.2.1 Power back-off simulation from company A (R4-2503176)

C.1.2.1.1 Assumption for power-back off simulation

The following are common assumptions for the power back-off simulation results for the NTN band n252:

- Tx power: PC3 (+23dBm)
- LO placement: always in the centre of the carrier
- Additional requirements: -40dBm/MHz and -30dBm/MHz protection criterion for band n25

C.1.2.1.2 Power back-off simulations for -40dBm/MHz

Figure C.1.2.1.2-1 and Figure C.1.2.1.2-2 below present simulation results for the 5MHz channel, which is placed in the following locations within the band: left edge of the band ($F_c=2002.5\text{MHz}$) and when the center frequency shifted further away from the left edge ($F_c=2007.5\text{MHz}$). For the sake of brevity, we present only DFT-s-OFDM and CP-OFDM results (DFT-s-OFDM PI/2 BPSK results are usually close to DFT-s-OFDM QPSK and sometimes even outperform the latter). So, referring to Figure C.1.2.1.2-1 and Figure C.1.2.1.2-2 the following general observations could be made for the 5MHz channel:

- If the 5MHz channel is right at the left edge of the band, then large UL allocations may need large power back-off of up to 5dB even for DFT-s-OFDM. On the other hand, if we consider small UL allocations, which are likely to be the case for the NTN channels, then no power back-off is needed. In fact, even medium size UL allocations of up to 10RBs are not likely to have any power back-off.
- If the 5MHz channel is shifted farther away from the band lower edge, then required power back-off reduces dramatically. Only large UL allocations may need power back-off.

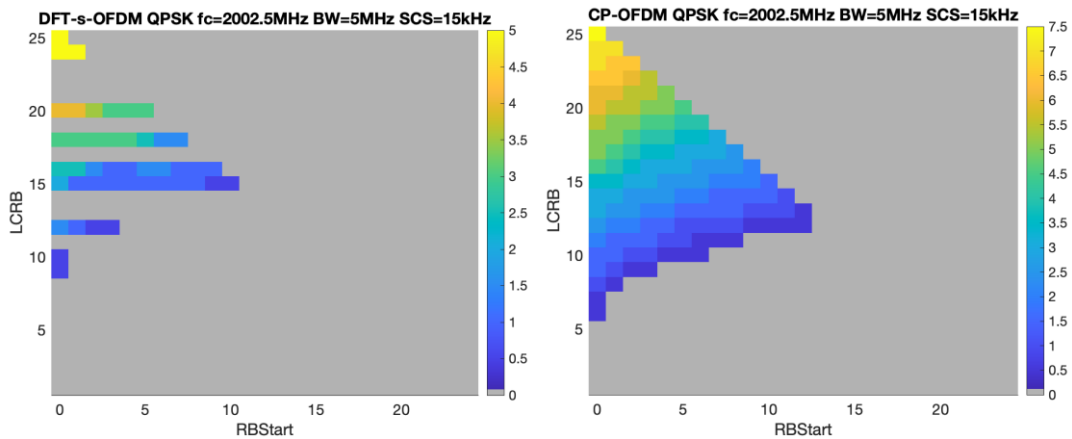


Figure C.1.2.1.2-1: Power back-off values for the 5MHz channel ($F_c=2002.5\text{MHz}$)

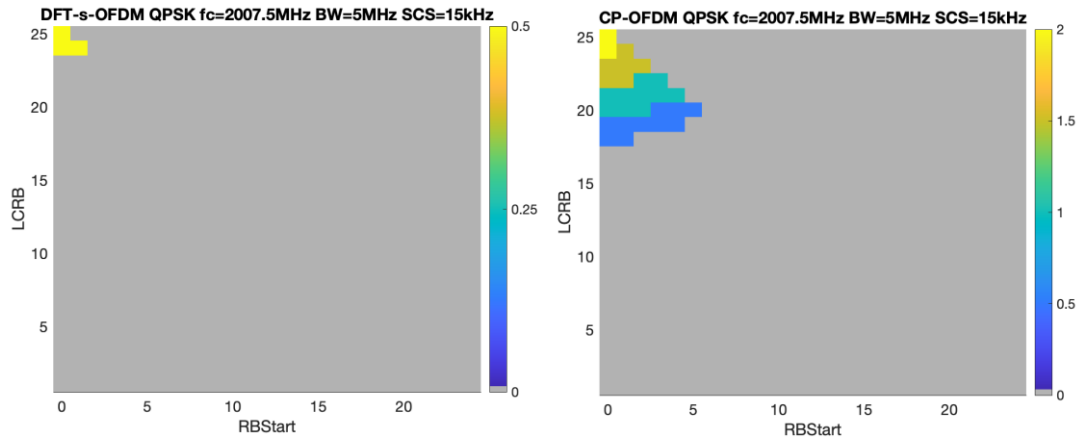


Figure C.1.2.1.2-2: Power back-off values for the 5MHz channel (Fc=2007.5MHz)

Figure C.1.2.1.2-3 and Figure C.1.2.1.2-4 present power back-off values for the 10MHz channel considering the following locations of the channel: right at the left edge of the band (Fc=2005) and shifted to the right so that the offset from the band edge is 5MHz (Fc=2010). The following general observations can be made:

- For the first case, i.e. when the 10MHz channel resides at the left edge of the band, large power back-off is needed even for small UL allocations.
- If the 10MHz channel is offset by at least the 5MHz gap from the left edge, then small UL allocations will require no power back-off backoff.

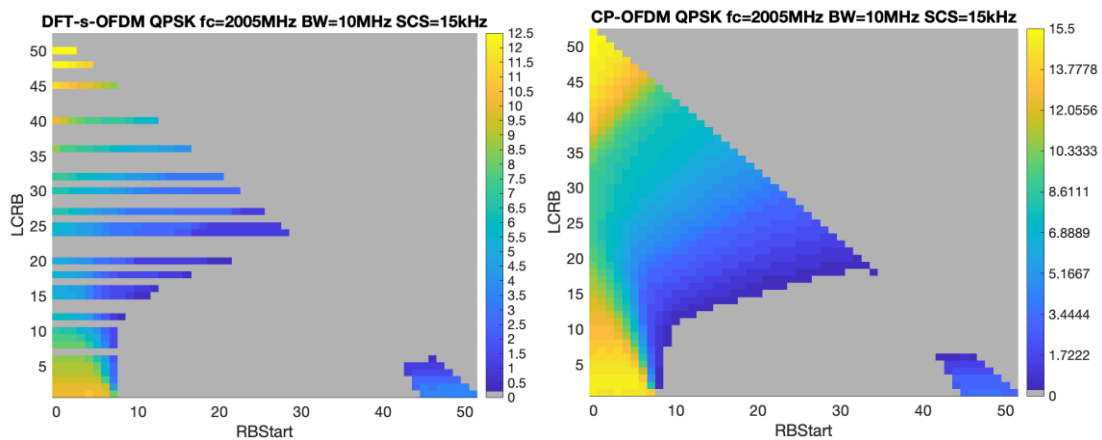


Figure C.1.2.1.2-3: Power back-off values for the 10MHz channel (Fc=2005MHz)

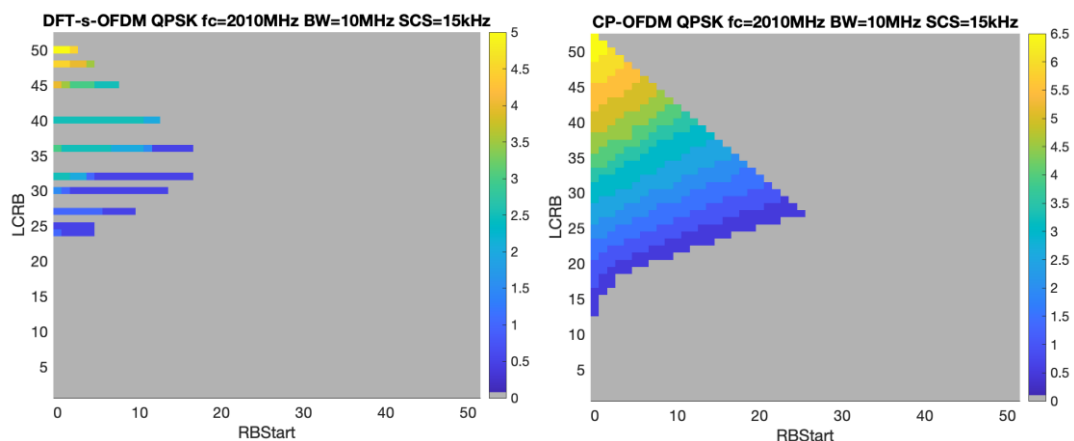


Figure C.1.2.1.2-4: Power back-off values for the 10MHz channel (Fc=2010MHz)

Next, Figure C.1.2.1.2-5 and C.1.2.1.2-6 present power back-off values for the 15MHz channel at the following locations: right at the left edge of the band ($F_c=2007.5$) and shifted to the right so that the offset from the band edge is 5MHz ($F_c=2012.5$). As can be seen from the figures, irrespective of the 15MHz channel location, edge allocations, even small ones, will require large power back-off.

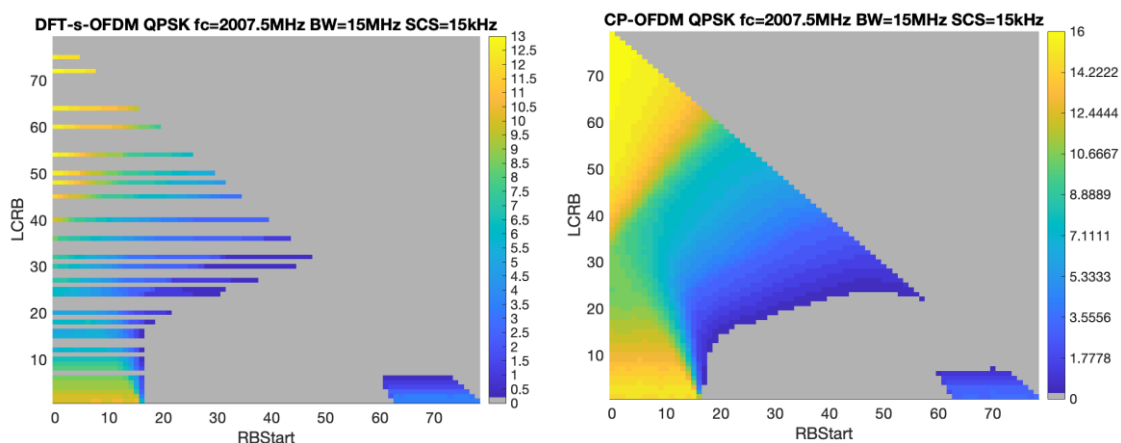


Figure C.1.2.1.2-5: Power back-off values for the 15MHz channel (Fc=2007.5MHz)

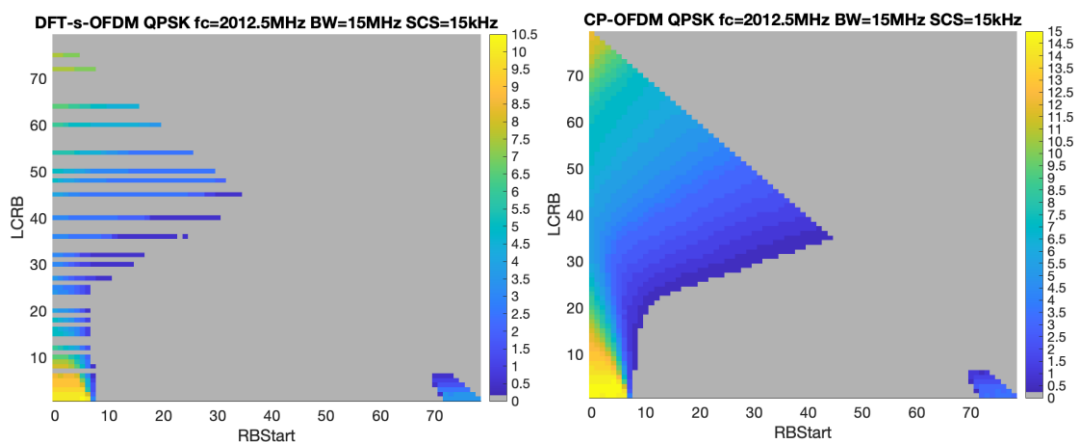


Figure C.1.2.1.2-6: Power back-off values for the 15MHz channel (Fc=2012.5MHz)

Figure C.1.2.1.2-7 presents power back-off results for the 20MHz channel, which as can be seen has even large back-off values comparing to the 15MHz channel bandwidth.

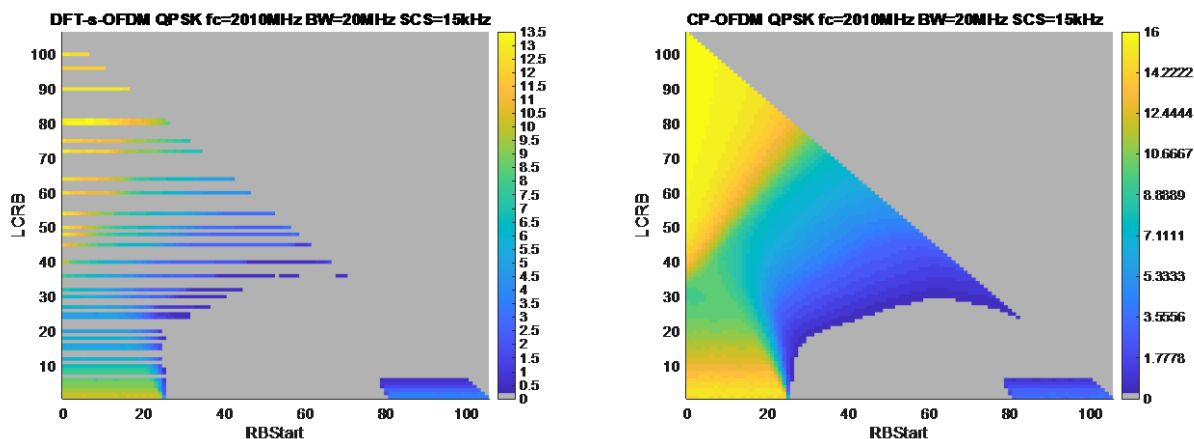


Figure C.1.2.1.2-7: Power back-off values for the 20MHz channel (Fc=2010MHz)

C.1.2.1.3 Power back-off simulations for -30dBm/MHz

In this section we present UE power back-off values for the same channel sizes and center frequencies as in the previous sub-section, but a more relaxed -30dBm/MHz protection criterion is used.

Figure C.1.2.1.3-1 below presents power back-off values for the 5MHz channel, which is right at the left edge of the band. As can be seen, only large allocations might need power back-off, which should be even within the existing MPR margins.

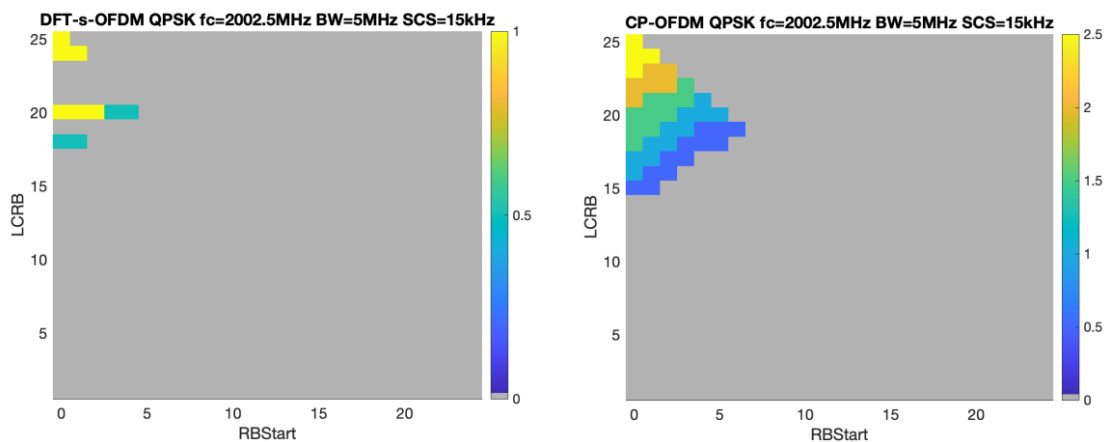


Figure C.1.2.1.3-1: Power back-off values for the 5MHz channel (Fc=2002.5MHz)

Figure C.1.2.1.3-2 and Figure C.1.2.1.3-3 present power back-off values for the 10MHz channel. General observations are the same as for the -40dBm/MHz protection limit, with the only noticeable difference that for the edge 10MHz channel bandwidth only the leftmost edge allocations will need high power back-off values.

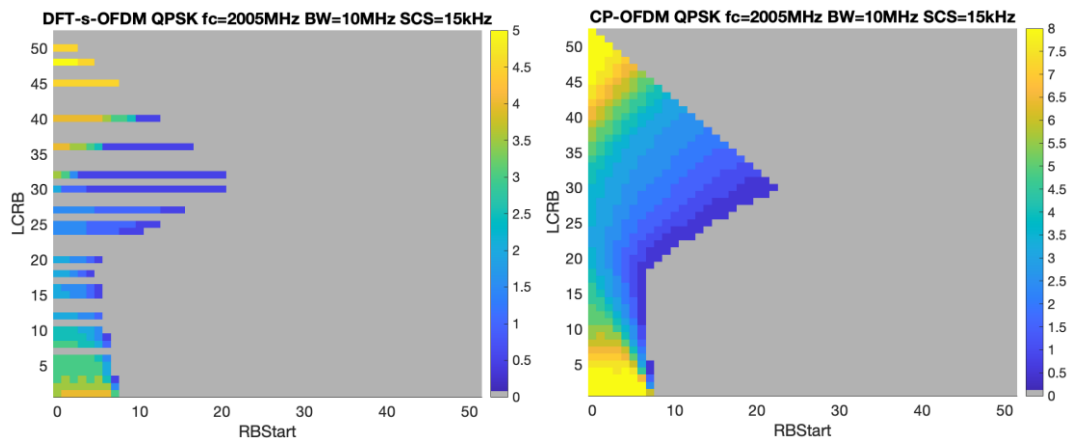


Figure C.1.2.1.3-2: Power back-off values for the 10MHz channel (Fc=2005MHz)

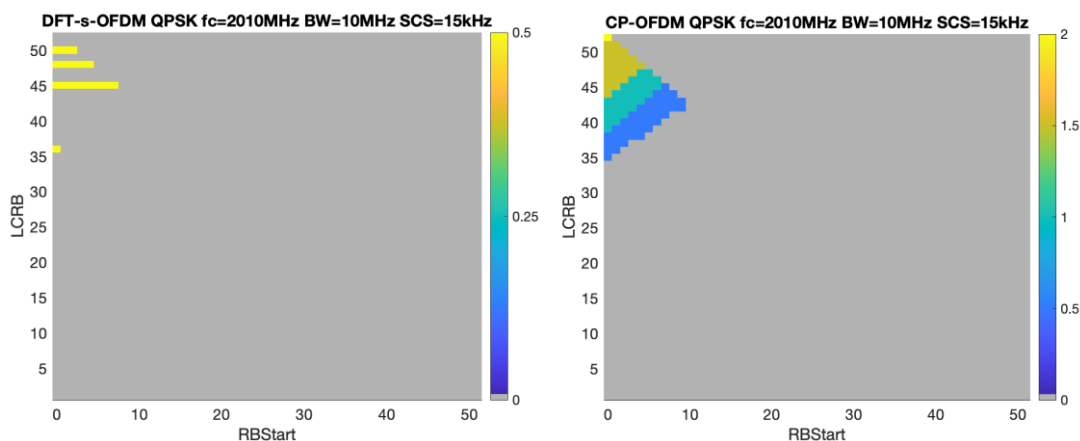


Figure C.1.2.1.3-3: Power back-off values for the 10MHz channel (Fc=2010MHz)

Figure C.1.2.1.3-4 and Figure C.1.2.1.3-5 present power back-off values for the 15MHz channel. Like in the previous simulation case of the -40dBm/MHz protection limit, the edge 15MHz channel will require large back-off values for a wide range of UL allocations. Once shifted to the right, large power back-off values are mostly constrained to the leftmost edge RBs.

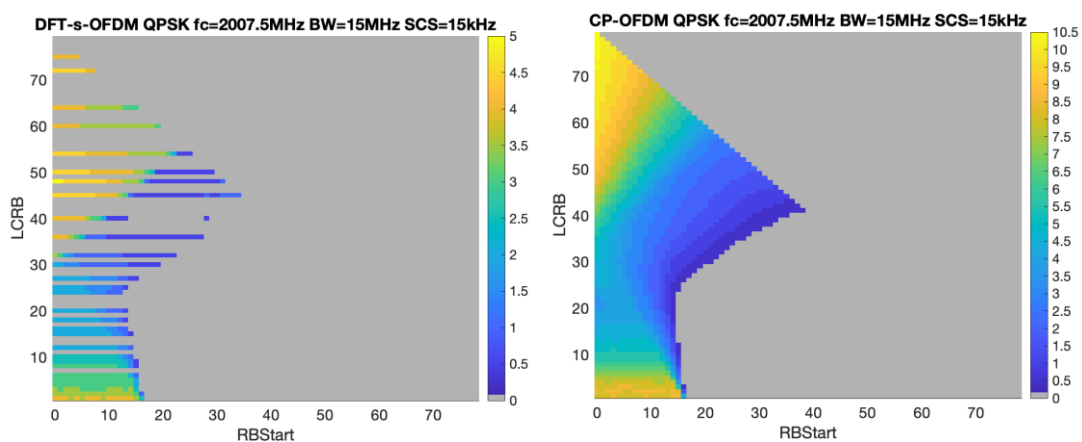


Figure C.1.2.1.3-4: Power back-off values for the 15MHz channel (Fc=2007.5MHz)

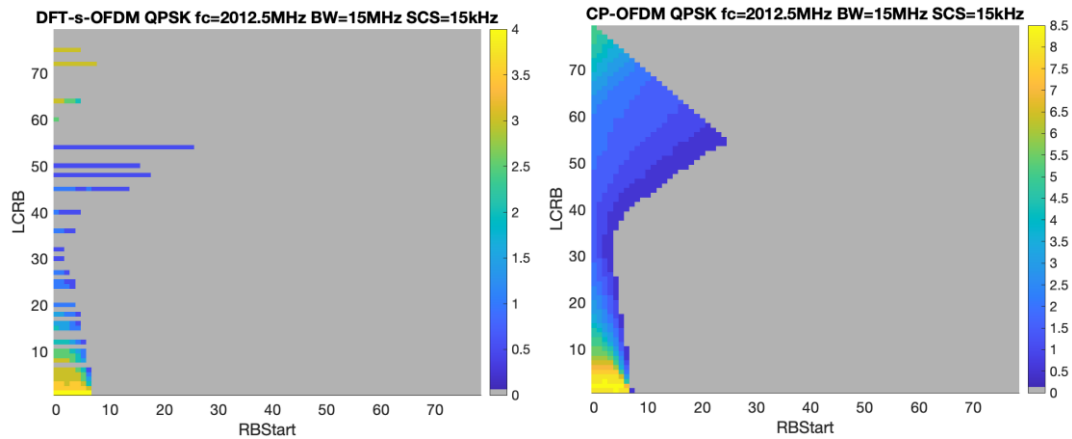


Figure C.1.2.1.3-5: Power back-off values for the 15MHz channel (Fc=2012.5MHz)

Figure C.1.2.1.3-6 presents power back-off results for the 20MHz channel, which follows the trend of large channels resulting in higher power back-off values for large allocations and allocations residing at the left edge of the channel.

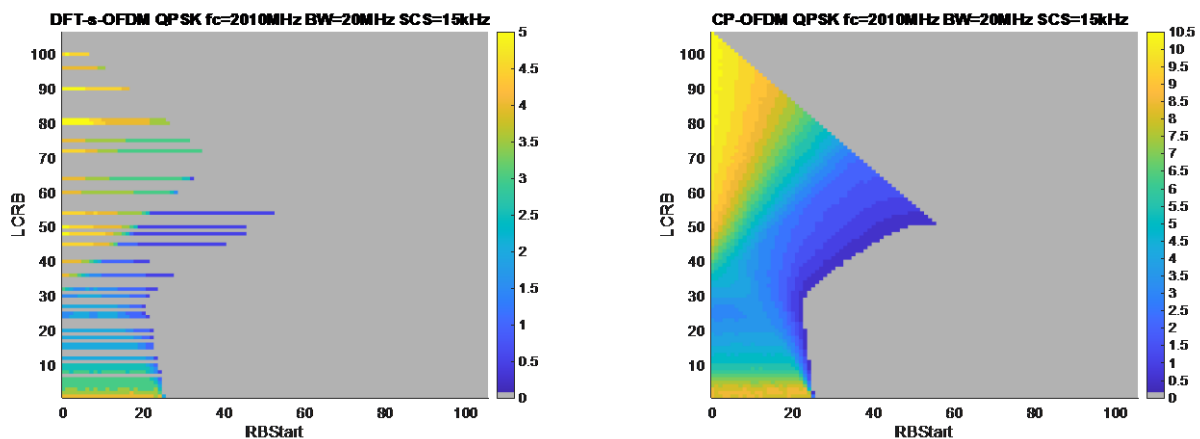


Figure C.1.2.1.3-6: Power back-off values for the 20MHz channel (Fc=2010MHz)

C.1.2.2 Power back-off simulation from company B (R4-2504544)

The need for A-MPR was studied using the following assumptions:

- PA calibrated to 30 dB ACLR with 100 RB DFT-s-OFDM QPSK waveform with 22 dBm at antenna connector
- -28 dBc IQ-image and DC-leakage
- -60 dBc CIM3

The proposed A-MPR is summarized in Tables C.1.2.2-1 to C.1.2.2-4.

Table C.1.2.2-1: A-MPR regions for -40 dBm/MHz

Channel BW	Carrier Frequency, F_c , MHz	$RB_{Start} \cdot 12 \cdot SCS$ (MHz)	$L_{CRB} \cdot 12 \cdot SCS$ (MHz)	A-MPR
5MHz	< 2007.5	≥ 0	≥ 2.7	A1
10MHz	< 2010	≤ 1.44	≥ 0	A2
		> 1.44	≥ 3.6	A1
		> 7.74	≤ 0.72	A1
	$2010 \leq F_c \leq 2015$	≥ 0	≥ 4.5	A1
15MHz	< 2012.5	≤ 3.06	≥ 0	A3
		> 3.06	$4.5 \leq L_{CRB} \cdot 12 \cdot SCS \leq 9$	A1
		> 3.06	> 9	A2
		> 10.98	≤ 0.72	A1
	2012.5	≤ 1.44	≥ 0	A2
		> 1.44	≥ 6.48	A1
		> 12.6	≤ 0.72	A1
20 MHz	2010	≤ 4.68	≥ 0	A3
		> 4.68	$4.5 \leq L_{CRB} \cdot 12 \cdot SCS \leq 9$	A1
		> 4.68	> 9	A2
		> 12.6	≤ 0.72	A1

Table C.1.2.2-2: A-MPR for -40 dBm/MHz

Modulation/Waveform	A1	A2	A3
	Outer/Inner	Outer/Inner	Outer/Inner
DFT-s-OFDM PI/2 BPSK	≤ 4	≤ 6	≤ 7
DFT-s-OFDM QPSK	≤ 4.5	≤ 6.5	≤ 7.5
DFT-s-OFDM 16 QAM	≤ 5	≤ 6.5	≤ 7.5
DFT-s-OFDM 64 QAM	≤ 5.5	≤ 7	≤ 8
CP-OFDM QPSK	≤ 6	≤ 8.5	≤ 9
CP-OFDM 16 QAM	≤ 6	≤ 8.5	≤ 9
CP-OFDM 64 QAM	≤ 6	≤ 8.5	≤ 9

Table C.1.2.2-3: A-MPR regions for -30 dBm/MHz

Channel BW	Carrier Frequency, F_c , MHz	$RB_{Start} \cdot 12 \cdot SCS$ (MHz)	$L_{CRB} \cdot 12 \cdot SCS$ (MHz)	A-MPR
5 MHz	-	-	-	-
10MHz	< 2010	≤ 1.44	≤ 2.7	A1
		≤ 1.44	$2.7 < L_{CRB} \cdot 12 \cdot SCS < 4.5$	A2
		≤ 1.44	≥ 4.5	A1
		≥ 1.44	≥ 5.4	A2
15MHz	< 2012.5	≤ 3.06	≤ 2.7	A1
		≤ 3.06	$2.7 < L_{CRB} \cdot 12 \cdot SCS < 4.5$	A2
		≤ 3.06	≥ 4.5	A1
		≥ 3.06	≥ 5.4	A2
	2012.5	≤ 1.44	≤ 2.7	A1
		≤ 1.44	$2.7 < L_{CRB} \cdot 12 \cdot SCS < 5.4$	A2
		≤ 1.44	≥ 5.4	A1
		≥ 1.44	≥ 7.2	A2
20 MHz	2010	≤ 4.68	≤ 2.7	A1
		≤ 4.68	$2.7 \leq L_{CRB} \cdot 12 \cdot SCS \leq 5.4$	A2
		≤ 4.68	> 5.4	A1
		> 4.68	≥ 7.2	A2

Table C.1.2.2-4: A-MPR for -30 dBm/MHz

Modulation/Waveform	A1	A2
	Outer/Inner	Outer/Inner
DFT-s-OFDM PI/2 BPSK	≤ 4	≤ 2.5
DFT-s-OFDM QPSK	≤ 4.5	≤ 3
DFT-s-OFDM 16 QAM	≤ 5	≤ 3.5
DFT-s-OFDM 64 QAM	≤ 5.5	≤ 4
CP-OFDM QPSK	≤ 6	≤ 4.5
CP-OFDM 16 QAM	≤ 6	≤ 4.5
CP-OFDM 64 QAM	≤ 6	≤ 4.5

C.1.2.3 Power back-off simulation from company C (R4-2506521)

C.1.2.3.1 Simulation parameters

The following are the parameters used for this simulation:

- Operating band n252 (UL 2000 MHz – 2020 MHz)
- 10 MHz, 15 MHz, and 20 MHz CBWs
- Power class PC3
- WOLA processing
- I/Q image -28 dBc
- Carrier leakage -28 dBc
- CIM3 -60 dBc
- Only additional spurious emission mask from NS_07N is simulated. Other gating factors (ACLR, SEM, etc.) are excluded because they are handled by MPR.

C.1.2.3.2 NS_07N mask

Table C.1.2.3.2-1: Additional out-of-band requirements for "NS_07N"

Frequency range (MHz)	Channel bandwidth / Spectrum emission limit (dBm)	Measurement bandwidth	NOTE
	5 MHz, 10 MHz, 15 MHz, 20MHz		
$1559 \leq f < 1605$	-50	700 Hz	Averaged over any 2 millisecond active transmission interval
$1559 \leq f < 1605$	-40	1MHz	
$1930 \leq f \leq 1995$	-40	1MHz	

Table C.1.2.3.2-1 shows the NS_07N mask. This table is aligned with the agreed emission mask as described in the Table 7.4.2.2.8-3. Only the $1930 \text{ MHz} \leq f \leq 1995 \text{ MHz}$ segment is considered because the other segments are very far from the channel.

C.1.2.3.3 Simulation result

Figure C.1.2.3.3-1 and C.1.2.3.3-2 present the examples of the A-MPR triangles for 10 MHz and 20 MHz CBWs, respectively. The maximum A-MPR at 20 MHz CBW is close to the average between the two results in C.1.2.1 and C.1.2.2. Details results of the A-MPR triangles can be seen from the appendix of R4-2506521.

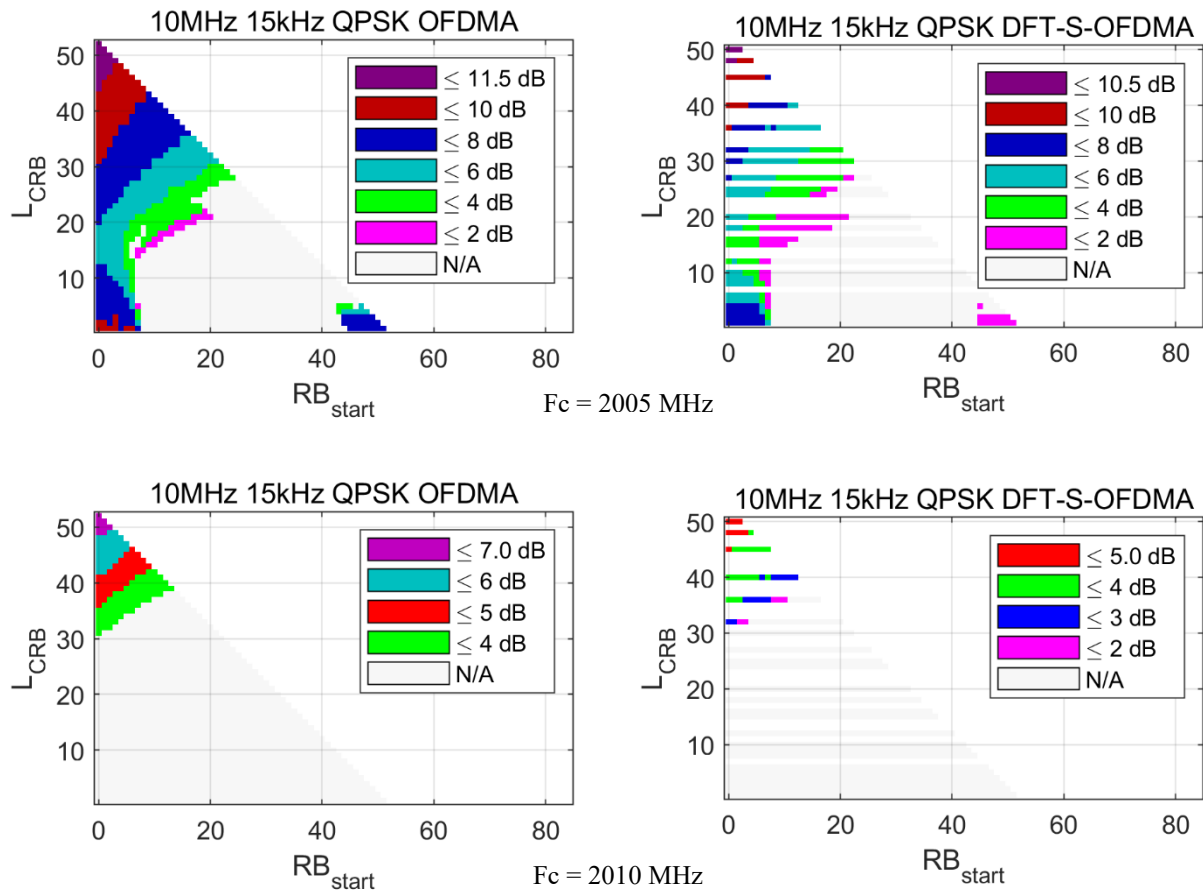


Figure C.1.2.3.3-1: Examples of the simulated A-MPR to satisfy the -40 dBm/MHz additional spurious emission limit for 10 MHz channel at $F_c = 2005$ MHz (upper row) and $F_c = 2010$ MHz (bottom row).

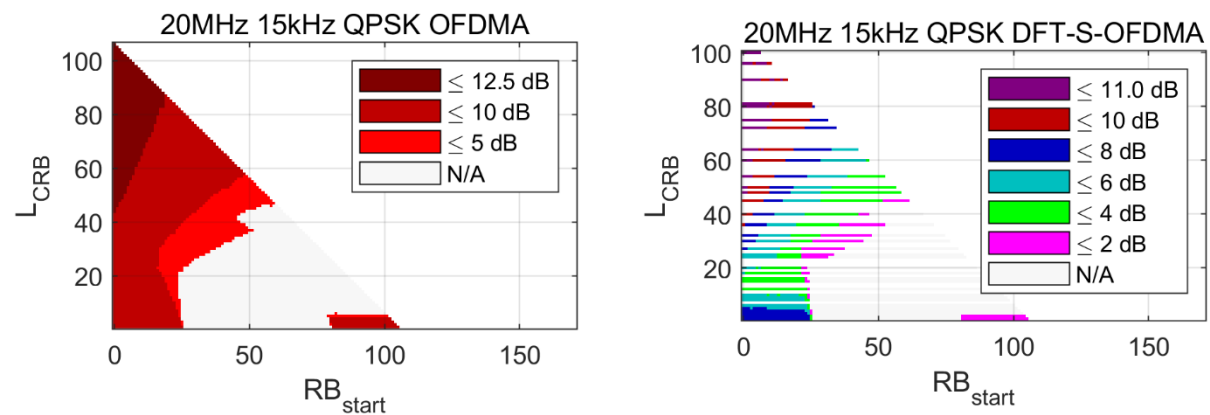


Figure C.1.2.3.3-2: Examples of the simulated A-MPR to satisfy the -40 dBm/MHz additional spurious emission limit for 20 MHz channel at $F_c = 2010$ MHz.

C.1.2.3.4 A-MPR proposal

Using with the result in C.1.2.1 and C.1.2.2 as a comparison, the result that differ to either of the two proposals are highlighted with yellow color. In this simulations the maximum A-MPR values at 10 MHz do not differ much from the corresponding ones at higher CBWs, therefore the A2 region at 10 MHz is changed to A3 to better optimize the remaining A2 values. If the region at 10 MHz CBW is kept as A2, then the A2 values need to be increased.

Table C.1.2.3.4-1: A-MPR regions for -40 dBm/MHz

Channel BW	Carrier Frequency, F_c , MHz	$RB_{Start} \cdot 12 \cdot SCS$ (MHz)	$L_{CRB} \cdot 12 \cdot SCS$ (MHz)	A-MPR
10MHz	< 2010	≤ 1.44	≥ 0	A3
		> 1.44	≥ 3.6	A1
		> 7.74	≤ 0.72	A1
	$2010 \leq F_c \leq 2015$	≥ 0	≥ 4.5	A1
15MHz	< 2012.5	≤ 3.06	≥ 0	A3
		> 3.06	$\geq 4.5, \leq 7.2$	A1
		> 3.06	> 7.2	A2
		> 10.98	≤ 0.72	A1
	2012.5	≤ 1.44	≥ 0	A2
		> 1.44	≥ 6.48	A1
		> 12.6	≤ 0.72	A1
20 MHz	2010	≤ 4.68	≥ 0	A3
		> 4.68	$\geq 4.5, \leq 9$	A1
		> 4.68	> 9	A2
		> 12.6	≤ 0.72	A1

Table C.1.2.3.4-2: A-MPR for -40 dBm/MHz

Modulation/Waveform	A1	A2	A3
	Outer/Inner	Outer/Inner	Outer/Inner
DFT-s-OFDM PI/2 BPSK	≤ 4	≤ 8	≤ 10.5
DFT-s-OFDM QPSK	≤ 4.5	≤ 8	≤ 11
DFT-s-OFDM 16 QAM	≤ 5	≤ 8	≤ 11.5
DFT-s-OFDM 64 QAM	≤ 5.5	≤ 8	≤ 11.5
DFT-s-OFDM 256 QAM	≤ 5.5	≤ 8	≤ 11.5
CP-OFDM QPSK	≤ 7	≤ 8.5	≤ 12.5
CP-OFDM 16 QAM	≤ 7	≤ 8.5	≤ 12.5
CP-OFDM 64 QAM	≤ 7	≤ 8.5	≤ 12.5
CP-OFDM 256 QAM	≤ 7	≤ 8.5	≤ 12.5

Annex D:

Summary of simulation results for the IoT derivation

In this annex, simulations results used to derive the IoT value were collected.

[To be updated]

Annex E: Examples of Delay and Doppler shift variation as a function of time for different satellite orbit types

The purpose of Annex E is only informative and is related to the distance variation between NTN UE and Satellite as captured in Section 5.3 of TR 38.811 [5] which presents methodology to characterise Doppler and propagation Delay.

Therefore, the information in current section shall not be interpreted as testing configuration. The purpose is only to provide a more realistic representation of Doppler and Delay with respect to a few constellations given as examples.

As described below, Delay and Doppler shift variation with time are presented as examples for different defined NGSO (LEO@600km, LEO@1200km) and GSO satellite representative orbits. The NTN UE location is assumed static for all graphs as follows:

- Latitude: 25.08439333 deg;
- Longitude: 121.56076999 deg;
- Altitude: 0 m.

Three representative cases are further defined in following sub-sections E.1, E.2 and E.3 respectively:

- Low Earth Orbit (LEO) at 600 km altitude case;
- Low Earth Orbit (LEO) at 1200 km altitude case;
- GSO orbit case with a satellite inclination of 7 degrees relative to the Equator.

E.1 LEO 600 km case

For this example, a satellite orbit inclination with respect of the Equator plane of 88° is considered.

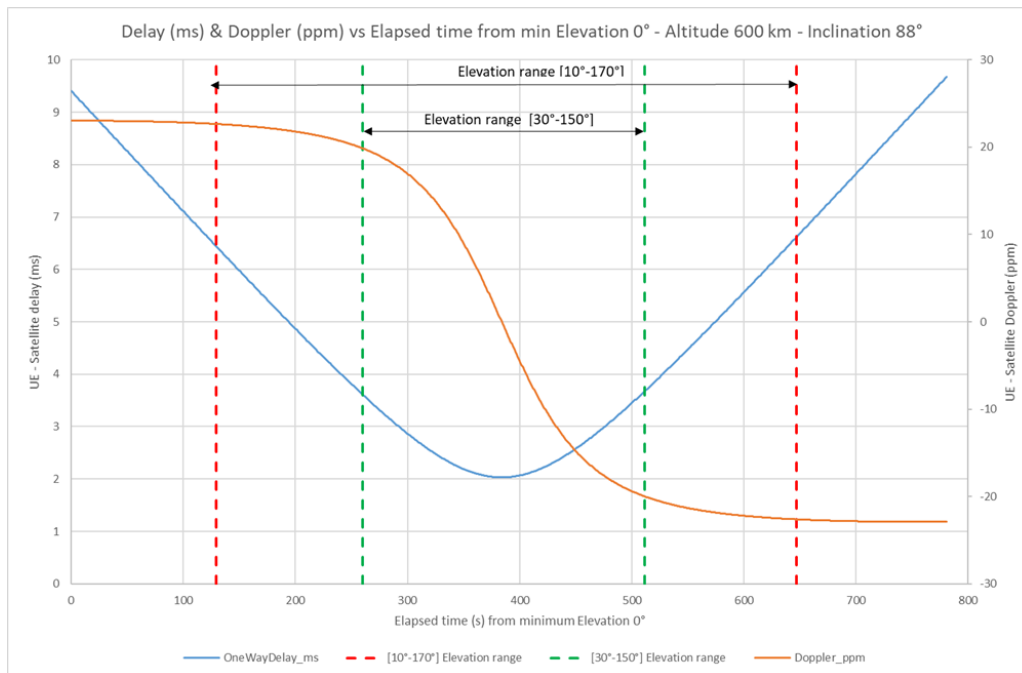


Figure E.1-1: LEO600km with 88° orbit inclination

E.2 LEO 1200 km case

For this example, a satellite orbit inclination with respect of the Equator plane of 88° is considered.

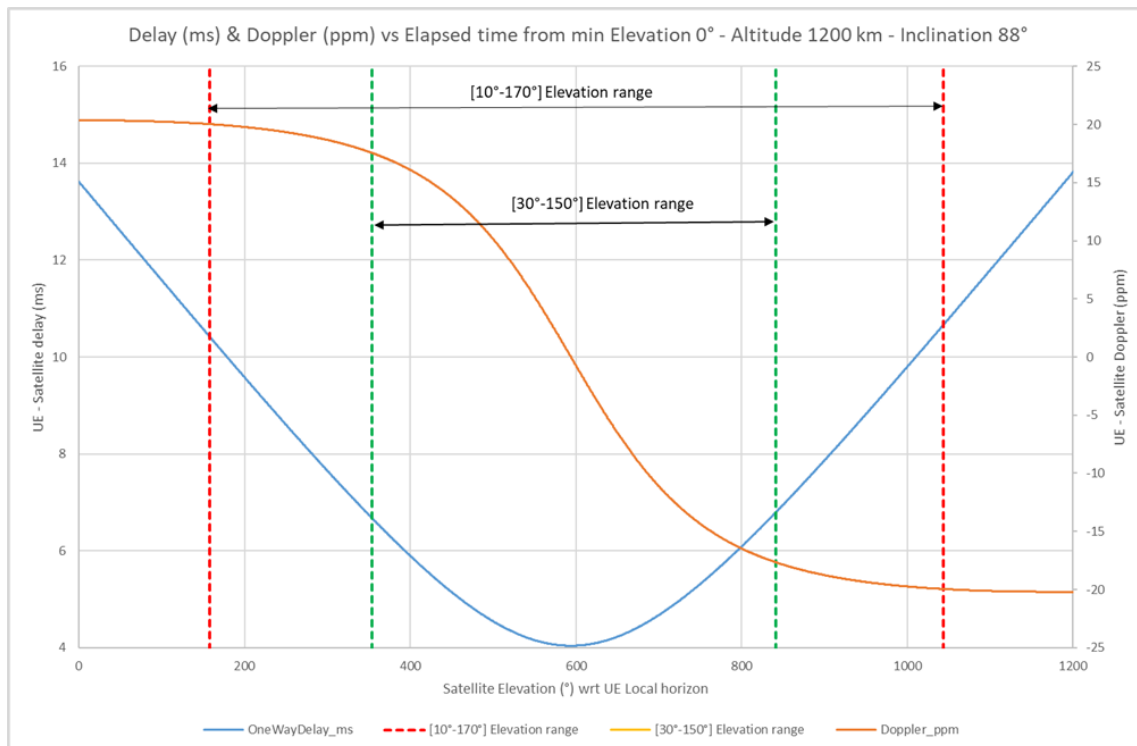


Figure E.2-1: LEO1200km with 88° orbit inclination

E.3 GSO orbit case

For this example, a satellite orbit inclination with respect of the Equator plane of 7° is considered.

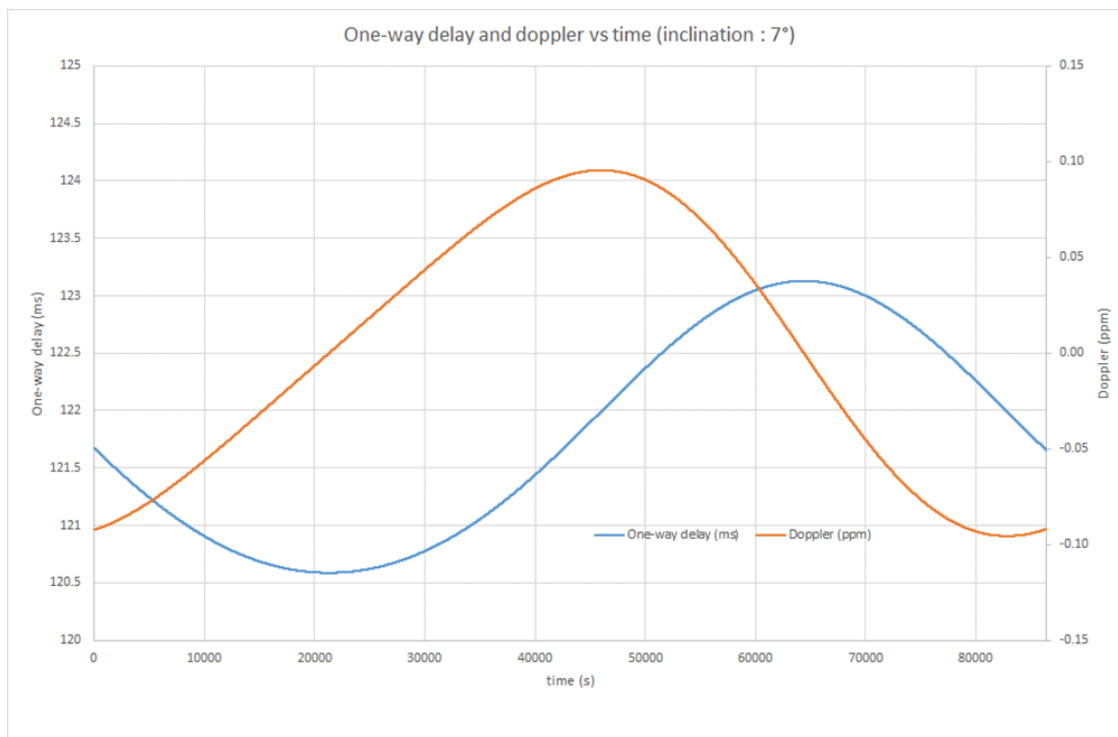


Figure E.3-1: GSO with 7° orbit inclination

E.4 Examples of Doppler and Delay for different elevation angles and for different satellite orbit types

One can represent the following few “selected” example scenarios computed for a min elevation angle target of 30°:

Table E.4-1: Example scenarios with values computed for 30° min elevation angle target

Frequency	Constellation	Elevation	Doppler shift (kHz)	OneWayDelay_ms
2,00 GHz	LEO@600 km	30,0	39,734	3,6
2,00 GHz	LEO@600 km	149,9	-39,899	3,7
2,00 GHz	LEO@600 km	89,6	0,026	2,0
2,00 GHz	LEO@1200 km	29,9	35,127	6,7
2,00 GHz	LEO@1200 km	149,9	-35,273	6,8
2,00 GHz	LEO@1200 km	89,5	0,007	4,0
30,00 GHz	LEO@600 km	30,0	596,011	3,6
30,00 GHz	LEO@600 km	149,9	-598,486	3,7
30,00 GHz	LEO@600 km	89,6	0,384	2,0
30,00 GHz	LEO@1200 km	29,9	526,908	6,7
30,00 GHz	LEO@1200 km	149,9	-529,095	6,8
30,00 GHz	LEO@1200 km	89,5	0,110	4,0
2,00 GHz	GEO	90,14	0,000	119,37
2,00 GHz	GSO	58,85	0,19	121,99
2,00 GHz	GSO	58,85	-0,19	121,99
2,00 GHz	GSO	68,87	0,000	120,59
30,00 GHz	GEO	90,14	0,000	119,37
30,00 GHz	GSO	58,85	2,87	121,99
30,00 GHz	GSO	58,85	-2,87	121,99
30,00 GHz	GSO	68,87	0,000	120,59

Similarly, one can represent the following few “selected” example scenarios computed for a min elevation angle target of 10°:

Table E.4-2: Example scenarios with values computed for 10° min elevation angle target

Frequency	Constellation	Elevation	Doppler shift (kHz)	OneWayDelay_ms
2,00 GHz	LEO@600 km	9,9	45,304	6,5
2,00 GHz	LEO@600 km	170,0	-45,239	6,6
2,00 GHz	LEO@600 km	89,6	0,026	2,0
2,00 GHz	LEO@1200 km	10,0	40,062	10,4
2,00 GHz	LEO@1200 km	170,0	-39,918	10,7
2,00 GHz	LEO@1200 km	89,5	0,007	4,0
30,00 GHz	LEO@600 km	9,9	679,567	6,458
30,00 GHz	LEO@600 km	170,0	-678,582	6,605
30,00 GHz	LEO@600 km	89,6	0,384	2,031
30,00 GHz	LEO@1200 km	10,0	600,927	10,441
30,00 GHz	LEO@1200 km	170,0	-598,774	10,656
30,00 GHz	LEO@1200 km	89,5	0,110	4,039
2,00 GHz	GEO	90,14	0,000	119,37
2,00 GHz	GSO	58,85	0,19	121,99
2,00 GHz	GSO	58,85	-0,19	121,99
2,00 GHz	GSO	68,87	0,000	120,59
30,00 GHz	GEO	90,14	0,000	119,37
30,00 GHz	GSO	58,85	2,87	121,99
30,00 GHz	GSO	58,85	-2,87	121,99
30,00 GHz	GSO	68,87	0,000	120,59

Note: UE position was considered as stationary and configured with Latitude: 25.08439333, Longitude: 121.56076999, Altitude: 0.

Annex F:

Power back-off simulation results for the S-band

F.1 PC3 power class

F.1.1 General aspects

NOTE: The purpose of the section is to capture general aspects, such as considered UE RF architectures, assumptions, considered regulations, etc.

F.1.1.1 NS_24N

F.1.1.1.1 Company A

F.1.1.1.1.1 Simulation assumptions

The power back-off simulations are performed for the PC3 power class and the 3MHz channel with the following common assumptions:

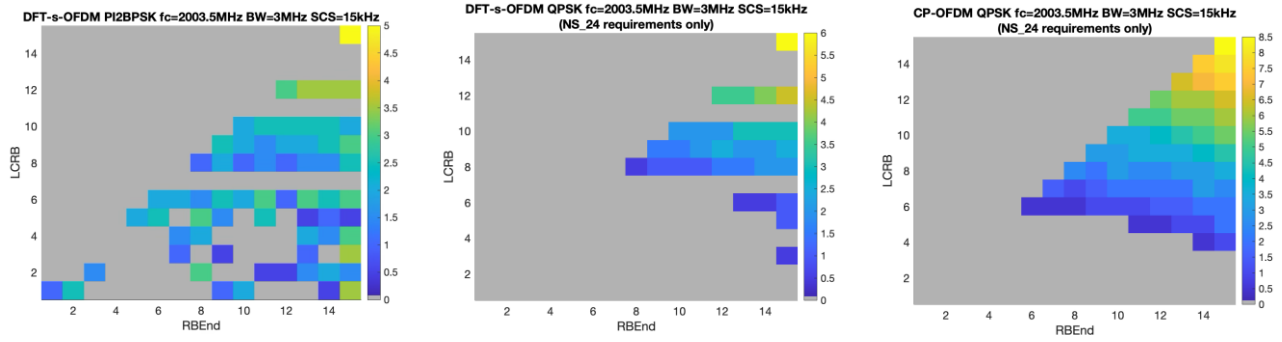
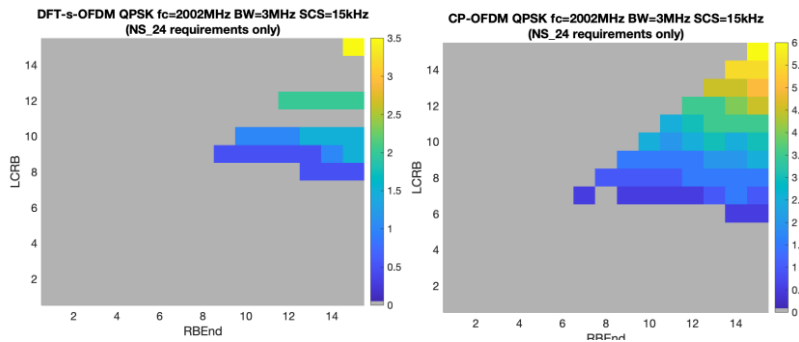
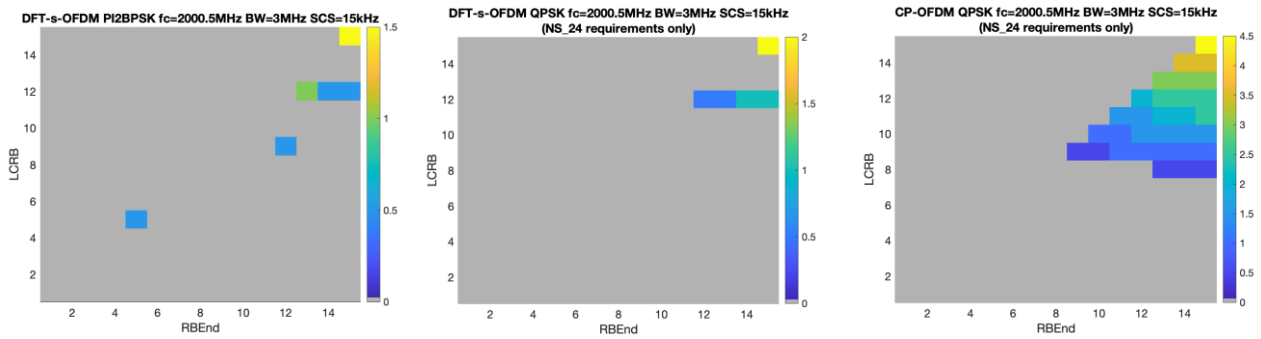
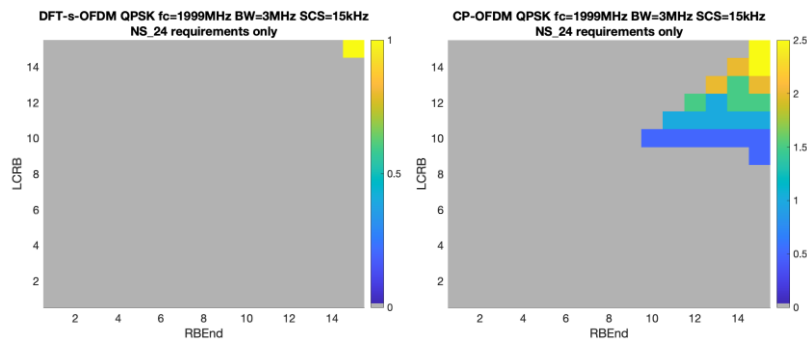
- Tx power: PC3 (+23dBm)
- Calibration: 1dB MPR with 30dBc ACLR and DFT-s-OFDM QPSK 20MHz 100RB0
- Carrier leakage: -28dBc
- Image: -28dBc
- CIM3: -60dBc
- CIM5: -70dBc
- LO placement: always in the centre of the carrier
- Additional requirements: -50dBm/MHz to protect the $2010 \leq f \leq 2025$ MHz frequency range

The power back-off simulations were conducted for the following center frequencies:

- the right-most channel with the 5MHz offset, as "guard" band, from the upper edge of the band ($F_c=2003.5$ MHz)
- the upper channel with the 6.5MHz offset from the upper edge ($F_c=2002$ MHz)
- the upper channel with the 8MHz offset from the upper edge ($F_c=2000.5$ MHz)
- the upper channel with the 9.5MHz offset from the upper edge ($F_c=1999$ MHz)

F.2.1.1.1.2 Simulation results

Figures below present the power back-off simulation results for the 3MHz channel at three different locations as mentioned in the previous sub-clause. As anticipated, the upper most channel has the highest power back-off values. For instance, the full channel allocation of 15RB might require 6dB power back-off. And as the channel shifts to the left from the upper edge of the band, the power back-off decreases as well.

Figure F.2.1.1.2-1: Power back-off results at $F_c=2003.5\text{MHz}$.Figure F.2.1.1.2-2: Power back-off results at $F_c=2002\text{MHz}$.Figure F.2.1.1.2-3: Power back-off results at $F_c=2000.5\text{MHz}$.Figure F.2.1.1.2-4: Power back-off results at $F_c=1999\text{MHz}$.

F.1.1.1.2 Company B

F.1.1.1.2.1 Simulation assumptions

For the power back-off simulations the PA was calibrated with the following assumptions:

- PC3 23dBm
- 1dB MPR: DFT-s-OFDM QPSK 20MHz, 100RB
- Carrier leakage of 28dBc, image of 28dBc, CIM3 of 60dBc, and CIM5 of 70dBc
- ASE and A-SEM, SEM and ACLR are considered as gating factors (EVM was excluded)

F.1.1.1.2.2 Simulation results

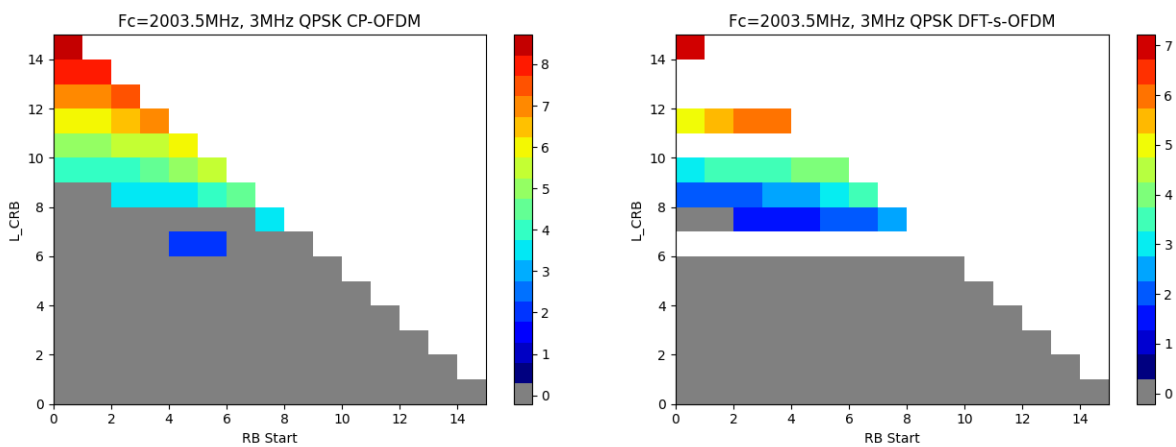


Figure F.2.1.1.2.2-1: A-MPR evaluations of the 3MHz channel at Fc=2003.5 MHz

F.1.1.1.3 Company C

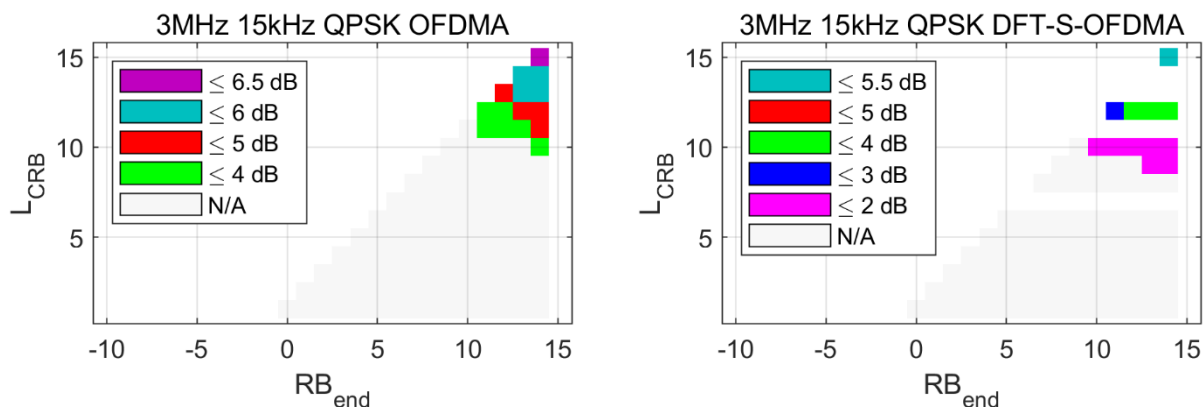
F.1.1.1.3.1 Simulation assumptions

The following assumptions are used for the power back-off simulations:

- Power class PC3
- WOLA processing
- I/Q image -28 dBc
- Carrier leakage -28 dBc
- CIM3 -60 dBc, CIM5 -70 dBc
- Additional requirements are -50dBm/MHz to protect the $2010 \leq f \leq 2025$ MHz frequency range

Only additional spurious emission mask from NS_24 is simulated. Other gating factors (ACLR, SEM, etc.) are excluded because they are covered by the existing MPR requirements.

F.1.1.1.3.2 Simulation results

Figure F.1.1.1.3.2-1: Simulated power back-off for the 3MHz channel at $F_c=2003.5$ MHz.

F.2 PC2 power class

F.2.1 General aspects

NOTE: The purpose of the section is to capture general aspects, such as considered UE RF architectures, assumptions, considered regulations, etc.

F.2.1.1 NS_24N

F.2.1.1.1 Company A (MediaTek/R4-2509854)

F.2.1.1.1.1 Simulation assumptions F.2.1.1.1.2 Simulation results

Table below summarises proposed A-MPR values for the existing A-MPR regions, i.e., same regions that already exist for the PC3 power class.

Table F.2.1.1.1.2-1: Proposed A-MPR regions and values

Modulation/Waveform	A1	A2	A3	A4	A5	A6	A7
	Outer/ Inner	Outer/ Inner	Outer/ Inner	Outer	Outer/ Inner	Outer/ Inner	Outer
DFT-s-OFDM PI/2 BPSK	≤ 12.5	≤ 7	≤ 4.5	≤ 8.5	≤ 19.6	≤ 11.7	≤ 3.5
DFT-s-OFDM QPSK	≤ 12.5	≤ 7	≤ 4.5	≤ 8.5	≤ 19.6	≤ 11.7	≤ 3.5
DFT-s-OFDM 16 QAM	≤ 13.3	≤ 7.1	≤ 5.2	≤ 8.5	≤ 21	≤ 13.0	≤ 3.5
DFT-s-OFDM 64 QAM	≤ 13.3	≤ 7.1	≤ 5.2	≤ 8.5	≤ 21	≤ 13.5	≤ 3.5
DFT-s-OFDM 256 QAM	≤ 13.3	≤ 7.1		≤ 8.5	≤ 21	≤ 13.5	
CP-OFDM QPSK	≤ 13.9	≤ 7.3	≤ 5.6	≤ 8.5	≤ 22.1	≤ 15.5	≤ 5.5
CP-OFDM 16 QAM	≤ 13.9	≤ 7.3	≤ 5.6	≤ 8.5	≤ 22.6	≤ 15.9	≤ 5.5
CP-OFDM 64 QAM	≤ 13.9	≤ 7.3	≤ 5.6	≤ 8.5	≤ 22.6	≤ 15.9	≤ 5.5
CP-OFDM 256 QAM	≤ 13.9	≤ 7.3		≤ 8.5	≤ 22.6	≤ 16.1	

F.2.1.1.2 Company B (Qualcomm/R4-2511400)

F.2.1.1.2.1 Simulation assumptions

The following simulation assumptions are used:

- Tx power: PC3 (23dBm)
- PA calibrated to 1dB MPR with 30dBc ACLR with DFT-s-OFDM QPSK 100 RB waveform
- -28 dBc IQ-image
- -28 dBc LO-leakage
- -60 dBc CIM3 and -70 dBc CIM5

F.2.1.1.2.2 Simulation results

In the table below, A-MPR regions A to C are the existing regions from the PC3 power, and region D is a new A-MPR region.

Table F.2.1.1.2.2-1: Proposed A-MPR regions and values

Channel Bandwidth, MHz	Carrier Centre Frequency, Fc, MHz	Region A			Region B			Region C			Region D		
		RB _{end} *12*SCS MHz	LCRB*12*SCS MHz	A-MPR	RB _{end} *12*SCS MHz	LCRB*12*SCS MHz	A-MPR	RB _{end} *12*SCS MHz	LCRB*12*SCS MHz	A-MPR	RB _{end} *12*SCS MHz	LCRB*12*SCS MHz	A-MPR
5MHz	1987.5 < Fc ≤ 1992.5		>3.24	A7									
5MHz	1992.5 < Fc ≤ 1997.5		>3.24	A4		≤ 3.24	A8						
5MHz	1997.5 < Fc ≤ 2002.5		>1.98	A1	>3.6	>1.08 ≤1.98	A2	≤3.6	≤1.98	A3			
						≤1.08	A6						
10MHz	Fc = 1985	>5.4		A4		≥ 4.5	A8						
10MHz	1985 ≤ Fc ≤ 1995		>4.32	A1	≥7.2	>1.08 ≤4.32	A2	<7.2	≤4.32	A3			
						≤1.08	A6						
10MHz	1995 < Fc ≤ 2000	≥5.76		A5	<3.06		A5	≥3.06 <5.76	>1.44	A6	≥3.06 <5.76	≤1.44	A8
15MHz	Fc = 1987.5		>6.84	A1	≥10.8	>1.08 ≤6.84	A2	<10.8	≤6.84	A3			
						≤1.08	A6						
15MHz	1987.5 < Fc ≤ 1997.5	≥8.64		A5	<4.04		A5	≥4.04 <8.64	>1.44	A6	≥4.04 <8.64	≤1.44	A8
20MHz	Fc = 1990	≥12.96		A5	<4.68		A5	≥4.68 <12.96	>2.16	A6	≥ 4.68 < 12.96	≤2.16	A8
20MHz	1990 < Fc ≤ 1995	≥11.52		A5	<5.58		A5	≥5.58 <11.52	>1.44	A6	≥ 5.58 < 11.52	≤1.44	A8

Modulation/Waveform	A1	A2	A3	A4	A5	A6	A7	A8
	Outer/ Inner	Outer/ Inner	Outer/ Inner	Outer/ Inner	Outer/ Inner	Outer/ Inner	Outer/ Inner	Outer/ Inner
DFT-s-OFDM PI/2 BPSK	≤ 14	≤ 8	≤ 5.5	≤ 8.5	≤ 21	≤ 13	≤ 4.5	≤ 3
DFT-s-OFDM QPSK	≤ 14	≤ 8	≤ 5.5	≤ 8.5	≤ 21	≤ 13	≤ 4.5	≤ 3
DFT-s-OFDM 16 QAM	≤ 14	≤ 8	≤ 5.5	≤ 8.5	≤ 21	≤ 13	≤ 4.5	≤ 4
DFT-s-OFDM 64 QAM	≤ 14	≤ 8	≤ 6	≤ 8.5	≤ 22	≤ 13	≤ 4.5	≤ 4.5
CP-OFDM QPSK	≤ 15	≤ 9	≤ 6.5	≤ 8.5	≤ 22	≤ 15	≤ 5.5	≤ 4.5
CP-OFDM 16 QAM	≤ 15	≤ 9	≤ 6.5	≤ 8.5	≤ 22	≤ 15	≤ 5.5	≤ 5
CP-OFDM 64 QAM	≤ 15	≤ 9	≤ 6.5	≤ 8.5	≤ 22	≤ 15	≤ 5.5	≤ 5

F.2.1.1.3 Company C (R4-2510521/OPPO)

F.2.1.1.3.1 Simulation assumptions

The simulation assumptions are as follows:

- PC2 power class with 1Tx
- 5,10, 15 and 20MHz channels residing at the higher edge of the uplink operating band n256 and keeping 5MHz protecting guard gap between the higher edge of the channel bandwidth and the highest edge of the uplink operating band.
- CP-OFDM and DFT-s-OFDM

F.2.1.1.3.2 Simulation results

To assess PC2 power back-off values and potential A-MPR regions, PC2 CP-OFDM QPSK power back-off values are compared to the corresponding results obtained for the PC3 power class under the existing A-MPR regions. Based on that the following observations are made:

- The PC3 power back-off simulations results have more margins compared the current region frame and related AMPR values in existing specification.
- Some regions for PC2 are extended compared with PC3 due to higher output power, i.e., A5 region for 10MHz, 15MHz and 20 MHz, A1 region for 5MHz and 10MHz, A4 region for 5MHI.
- The power back-off value of A6 region in F.2.1.1.3.2-1 is 5dB smaller than the value of A6 region in F.2.1.1.3.2-2.

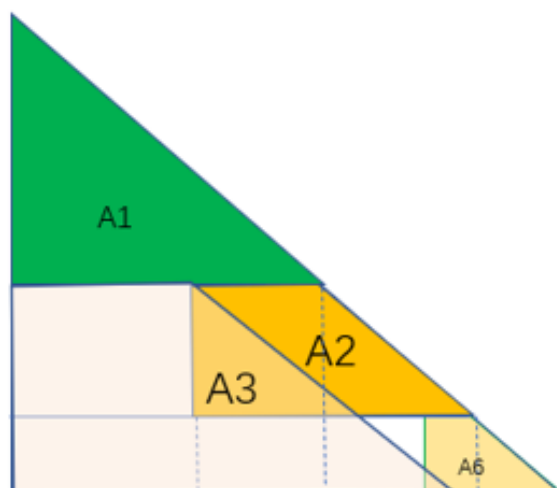


Figure F.2.1.1.3.2-1: A-MPR regions for the for 5MHz channel at $1997.5 < F_c \leq 2002.5$ and 10MHz channel at $1985 < F_c \leq 1995$

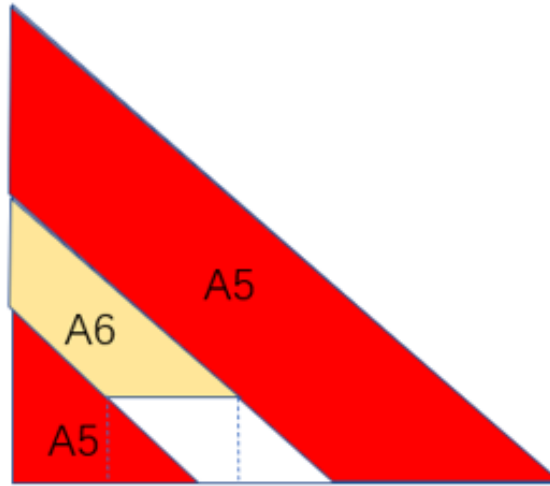


Figure F.2.1.1.3.2-2: A-MPR regions for the for 10MHz at $1995 < F_c \leq 2000$, 15MHz and 20MHz.

According to the simulation results, A-MPR regions for NS_24N can reuse same regions as the existing PC3 regions. However, some region ranges should be extended compared to PC3, i.e., A5 region for 10MHz, 15MHz and 20 MHz, A1 region for 5MHz and 10MHz, A4 region for 5MHz. It is also necessary to a new region name for the A6 region in Figure F.2.1.1.3.2-2 for 10MHz, 15MHz, and 20MHz to distinguish with the A6 region in F.2.1.1.3.2-1 for 5MHz and 10MHz, i.e., A8 replace A6.

In the figures below further results are presented for NS_24N with PC2 10MHz at $F_c=2000\text{MHz}$ under different modulations with CP-OFDM and DFT-s-OFDM waveforms, whereupon the simulation results are based on fixed bias PA.

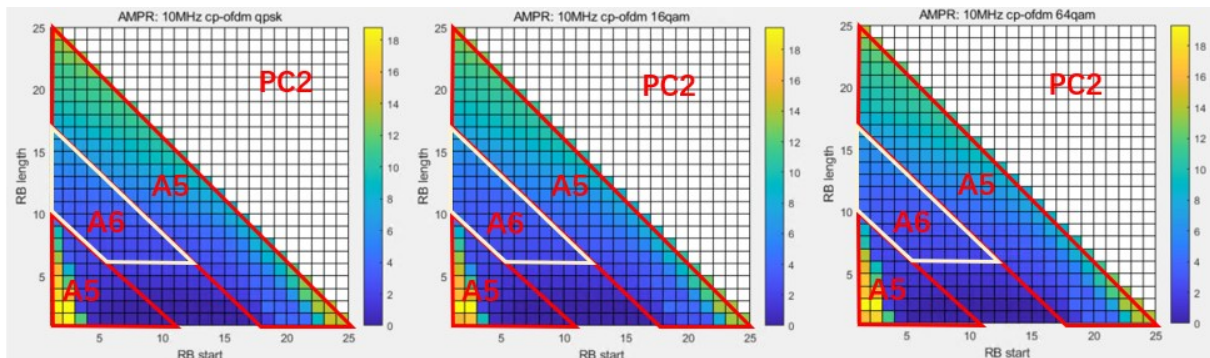


Figure F.2.1.1.3.2-3: AMPR for PC2 10MHz CP-OFDM under different modulations.

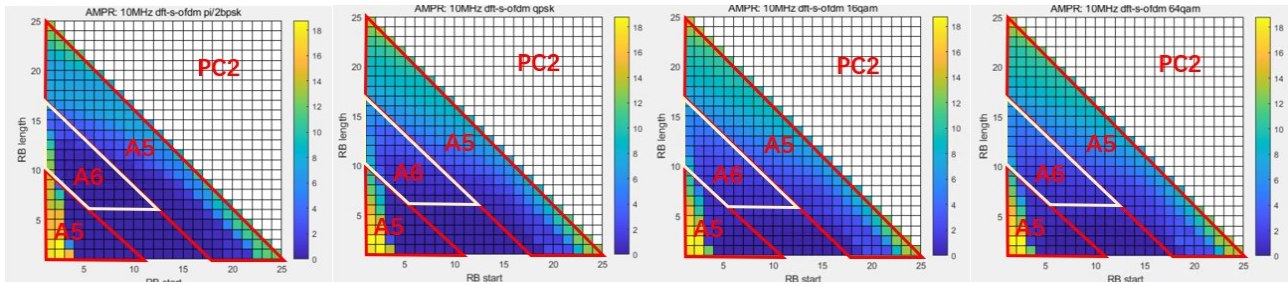


Figure F.2.1.1.3.2-4: AMPR for PC2 10MHz DFT-s-OFDM under different modulations.

Table F.2.1.1.3.2-1 proposes A-MPR regions and power back-off values, which are based on the existing PC3 regions.

Table F.2.1.1.3.2-1: Proposed A-MPR values and regions based on existing PC3 A-MPR regions

Channel Bandwidth, MHz	Carrier Centre Frequency, F_c , MHz	Region A			Region B			Region C		
		$RB_{end} \cdot 12^*$ SCS MHz	$LCRB \cdot 12^*$ SCS MHz	A-MPR	$RB_{end} \cdot 12^* S$ CS MHz	$LCRB \cdot 12^*$ SCS MHz	A-MPR	$RB_{end} \cdot 12^* S$ CS MHz	$LCRB \cdot 12^*$ SCS MHz	A-MPR
5MHz	$1987.5 < F_c \leq 1992.5$		>3.24	A7						
5MHz	$1992.5 < F_c \leq 1997.5$		>3.24	A4						
5MHz	$1997.5 < F_c \leq 2002.5$		>1.98	A1	>3.6	>1.08 ≤ 1.98	A2	≤ 3.6	≤ 1.98	A3
						≤ 1.08	A6			
10MHz	$1985 < F_c \leq 1995$		>4.32	A1	≥ 7.2	>1.08 ≤ 4.32	A2	< 7.2	≤ 4.32	A3
						≤ 1.08	A6			
10MHz	$1995 < F_c \leq 2000$	≥ 5.76		A5	< 3.06		A5	≥ 3.06 < 5.76	> 1.44	A6
15MHz	$1987.5 < F_c \leq 1997.5$	≥ 8.64		A5	< 3.78		A5	≥ 3.78 < 8.64	> 1.44	A6
20MHz	$1990 < F_c \leq 1995$	≥ 11.52		A5	< 5.58		A5	≥ 5.58 < 11.52	> 1.44	A6

Modulation/ Waveform	A1	A2	A3	A4	A5	A6	A7
	Outer/Inner	Outer/Inner	Outer/Inner	Outer	Outer/Inner	Outer/Inner	Outer
CP-OFDM QPSK	16	12	11	9.5	26	21	8.5
CP-OFDM 16QAM	16	12	11	9.5	26	21	8.5
CP-OFDM 64QAM	16	12	11	9.5	26	21	8.5
DFT-s-OFDM pi/2 BPSK	15	11	10	8.5	26	20	7.5
DFT-s-OFDM QPSK	15	11	10	8.5	26	20	7.5
DFT-s-OFDM 16QAM	15	11	10	8.5	26	20	7.5
DFT-s-OFDM 64QAM	15	11	10	8.5	26	20	7.5

If we use the modified regions as shown in Table F.2.1.1.3.2-2 for PC2 AMPR, we can see that the regions of A1, A4 and A5 are extended, but regions A2, A3 and A8 are reduced.

Table F.2.1.1.3.2-2: Proposed values based on modified AMPR regions

Channel Bandwidth, MHz	Carrier Centre Frequency, Fc, MHz	Region A			Region B			Region C		
		RB _{end} *12* SCS MHz	LCRB*12* SCS MHz	A-MPR	RB _{end} *12*S CS MHz	LCRB*12* SCS MHz	A-MPR	RB _{end} *12*S CS MHz	LCRB*12* SCS MHz	A-MPR
5MHz	1987.5 < Fc ≤ 1992.5		>3.24	A7						
5MHz	1992.5 < Fc ≤ 1997.5		> 2.16	A4						
5MHz	1997.5 < Fc ≤ 2002.5		> 1.44	A1	> 2.7	>1.08 ≤ 1.44	A2	≤2.7	≤ 1.44	A3
						≤1.08	A6			
10MHz	1985 < Fc ≤ 1995		> 3.24	A1	≥ 5.4	> 2.16 ≤ 3.24	A2	< 5.4	≤ 3.24	A3
						≤ 2.16	A6			
10MHz	1995 < Fc ≤ 2000	≥ 5.4		A5	< 3.42		A5	≥ 3.42 <5.4	>1.08	A8
15MHz	1987.5 < Fc ≤ 1997.5	≥8.64		A5	< 5.94		A5	≥ 5.94 <8.64	>1.44	A8
20MHz	1990 < Fc ≤ 1995	≥11.52		A5	<7.74		A5	≥7.74 <11.52	>1.44	A8

Modulation/ Waveform	A1	A2	A3	A4	A5	A6	A8	A7
	Outer/Inner	Outer/Inner	Outer/Inner	Outer	Outer/Inner	Outer/Inner	Outer	Outer
CP-OFDM QPSK /16QAM/64QAM	≤ 16	≤ 10	≤ 10	≤ 9.5	≤ 26	≤ 12	≤ 17	≤ 8.5
DFT-s-OFDM pi/2 BPSK /QPSK/16QAM/64QAM	≤ 15	≤ 9	≤ 9	≤ 8.5	≤ 26	≤ 11	≤ 16	≤ 7.5

F.2.1.1.4 Company D (R4-2511269/Nokia)

F.2.1.1.4.1 Simulation assumptions

The following simulation assumptions are made:

- Operating band n256 (UL 1980 MHz – 2010 MHz)
- 5MHz, 10 MHz, 15 MHz, and 20 MHz channel bandwidth
- Power class PC2
- WOLA processing
- I/Q image -28 dBc (default according to 38.101-1, Table 6.4.2.3-1) for other modulations than 256QAM
- For 256QAM the I/Q image is -33.7 dBc as in MPR simulations.
- Carrier leakage -28 dBc (default according to 38.101-1, Table 6.4.2.3-1)
- CIM3 -60 dBc
- CIM5 -70 dBc
- Channel noise 38.5 dBc
- PA noise -75 dBm/MHz
- Only additional spurious emission mask from NS_24 is simulated. Other gating factors (ACLR, SEM, etc.) are excluded because they are handled by MPR.

- Phase noise and channel noise are only added for QAM256, following LTE 256-QAM MPR simulations.

F.2.1.1.4.2 Simulation results

The following figures provide simulation results for different channel sizes and different locations for PI/2 BPSK, DFT-s-OFDM QPSK and CP-OFDM QPSK.

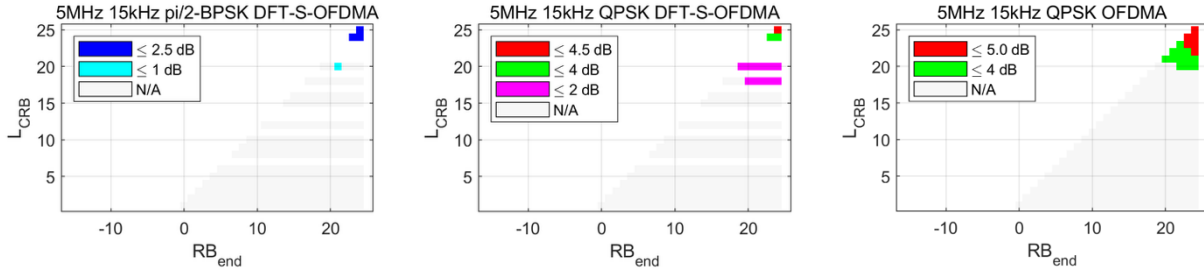


Figure F.2.1.1.4.2-1: 5MHz results at $F_c=1997.5\text{MHz}$

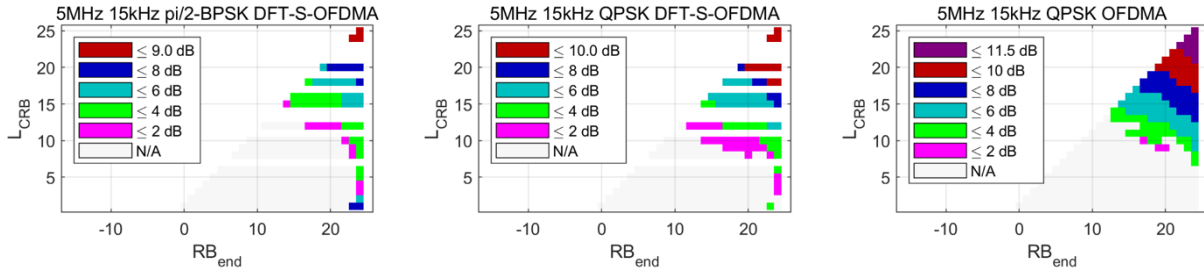


Figure F.2.1.1.4.2-2: 5MHz results at $F_c=2002.5\text{MHz}$

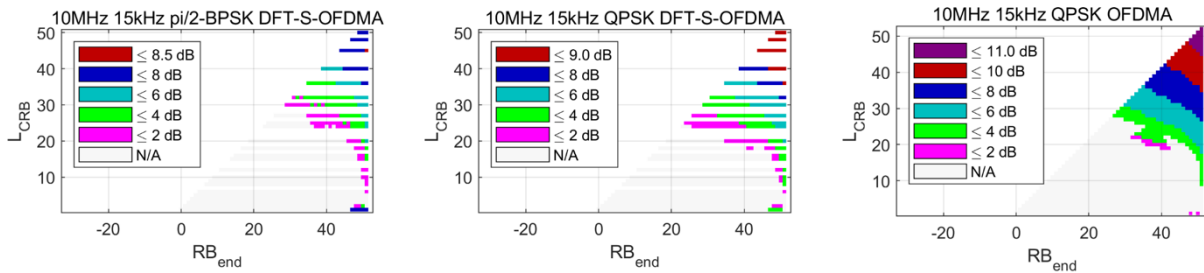


Figure F.2.1.1.4.2-3: 10MHz results at $F_c=1995\text{MHz}$

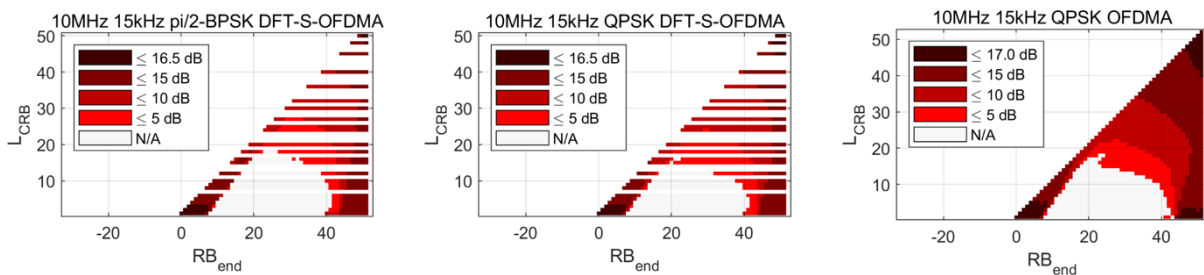
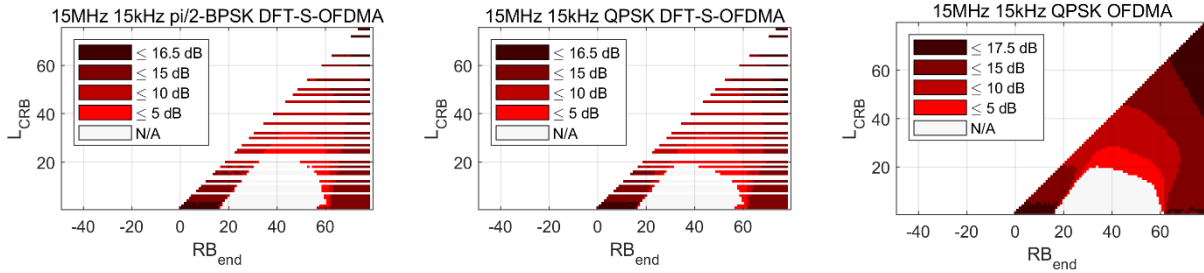
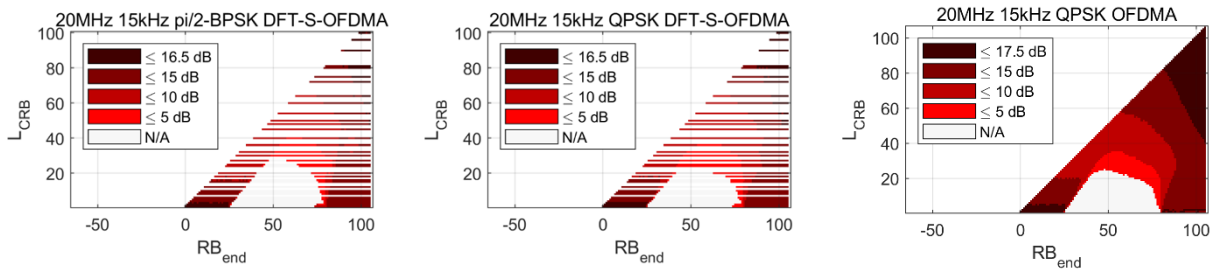


Figure F.2.1.1.4.2-4: 10MHz results at $F_c=2000\text{MHz}$ **Figure F.2.1.1.4.2-5: 15MHz results at $F_c=2000\text{MHz}$** **Figure F.2.1.1.4.2-6: 20MHz results at $F_c=1995\text{MHz}$**

Based on conducted simulations the follow A-MPR values and regions can be defined. The simulation results based on the A-MPR table of PC3, but some regions are different from PC3. Empty cells indicate that MPR is already enough according to our results. In some cases, region B needs to be slightly extended to cover all the allocations that need A-MPR, which also affects region C. This extension is likely needed due to the tail of the CIM3 peak. An alternative way to cover the A-MPR allocations is to decrease the upper LCRB limit of region C or create a new region. We also found that in case of PC2 the region A at 5MHz needs to be extended a bit due to regrowth of the allocation. and regions B and C need to be modified for 15MH. The reason we think why 5MHz regions for PC1 can be the same as PC3's is because the linearity of the PC1 PAs is better due to the difference of form factors from PC2 and PC2 PAs.

Table F.2.1.1.4.2-1: Proposed A-MPR regions and values for PC2

Channel Bandwidth, MHz	Carrier Centre Frequency, Fc, MHz	Region A			Region B			Region C		
		RB _{end} *12*S CS MHz	LCRB*12*S CS MHz	A-MP R	RB _{end} *12*S CS MHz	LCRB*12*S CS MHz	A-MP R	RB _{end} *12*S CS MHz	LCRB*12*S CS MHz	A-MP R
5MHz	1987.5 < Fc ≤ 1992.5		>3.24	A7						
5MHz	1992.5 < Fc ≤ 1997.5		>3.06	A4						
5MHz	1997.5 < Fc ≤ 2002.5		>1.98	A1	>3.6	>1.08 ≤1.98	A2	≤3.6	≤1.98	A3
						≤1.08	A6			
10MHz	1975 < Fc ≤ 1985	>5.4		A4						
10MHz	1985 < Fc ≤ 1995		>4.32	A1	≥7.2	>1.08 ≤4.32	A2	<7.2	≤4.32	A3
						≤1.08	A6			
10MHz	1995 < Fc ≤ 2000	≥5.76		A5	<3.06		A5	≥3.06 <5.76	>1.44	A6
15MHz	1972.5 < Fc ≤ 1987.5		>6.84	A1	≥10.8	>1.08 ≤6.84	A2	<10.8	≤6.84	A3
						≤1.08	A6			
15MHz	1987.5 < Fc ≤ 1997.5	≥8.64		A5	<4.14		A5	≥4.14 <8.64	>1.44	A6
20MHz	1970 < Fc ≤ 1990	≥12.96		A5	<4.68		A5	≥4.68 <12.96	>2.16	A6
20MHz	1990 < Fc ≤ 1995	≥11.52		A5	<5.58		A5	≥5.58 <11.52	>1.44	A6

Modulation/Waveform	A1	A2	A3	A4	A5	A6	A7
	Outer/Inner	Outer/Inner	Outer/Inner	Outer	Outer/Inner	Outer/Inner	Outer
DFT-s-OFDM PI/2 BPSK	9	5	0.5	2.5	16.5	11.5	
DFT-s-OFDM QPSK	10	5	2	4.5	16.5	11.5	
DFT-s-OFDM 16 QAM	11	5	2.5	4.5	16.5	11.5	
DFT-s-OFDM 64 QAM	11	5		4.5	17	11.5	
DFT-s-OFDM 256 QAM	11	5			17	11.5	
CP-OFDM QPSK	11.5	5.5	3.5	5	17.5	12.5	
CP-OFDM 16 QAM	11.5	5.5	3.5	5.5	17.5	12.5	
CP-OFDM 64 QAM	11.5	5.5		5.5	17.5	12.5	
CP-OFDM 256 QAM	11.5				17.5	12.5	

F.2.1.1.5 Company E (R4-2509537/Apple)

F.2.1.1.5.1 Simulation assumptions

PC2 power class simulations are conducted with the common assumptions as presented below:

- Tx power: PC2 (+26dBm)
- Calibration: 1dB MPR with 31dBc ACLR and DFT-s-OFDM QPSK 20MHz 100RB0
- Carrier leakage: -28dBc
- Image: -28dBc
- CIM3: -60dBc
- CIM5: -70dBc
- LO placement: always in the centre of the carrier
- Regulations: Requirements from NS_24

For the sake of completeness, the channel bandwidths and the corresponding center frequencies are summarised below:

- 5MHz: Fc=2002.5MHz, Fc=1997.5MHz, Fc=1992.5MHz
- 10MHz: Fc=2000MHz, Fc=1995MHz, Fc=1985MHz
- 15MHz: Fc=1997.5MHz, Fc=1987.5MHz
- 20MHz: Fc=1995MHz, Fc=1990MHz

F.2.1.1.5.2 Simulation results

Figure F.2.1.1.5.2-1, Figure F.2.1.1.5.2-2 and Figure F.2.1.1.5.2-3 present results for the 5MHz channel with the PC2 power class. The results are in line with the existing 5MHz channel A-MPR values for PC3 as defined by NS_24. For instance, while the large allocation with CP-OFDM and PC3 would need up to 13dB A-MPR, in case of PC2 the maximum power back-off increases up to 15dB. Nevertheless, it is worth noting that large power back-off is needed only for the full channel allocations, which are not likely to be used with the NTN deployments. Once the 5MHz channel is shifted father away from the upper band edge, one can notice an improvement in the required power back-off values. For instance, for the 5MHz channel residing at 1992.5MHz A-MPR would be need only for the large allocations, whereas small allocations do not even need MPR.

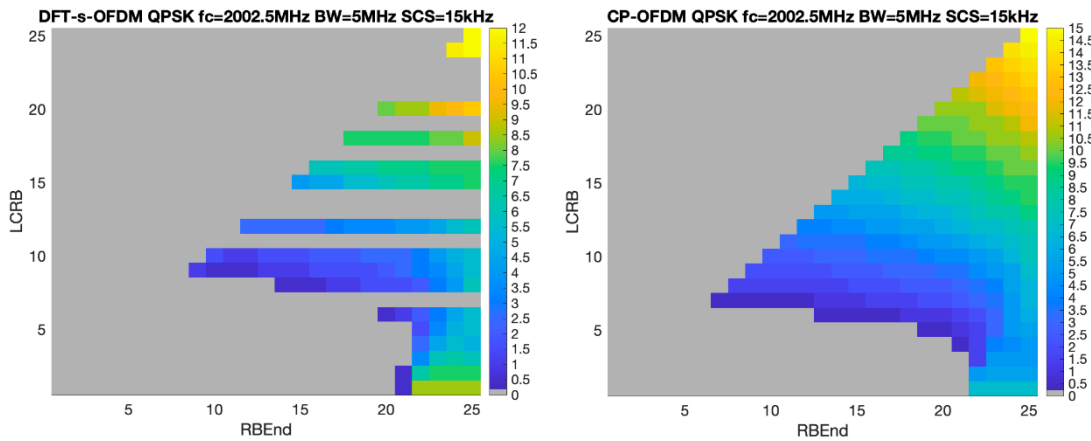


Figure F.2.1.1.5.2-1: Power back-off results for 5MHz at Fc=2002.5MHz.

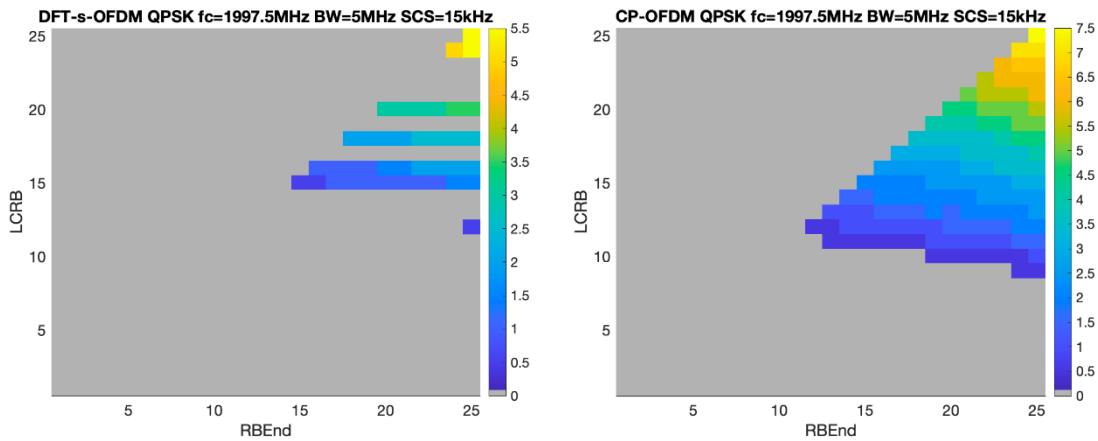


Figure F.2.1.1.5.2-2: Power back-off results for 5MHz at Fc=1997.5MHz.

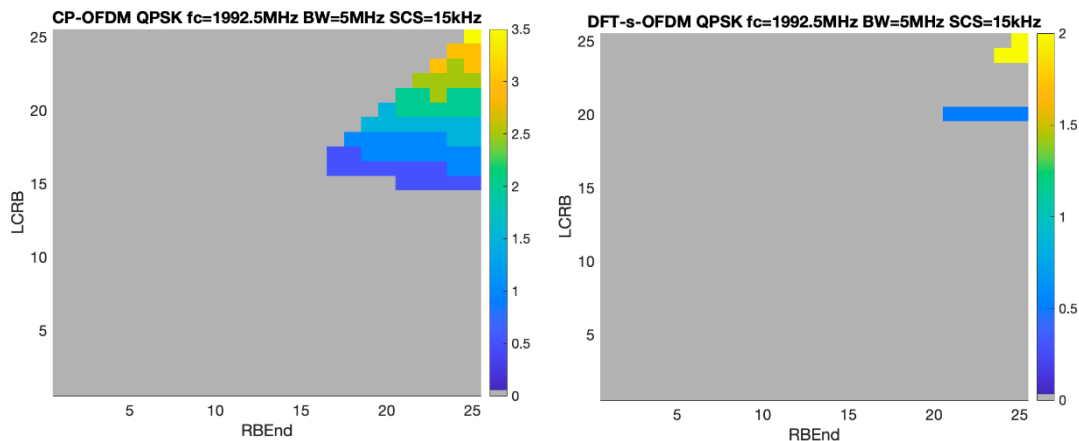


Figure F.2.1.1.5.2-3: Power back-off results for 5MHz at Fc=1992.5MHz.

Next, Figure F.2.1.1.5.2-4 to F.2.1.1.5.2-6 present the 10MHz simulation results, from which as anticipated we can see even higher power back-off required for the full channel and upper band edge allocations. In fact, the same trend can be observed further for the 15MHz and 20MHz channel bandwidths presented at figures below.

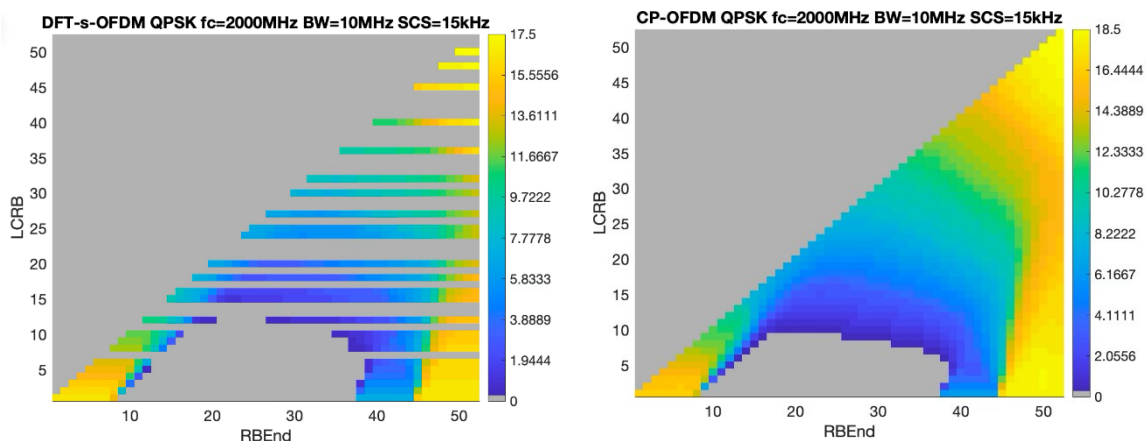


Figure F.2.1.1.5.2-4: Power back-off results for 10MHz at Fc=2000MHz

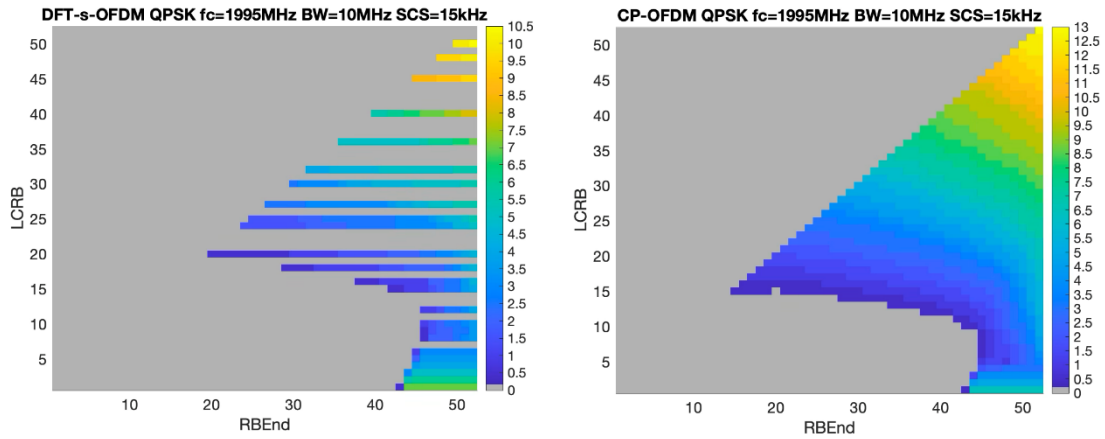


Figure F.2.1.1.5.2-4: Power back-off results for 10MHz at Fc=1995MHz

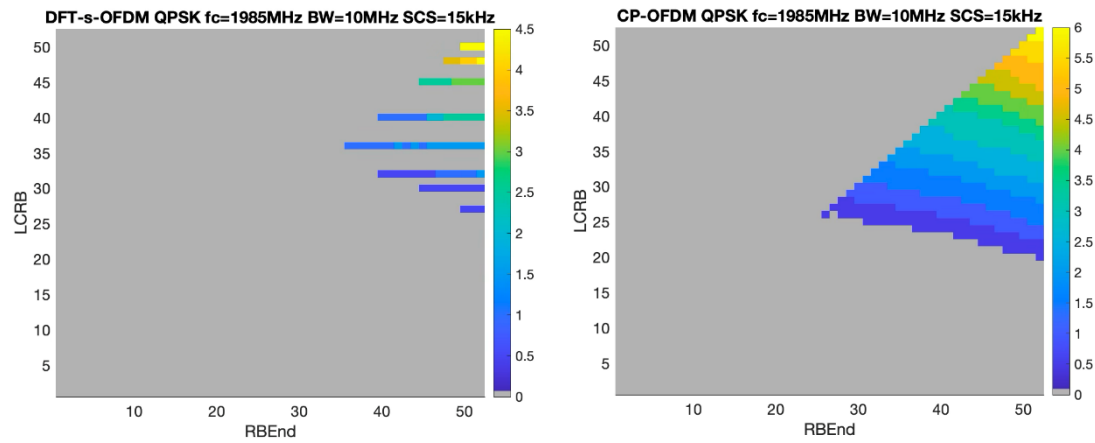


Figure F.2.1.1.5.2-5: Power back-off results for 10MHz at Fc=1985MHz

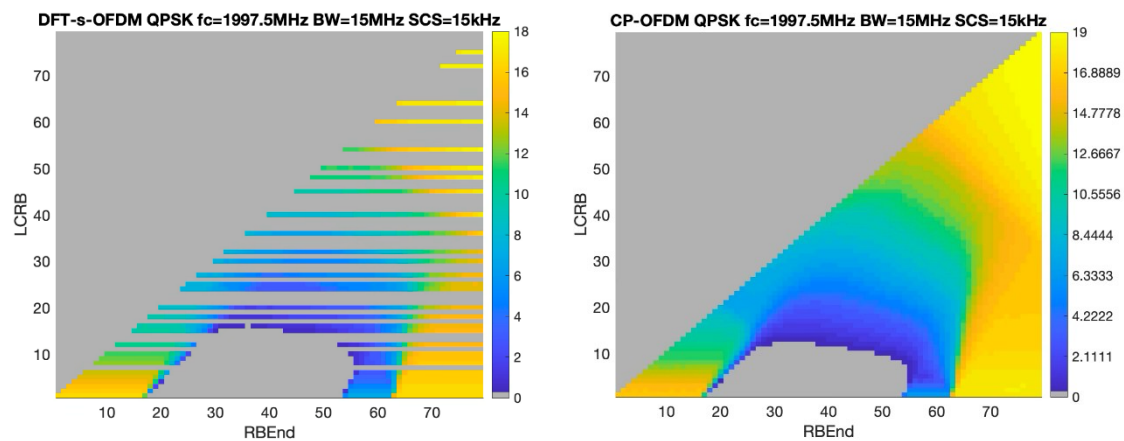
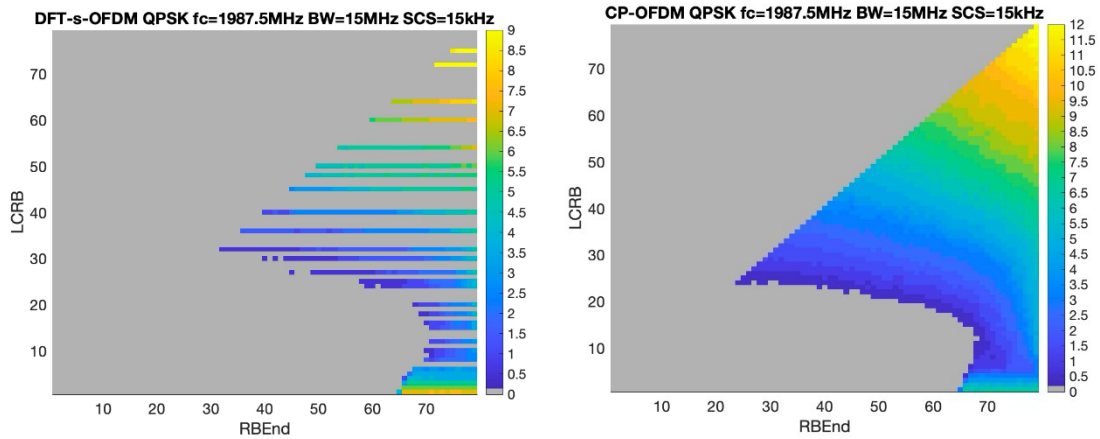
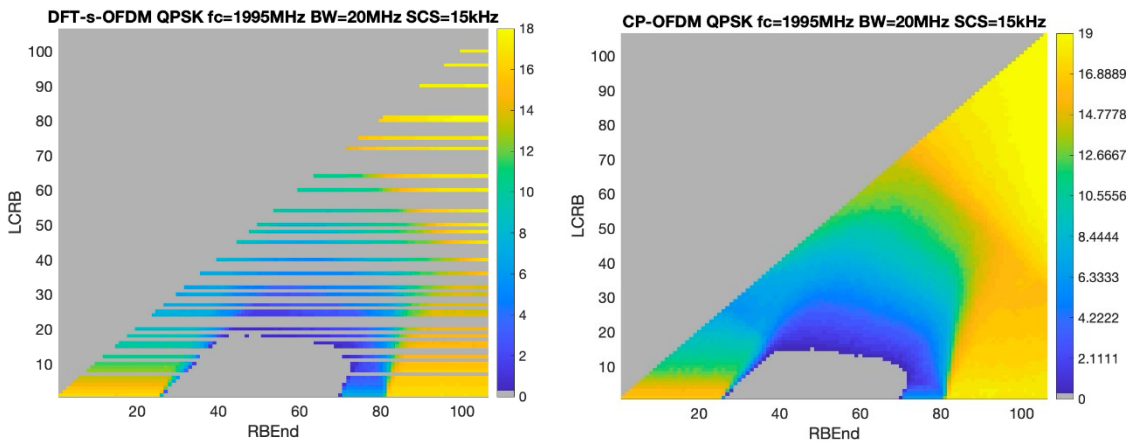
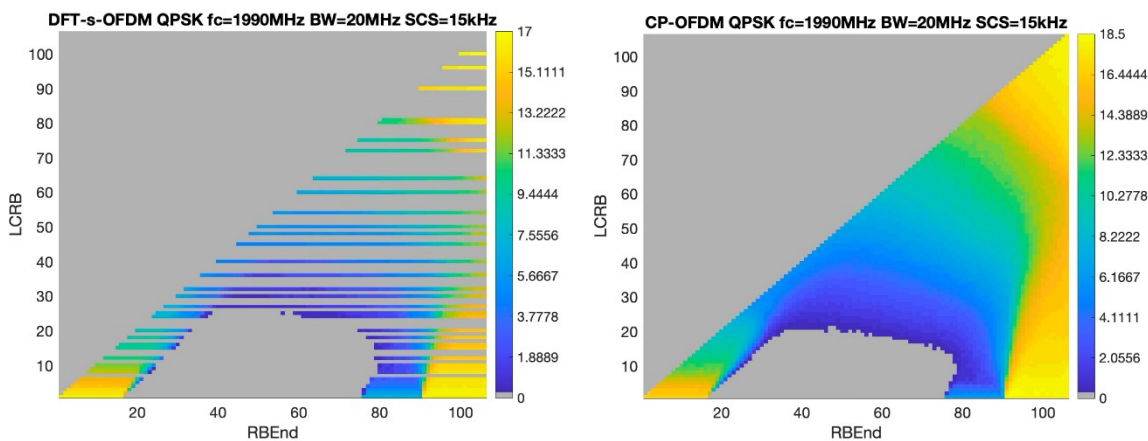


Figure F.2.1.1.5.2-6: Power back-off results for 15MHz at Fc=1997.5MHz

Figure F.2.1.1.5.2-7: Power back-off results for 15MHz at $F_c=1987.5\text{MHz}$ Figure F.2.1.1.5.2-8: Power back-off results for 20MHz at $F_c=1995\text{MHz}$ Figure F.2.1.1.5.2-9: Power back-off results for 20MHz at $F_c=1990\text{MHz}$

F.3 PC1.5 power class

F.3.1 General aspects

NOTE: The purpose of the section is to capture general aspects, such as considered UE RF architectures, assumptions, considered regulations, etc.

F.4 PC1 power class

F.4.1 General aspects

NOTE: The purpose of the section is to capture general aspects, such as considered UE RF architectures, assumptions, considered regulations, etc.

F.4.1.1 NS_24N

F.4.1.1.1 Company A (R4-2511269/Nokia)

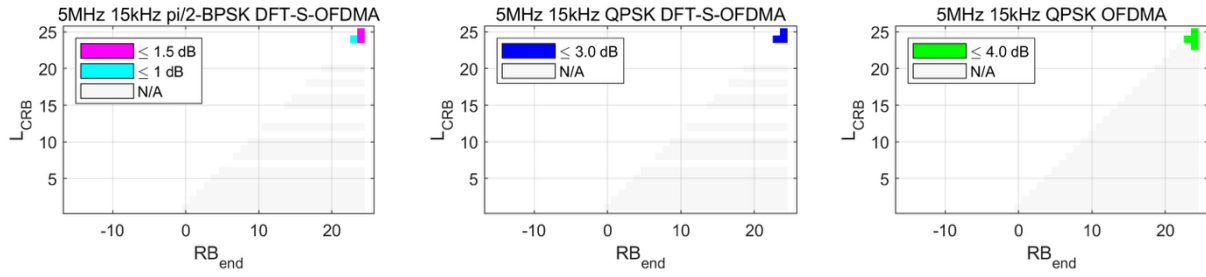
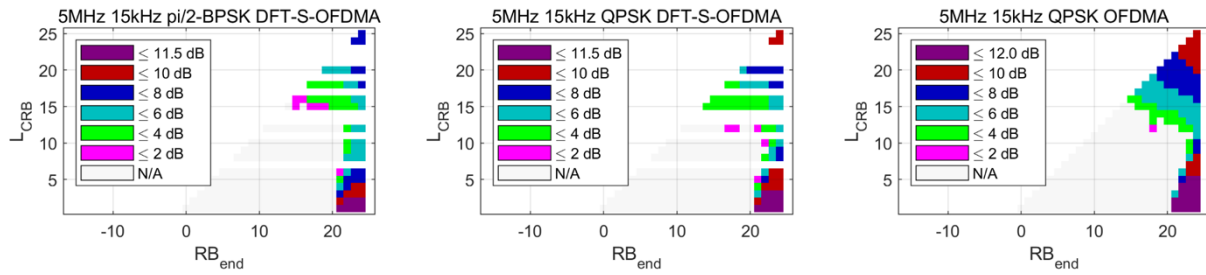
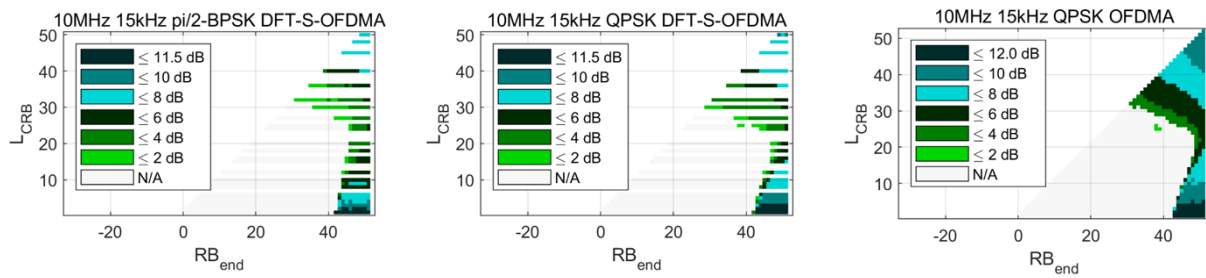
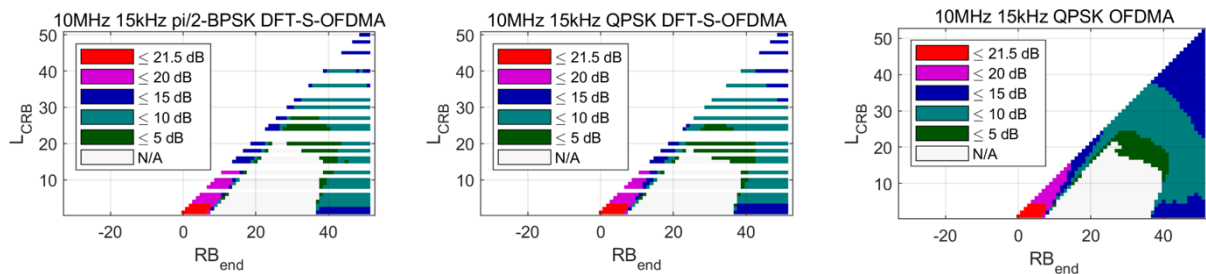
F.4.1.1.1.1 Simulation assumptions

The following simulation assumptions are made:

- Operating band n256 (UL 1980 MHz – 2010 MHz)
- 5MHz, 10 MHz, 15 MHz, and 20 MHz channel bandwidth
- Power class PC1
- WOLA processing
- I/Q image -28 dBc (default according to 38.101-1, Table 6.4.2.3-1) for other modulations than 256QAM
- For 256QAM the I/Q image is -33.7 dBc as in MPR simulations.
- Carrier leakage -28 dBc (default according to 38.101-1, Table 6.4.2.3-1)
- CIM3 -60 dBc
- CIM5 -70 dBc
- Channel noise 38.5 dBc
- PA noise -75 dBm/MHz
- Only additional spurious emission mask from NS_24 is simulated. Other gating factors (ACLR, SEM, etc.) are excluded because they are handled by MPR.
- Phase noise and channel noise are only added for QAM256, following LTE 256-QAM MPR simulations.

F.4.1.1.1.2 Simulation results

The following figures provide simulation results for different channel sizes and different locations for PI/2 BPSK, DFT-s-OFDM QPSK and CP-OFDM QPSK. It should be noted that power back-off is presented only for those cases when the corresponding value exceeds existing MPR margins.

Figure F.4.1.1.2-1: 5MHz results at $F_c = 1997.5\text{MHz}$ Figure F.4.1.1.2-2: 5MHz results at $F_c = 2002.5\text{MHz}$ Figure F.4.1.1.2-3: 10MHz results at $F_c = 1995\text{MHz}$ Figure F.4.1.1.2-4: 10MHz results at $F_c = 2000\text{MHz}$

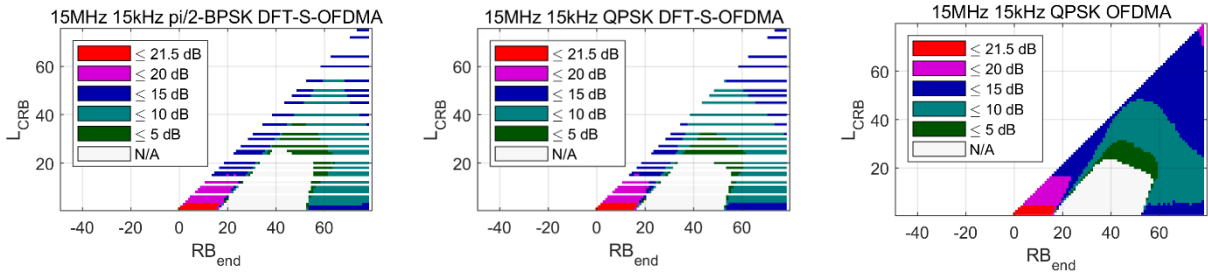


Figure F.4.1.1.2-5: 15MHz results at $F_c=2000\text{MHz}$

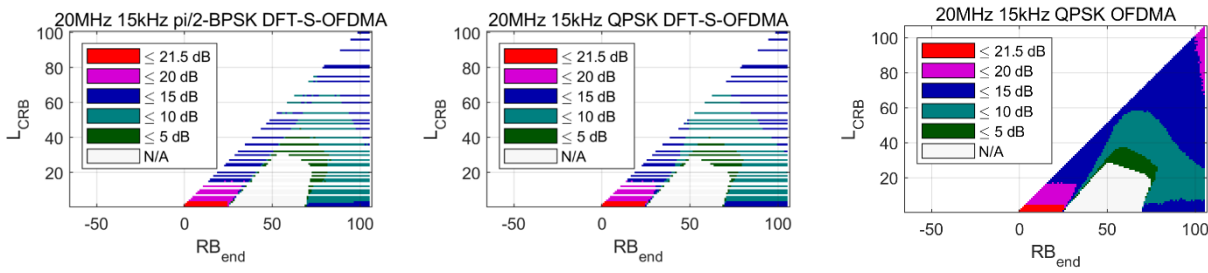


Figure F.4.1.1.2-6: 20MHz results at $F_c=1995\text{MHz}$

Based on the presented results PC1 A-MPR regions and the corresponding A-MPR values can be as summarised below in Table F.4.1.1.2-1.

Table F.4.1.1.2-1: Proposed A-MPR regions and values for PC1

Channel Bandwidth, MHz	Carrier Centre Frequency, Fc, MHz	Region A			Region B			Region C		
		RB _{end} *12*S CS MHz	LCRB*12* SCS MHz	A-MP R	RB _{end} *12*S CS MHz	LCRB*12* SCS MHz	A-MP R	RB _{end} *12*S CS MHz	LCRB*12* SCS MHz	A-MP R
5MHz	1987.5 < Fc ≤ 1992.5		>3.24	A7						
5MHz	1992.5 < Fc ≤ 1997.5		>3.24	A4						
5MHz	1997.5 < Fc ≤ 2002.5		>1.98	A1	>3.6	>1.08 ≤1.98	A2	≤3.6	≤1.98	A3
						≤1.08	A6			
10MHz	1975 < Fc ≤ 1985	>5.4		A4						
10MHz	1985 < Fc ≤ 1995		>4.32	A1	≥7.2	>1.08 ≤4.32	A2	<7.2	≤4.32	A3
						≤1.08	A6			
10MHz	1995 < Fc ≤ 2000	≥5.76		A5	<3.06		A5	≥3.06 <5.76	>1.44	A6
15MHz	1972.5 < Fc ≤ 1987.5		>6.84	A1	≥10.8	>1.08 ≤6.84	A2	<10.8	≤6.84	A3
						≤1.08	A6			
15MHz	1987.5 < Fc ≤ 1997.5	≥8.64		A5	<4.14		A5	≥4.14 <8.64	>1.44	A6
20MHz	1970 < Fc ≤ 1990	≥12.96		A5	<4.86		A5	≥4.86 <12.96	>2.16	A6
20MHz	1990 < Fc ≤ 1995	≥11.52		A5	<5.76		A5	≥5.76 <11.52	>1.44	A6

Modulation/Waveform	A1	A2	A3	A4	A5	A6	A7
	Outer/Inner	Outer/Inner	Outer/Inner	Outer	Outer/Inner	Outer/Inner	Outer
DFT-s-OFDM PI/2 BPSK	8	6.5		1.5	21.5	16.5	
DFT-s-OFDM QPSK	9	8		3	21.5	16	
DFT-s-OFDM 16 QAM	9.5	8		3.5	21.5	16	
DFT-s-OFDM 64 QAM	9.5	8.5		3.5	21.5	16	
DFT-s-OFDM 256 QAM	9.5	8.5			21.5	16	
CP-OFDM QPSK	10	10		4	21.5	16.5	
CP-OFDM 16 QAM	10	10		4	21.5	16.5	
CP-OFDM 64 QAM	10	10		4	21.5	16.5	
CP-OFDM 256 QAM	10	10			21.5	16.5	

F.4.1.1.2 Company B (R4-2510521/OPPO)

F.4.1.1.2.1 Simulation assumptions

Simulation assumptions are as follows:

- PC1 power class with 4Tx
- 5,10, 15 and 20MHz channels residing at the higher edge of the uplink operating band n256 and keeping 5MHz protecting guard gap between the higher edge of the channel bandwidth and the highest edge of the uplink operating band.
- CP-OFDM and DFT-s-OFDM

F.4.1.1.2.2 Simulation results

Figure F.4.1.1.2.2-1 and Figure F.4.1.1.2.2-2 present a comparison of the NS_24 power back-off results for 5MHz ($F_c=2002.5$) and 10MHz ($F_c=2000$ MHz) under PC2 1Tx, PC1.5 2Tx and PC1 4Tx with CP-OFDM QPSK. The simulation results are based on fixed bias PA.

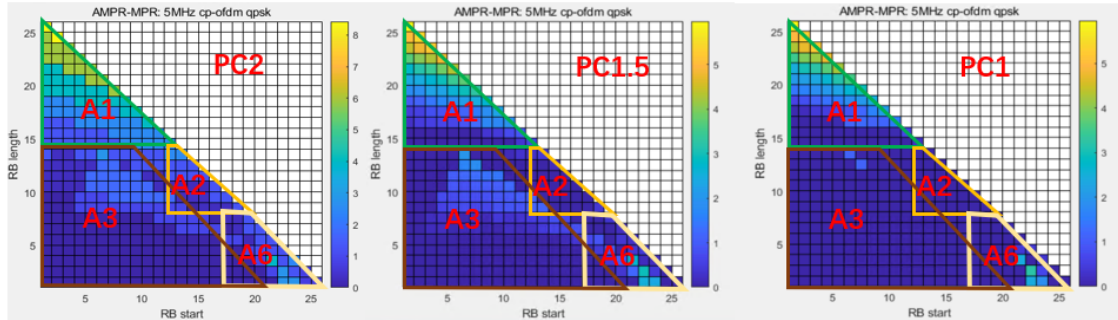


Figure F.4.1.1.2.2-1 AMPR for 5MHz CP-OFDM QPSK under different power classes

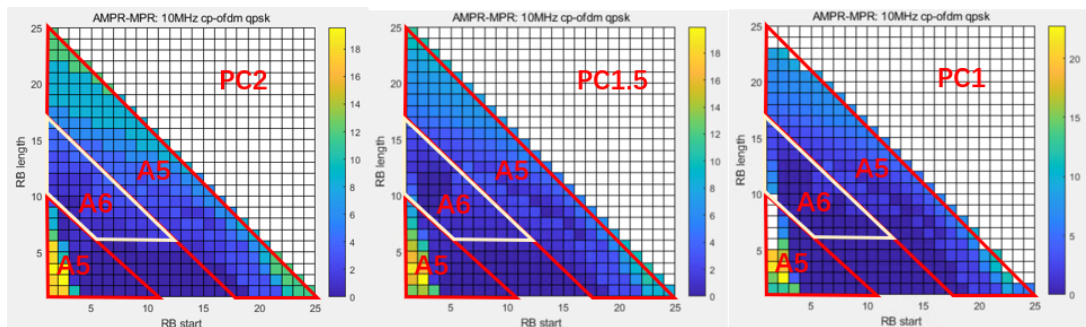


Figure F.4.1.1.2.2-2 AMPR for 10MHz CP-OFDM QPSK under different power classes

In the AMPR simulation for PC2 1Tx, PC1.5 2Tx and PC1 4Tx, the same PC2 PA with fixed bias is assumed for these three power classes. From Figure F.4.1.1.2.2-1 and Figure F.4.1.1.2.2-2 it can be seen that:

- The PC2 and PC1.5 have the similar AMPR region frame and values due to the maximum output power of each PA for PC2 and PC1.5 is 26dBm;
- The AMPR region frame for PC1 has more margin than PC2 and PC1.5, since the maximum output power of each PA for PC1 is 25dBm.

Therefore, PC1 can use same AMPR regions as PC2. Table F.4.1.1.2.2-1 proposes power back-off values, which are based on the existing PC3 regions.

Table F.4.1.1.2.2-1: Proposed PC1 4Tx AMPR values based on PC3 A-MPR regions

Modulation/ Waveform	A1	A2	A3	A4	A5	A6	A7
	Outer/Inner	Outer/Inner	Outer/Inner	Outer	Outer/Inner	Outer/Inner	Outer
CP-OFDM QPSK	≤ 20.5	≤ 16	≤ 16	≤ 15	≤ 31	≤ 23	≤ 10
CP-OFDM 16QAM	≤ 20.5	≤ 16	≤ 16	≤ 15	≤ 31	≤ 23	≤ 10
CP-OFDM 64QAM	≤ 20.5	≤ 16	≤ 16	≤ 15	≤ 31	≤ 23	≤ 10
DFT-s-OFDM $\pi/2$ BPSK	≤ 18.5	≤ 15	≤ 15	≤ 13.5	≤ 31	≤ 22	≤ 9
DFT-s-OFDM QPSK	≤ 18.5	≤ 15	≤ 15	≤ 13.5	≤ 31	≤ 22	≤ 9
DFT-s-OFDM 16QAM	≤ 18.5	≤ 15	≤ 15	≤ 13.5	≤ 31	≤ 22	≤ 9
DFT-s-OFDM 64QAM	≤ 18.5	≤ 15	≤ 15	≤ 13.5	≤ 31	≤ 22	≤ 9

If the modified regions are used as shown in Table F.2.1.1.3.2-2, regions A1, A4 and A5 are extended, but regions A2, A3 and A8 are reduced as shown in Table F.4.1.1.2.2-2 below

Table F.4.1.1.2.2-2: PC1 4Tx AMPR values based on modified A-MPR regions as PC2

Modulation/ Waveform	A1	A2	A3	A4	A5	A6	A8	A7
	Outer/Inner	Outer/Inner	Outer/Inner	Outer	Outer/Inner	Outer/Inner		Outer
CP-OFDM QPSK	≤ 20.5	≤ 15	≤ 15	≤ 10	≤ 31	≤ 15.5	≤ 23	≤ 10
CP-OFDM 16QAM	≤ 20.5	≤ 15	≤ 15	≤ 10	≤ 31	≤ 15.5	≤ 23	≤ 10
CP-OFDM 64QAM	≤ 20.5	≤ 15	≤ 15	≤ 10	≤ 31	≤ 15.5	≤ 23	≤ 10
DFT-s-OFDM $\pi/2$ BPSK	≤ 18.5	≤ 14	≤ 14	≤ 9	≤ 31	≤ 14	≤ 22	≤ 9
DFT-s-OFDM QPSK	≤ 18.5	≤ 14	≤ 14	≤ 9	≤ 31	≤ 14	≤ 22	≤ 9
DFT-s-OFDM 16QAM	≤ 18.5	≤ 14	≤ 14	≤ 9	≤ 31	≤ 14	≤ 22	≤ 9
DFT-s-OFDM 64QAM	≤ 18.5	≤ 14	≤ 14	≤ 9	≤ 31	≤ 14	≤ 22	≤ 9

Annex G: Power back-off simulation results for the L-band

G.1 PC3 power class

G.1.1 General aspects

NOTE: The purpose of the section is to capture general aspects, such as considered UE RF architectures, assumptions, considered regulations, etc.

G.1.1.1 NS_02N

G.1.1.1.1 Company A (OPPO/R4-2510521)

G.1.1.1.1.1 Simulation assumptions

Since filter rejection at the frequencies 1559-1610MHz is not a constant 10dB attenuation, but rather has a slope, it is hard to add the filter rejection without the slope factor. Thus, to evaluate the rejection effect of the filter, required power back-off is simulated under the following assumptions:

- no filter rejection is assumed;
- PC3 CP-OFDM QPSK using fixed bias PA and APT PA;
- 5, 10, 15 and 20MHz channels residing at the lowest edge of the uplink operating band n255

G.1.1.1.1.2 Simulation results

For PC3 CP-OFDM QPSK using fixed bias PA:

- Power back-off 0dB is needed for 5MHz, 10MHz and 15MHz without additional filter rejection.
- Power back-off 3dB is needed for 20MHz without additional filter rejection.

For PC3 CP-OFDM QPSK using APT PA:

- Power back-off 0dB is needed for 5MHz and 10MHz without additional filter rejection.
- Power back-off 2dB for 15MHz and 10.5dB for 20MHz without additional filter rejection.

Based on the simulation results, no matter which PA model is used, the filter attenuation is needed for NS_02N for PC3 to be able to operate within existing MPR margins.

G.1.1.2 NS_09N

G.1.1.2.1 Company A

G.1.1.2.1.1 Simulation assumptions

The following simulation assumptions are used:

- Considered Tx frequency: 1668 – 1675MHz
- Tx power: PC3 (23dBm)
- Calibration: 1dB MPR with DFT-s-OFDM QPSK 20MHz, 100RB0
- Carrier Leakage: 28dBc
- Image: 28dBc
- CIM3: 60dBc
- CIM5: 70dBc
- Additional requirements: out-of-band unwanted emission limits according to ETSI EN 301681

G.1.1.2.1.2 Simulation results

Figure G.1.1.2.1.2-1 presents power back-off values for the 5MHz channel at the lower edge of the band. As can be seen most allocations do not need A-MPR, except the case(s) with a single RB allocated at the right edge of the 5MHz

channel. The reason for that is the upper edge out-of-band emission requirement limits, which would require some A-MPR. As presented in Figure G.1.1.2.1.2-2 below, CP-OFDM slightly exceeds the limit of -30dBm/30kHz and DFT-s-OFDM is also hitting that limit.

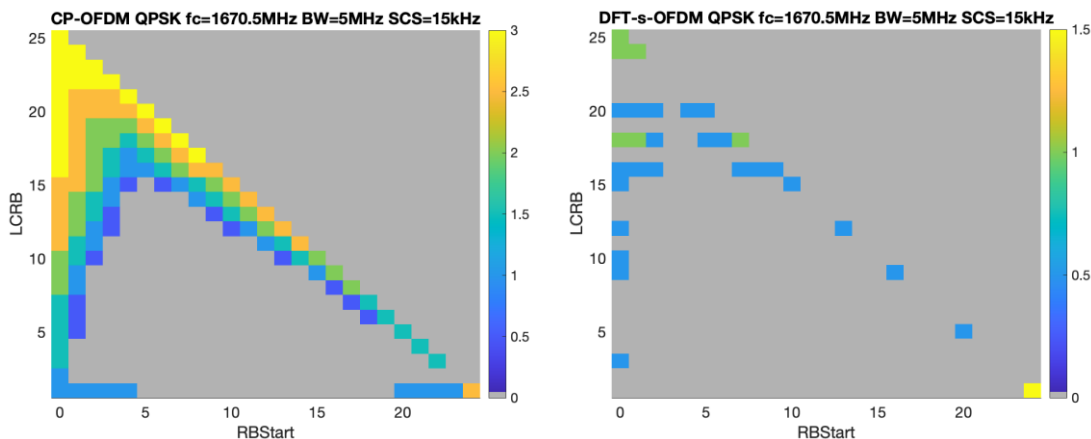


Figure G.1.1.2.1.2-1: Power back-off for CP-OFDM and DFT-s-OFDM ($F_c=1670.5\text{MHz}$).

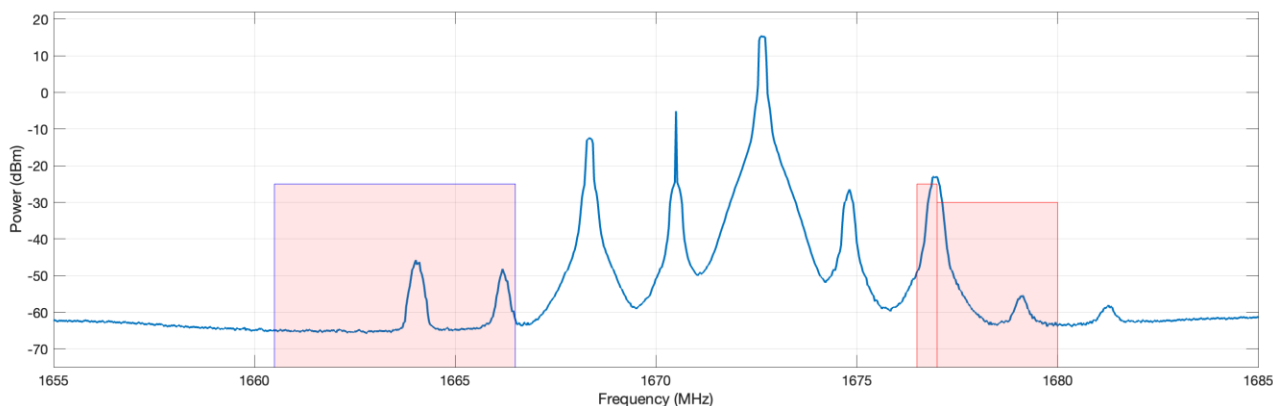


Figure G.1.1.2.1.2-2: Spectrum plot for the 5MHz channel and 1RB allocation ($F_c=1670.5\text{MHz}$).

Once the 5MHz channel is shifted farther towards the upper edge of the band, even higher A-MPR might be needed, which is however mostly concerns single RB allocations at the upper edge of the configured channel.

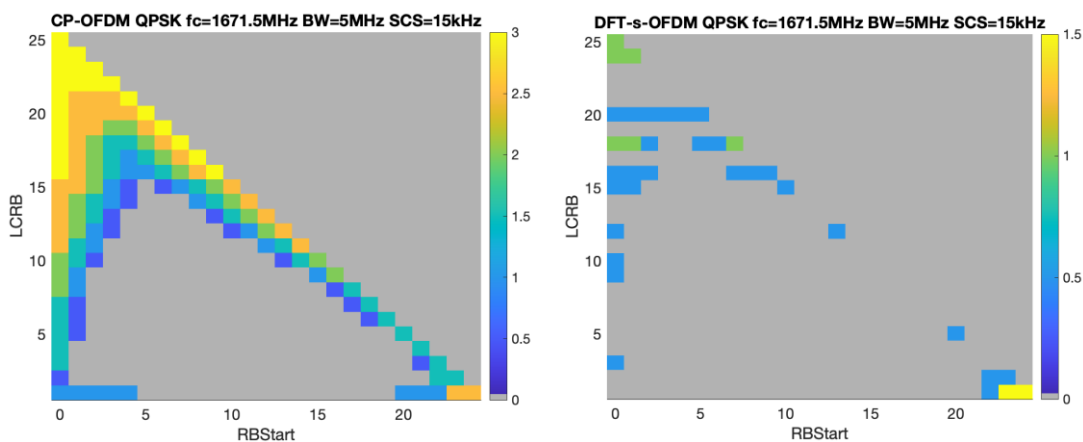


Figure G.1.1.2.1.2-3: Power back-off for CP-OFDM and DFT-s-OFDM ($F_c=1671.5\text{MHz}$).

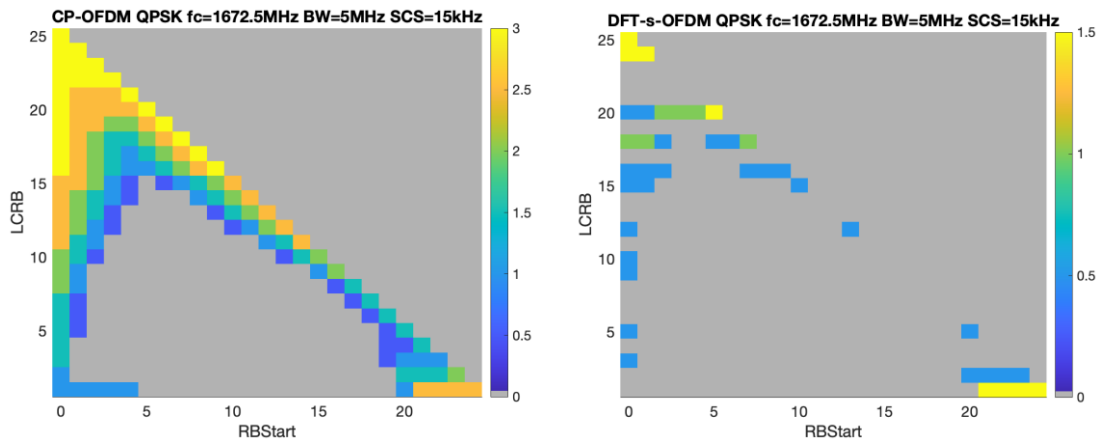


Figure G.1.1.2.1.2-4: Power back-off for CP-OFDM and DFT-s-OFDM (Fc=1672.5MHz).

G.1.1.3 NS_10N

G.1.1.3.1 Company A

G.1.1.3.1.1 Simulation assumptions

The following simulation assumptions are used:

- Considered Tx frequency: 1626.5 – 1660.5MHz
- Tx power: PC3 (23dBm)
- Calibration: 1dB MPR with DFT-s-OFDM QPSK 20MHz, 100RB0
- Carrier Leakage: 28dBc
- Image: 28dBc
- CIM3: 60dBc
- CIM5: 70dBc
- Addition requirements: out-of-band unwanted emission limits according to ETSI EN 301681

G.1.1.3.1.2 Simulation results

As can be seen from the Figure G.1.1.3.1.2-1 below, when the 5MHz NR channel is in the middle of the band, no additional AMPR is anticipated: the out-of-band emission requirements are not relevant because of the size of the band and the in-band ETSI emission mask does not create any additional restrictions either.

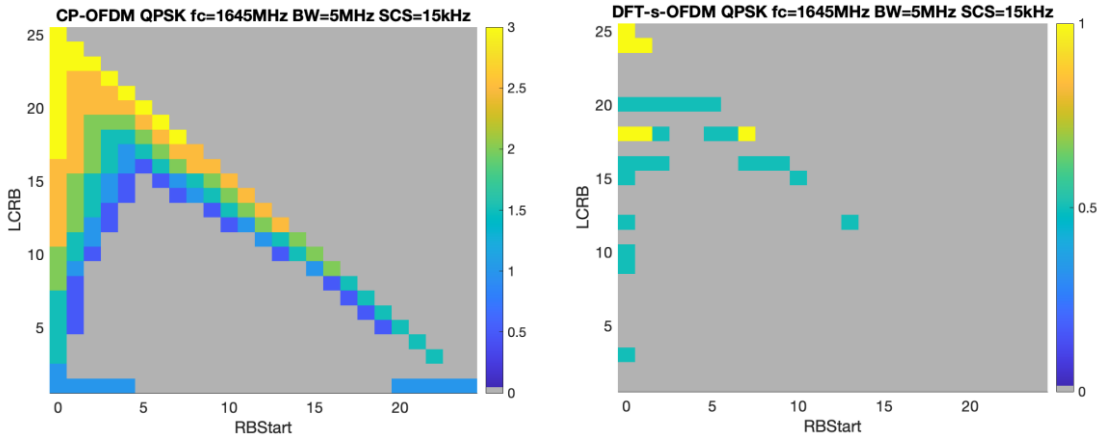


Figure G.1.1.3.1.2-1: Power back-off for CP-OFDM and DFT-s-OFDM (5MHz at $F_c=1645\text{MHz}$).

As for the channels either at the lower or the upper edge of the band, extra A-MPR is needed for larger allocations with DFT-s-OFDM and the right-most channel edge allocations as presented in Figure G.1.1.3.1.2-2 below. And the same observations can be done for the 5MHz channel at the upper edge of the band: left-most channel allocations will need extra A-MPR to comply with the out-of-band emission requirements.

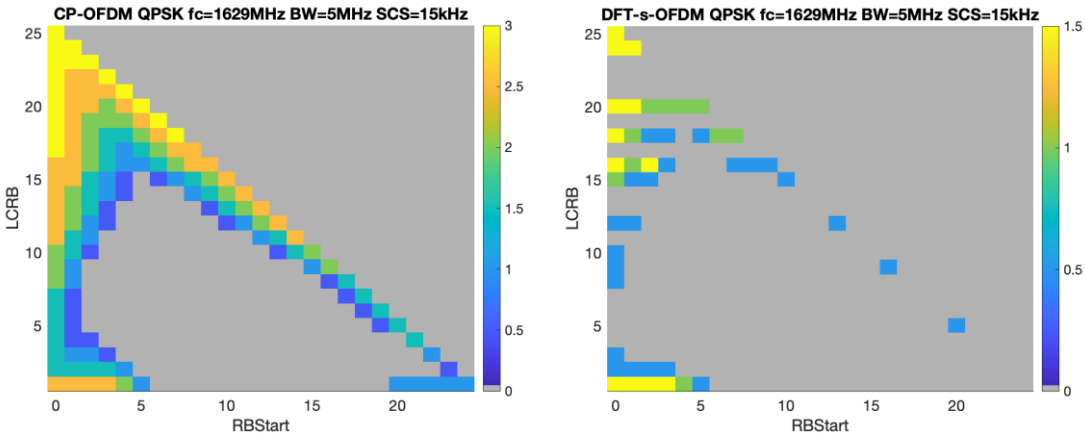


Figure G.1.1.3.1.2-2: Power back-off for CP-OFDM and DFT-s-OFDM (5MHz at $F_c=1629\text{MHz}$).

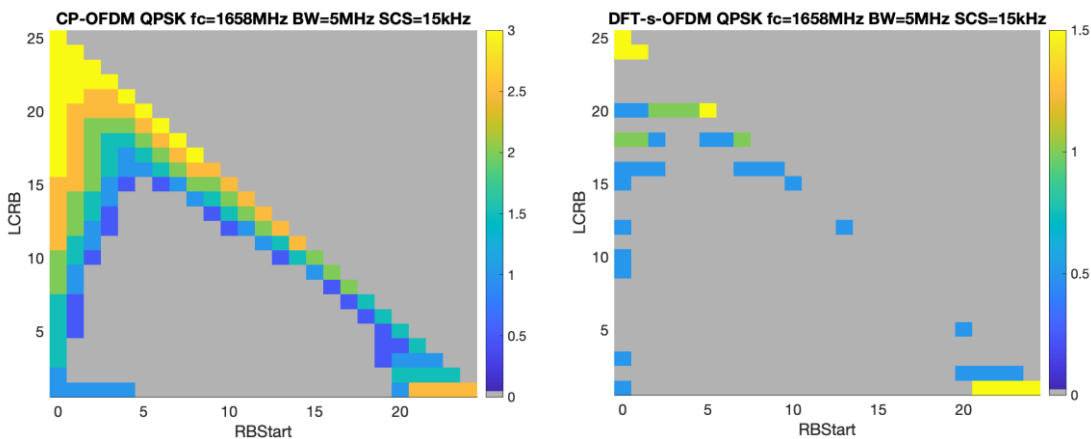


Figure G.1.1.3.1.2-3: Power back-off for CP-OFDM and DFT-s-OFDM (5MHz at $F_c=1658\text{MHz}$).

One of the reasons why the left- and right-most channel power back-off look identical despite the fact that out-of-band emission requirements are not the same is the fact that the limiting factor is $-30\text{dBm}/30\text{kHz}$. Two figures below present the spectrum plot for the 5MHz channels at the left and right edges of the band, from which one can see that the most limiting factor is the $-30\text{dBm}/30\text{kHz}$ emission limit at the offset of 2MHz from the band edge.

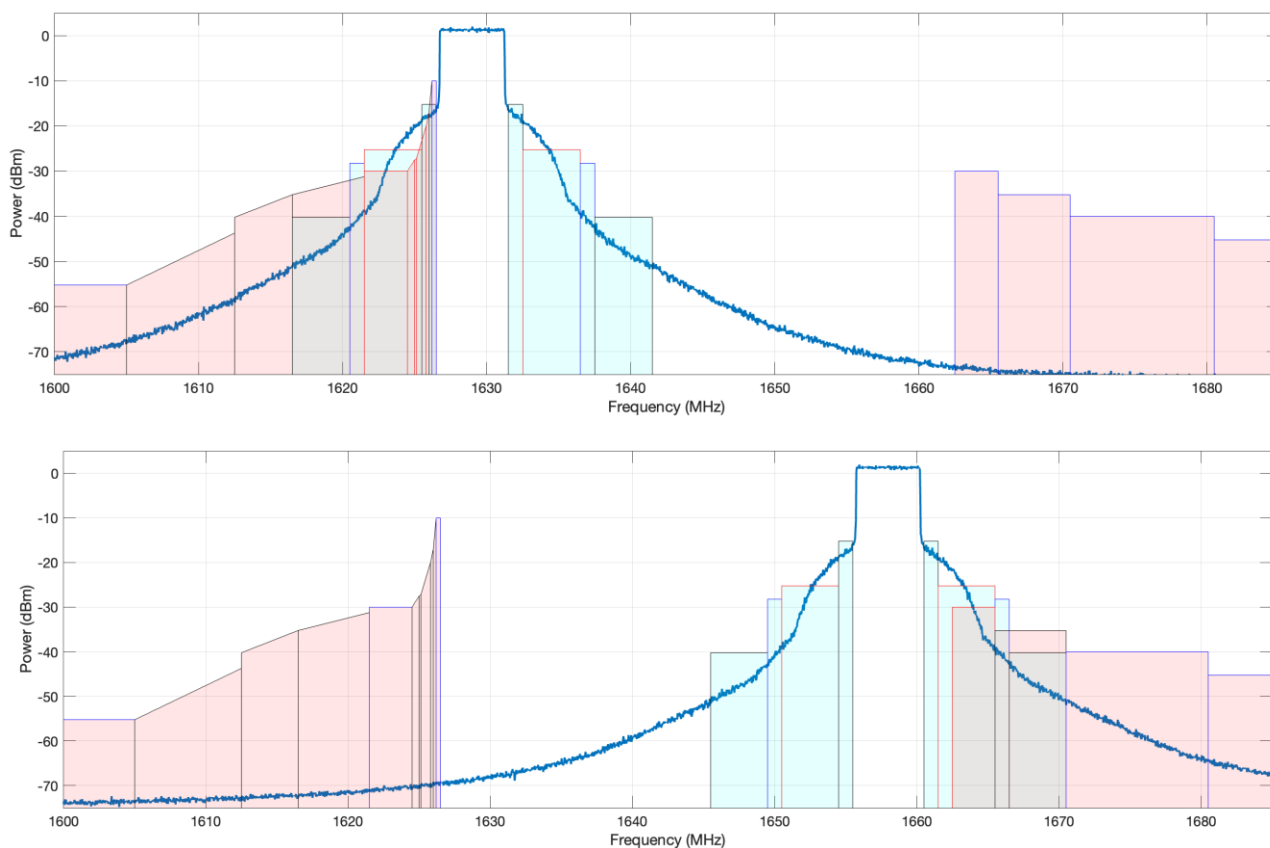


Figure G.1.1.3.1.2-4: Spectrum plot for the 5MHz channel at the left edge (top figure) and right edge (bottom figure) of the band.

For the 10MHz channel, G.1.1.3.1.2-4 presents results for a case when the 10MHz channel resides in the middle of the band; and as can be seen from the figure, no extra A-MPR is needed, i.e. power back-off values are within the MPR margin.

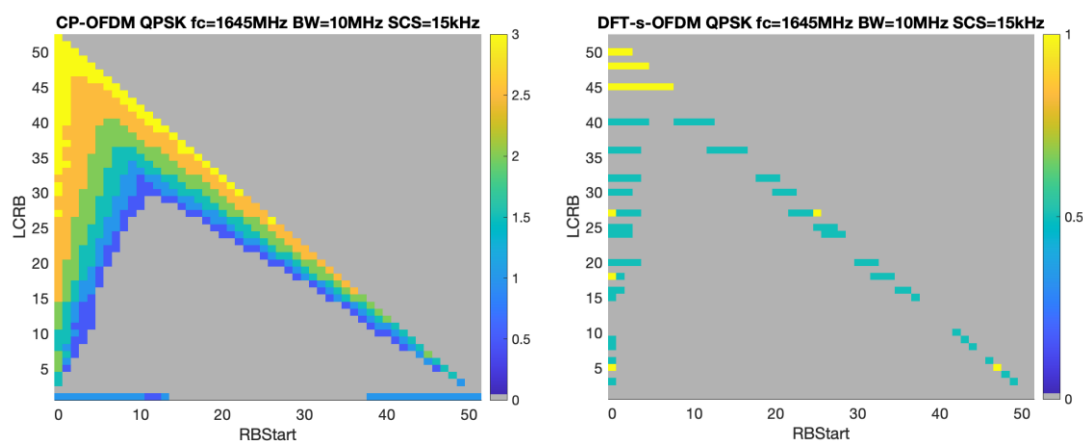


Figure G.1.1.3.1.2-4: Power back-off for CP-OFDM and DFT-s-OFDM (10MHz at Fc=1645MHz)

Figure G.1.1.3.1.2-5 and G.1.1.3.1.2-6 present results for channels residing at the left and the right edges of the band. General observations are similar to the 5MHz channel, the only noticeable difference is that the 10MHz channel at the upper edge of the band requires slightly higher power back-off due to more stringent out-of-band emission requirements.

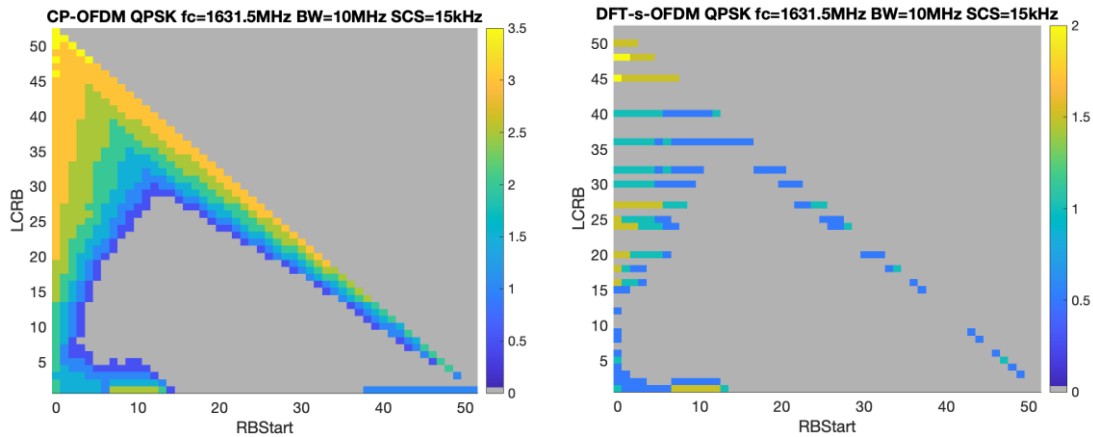


Figure G.1.1.3.1.2-5: Power back-off for CP-OFDM and DFT-s-OFDM (10MHz at $F_c=1631.5\text{MHz}$).

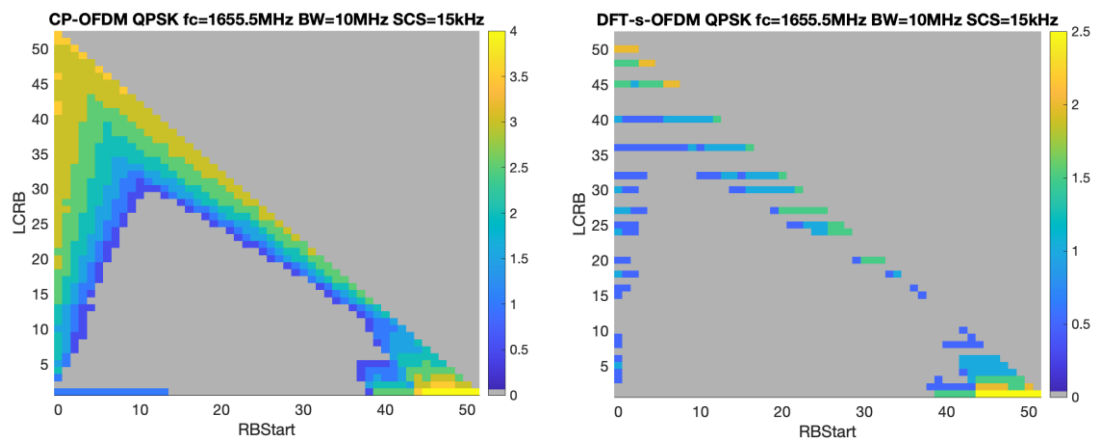


Figure G.1.1.3.1.2-6: Power back-off for CP-OFDM and DFT-s-OFDM (10MHz at $F_c=1655.5\text{MHz}$).

Figure 2.3.1.3-1 presents results for the 15MHz channel located in the middle of the band. Unlike previous cases for the 5 and 10MHz channels, this time extra A-MPR is need because the power back-off values exceed the MPR margin for edge RB allocations.

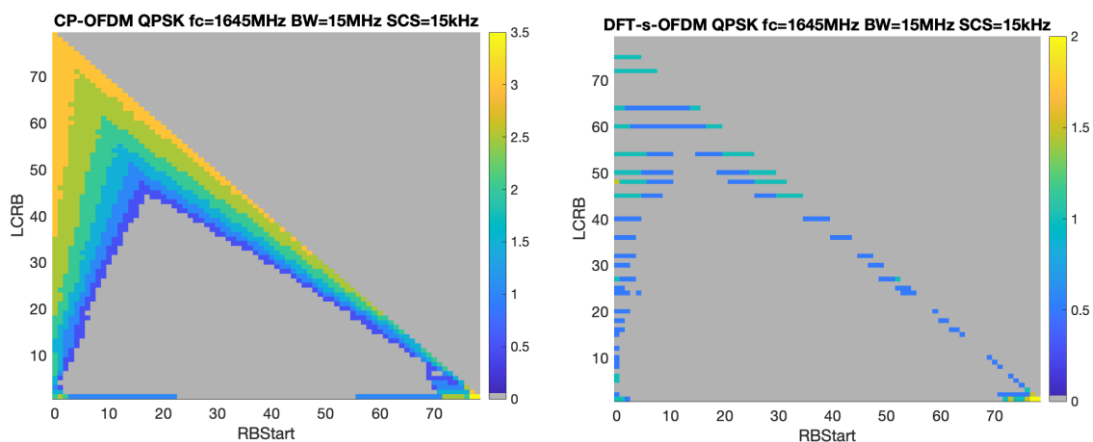


Figure G.1.1.3.1.2-7: Power back-off for CP-OFDM and DFT-s-OFDM (15MHz at $F_c=1645\text{MHz}$).

Power back-off simulation results for the 15MHz channels residing at the band edges are logically identical to the 10MHz results presented above. The 15MHz channel at the upper edge of the band requires even higher power back-off for large allocations and the for the right-most allocations within the band.

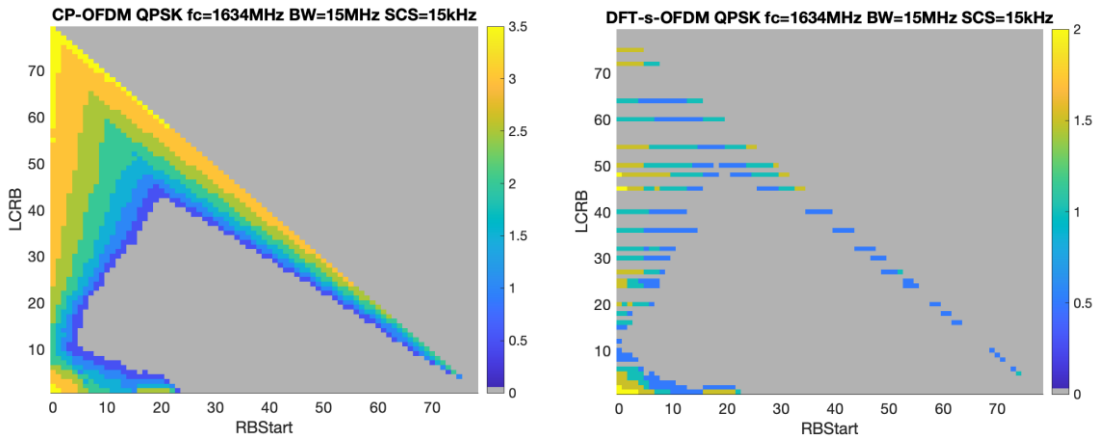


Figure G.1.1.3.1.2-8: Power back-off for CP-OFDM and DFT-s-OFDM (15MHz at Fc=1634MHz).

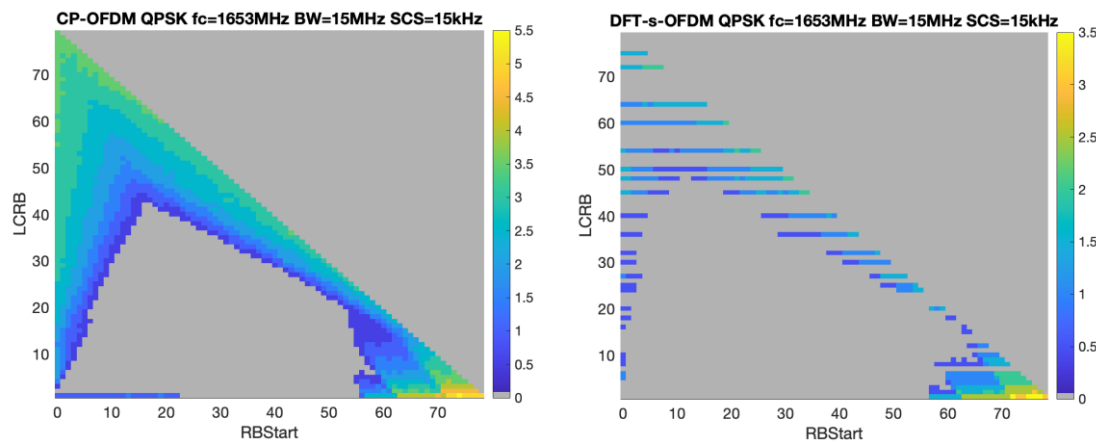


Figure G.1.1.3.1.2-9: Power back-off for CP-OFDM and DFT-s-OFDM (15MHz at Fc=1653MHz).

The 20MHz channel will require even higher power back-off. As can be seen from Figure G.1.1.3.1.2-10, when the 20MHz channel is in the middle of the band, only edge RB allocations need extra A-MPR.

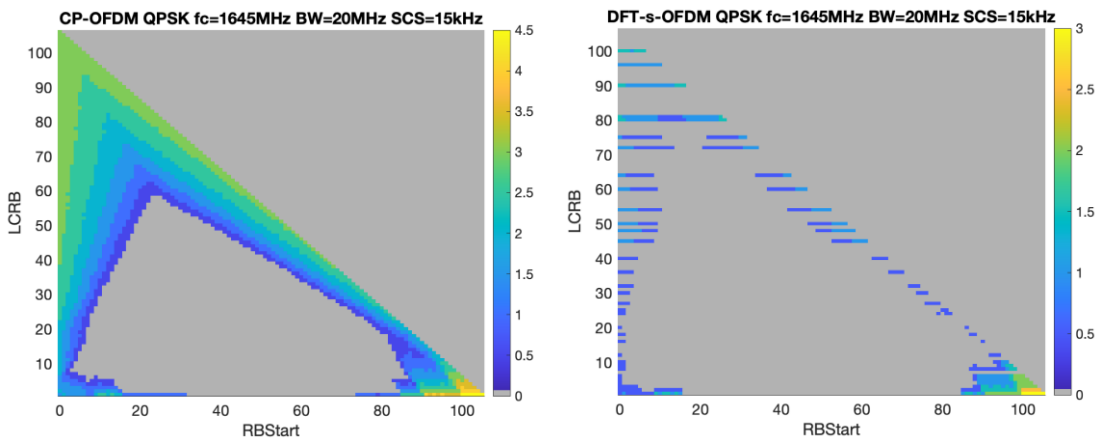


Figure G.1.1.3.1.2-10: Power back-off for CP-OFDM and DFT-s-OFDM (20MHz at Fc=1645MHz).

For the channel at the lower edge of the band, the left-most allocations will need the highest power back-off reaching 5-6.5dB. And similarly, the right-most allocations within the channel at the upper edge of the band will require power back-off in the order of 5.5dB.

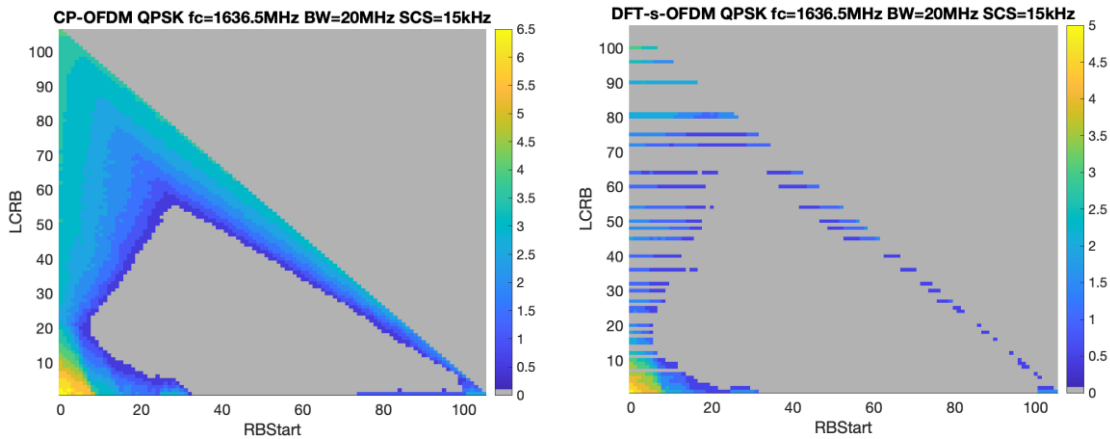


Figure G.1.1.3.1.2-11: Power back-off for CP-OFDM and DFT-s-OFDM (20MHz at $F_c=1636.5\text{MHz}$).

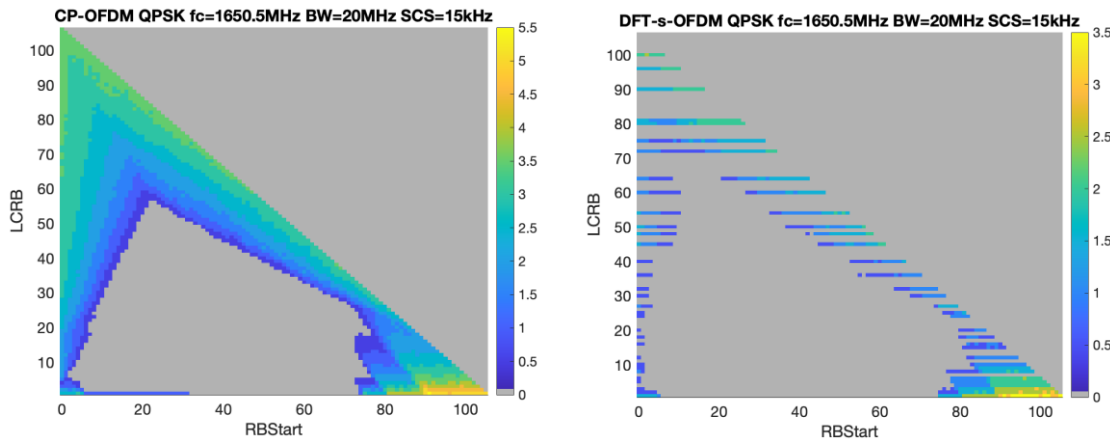


Figure G.1.1.3.1.2-12: Power back-off for CP-OFDM and DFT-s-OFDM (20MHz at $F_c=1650.5\text{MHz}$).

G.1.1.4 NS_14N

G.1.1.4.1 Company A

G.1.1.4.1.1 Simulation assumptions

The following simulation assumptions are used:

- PA calibrated to 1 dB MPR with 30 dBc ACLR with DFT-s-OFDM QPSK 100 RB waveform
- -28 dBc IQ-image
- -28 dBc LO-leakage
- -60 dBc CIM3 and -70 dBc CIM5

Additional emission limits considered in Table G.1.1.4.1.1-1. Only backoff due to additional emissions limits is shown in figures in clause G.1.1.4.1.2.

Table G.1.1.4.1.1-1: Additional emission requirements

Frequency range (MHz)	Spectrum emission limit ¹ (dBm)	Measurement bandwidth	NOTE (measurement method)
1559 to 1605	-40	1 MHz	Average
1605 to 1612.5	-40 to -28.5	1 MHz	Average
1612.5 to 1616.5	-25 to -20	1 MHz	Average
1616.5 to 1621.5	-20 to -16	1 MHz	Average
1621.5 to 1624.5	-30 to -27.5	30 kHz	Average
1624.5 to 1625.125	-27.5 to -27.2	30 kHz	Average
1625.125 to 1625.8	-27.2 to -20	30 kHz	Average
1625.8 to 1626	-20 to -17	30 kHz	Average
1626 to 1626.2	-17 to -10	30 kHz	Average
1626.2 to 1626.5	-10	30 kHz	Average

NOTE 1: Spectrum emissions are linearly interpolated versus frequency offset.

G.1.1.4.1.2 Simulation results

Only backoff due to additional emissions limits is shown.

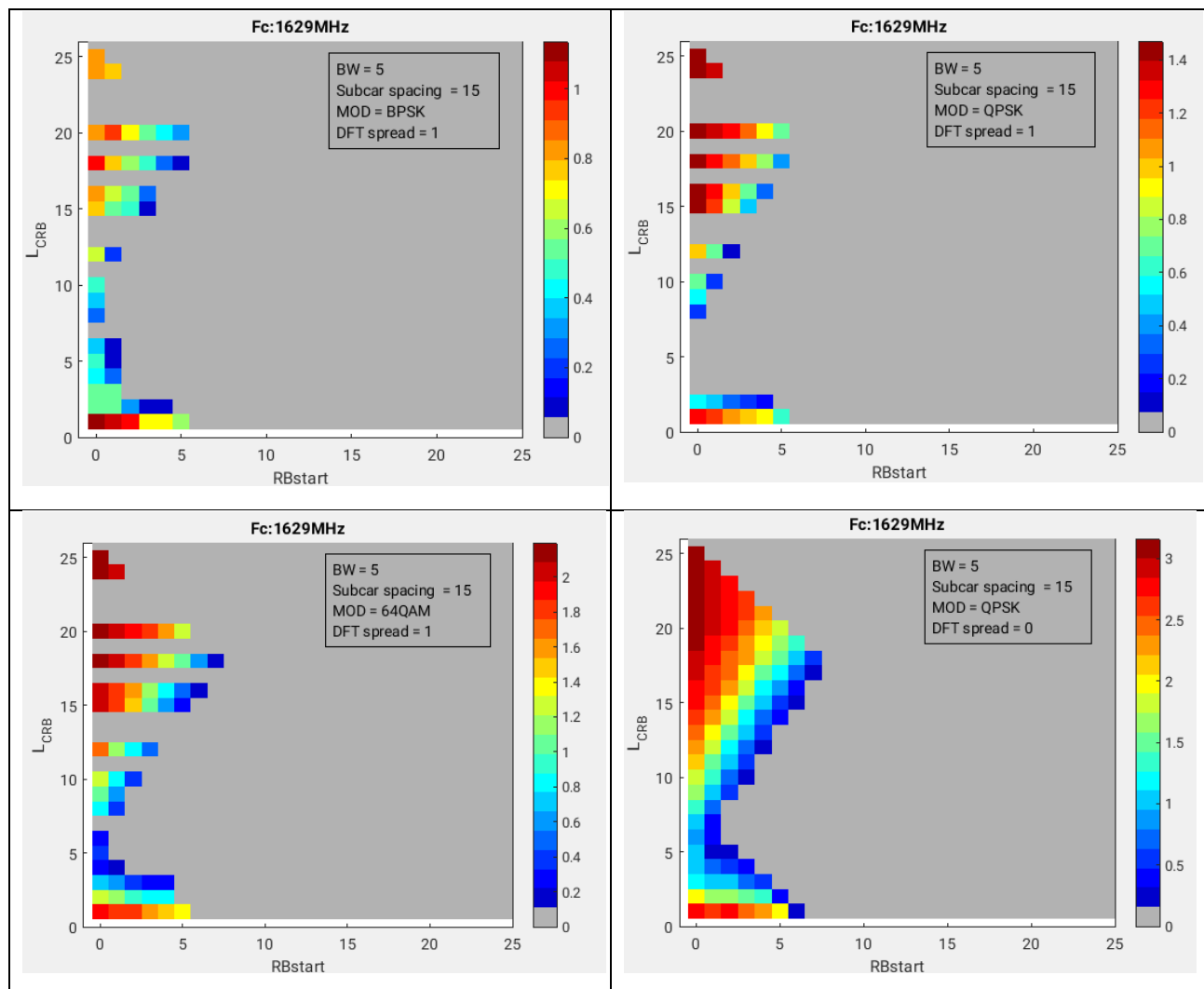
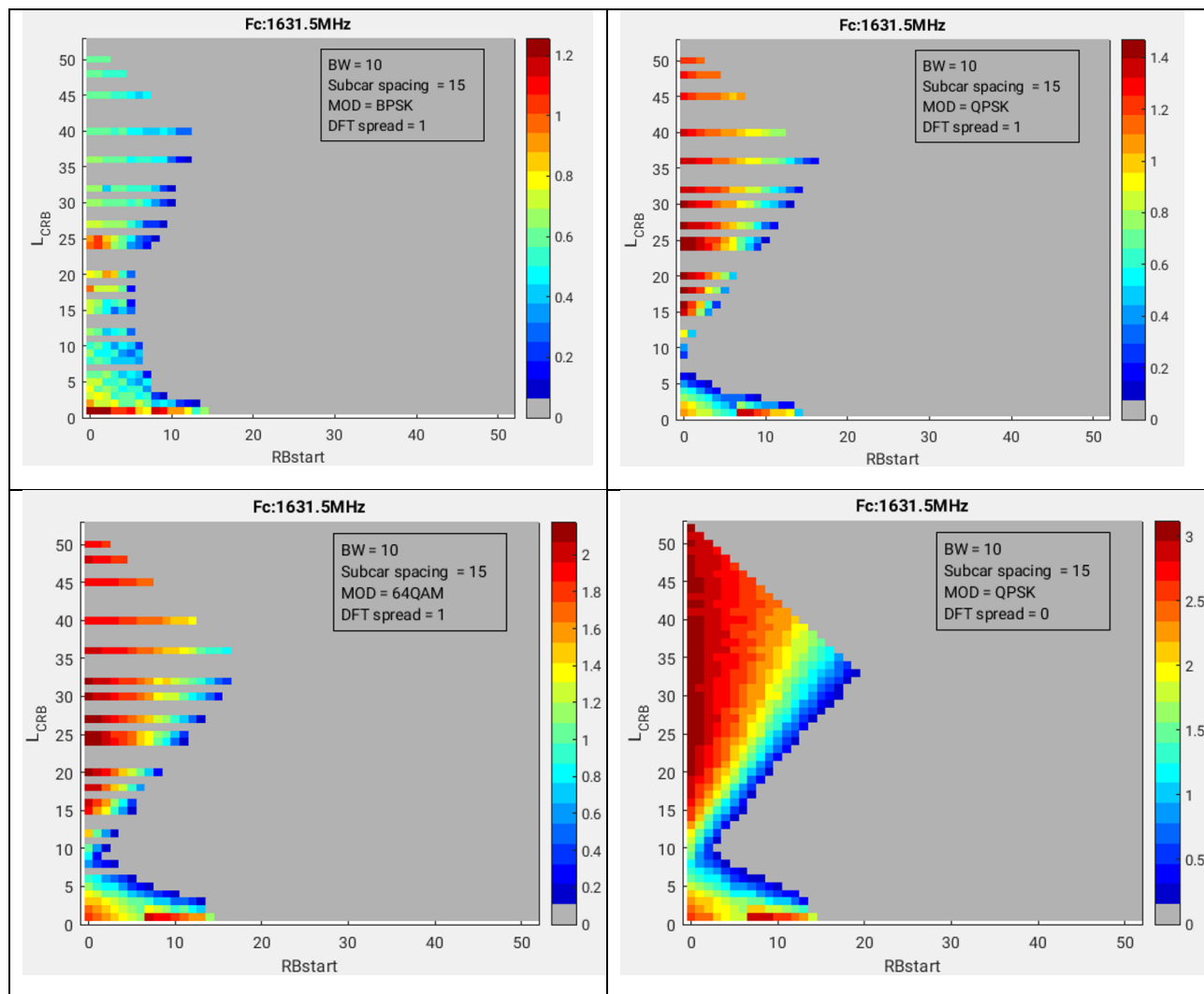
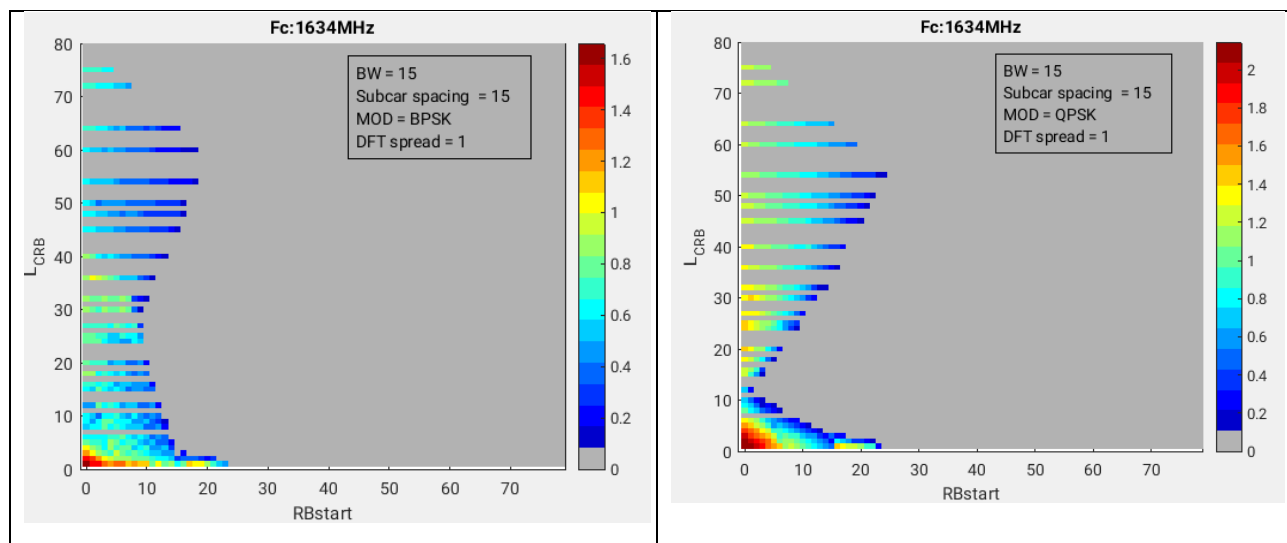
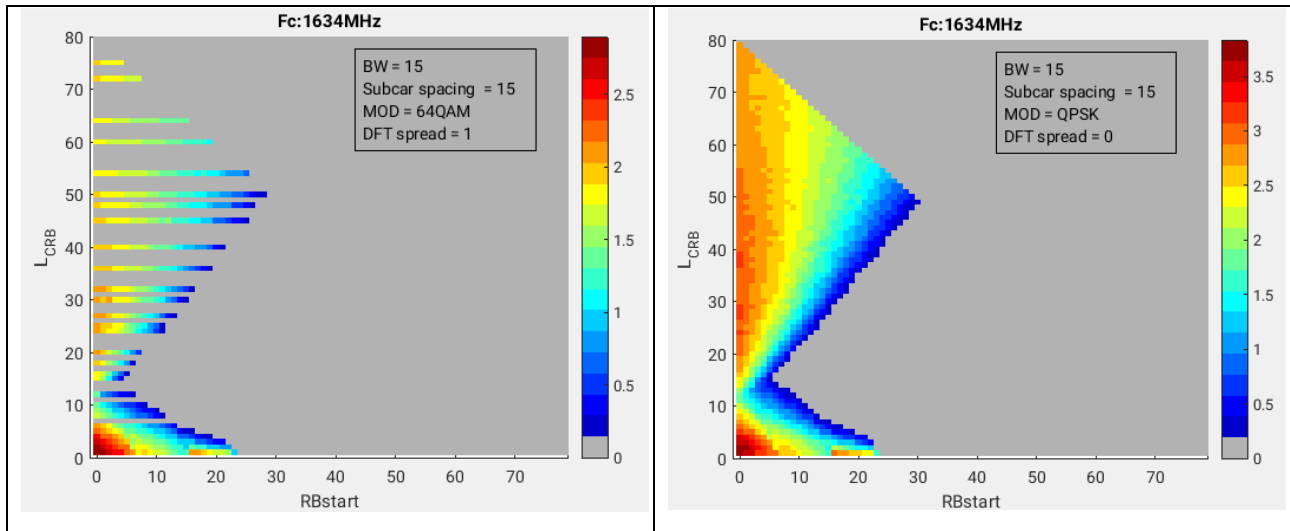
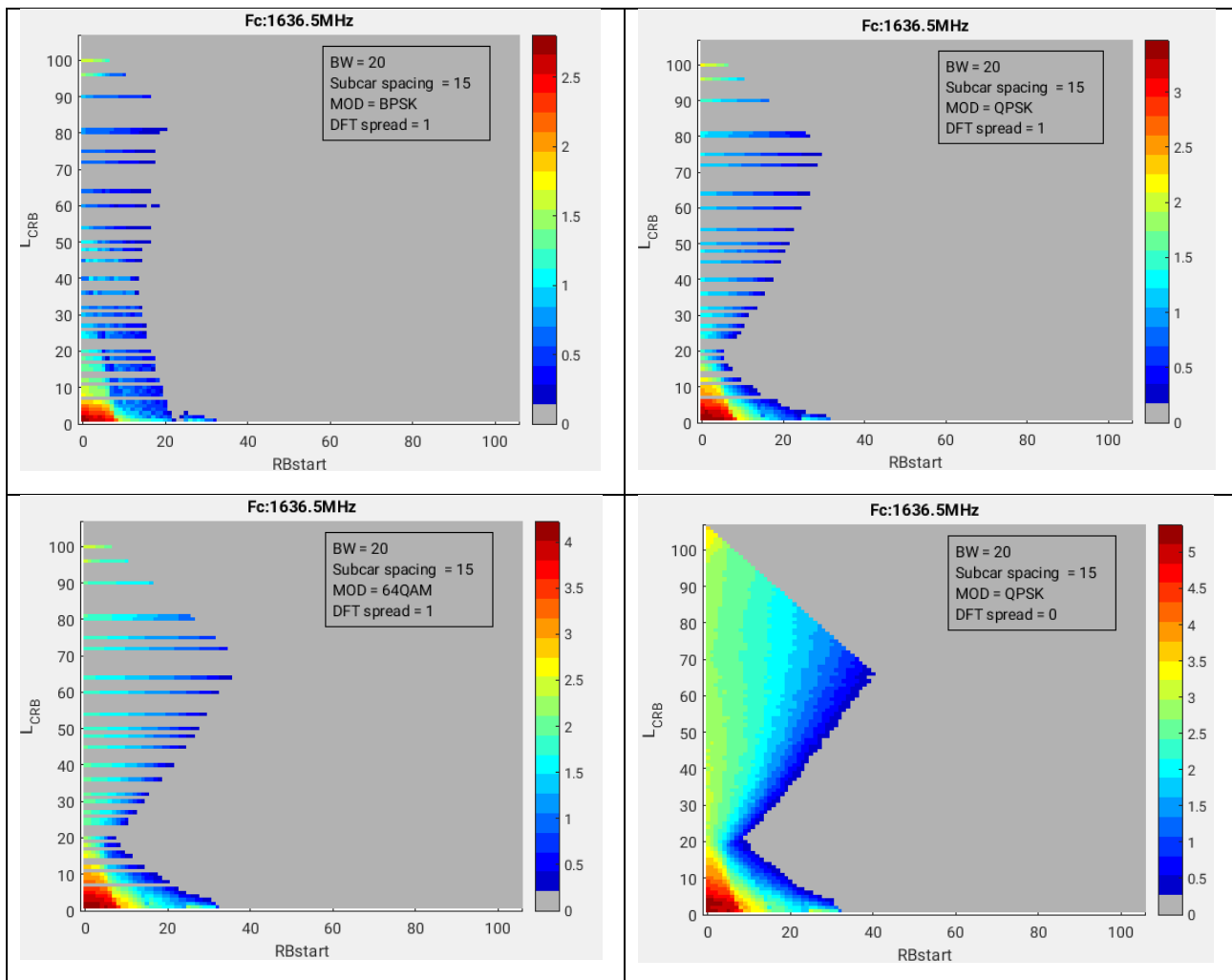


Figure G.1.1.4.1.2-1: 5MHz at Fc=1629MHz.

Figure G.1.1.4.1.2-2: 10MHz at $F_c=1631.5\text{MHz}$.

Figure G.1.1.4.1.2-3: 15MHz at $F_c=1634\text{MHz}$.Figure G.1.1.4.1.2-4: 20MHz at $F_c=1636.5\text{MHz}$.

G.2 PC2 power class

G.2.1 General aspects

NOTE: The purpose of the section is to capture general aspects, such as considered UE RF architectures, assumptions, considered regulations, etc.

G.2.1.1 NS_02N

G.2.1.1.1 Company A (Qualcomm/R4-2511400)

G.2.1.1.1.1 Simulation assumptions

The following simulation assumptions are used:

- Tx power: PC2 (26dBm)
- PA calibrated to 1dB MPR with 31dBc ACLR with DFT-s-OFDM QPSK 100 RB waveform
- -28 dBc IQ-image
- -28 dBc LO-leakage
- -60 dBc CIM3 and -70 dBc CIM5

G.2.1.1.1.2 Simulation results

If the 10dB filter rejection is assumed for the frequency range 1559-1610MHz, then a small A-MPR is needed to meet emission requirements. As presented in Table below A-MPR is needed only for the 20MHz channel.

Table G.2.1.1.2-1: Proposed A-MPR regions and values.

Channel Bandwidth, MHz	Carrier Centre Frequency, Fc, MHz	RB _{start} *12*SCS MHz	LCRB*12*SCS MHz	A-MPR
20MHz	1636.5 ≤ Fc ≤ 1650.5	≤ 0.54	< 1.08	A1
			> 14.4	A1

Modulation/Waveform	A1
	Outer/Inner
DFT-s-OFDM PI/2 BPSK	≤ 2.5
DFT-s-OFDM QPSK	≤ 2.5
DFT-s-OFDM 16 QAM	≤ 3
DFT-s-OFDM 64 QAM	≤ 3.5
CP-OFDM QPSK	≤ 4
CP-OFDM 16 QAM	≤ 4
CP-OFDM 64 QAM	≤ 4

G.2.1.1.2 Company B (OPPO/R4-2510521)

G.2.1.1.2.1 Simulation assumptions

Simulation assumptions are as follows:

- PC2 power class
- 10, 15 and 20MHz channels residing at the lowest edge of the uplink operating band n255
- CP-OFDM and DFT-s-OFDM

- both cases, with and without 10dB filter attenuation are considered

G.2.1.1.2.2 Simulation results

To check the effect of filter rejection, power back-off results are presented for the 10, 15 and 20MHz channels with PC2 CP-OFDM QPSK using APT PA w/o filter rejection and with a constant 10 dB filter attenuation at the frequencies 1559-1610 MHI. In the following figures, A-MPR and "A-MPR – MPR" simulation results are provided separately, i.e. "A-MPR – MPR" figures show the difference between required power back-off and existing MPR margins.

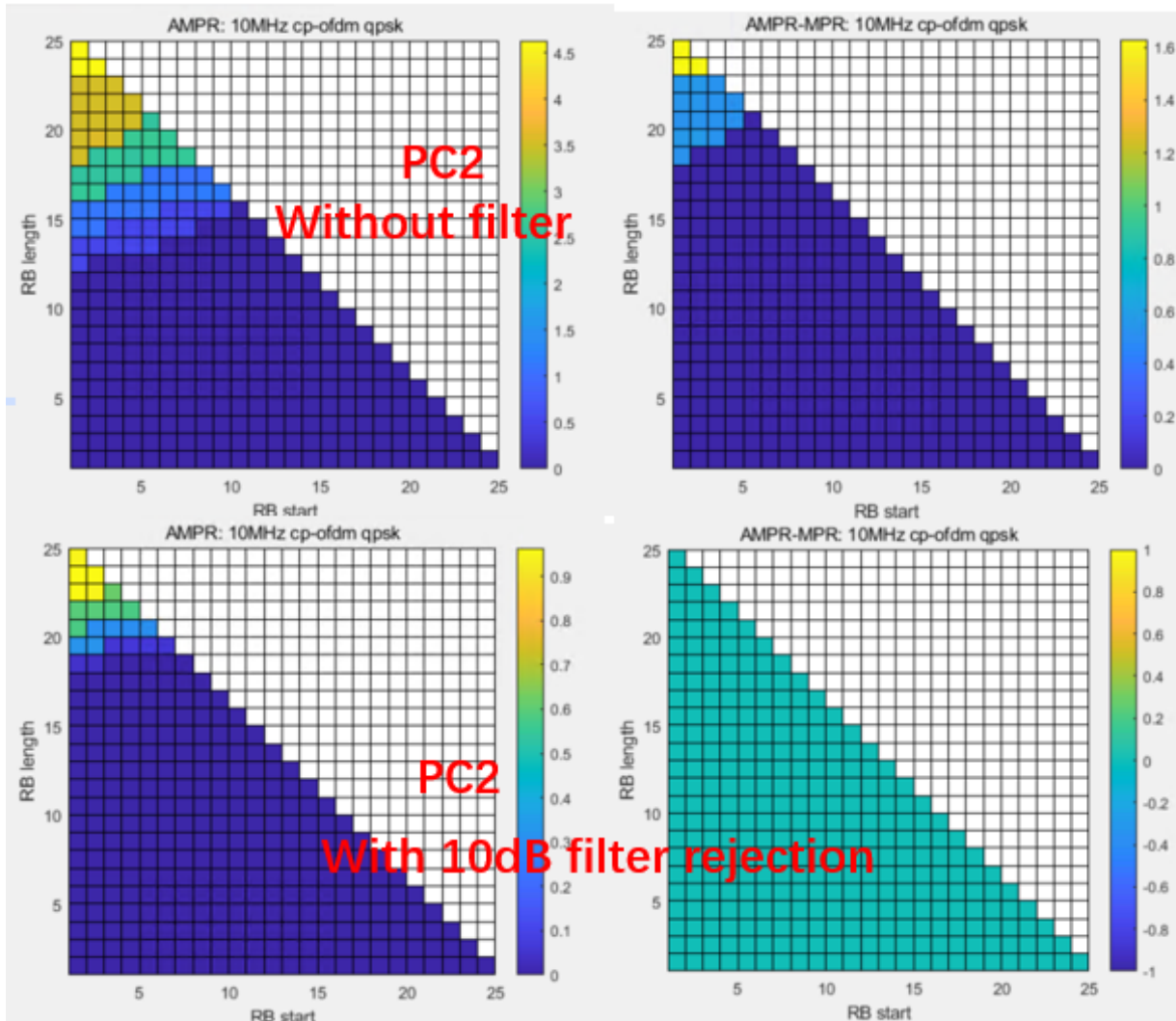


Figure G.2.1.1.2.2-1: Power back-off change for 10MHz based on a constant 10dB filter rejection using APT PA

From Figure G.2.1.1.2.2-1 for the 10MHz channel it can be seen that:

- For PC2 without filter isolation, the AMPR region starts at $LCRB > 16RB$ and the largest AMPR value is around 4.5dB and even larger.
- For PC2 with 10dB filter isolation, we can see that required power back-off is smaller than the existing MPR margin, i.e. there is no need to define A-MPR for 10MHz channel bandwidth.

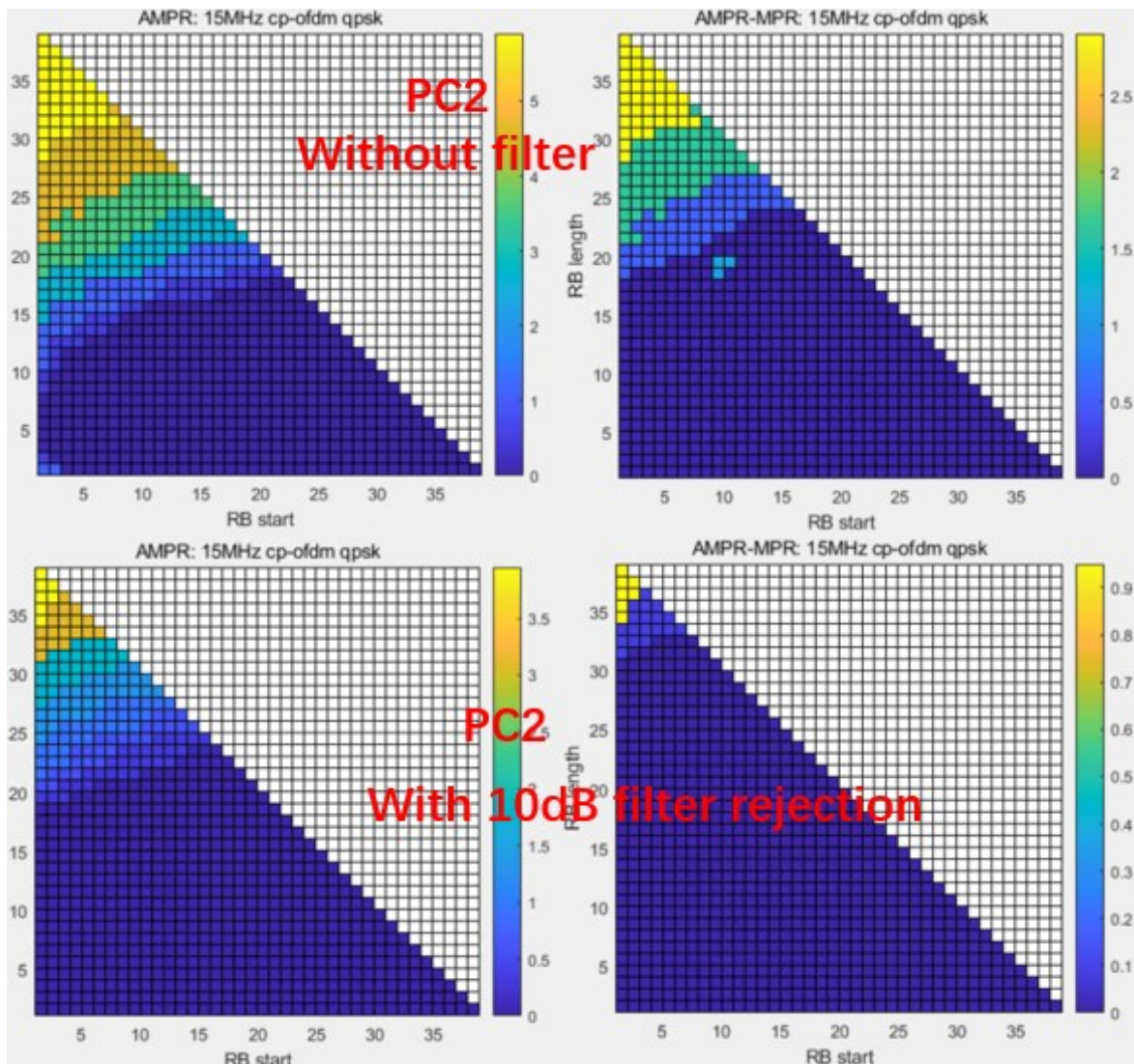


Figure G.2.1.1.2.2-2: Power back-off change for 15MHz based on a constant 10dB filter rejection using APT PA

From Figure G.2.1.1.2.2-2 for the 15MHz channel it can be seen that:

- The A-MPR regions with the 10dB filter isolation becomes smaller than the A-MPR regions without filter isolation, but the overall required power back-off is still larger than MPR.
- For PC2 with the 10dB filter isolation, the AMPR region starts building at $LCRB > 30RB$, whereupon the largest power back-off is larger than 3.5 dB.

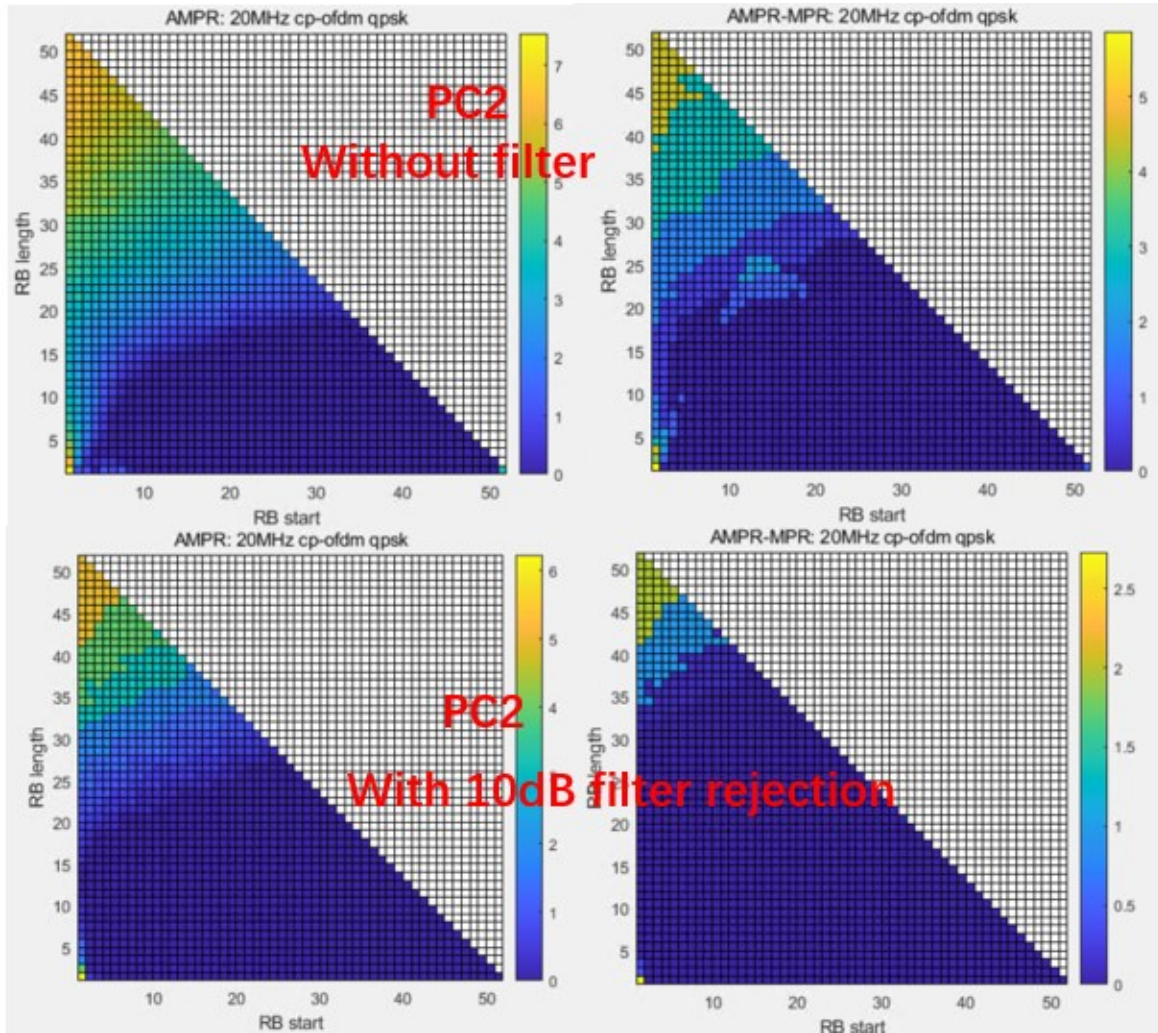


Figure G.2.1.1.2.2-3: Power back-off change for 20MHz based on a constant 10dB filter rejection using APT PA

From Figure G.2.1.1.2.2-3 for the 20MHz channel it can be seen that:

- The A-MPR region with the 10dB filter isolation becomes smaller than the AMPR region without filter isolation, but the required power back-off is larger than MPR.
- For PC2 with the 10dB filter isolation, the AMPR region starts at $LCRB > 34RB$, and the required power back-off is larger than 6dB.

In addition to the above results, Figure G.2.1.1.2.2-4 and G.2.1.1.2.2-5 below presents additional results for the CP-OFDM and DFT-s-OFDM for the 15MHz channel with different modulations.

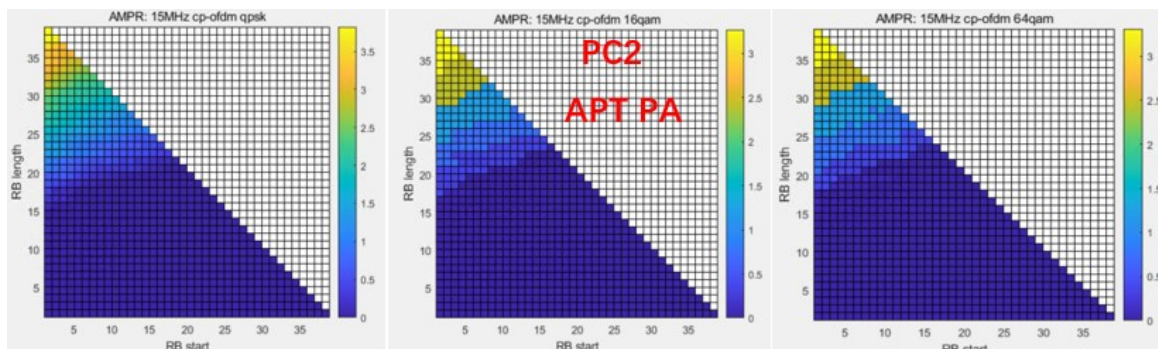
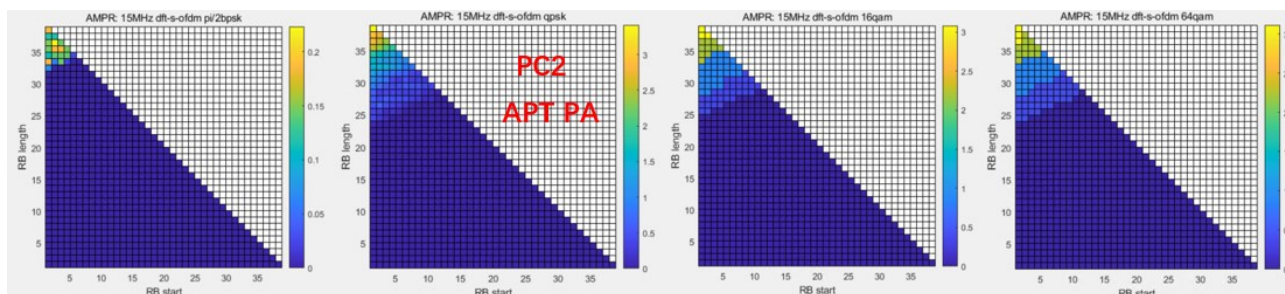


Figure G.2.1.1.2.2-4: Power back-off for 15MHz CP-OFDM QPSK, 16QAM and 64QAM.**Figure G.2.1.1.2.2-5: Power back-off for 15MHz DFT-s-OFDM PI/2 BPSK, QPSK, 16QAM and 64QAM.**

From Figure G.2.1.1.2.2-4 and Figure G.2.1.1.2.2-5 it can be seen that:

- The A-MPR regions and the A-MPR values are almost the same for different modulations for both of CP-OFDM and DFT-s-OFDM.

Considering presented results and the 10dB filter rejection, PC2 A-MPR may look as presented in the table below.

Table G.2.1.1.2.2-1: Proposed A-MPR regions and values

Channel Bandwidth, MHz	Carrier Centre Frequency, F_c , MHz	RBstart*12*SCS MHz	LCRB*12*SCS MHz	A-MPR
15MHz	$1634 < F_c \leq 1649$		>10.8	A1
20MHz	$1636.5 < F_c \leq 1650.5$		>10.8	A2
		<0.54	<1.8	

Modulation		A1	A2
DFT-s-OFDM	Pi/2 BPSK	≤ 0.5	≤ 3
	QPSK/16QAM/64QAM	≤ 3.5	≤ 6
CP-OFDM	QPSK/16QAM/64QAM	≤ 4	≤ 6.5

G.2.1.1.3 Company C (LGE/R4-2510776)

G.2.1.1.3.1 Simulation assumptions

Simulation assumptions are as follows:

- PC2 power class
- 5, 10, 15 and 20MHz channels in the frequency range of 1629-1654MHz
- DFT-s-OFDM QPSK
- both cases, with and without 10dB filter attenuation are considered

G.2.1.1.3.2 Simulation results

Table below presents required power back-off values for two cases with and without 10dB filter rejection. According to the simulation results, there is only one case without the 10dB filter at 20 MHz bandwidth and center frequency of 1636.5MHz (RBstart: 0, LCRB: 51) requires A-MPR driven by the NS_02N limit. In all other cases, power back-off is determined by factors other than NS_02N, such as ACLR or SEM. Furthermore, when the 10 dB filtering is applied, none of the scenarios require A-MPR driven by NS_02N.

Table G.2.1.1.3.2-1: Summary of A-MPR simulations for DFT-s-OFDM QPSK.

Channel bandwidth	Frequency range	(RB_start, LCRB)			
5MHz	$1629 < F_c \leq 1654$	(0,11)	(0,1)		
PC2 W/O 10dB		0	1.47		
PC2 W 10dB		0	1.47		
10MHz	$1631.5 < F_c \leq 1641.5$	(0,24)	(0,1)		
PC2 W/O 10dB		0	2.09		
PC2 W 10dB		0	2.09		
10MHz	$1641.5 < F_c \leq 1655.5$	(0,24)	(0,1)		
PC2 W/O 10dB		0	2.03		
PC2 W 10dB		0	2.03		
15MHz	$1634 < F_c \leq 1649$	(0,38)	(0,1)		
PC2 W/O 10dB		0.31	4.03		
PC2 W 10dB		0.31	4.03		
15MHz	$1649 < F_c \leq 1653$	(0,38)	(0,1)		
PC2 W/O 10dB		0.34	4.16		
PC2 W 10dB		0.34	4.16		
20MHz	$1636.5 < F_c \leq 1640.5$	(0,19)	(0,51)	(0,1)	(50,1)
PC2 W/O 10dB		0	1.19	3.69	3.28
PC2 W 10dB		0	0.38	3.69	3.28
20MHz	$1640.5 < F_c \leq 1650.5$	(0,38)	(0,1)	(0,1)	(50,1)
PC2 W/O 10dB		0	0.38	3.72	3.09
PC2 W 10dB		0	0.38	3.72	3.09

Additionally, further simulations are conducted as shown below to determine the required A-MPR region for the 20MHz channel bandwidth. According to these results A-MPR would be needed for scenarios from (0,51) to (0,45).

Table G.2.1.1.3.2-2: Further results for the 20MHz channel at $1636.5 < F_c \leq 1640.5$.

Channel	(RB_start, LCRB)							
20MHz	(0,51)	(0,49)	(0,47)	(0,45)	(0,43)	(0,41)	(0,39)	(0,37)
DFT-s-OFDM QPSK	1.19	0.97	0.72	0.41	0.34	0.38	0.41	0.41

Based on the presented results potential A-MPR definitions may look as follows.

Table G.2.1.1.3.2-3: Proposed A-MPR regions and values

Channel Bandwidth, MHz	Carrier Centre Frequency, F_c , MHz	Region A		
		RBend*12*SCS MHz	LCRB*12*SCS MHz	A-MPR
20MHz	$1636.5 < F_c \leq 1640.5$		>16.2	A1

Modulation/Waveform	A1
	Outer/Inner
DFT-s-OFDM PI/2 BPSK	1.19
DFT-s-OFDM QPSK	1.19
DFT-s-OFDM 16 QAM	1.41
DFT-s-OFDM 64 QAM	1.44
CP-OFDM QPSK	1.91
CP-OFDM 16 QAM	1.94
CP-OFDM 64 QAM	1.91

G.2.1.1.4 Company D (Apple/R4-2509536)

G.2.1.1.4.1 Simulation assumptions

Simulation assumptions are as follows:

- Tx power: PC2 (+26dBm)
- Calibration: 1dB MPR with 31dBc ACLR and DFT-s-OFDM QPSK 20MHz 100RB0
- Carrier leakage: -28dBc
- Image: -28dBc
- CIM3: -60dBc
- CIM5: -70dBc
- LO placement: always in the centre of the carrier
- Regulations: Requirements from NS_02N
- 10dB filter rejection is assumed

For the sake of brevity, simulations are considered for the 5, 10, 15 and 20MHz channel bandwidths, whereupon the corresponding channel is always located at the lower edge of the n255 band:

- 5MHz: $F_c=1629\text{MHz}$
- 10MHz: $F_c=1631.5\text{MHz}$
- 15MHz: $F_c=1634\text{MHz}$
- 20MHz: $F_c=1636.5\text{MHz}$

G.2.1.1.4.2 Simulation results

According to the preliminary simulation results, no additional power back-off is needed for the 5 and 10MHz channel bandwidths. Thus, the corresponding figures for these channel bandwidths will not be presented.

As for the 15 and 20MHz channels, some power back-off is needed as can be seen from figures below. Nevertheless, the 15MHz channel power back-off is quite marginal and is within the existing MPR margins. As for the 20MHz channel, it naturally experiences higher power back-off, which in some cases reaches existing MPR margins, especially for the full channel and single RB edge allocations.

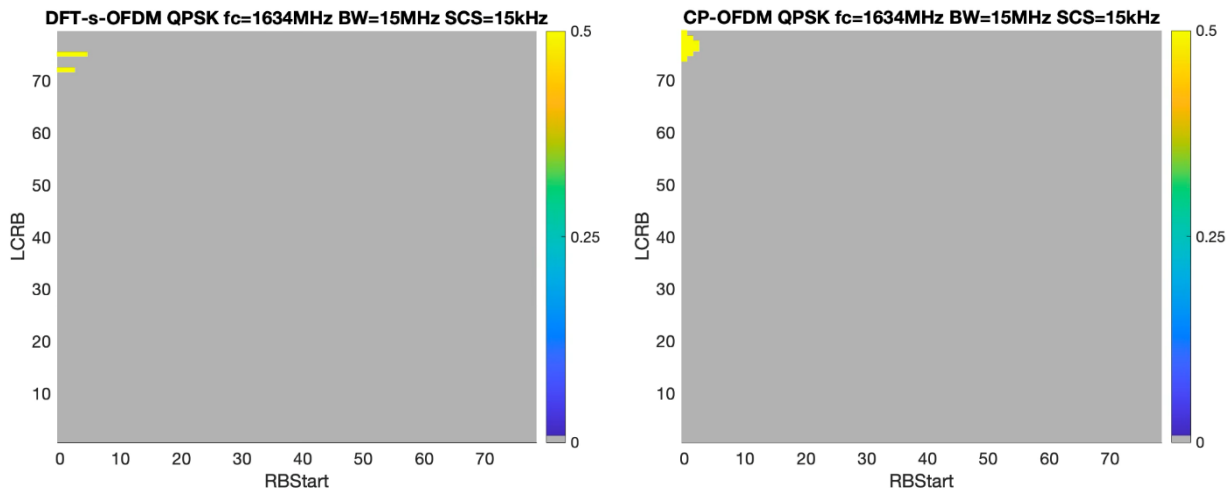


Figure G.2.1.1.4.2-1: Power back-off results for 15MHz at $F_c=1634\text{MHz}$.

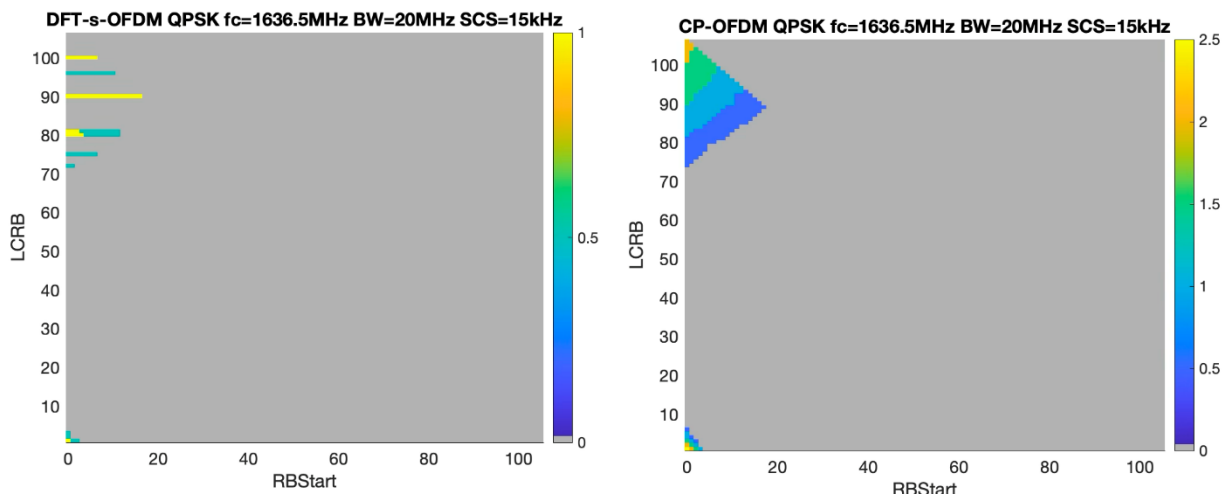


Figure G.2.1.1.4.2-2: Power back-off results for 20MHz at Fc=1636.5MHz.

Based on the presented results it is not likely that 15MHz channel needs A-MPR. As for the 20MHz channel, small A-MPR can be considered for large allocations.

G.3 PC1.5 power class

G.3.1: General aspects

NOTE: The purpose of the section is to capture general aspects, such as considered UE RF architectures, assumptions, considered regulations, etc.

G.3.1.1 NS_02N

G.3.1.1.1 Company B (OPPO/ R4-2510521)

G.3.1.1.1.1 Simulation assumptions

The following simulation assumptions are used:

- PC1.5 power class with 2Tx
- 10, 15 and 20MHz channels residing at the lowest edge of the uplink operating band n255
- CP-OFDM and DFT-s-OFDM
- without 10dB filter attenuation

G.3.1.1.1.2 Simulation results

Figure G.3.1.1.1.2-1 and Figure G.3.1.1.1.2-2 show the AMPR and AMPR-MPR simulation results for CP-OFDM and DFT-s-OFDM with QPSK for 15MHz and 20MHz separately.

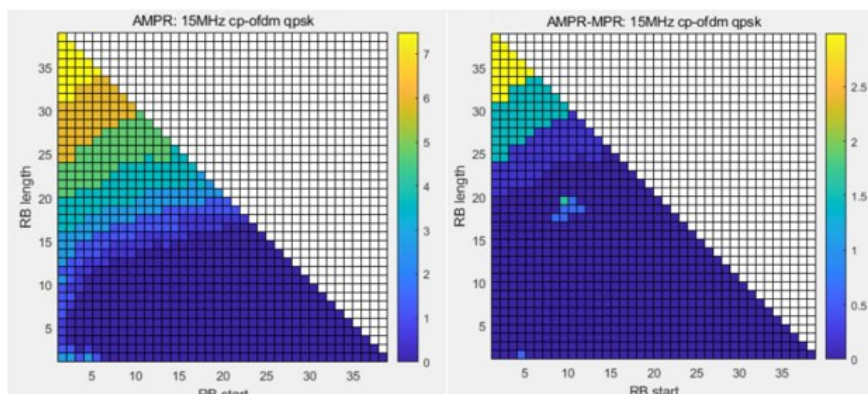


Figure G.3.1.1.2-1: Power back-off change for 15MHz without filter rejection using APT PA

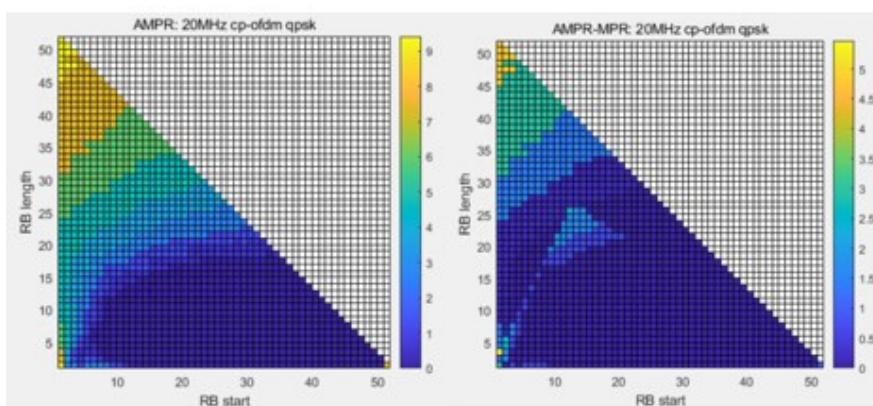


Figure G.3.1.1.2-2: Power back-off change for 20MHz without filter rejection using APT PA

From Figure G.3.1.1.2-1 for the 15MHz channel it can be seen that:

- For PC1.5 without filter isolation, the AMPR region is at $LCRB > 15RB$, and the largest AMPR is almost 7.5 dB.

From Figure G.3.1.1.2-2 for the 20MHz channel we can find that:

- For PC1.5 without filter isolation, the largest AMPR is almost 9.5 dB for the AMPR region at $LCRB > 15RB$ s. And the AMPR region of $LCRB < 15RB$ s and $RBstart < 3RB$ s.

In additional, due to time limitation, we didn't do the simulation with 10dB filter isolation totally, we just simulated the AMPR for the point of ($RBstart=0$, $LCRB=52$) with 10dB filter isolation, the AMPR with 10dB filter isolation is smaller 1.5dB than the AMPR value without filter isolation.

If the 10dB filter rejection is assumed for the frequency range 1559-1610MHz, PC1.5 A-MPR presents in the table below:

Table G.3.1.1.2-1: Proposed A-MPR regions and values.

Channel Bandwidth, MHz	Carrier Centre Frequency, F_c , MHz	$RBstart \cdot 12 \cdot SCS$ MHz	$LCRB \cdot 12 \cdot SCS$ MHz	A-MPR
20MHz	$1636.5 \leq F_c \leq 1650.5$	≤ 0.54	< 1.08	A1
			> 14.4	A1
			$> 10.8, \leq 14.4$	A2
15MHz	$1634 \leq F_c \leq 1645$		> 10.8	A2

Modulation/Waveform	A1	A2
	Outer/Inner	Outer/Inner
DFT-s-OFDM PI/2 BPSK	≤ 8	≤ 4
DFT-s-OFDM QPSK	≤ 8	≤ 4
DFT-s-OFDM 16 QAM	≤ 8	≤ 4
DFT-s-OFDM 64 QAM	≤ 8	≤ 4
CP-OFDM QPSK	≤ 8	≤ 6
CP-OFDM 16 QAM	≤ 8	≤ 6
CP-OFDM 64 QAM	≤ 8	≤ 6

G.4 PC1 power class

G.4.1 General aspects

NOTE: The purpose of the section is to capture general aspects, such as considered UE RF architectures, assumptions, considered regulations, etc.

G.4.1.1 NS_02N

G.4.1.1.1 Company A (Qualcomm/R4-2511400)

G.4.1.1.1.1 Simulation assumptions

The following simulation assumptions are used:

- Tx power: PC1 (31dBm) with 4TX
- PA calibrated to 1dB MPR with 34dBc ACLR with DFT-s-OFDM QPSK 100 RB waveform
- -28 dBc IQ-image
- -28 dBc LO-leakage
- -60 dBc CIM3 and -70 dBc CIM5

G.4.1.1.1.2 Simulation results

If the 10dB filter rejection is assumed for the frequency range 1559-1610MHz, then a small A-MPR is needed to meet emission requirements. As presented in Table below A-MPR is needed only for the 20MHz channel.

Table G.4.1.1.1.2-1: Proposed A-MPR regions and values.

Channel Bandwidth, MHz	Carrier Centre Frequency, F_c , MHz	RBstart*12*SCS MHz	LCRB*12*SCS MHz	A-MPR
20MHz	$1636.5 \leq F_c \leq 1650.5$	≤ 0.54	< 1.08	A1
			> 14.4	A1
			$> 10.8, \leq 14.4$	A2
15MHz	$1634 \leq F_c \leq 1645$		> 10.8	A2

Modulation/Waveform	A1	A2
	Outer/Inner	Outer/Inner
DFT-s-OFDM PI/2 BPSK	≤ 5	≤ 3
DFT-s-OFDM QPSK	≤ 5	≤ 3
DFT-s-OFDM 16 QAM	≤ 5	≤ 3.5
DFT-s-OFDM 64 QAM	≤ 5.5	≤ 4
CP-OFDM QPSK	≤ 6	≤ 4
CP-OFDM 16 QAM	≤ 6	≤ 4
CP-OFDM 64 QAM	≤ 6	≤ 4

Annex H:

Summary of NTN HPUE simulation results

All NTN HPUE co-existence study results have been captured in the list. Please see attachment Summary of NTN HPUE simulation results.

Annex I (informative): Change history

Change history						
Date	Meeting	TDoc	CR	Rev	Cat	Subject/Comment
2021-11	RAN4#101-e	R4-2120776				Added approved TPs in RAN4#101-e including: R4-2120759, draft TP to TR 38.863: Operating bands and channel arrangements R4-2120760, TP for 38.863 on system parameters on satellite bands R4-2120761, TP to TR 38.863 – Regulatory aspects R4-2120762, TP to TR 38.863: node class, RF RX (6.2) R4-2120763, TP for 38.863 on NTN UE transmission characteristics R4-2120772, Draft text proposal to update TR 38.863 NTN related RF and co-existence aspects
2022-01	RAN4#101-bis-e					Added approved TPs in RAN4#101-bis-e including: R4-2201257, TP for 38.863 on system parameters to clarify “NTN satellite bands” R4-2201838, Draft proposal to update TR 38.863 NTN related RF and co-existence aspects R4-2202988, Draft Text Proposal for TR 38.863 R4-2203037, TP for 38.863 on UE transmission characteristics for satellite access R4-2203038, TP for 38.863 on maximum input level for NTN UE R4-2203039, TP for 38.863 on UE Receiver characteristics for satellite access R4-2203040, TP to TR 38.863 on transmitter characteristics for satellite access node R4-2203081, Draft text proposal to update TR 38.863 Chapter 3 R4-2203082, TP to TR 38.863 on channel raster and sync raster R4-2203084, TP to TR 38.863 Regulatory aspects for HAPS R4-2203085, TP to TR 38.863 on general aspects R4-2203129, TP to TR 38.863 – Regulatory aspects
2022-03	RAN4#102-e					Added approved TPs in RAN4#102-e including: R4-2205557, R4-2207330, TP TR 38.863 7.4.1 NTN UE Requirement (General) R4-2207333, TP to TR 38.863 on Section 5.2 NTN Satellite band R4-2207338, TP for TR 38.863: Regulatory aspects for NTN satellite access nodes and Ues operating in UL1626.5-1660.5 MHz and DL 1525-1559 MHz frequencies ranges R4-2207339, TP to TR 38.863 Regulatory aspects for HAPS R4-2207345, Draft text proposal for Clauses 7, 7.1, 7.2, 7.3 in TR 38.863 R4-2207351, Draft text proposal to update TR 38.863 Chapter 6 R4-2207353, Draft text proposal for Clauses 6.4 and 6.5 in TR 38.863 to include simulation results based on Non-AAS antenna assumption R4-2207360, Draft text proposal for Clause 7.3.4.7.3 OTA ACLR in TR 38.863 R4-2207367, Draft text proposal for Clause 7.3.5.6 OTA Out-of-band blocking in TR 38.863 R4-2207379, Draft text proposal for Clause 7.3.3.2.4 Out-of-band blocking in TR 38.863 R4-2207381, Draft text proposal for Clauses 7.3.3.2.3.1 Adjacent Channel Selectivity (ACS) and 7.3.3.2.3.2 In-band blocking in TR 38.863 R4-2207397, TP on TR 38.863 for NTN UE Tx requirements R4-2207398, Draft TP to update TR 38.863 clause 7.4.3.2 on NTN UE ACS R4-2207399, Draft TP to update TR 38.863 clause 7.4.3.2 on Blocking characteristics R4-2207401, TP for TR38.863 on Intermodulation characteristics for NTN UE R4-2207402, TP for 38.863 on spurious response for NTN UE R4-2207403, TP for TR 38.863: Unwanted emissions for NTN satellite Ues transmitting in 1626.5 to 1660.5 MHz R4-2207406, TP for TR 38.863: Updates to UE Maximum Output Power for n255 R4-2207414, TP for 38.863 on UE Receiver characteristics for satellite access R4-2207460, Draft text proposal for Clause 7.3.2.2.4.1 ACLR in TR 38.863

2022-05	RAN4#10 3-e					Added approved TPs in RAN4#103e including: R4-2209089, TP for 38.863 on general part for NTN UE conducted receiver characteristics R4-2209364, TP for 38.863 on UE Rx spurious emission requirements for satellite access R4-2209365, TP for 38.863 on UE antenna characteristics for satellite access R4-2210228, Draft text proposal for Clauses 6.4 and 6.5 Corrections Typos - TR 38.863 R4-2210636, TP to TR 38.863: coexistence issues between NTN and TN from Rel-17 RAN4 study. R4-2210847, TP to TR 38.863 - Updates R4-2210848, Draft text proposal for Clause 6.1 Coexistence Figures - TR 38.863 R4-2210853, TP for 38.863: clause 7.3.2 Conducted transmission characteristics R4-2210858, TP to TR 38.863: Conducted reference sensitivity R4-2210859, TP to TR 38.863: Conducted Rx dynamic range R4-2210860, Draft text proposal for Clause 7.3.2.2.5 Transmitter spurious emissions - TR38.863 R4-2210865, Tentative draft pCR for Clause 7.3.2.2.4.2 Operating band unwanted emissions - TR 38.863 R4-2210866, Draft text proposal for Clause 7.3.2.2.1 SAN output power - TR 38.863 R4-2210867, Draft text proposal for Clause 7.3.3.3.1 OTA sensitivity - TR 38.863 R4-2210868, Draft text proposal for Clause 7.3.3.3.2 OTA reference sensitivity - TR 38.863 R4-2210869, Draft text proposal for Clause 7.3.3.3.3 OTA dynamic range - TR 38.863 R4-2210870, Draft text proposal for Clause 7.3.3.3.7 OTA receiver intermodulation - TR 38.863 R4-2210871, Draft text proposal for Clause 7.3.3.3.8 OTA in-channel selectivity - TR 38.863 R4-2210872, Draft text proposal for Clause 7.3.3.3.4 OTA in-band selectivity and blocking - TR 38.863 R4-2210875, TP for TR 38.863: Updates to n255 A-MPR Clause R4-2211133, Draft Text Proposal to Update TR 38.863 Chapter 3,6 and 8 R4-2211134, Draft Text Proposal to Update TR 38.863 structure R4-2211146, Draft text proposal for Clause 7.3.3.3.2 OTA reference sensitivity - TR 38.863	0.4.0
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Change history							
Date	Meeting	TDoc	CR	Rev	Cat	Subject/Comment	New version
2022-06	RAN#96					Approved by plenary – Rel-17 spec under change control	17.0.0
2022-09	RAN#97	RP-222035	0001		F	CR for 38.863 to maintain UE RF parts	17.1.0
2022-09	RAN#97	RP-222035	0002	1	F	Correction to TR 38.863 on Regulatory aspects for HAPS	17.1.0
2023-03	RAN#99	RP-230516	0005		F	CR for TR 38.863 to maintain SAN parts	17.2.0
2023-12	RAN#102	RP-233349	0009	1	F	CR for TR 38.863, Correction on Satellite and UE Antenna and beam forming pattern modelling	17.3.0

Change history							
Date	Meeting	TDoc	CR	Rev	Cat	Subject/Comment	New version
2023-12	RAN#102	RP-233362	0010		B	NTN enhancement: CR to TR 38.863 NTN Ka-band Regulatory aspects	18.0.0
2024-03	RAN#103	RP-240589	0012		B	CR on TR38.863 Addition of simulation assumptions in above 10GHz	18.1.0
2024-03	RAN#103	RP-240589	0013		B	CR proposal to add Doppler and Delay variation examples as a function of time for NGSO and GSO in a new Annex	18.1.0
2024-06	RAN#104	RP-241429	0015		B	CR on TR 38.863 Addition of simulation results in above 10GHz	18.2.0
2024-06	RAN#104	RP-241429	0016	1	B	CR on TR38.863 Summary of coexistence results in above 10GHz	18.2.0
2024-06	RAN#104	RP-241429	0017		F	CR for TR 38.863 to introduce some technical background for R18 NTN VSAT UE Tx requirements	18.2.0
2024-06	RAN#104	RP-241429	0018		F	CR for TR 38.863 to introduce some technical background for R18 NTN VSAT UE Rx requirements	18.2.0
2024-09	RAN#105	RP-242178	0020	1	F	Maintenance CR for Ka-band coexistence results to TR 38.863	18.3.0
2024-12	RAN#106	RP-243056	0021		F	(NR_NTN_enh-Core) CR for TR 38.863, On NTN system parameters for Ka-band co-existence	18.4.0
2025-03	RAN#107	RP-250593	0023	1	F	(NR_NTN_solutions-Core) Corrections to TR 38.863 to align with TS 38.101-5	18.5.0
2025-06	RAN#108	RP-250920	0025		A	CR for TR 38.863, Correction on tolerance for UE Transmission characteristics for satellite access	18.6.0

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Date	Meeting	TDoc	CR	Rev	Cat	Subject/Comment	New version
2025-06	RAN#108	RP-250949	0028	1	B	CR to TR 38.863: Introduction of NR-NTN band n252	19.0.0
2025-06	RAN#108					Minor editorial update	19.0.1
2025-09	RAN#109	RP-252422	0029		B	CR to TR 38.863 on SAN RF requirements for less than 5MHz	19.1.0
2025-09	RAN#109	RP-252386	0032		A	(NR_NTN_solutions-Core) Correction of the NR NTN band n256 REFSSENS values	19.1.0
2025-09	RAN#109	RP-252386	0035		A	(NR_NTN_solutions-Core) Clarification of the NR NTN band n256 out-of-band blocking requirements	19.1.0
2025-09	RAN#109	RP-252422	0037		B	Formal BigCR to TR 38.863 for NTN HPUE	19.1.0
2025-09	RAN#109	RP-252428	0039	1	B	CR to TR 38.863: Introduction of combined L-bands n253, n251, n250	19.1.0
2025-09	RAN#109	RP-252422	0036	1	B	Addition of the 3MHz channel to TR 38.863	19.1.0
2025-09	RAN#109	RP-252897	0042		B	Big CR for TR 38.863 Introduction of Ku Bands	19.1.0
2025-12	RAN#110	RP-253630	0043		F	Addition of power back-off simulation results for the NTN L-bands	19.2.0
2025-12	RAN#110	RP-253631	0045		F	(NR_NTN_Ku_bands-Core) CR to TR 38.863 - Regulatory context for the NTN Ku-bands	19.2.0
2025-12	RAN#110	RP-253664	0044	1	F	Clarification for the simulation assumptions for NTN L-band and S-band power back-off results	19.2.0